

13. An optical device which always produces images of $m = +1$ is :

- (A) Plane mirror
- (B) Convex mirror
- (C) Concave mirror
- (D) Convex lens

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(A)/ Plane mirror

- 15.** Concave mirror finds its application in which of the following cases ?
- (A) A solar furnace
 - (B) A rear-view mirror in vehicles
 - (C) To view full size erect image of distant tall buildings
 - (D) To produce divergent beam of light

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(A) / A solar furnace

- 25.** An object is placed 18 cm in front of a concave mirror of focal length 12 cm. Use mirror formula to determine the position of the image formed in this case.

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25. An object is placed 18 cm in front of a concave mirror of focal length 12 cm. Use mirror formula to determine the position of the image formed in this case.

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2

$$u = -18 \text{ cm}, f = -12 \text{ cm}, v = ?$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18}$$

$$\frac{1}{v} = \frac{-3 + 2}{36}$$

$$\frac{1}{v} = \frac{-1}{36}$$

$v = -36 \text{ cm}$ (image is formed at a distance 36 cm in front of the mirror)

$\frac{1}{2}$

$\frac{1}{2}$

1

16. Refractive index of diamond with respect to glass is 1.6. If the absolute refractive index of glass is 1.5, the absolute refractive index of diamond is :

(A) 1.09

(B) 1.96

(C) 2.4

(D) 2.5

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16. Refractive index of diamond with respect to glass is 1.6. If the absolute refractive index of glass is 1.5, the absolute refractive index of diamond is :

(A) 1.09

(B) 1.96

(C) 2.4

(D) 2.5

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(C)/ 2.4

- 25.** An object is placed 25 cm in front of a concave mirror of focal length 15 cm. Use mirror formula and new cartesian sign convention to determine the position of the image formed.

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25. An object is placed 25 cm in front of a concave mirror of focal length 15 cm. Use mirror formula and new cartesian sign convention to determine the position of the image formed.

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2

$u = -25 \text{ cm}$ $f = -15 \text{ cm}$ $v = ?$	$\frac{1}{2}$
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	$\frac{1}{2}$
$\frac{1}{v} - \frac{1}{25} = \frac{-1}{15}$	
$\frac{1}{v} = \frac{-1}{15} + \frac{1}{25}$	
$= \frac{-5 + 3}{75} = \frac{-2}{75}$	
$v = \frac{-75}{2} = -37.5 \text{ cm}$ (Image is formed 37.5 cm in front of the mirror.)	1

11. A 5 cm long rod is placed vertically in front of a concave mirror. If its 5 cm long image is formed at 20 cm in front of the mirror, the focal length of the mirror is :

- (A) – 20 cm
- (B) – 10 cm
- (C) – 5 cm
- (D) 20 cm

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- (B) – 10 cm
- (C) – 5 cm
- (D) 20 cm

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(B)/ –10 cm

20. *Assertion (A) :* The magnification of the virtual images produced by concave mirrors is always more than +1.

Reason (R) : The virtual images formed by concave mirrors are erect and magnified.

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Reason (R) : The virtual images formed by concave mirrors are erect and magnified.

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(A)/ Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

24. List four characteristics of the images formed by convex mirrors.

2

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2

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• Virtual • Erect • Diminished • Behind the mirror.	$\frac{1}{2} \times$ 4
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39. A highly polished surface which reflects most of the light falling on it is called a mirror. Mirrors may be plane as well as spherical. The reflecting surface of a spherical mirror may be curved inwards or outwards. A spherical mirror where reflecting surface is curved inwards is called concave mirror, whereas spherical mirrors where reflecting surfaces are curved outwards are known as convex mirrors.

- (a) State the laws of reflection of light. 1
- (b) Define the term 'Principal axis of a concave mirror.' 1
- (c) (i) Write the path of the following light rays after reflection from a concave mirror :
 - (I) A ray of light moving parallel to the principal axis of a concave mirror.
 - (II) A ray incident obliquely to the principal axis towards the pole of the concave mirror making angle θ . 2

OR

- (c) (ii) An object is placed at 18 cm in front of a concave mirror of focal length 12 cm. Use mirror formula to determine the position of the image. 2

(a)

Laws of Reflection

- Incident ray, reflected ray and the normal to the surface at the point of incidence all lie in the same plane.

$\frac{1}{2}$

- Angle of incidence ($\angle i$) = Angle of reflection ($\angle r$)

$\frac{1}{2}$

(b) Principal Axis is an imaginary straight line passing through the pole and centre of curvature of a concave mirror.

1

(c) (i) (I) passes through the focus of the concave mirror
(II) is reflected obliquely making equal angle (θ) with the principal axis

1

1

OR

(c) (ii) Here $f = -12$ cm $u = -18$ cm $v = ?$

$\frac{1}{2}$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \text{ or } \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$\frac{1}{2}$

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18} = \frac{-3+2}{36} = \frac{-1}{36}$$

$$v = -36 \text{ cm}$$

1

Position: Image is formed at 36 cm in front of the mirror

14. If the absolute refractive indices of two media X and Y are $\frac{6}{5}$ and $\frac{4}{3}$ respectively, then the refractive index of Y with respect to X will be :

(A) $\frac{10}{9}$

(B) $\frac{9}{10}$

(C) $\frac{9}{8}$

(D) $\frac{8}{9}$

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(B) $\frac{9}{10}$

(C) $\frac{9}{8}$

(D) $\frac{8}{9}$

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(A) / $\frac{10}{9}$

15. An object is placed at a distance of 30 cm from the pole of a concave mirror. If its real and inverted image is formed at 60 cm in front of the mirror, the focal length of the mirror is :

(A) -15 cm

(B) -20 cm

(C) $+20$ cm

(D) $+15$ cm

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15. An object is placed at a distance of 30 cm from the pole of a concave mirror. If its real and inverted image is formed at 60 cm in front of the mirror, the focal length of the mirror is :

(A) -15 cm

(B) -20 cm

(C) $+20$ cm

(D) $+15$ cm

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(B) / -20 cm

- 25.** (a) Out of the two lenses, one concave and the other convex, state which one will diverge a parallel beam of light falling on it. Draw a ray diagram to show the principal focus of the lens.

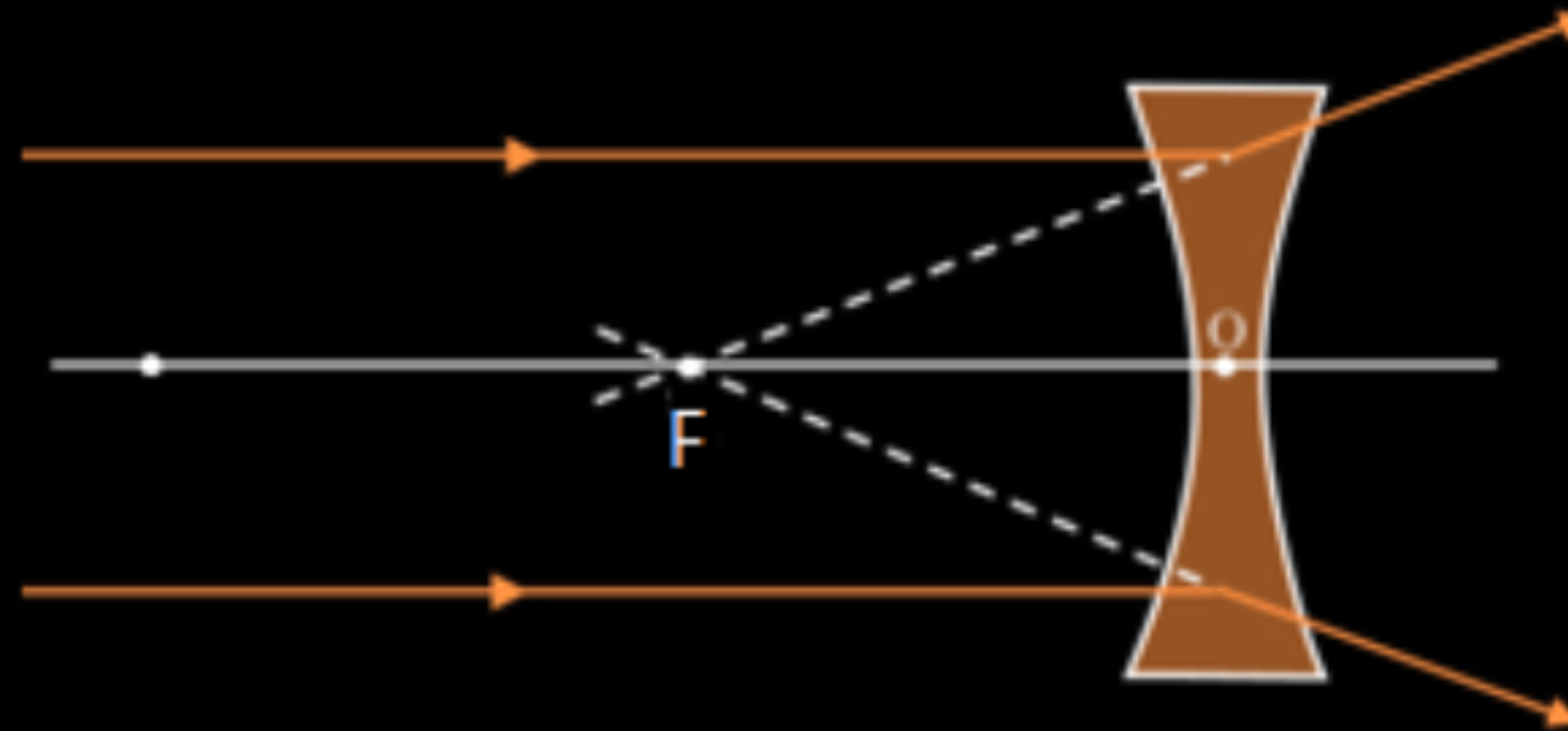
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25. (a) Out of the two lenses, one concave and the other convex, state which one will diverge a parallel beam of light falling on it. Draw a ray diagram to show the principal focus of the lens.

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2

(a) Concave lens



1

1

(b) A ray of light after refraction from a convex lens emerges parallel to its principal axis.

2

(i) Draw a labelled ray diagram to show it.

(ii) In this case, the incident ray before refraction from the lens passes through a point on its principal axis. Name the point.

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(b) A ray of light after refraction from a convex lens emerges parallel to its principal axis.

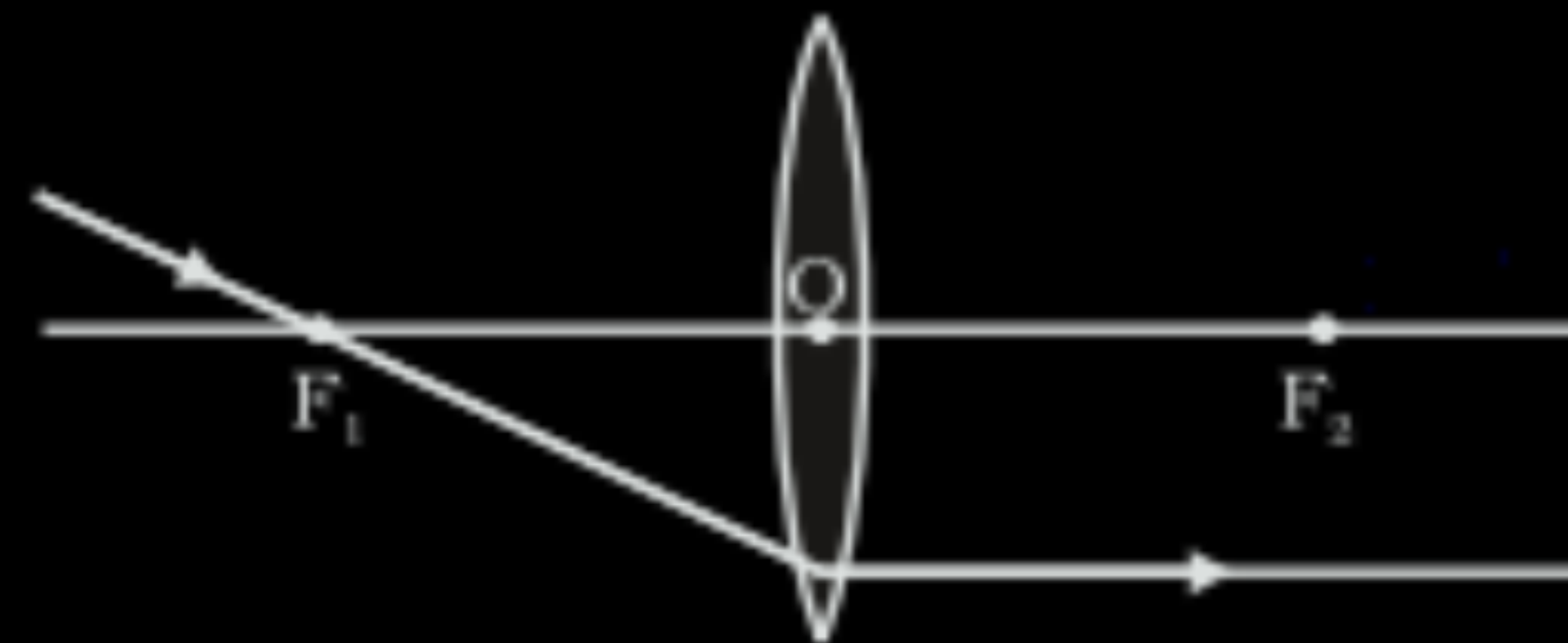
2

- (i) Draw a labelled ray diagram to show it.
- (ii) In this case, the incident ray before refraction from the lens passes through a point on its principal axis. Name the point.

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(b)

(i)



1

(ii)

Principal focus /Focus

1

- 31.** A convex mirror used for rear view on an automobile has a focal length of 1.5 m. If a 3 m high bus is located at 6.0 m from the mirror, use mirror formula to determine the position and size of the image of the bus as seen in the mirror.

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- 31.** A convex mirror used for rear view on an automobile has a focal length of 1.5 m. If a 3 m high bus is located at 6.0 m from the mirror, use mirror formula to determine the position and size of the image of the bus as seen in the mirror.

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3

Here $f = +1.5$ m; $u = -6.0$ m; $h = 3$ m

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-6} = \frac{1}{1.5}$$

$$\frac{1}{v} = \frac{1}{6} + \frac{1}{1.5}$$

$$v = \frac{6}{5} = 1.2 \text{ m}$$

$$\frac{h'}{h} = \frac{-v}{u}$$

$$\frac{h'}{3} = \frac{1.2}{6}$$

$$h' = 0.6 \text{ m}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

- 31.** An object is placed at a distance of 20 cm from the optical centre of a concave lens and its image is formed on the same side of the lens as the object. If the distance of the image from optical centre of the lens is 10 cm, use lens formula to determine (i) focal length, and (ii) power of the lens in new Cartesian sign conventions.

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- 31.** An object is placed at a distance of 20 cm from the optical centre of a concave lens and its image is formed on the same side of the lens as the object. If the distance of the image from optical centre of the lens is 10 cm, use lens formula to determine (i) focal length, and (ii) power of the lens in new Cartesian sign conventions.

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3

(i) $u = -20 \text{ cm}, v = -10 \text{ cm}$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-10} - \frac{1}{-20} = \frac{1}{f}$$

$$\frac{-1}{10} + \frac{1}{20} = \frac{1}{f}$$

$$\frac{-2+1}{20} = \frac{1}{f}$$

$$\frac{-1}{20} = \frac{1}{f}$$

$$f = -20 \text{ cm}$$

(ii) $P = \frac{1}{f(m)}$

$$= \frac{1}{-0.2} = -5D$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- 31.** A convex lens forms an 8.0 cm long image of a 2.0 cm long object which is kept at a distance of 6.0 cm from the optical centre of the lens. If the object and the image are on the same side of the lens, find (i) the nature of the image, (ii) the position of the image, and (iii) the focal length of the lens.

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- 31.** A convex lens forms an 8.0 cm long image of a 2.0 cm long object which is kept at a distance of 6.0 cm from the optical centre of the lens. If the object and the image are on the same side of the lens, find (i) the nature of the image, (ii) the position of the image, and (iii) the focal length of the lens.

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3

(i) Nature: Virtual and erect

1

(ii) Given $h' = + 8.0$ cm, $h = + 2.0$ cm, $u = - 6$ cm

$$m = \frac{h'}{h} = \frac{v}{u}$$

$$= \frac{8.0 \text{ cm}}{2.0 \text{ cm}} = \frac{v}{-6 \text{ cm}}$$

$\frac{1}{2}$

$$\text{or } v = - 24 \text{ cm}$$

$\frac{1}{2}$

Thus, the image is at a distance of 24 cm from the lens.

(iii) Lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{2}$

$$\frac{1}{-24} - \frac{1}{-6} = \frac{1}{f}$$

$$\frac{-1}{24} + \frac{1}{6} = \frac{1}{f}$$

$$\frac{1}{8} = \frac{1}{f}$$

$$f = 8 \text{ cm}$$

Thus the focal length of the lens = 8 cm

$\frac{1}{2}$

13. Mirror 'X' is used to concentrate sunlight in solar furnace and Mirror 'Y' is fitted on the side of the vehicle to see the traffic behind the driver. Which of the following statements are true for the two mirrors ?

- (i) The image formed by mirror 'X' is real, diminished and at its focus.
- (ii) The image formed by mirror 'Y' is virtual, diminished and erect.
- (iii) The image formed by mirror 'X' is virtual, diminished and erect.
- (iv) The image formed by mirror 'Y' is real, diminished and at its focus.

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (iii) and (iv)

(D) (i) and (iv)

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(A) (i) and (ii)

(B) (ii) and (iii)

(C) (iii) and (iv)

(D) (i) and (iv)

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- 24.** An object is placed at a distance of 60 cm from a concave lens of focal length 30 cm. Use lens formula to find the position of the image formed in this case.

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24. An object is placed at a distance of 60 cm from a concave lens of focal length 30 cm. Use lens formula to find the position of the image formed in this case.

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2

$$u = -60 \text{ cm} \quad f = -30 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{-30 \text{ cm}} = \frac{1}{v} - \frac{1}{-60 \text{ cm}}$$

$$\frac{1}{v} = \frac{1}{-30} - \frac{1}{60}$$

$$\frac{-3}{60 \text{ cm}} = \frac{1}{v} \Rightarrow v = -20 \text{ cm}$$

Position of image is 20 cm from a concave lens.

$\frac{1}{2}$

$\frac{1}{2}$

1

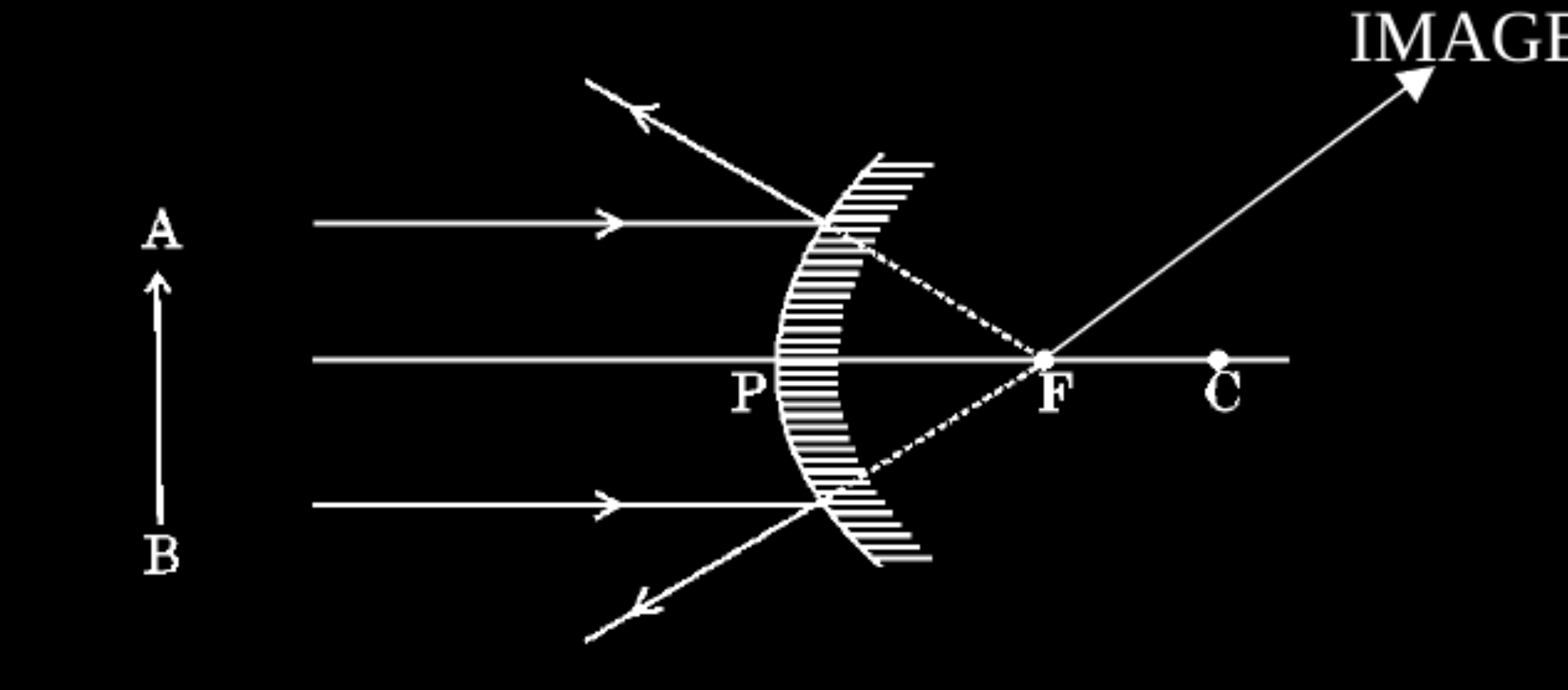
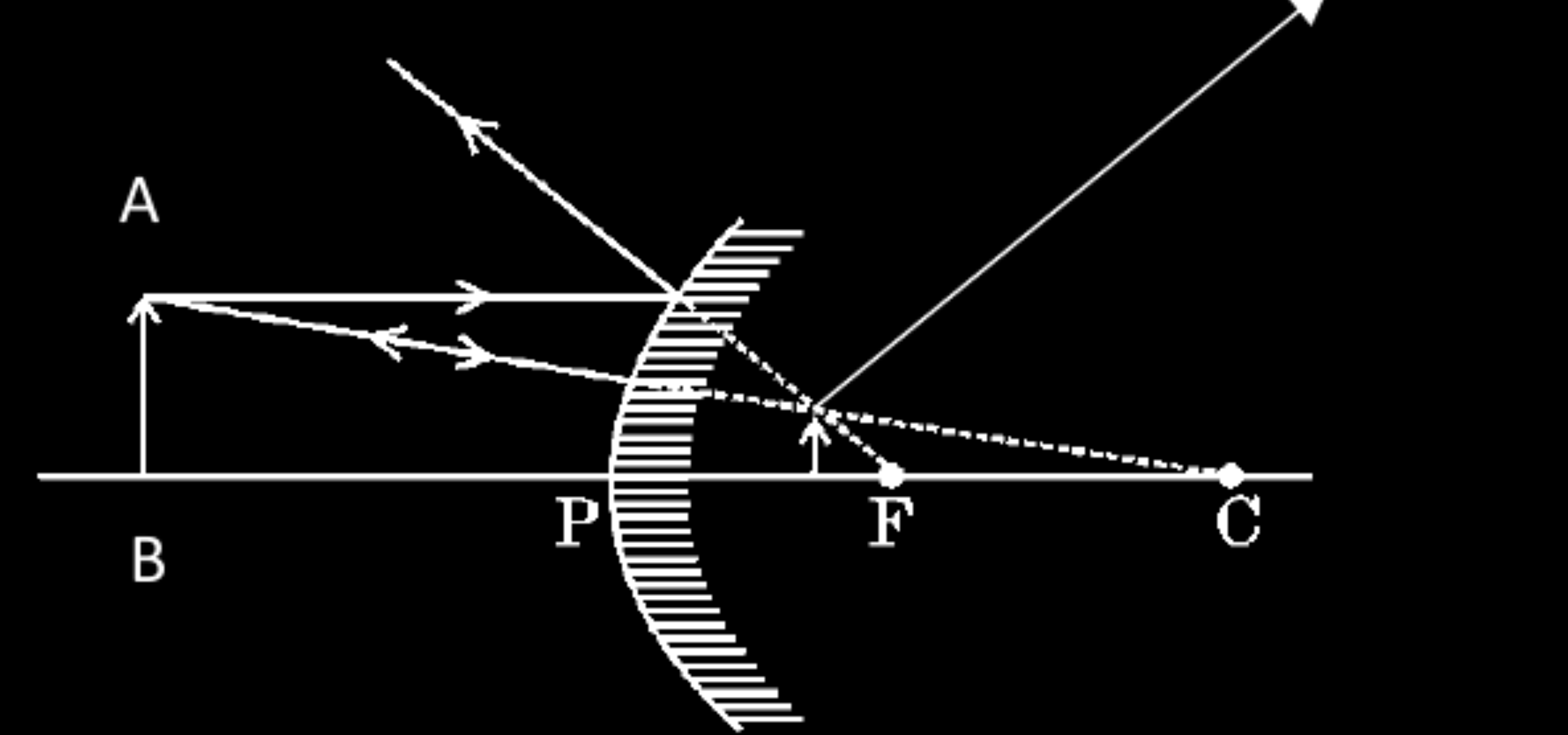
- 31.** Draw ray diagrams to show the nature, position and relative size of the image formed by a convex mirror when the object is placed (i) at infinity and (ii) between infinity and pole P of the mirror.

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3

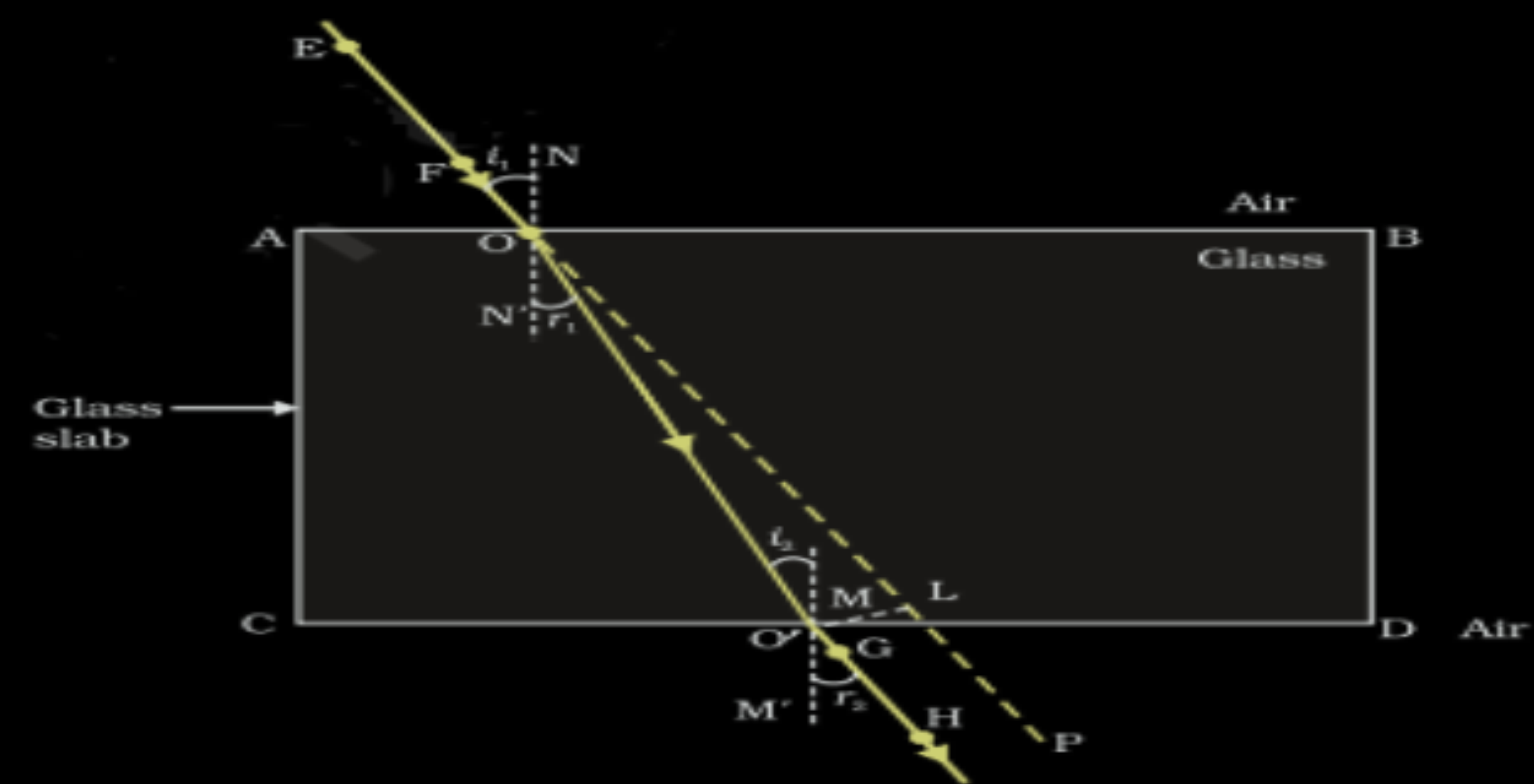
(i)  OBJECT	1 ½
(ii)  OBJECT	1 ½
(Note: Deduct ½ mark if arrows are not drawn.)	

- 36.** (a) (i) The power of a lens 'X' is -2.5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens ?
- (ii) "The value of magnification 'm' for a lens is -2 ." Using new Cartesian Sign Convention and considering that an object is placed at a distance of 20 cm from the optical centre of this lens, state :
- (I) the nature of the image formed;
 - (II) size of the image compared to the size of the object;
 - (III) position of the image, and
 - (IV) sign of the height of the image.
- (iii) The numerical values of the focal lengths of two lenses A and B are 10 cm and 20 cm respectively. Which one of the two will show higher degree of convergence/divergence ? Give reason to justify your answer.

(a) (i)	• Concave lens • $P = \frac{1}{f(m)}$ $-2.5 = \frac{1}{f}$ $f = \frac{10}{-2.5} = -0.4 \text{ m} = -40 \text{ cm}$ • Myopia	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
(ii)	(I) Real and inverted (II) magnified image /size of image is double the size of object (III) beyond 2F/ on the other side as that of object (IV) Negative	$\frac{1}{2} \times 4$
(iii)	• The lens with focal length 10 cm • less focal length, more converging/diverging power	$\frac{1}{2}$ $\frac{1}{2}$

- (b) (i) Draw a ray diagram to show the refraction of a ray of light through a rectangular glass slab when it falls obliquely from air into glass.
- (ii) State Snell's law of refraction of light.
- (iii) Differentiate between the virtual images formed by a convex lens and a concave lens on the basis of :
- (I) object distance, and
- (II) magnification.

(b) (i)



(if arrows not marked , deduct half mark)

(ii) The ratio of sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for a given pair of media. /

$$\frac{\sin i}{\sin r} = \text{constant}$$

(iii)

Convex Lens	Concave Lens
(I) Object to be placed between O and F	Object can be placed anywhere in front of the lens
(II) Magnified image	Diminished image

2

1

1

1

- 23.** An object is placed at a distance of 30 cm in front of a concave mirror of focal length 20 cm. Use mirror formula to determine the position of the image formed in this case.

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2

$$f = -20 \text{ cm} ; \quad u = -30 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{-20} - \frac{1}{-30}$$

$$\frac{1}{v} = \frac{1}{-20} + \frac{1}{30}$$

$$v = -60 \text{ cm}$$

Position of image is 60 cm in front of the mirror.

$\frac{1}{2}$

$\frac{1}{2}$

1

- 21.** An object is placed at a distance of 10 cm in front of a concave mirror of focal length 15 cm. Use mirror formula to determine the position of the image formed by this mirror.

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21. An object is placed at a distance of 10 cm in front of a concave mirror of focal length 15 cm. Use mirror formula to determine the position of the image formed by this mirror.

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2

Here $u = -10$ cm; $f = -15$ cm ;

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{-15} = \frac{1}{v} + \frac{1}{-10}$$

$$\frac{1}{v} = \frac{1}{-15} + \frac{1}{10}$$

$$v = +30 \text{ cm}$$

Position of image is 30 cm from the mirror

$\frac{1}{2}$

$\frac{1}{2}$

1

11. Absolute refractive index of water and glass is $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is :

(A) $\frac{9}{4}$ m/s

(B) $\frac{7}{3}$ m/s

(C) $\frac{16}{9}$ m/s

(D) $\frac{9}{8}$ m/s

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(B) $\frac{7}{3}$ m/s

(C) $\frac{16}{9}$ m/s

(D) $\frac{9}{8}$ m/s

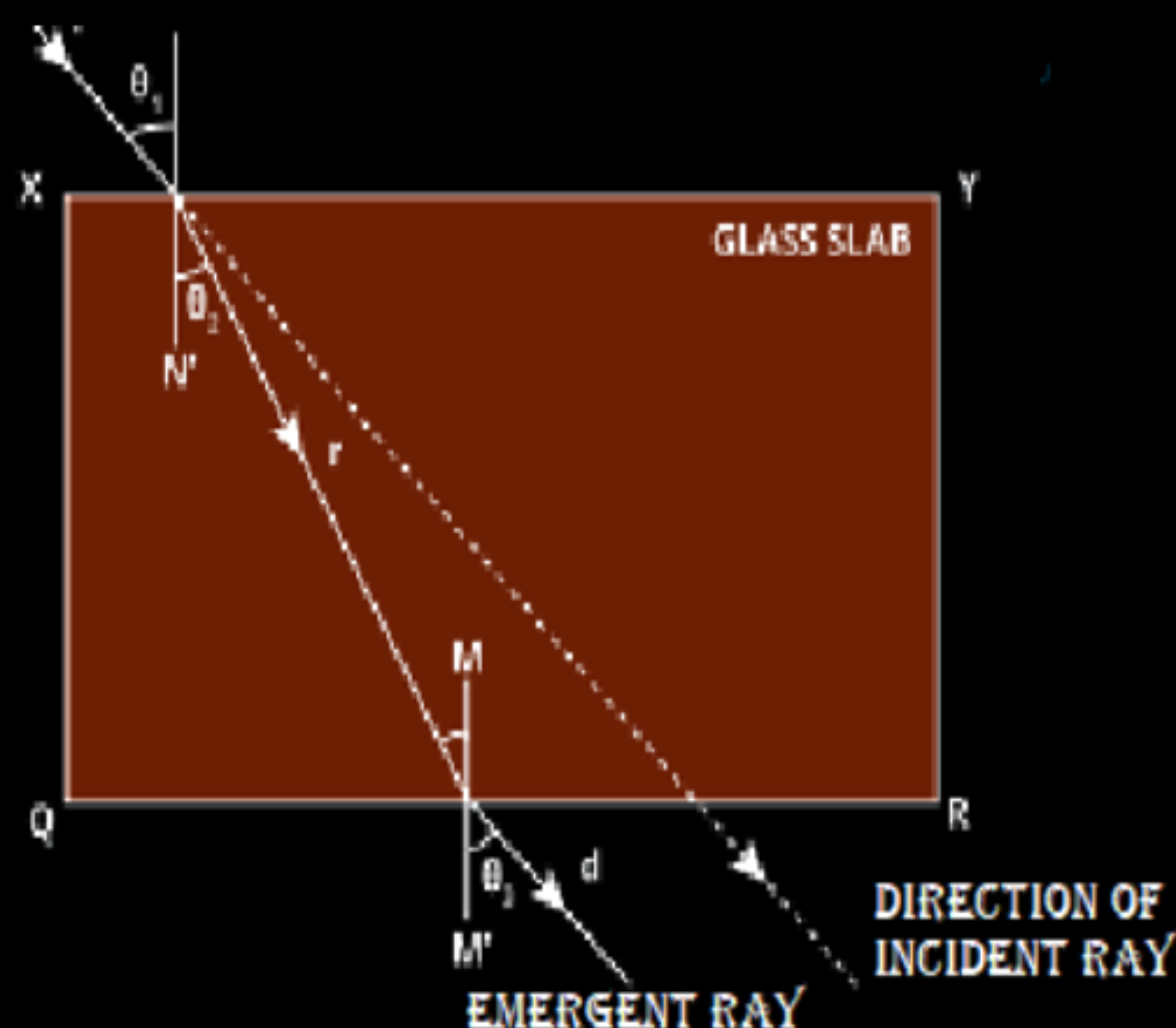
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NOTE: - Full credit of one mark to all the students.

- 36.** (a) (i) “In refraction of light through a rectangular glass slab, the emergent ray is always parallel to the direction of the incident ray.” Why ? Explain with the help of a ray diagram. What happens when a ray of light falls normally on one of the faces of a rectangular glass prism ? Draw diagram.
- (ii) An object is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 20 cm. Use Lens formula to determine the position of the image formed in this case.

(a) (i)

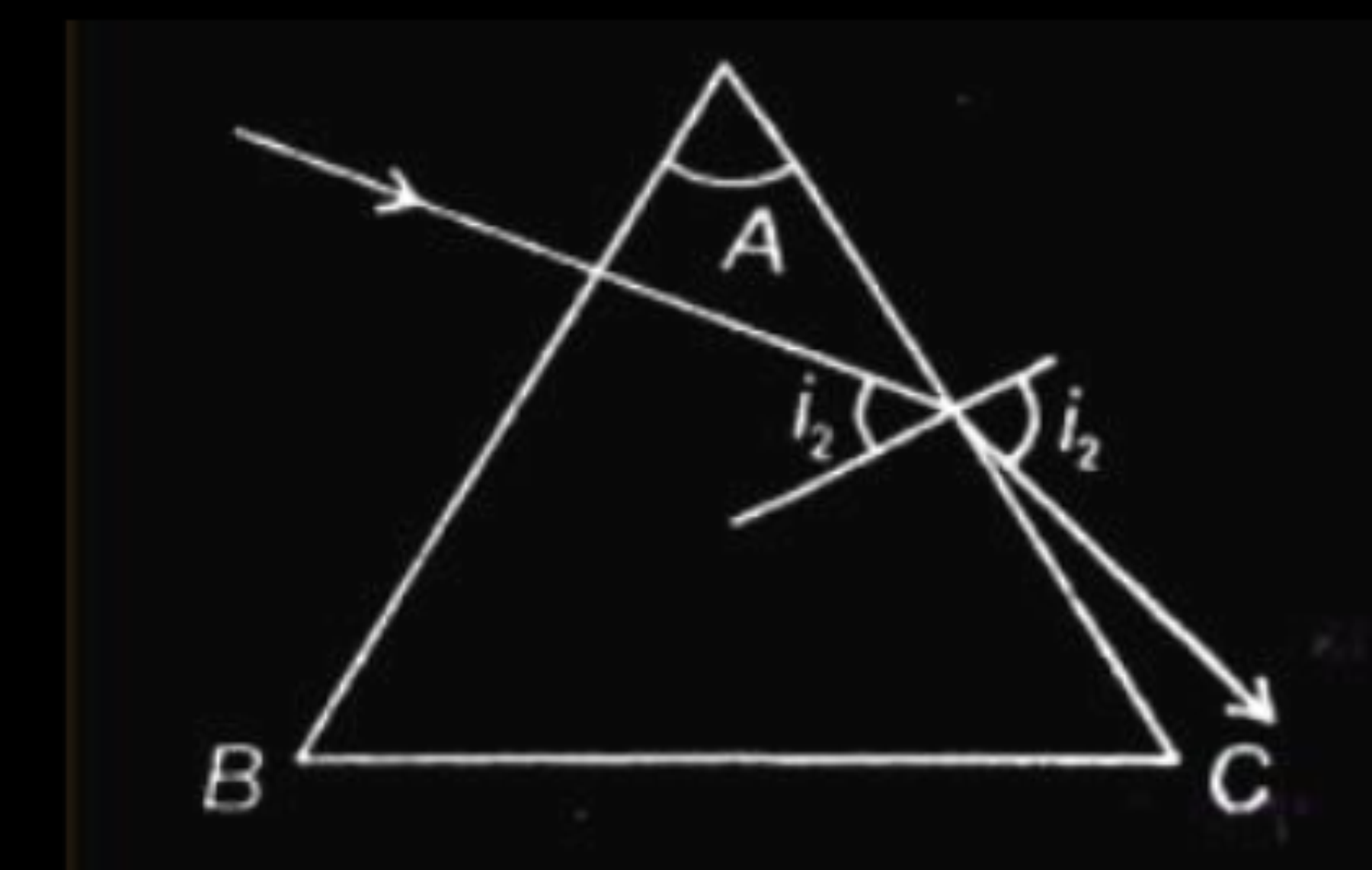
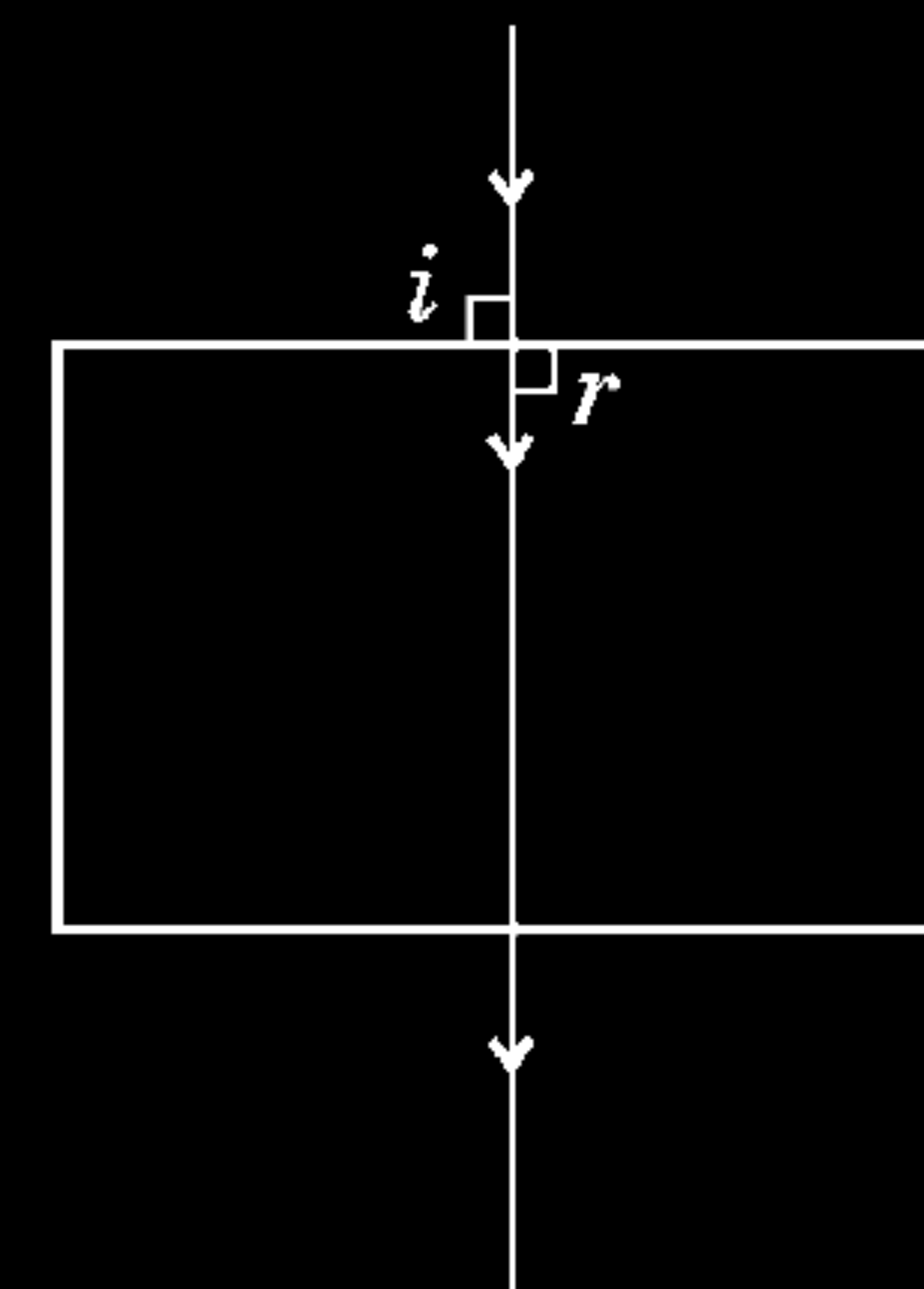
The extent of bending of the ray of light at the opposite parallel faces of the rectangular glass slab is equal and opposite. This is why the ray emerges parallel to the incident ray.



1

1

NORMAL INCIDENCE



1

(ii) $u = -30 \text{ cm}$, $f = -20 \text{ cm}$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{v} = \frac{1}{u} + \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{-30 \text{ cm}} + \frac{1}{-20 \text{ cm}}$$

$$= \frac{1}{-12 \text{ cm}} \quad \text{or} \quad v = -12 \text{ cm}$$

The image is at -12 cm on the same side of the lens.

$\frac{1}{2}$

$\frac{1}{2}$

1

- (b) (i) A student wishes to study the image formation by a concave mirror using candle flame as object. State the type of the image formed by the mirror and mention the change in the image formed, if any, that he observes when the candle flame is gradually moved away from the pole of the mirror. Draw a ray diagram to show the image formation when the object distance is nearly equal to the radius of curvature of the mirror.
- (ii) A convex mirror used for rear-view on an automobile has a focal length of 3.0 m. If a bus is located at 6.0 m from this mirror, use mirror formula to find the position of the image of the bus as seen in the mirror.

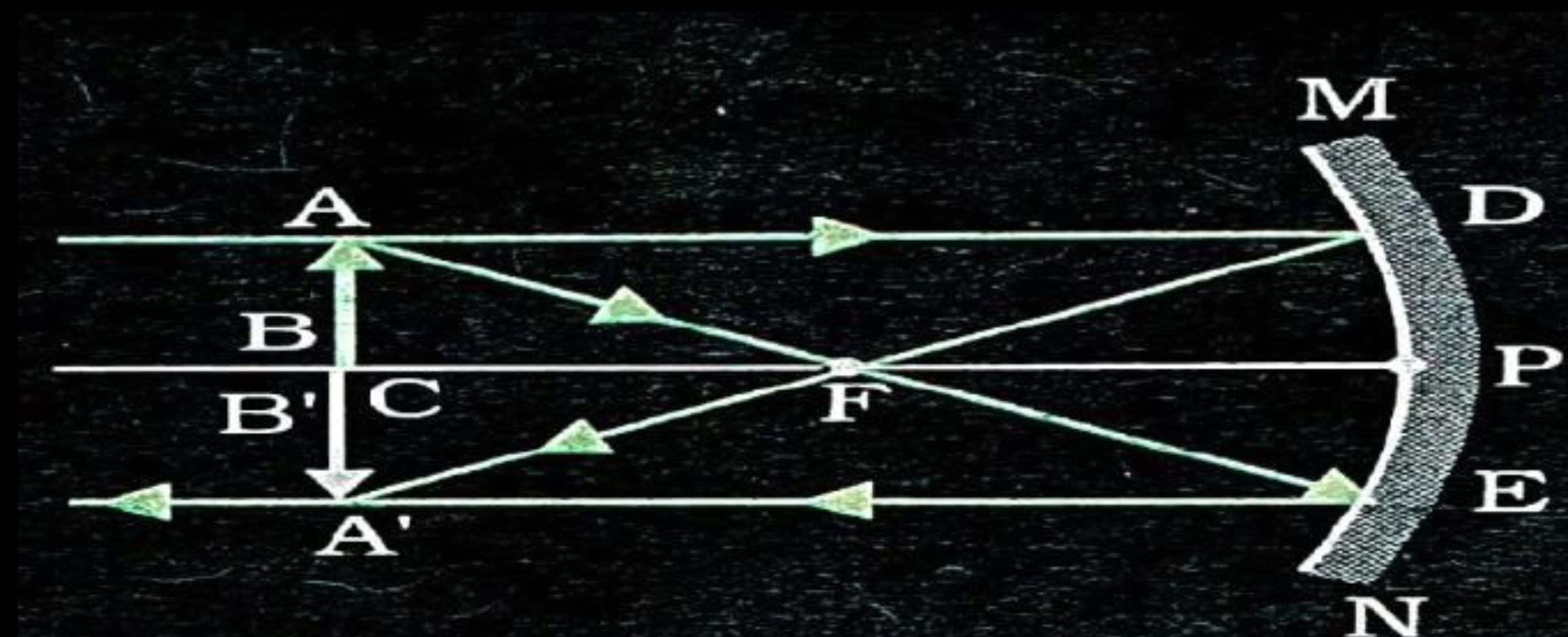
(b)

(i)

Type of Image

Real and Inverted (when candle is beyond F)/ Virtual and Erect (when candle is between P and F)

As the object is moved gradually away from the pole of the mirror, the image gets diminished.



(ii) Object distance, $u = -6.00 \text{ m}$

Image distance, $v = ?$

Focal length, $f = 3.00 \text{ m}$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{+3.00 \text{ m}} - \frac{1}{(-6.00 \text{ m})}$$

$$= \frac{1}{+3.00 \text{ m}} + \frac{1}{6.00 \text{ m}} = \frac{2+1}{6 \text{ m}}$$

$$\text{or } v = \frac{6}{3} = 2.0 \text{ m}$$

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

10. To get an image of magnification -1 on a screen using a lens of focal length 20 cm, the object distance must be :

1

- (a) Less than 20 cm
- (b) 30 cm
- (c) 40 cm
- (d) 80 cm

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10. To get an image of magnification -1 on a screen using a lens of focal length 20 cm, the object distance must be : 1

- (a) Less than 20 cm
- (b) 30 cm
- (c) 40 cm
- (d) 80 cm

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(c)/40cm

11. An optical device 'X' is placed obliquely in the path of a narrow parallel beam of light. If the emergent beam gets displaced laterally, the device 'X' is :

1

- (a) plane mirror
- (b) convex lens
- (c) glass slab
- (d) glass prism

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1

- (a) plane mirror
- (b) convex lens
- (c) glass slab
- (d) glass prism

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(c)/glass slab

- 31.** If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer. CBSE 2025 **3**

- 31.** If we want to obtain a virtual and magnified image of an object by using a concave mirror of focal length 18 cm, where should the object be placed? Use mirror formula to determine the object distance for an image of magnification +2 produced by this mirror to justify your answer. CBSE 2025 **3**

Object should be placed between F and P / At less than 18cm distance from the mirror.

$$\text{Mirror formula} = \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Magnification $m = +2$

$$f = -18 \text{ cm}$$

$$m = -\frac{v}{u} = +2$$

$$\therefore v = -2u$$

$$\frac{1}{-2u} + \frac{1}{u} = \frac{1}{-18 \text{ cm}}$$

$$\therefore \frac{1}{2u} = \frac{1}{-18 \text{ cm}}$$

$$u = -9 \text{ cm}$$

1

$\frac{1}{2}$

1

$\frac{1}{2}$

- 31.** If we want to obtain a real and magnified image of an object by using a concave mirror of focal length 18 cm. Where should the object be placed? Use mirror formula to determine the object distance for an image of magnification – 2 by this mirror to justify your answer.

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- 31.** If we want to obtain a real and magnified image of an object by using a concave mirror of focal length 18 cm. Where should the object be placed? Use mirror formula to determine the object distance for an image of magnification – 2 by this mirror to justify your answer.

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3

- Object should be placed between C and F / between 18 cm to 36 cm from the mirror.

- Mirror formula $= \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

- Magnification $m = -2$

$$f = -18 \text{ cm}$$

$$m = -\frac{v}{u} = -2$$

$$\therefore v = 2u$$

- $$\frac{1}{2u} + \frac{1}{u} = \frac{1}{-18 \text{ cm}}$$

$$\therefore \frac{3}{2u} = \frac{1}{-18 \text{ cm}}$$

$$u = -27 \text{ cm}$$

1

$\frac{1}{2}$

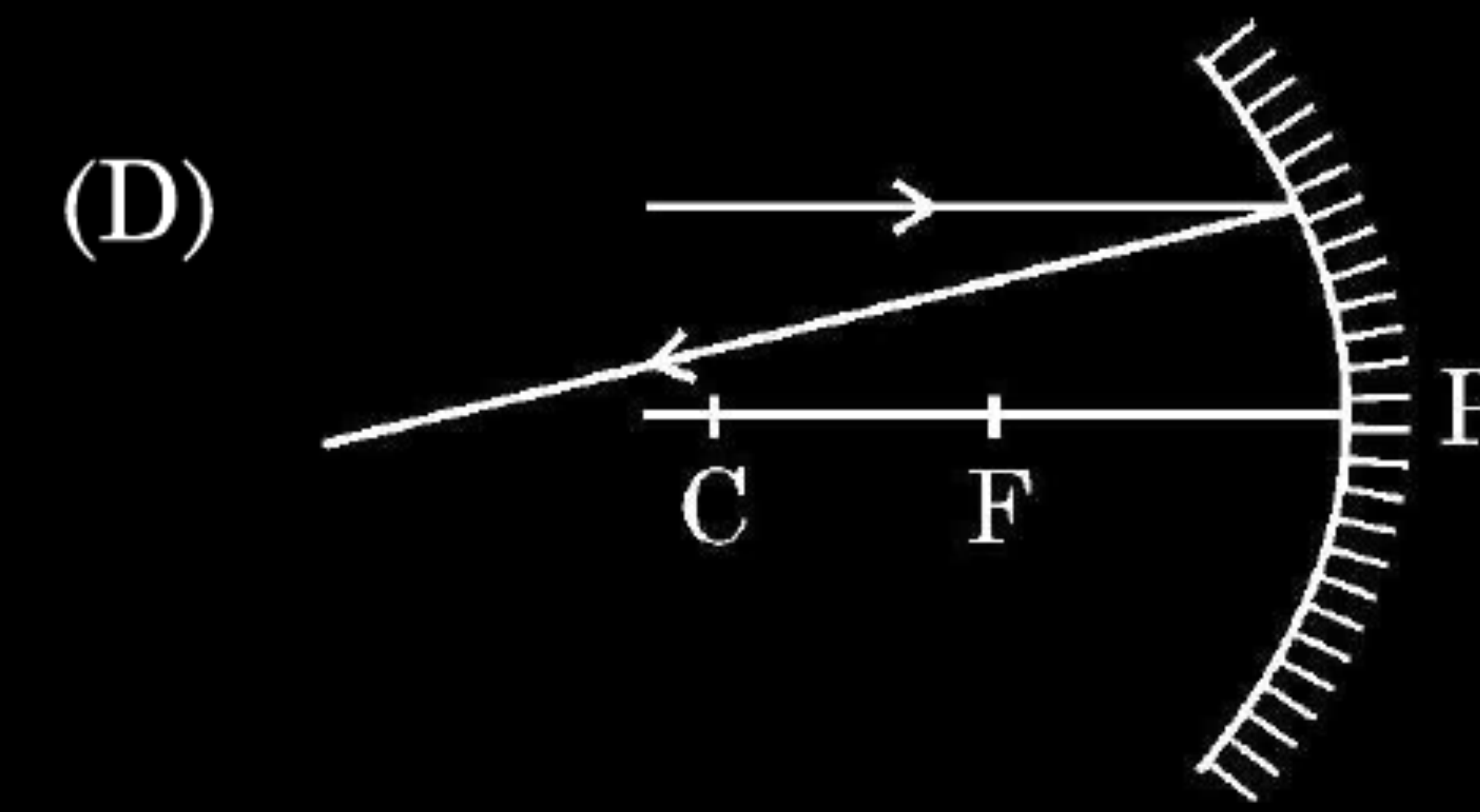
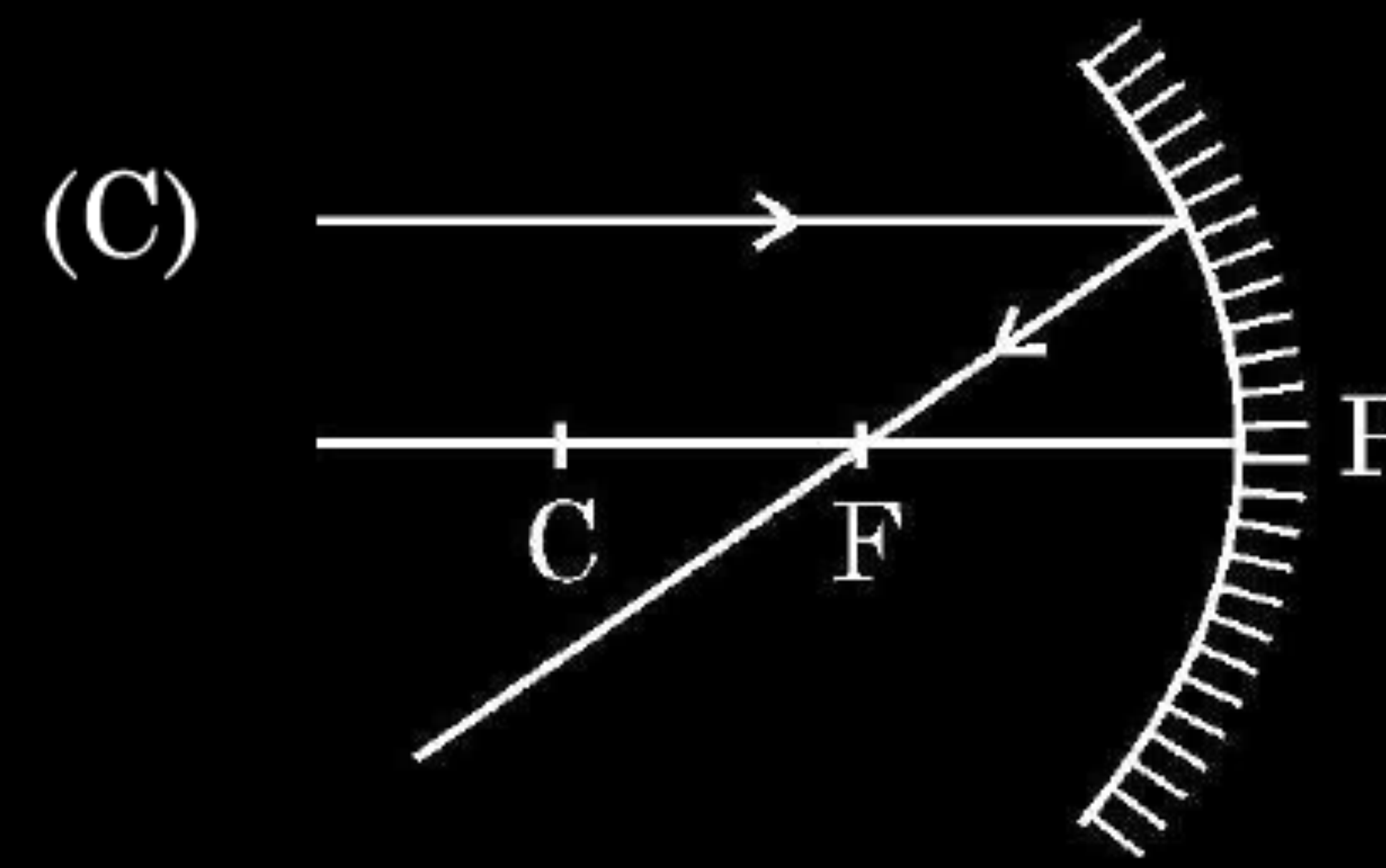
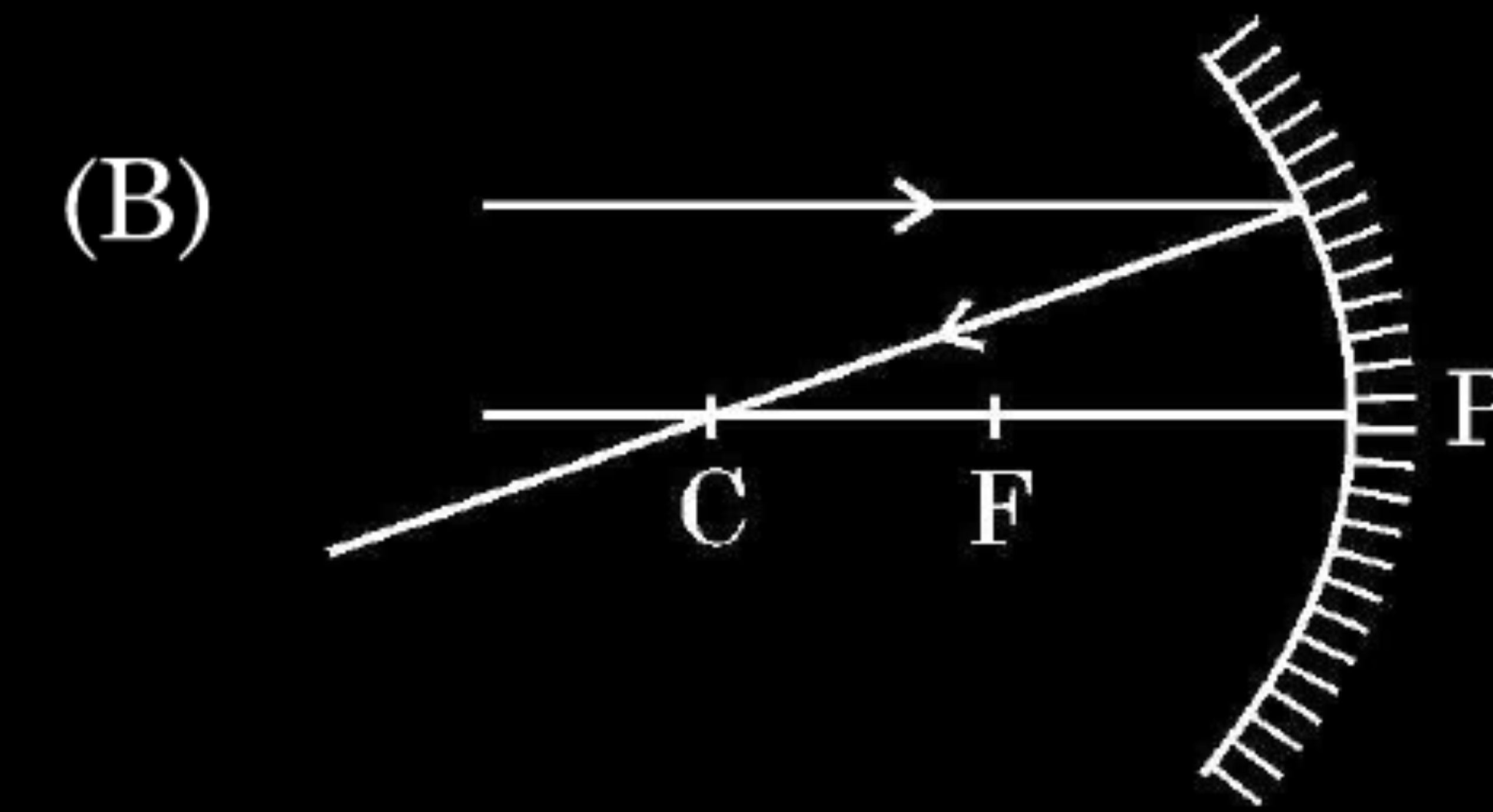
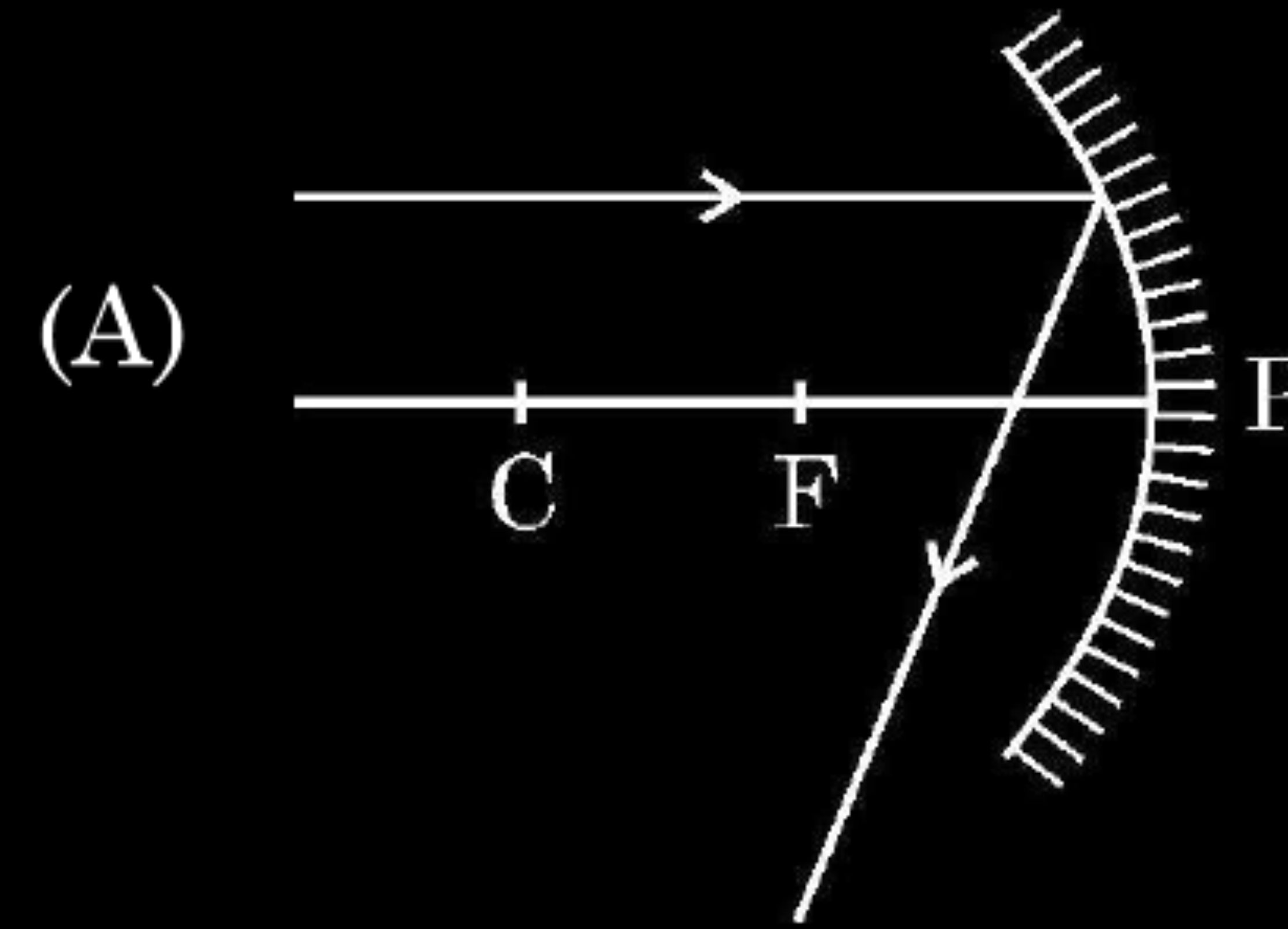
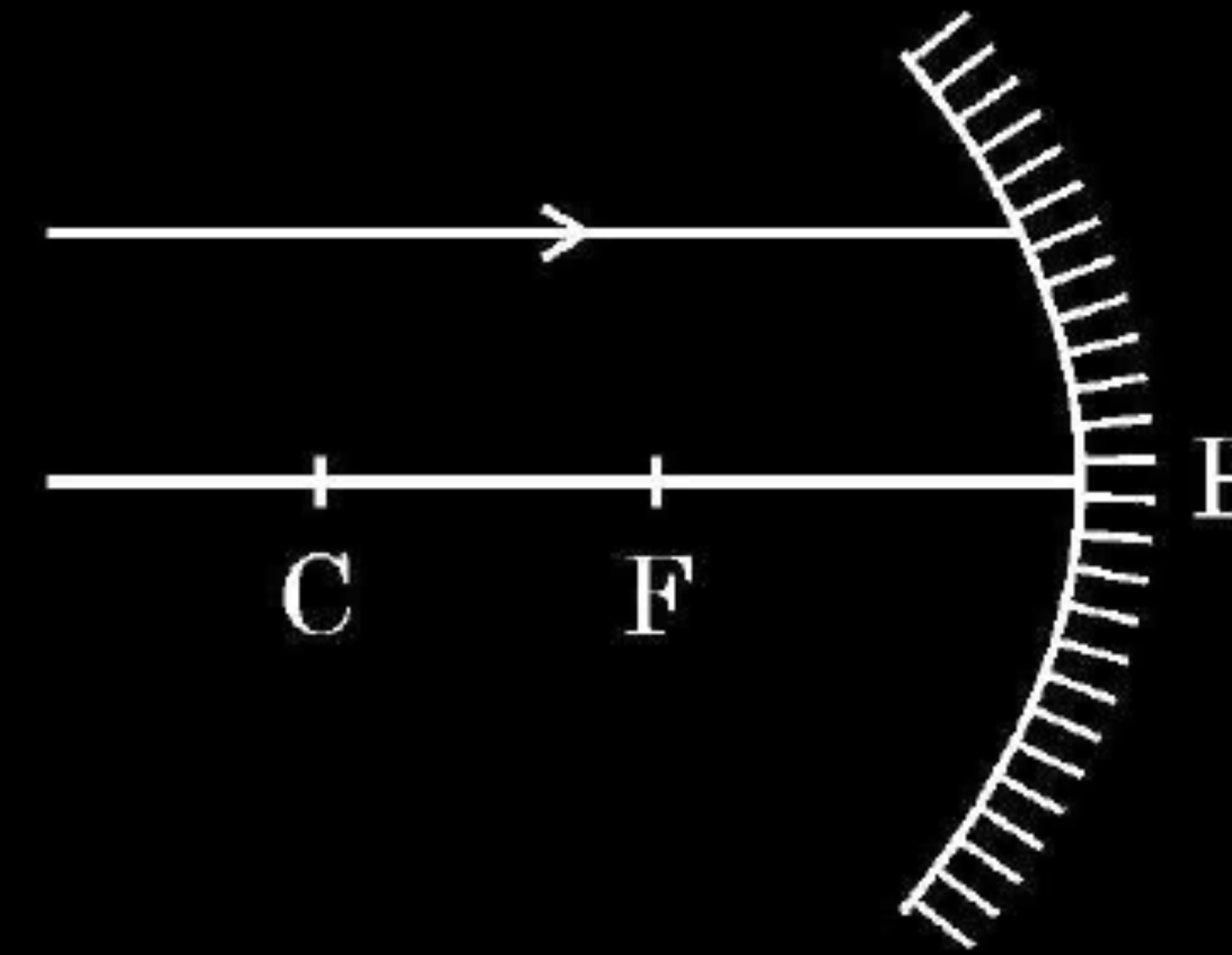
1

$\frac{1}{2}$

14. Identify from the following the ray diagram which shows the correct path of the reflected ray for the ray incident on a concave mirror as shown :

1

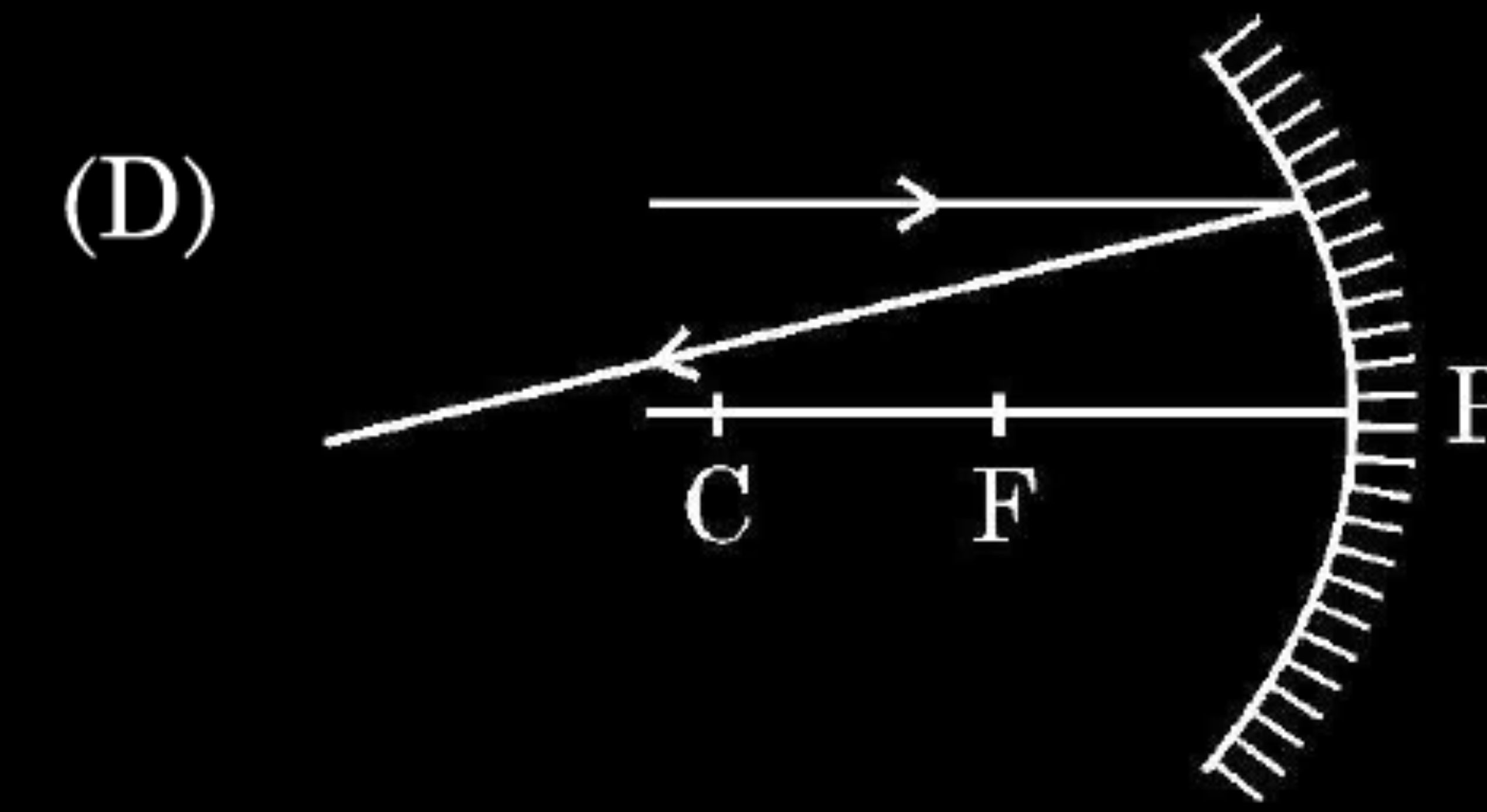
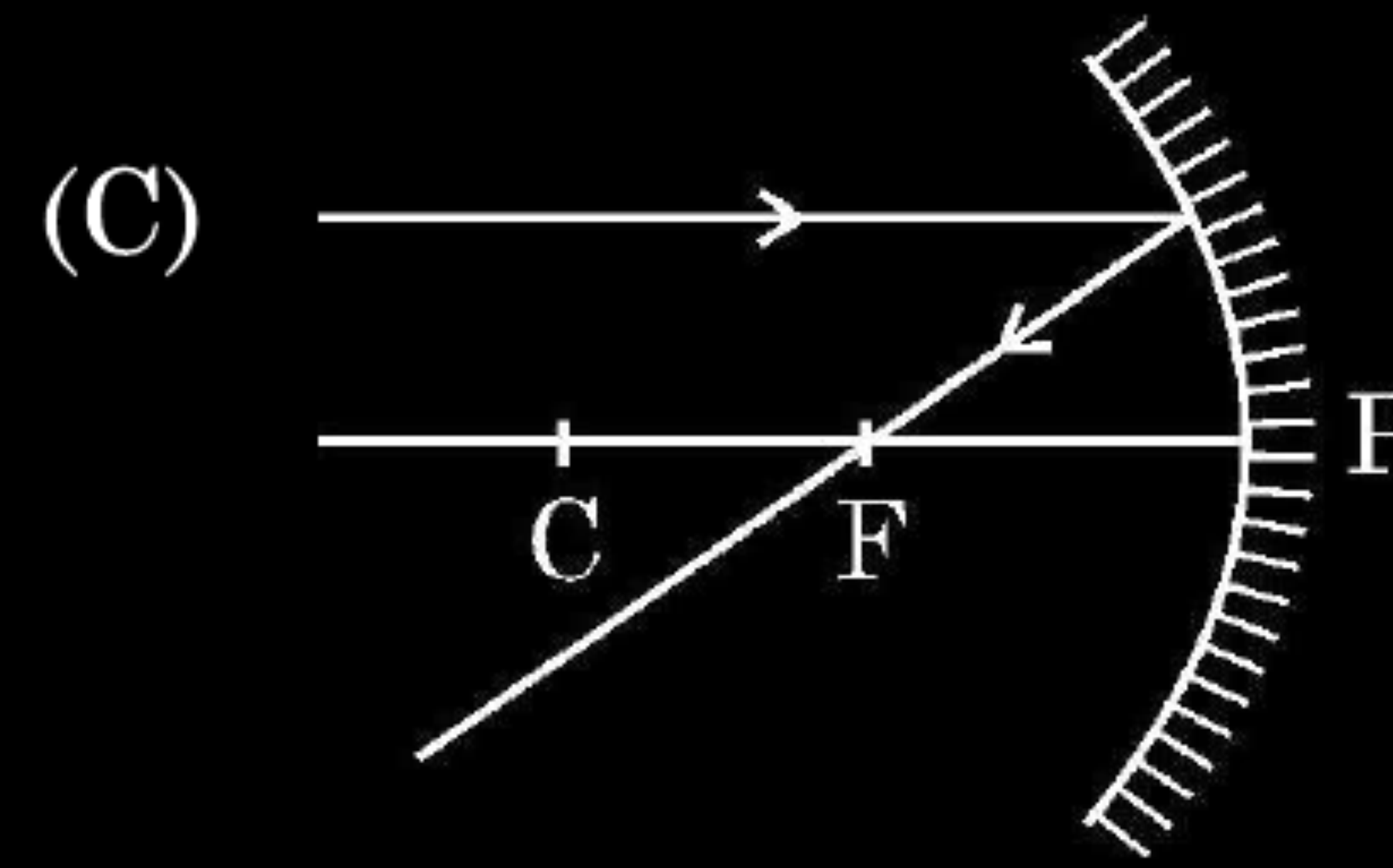
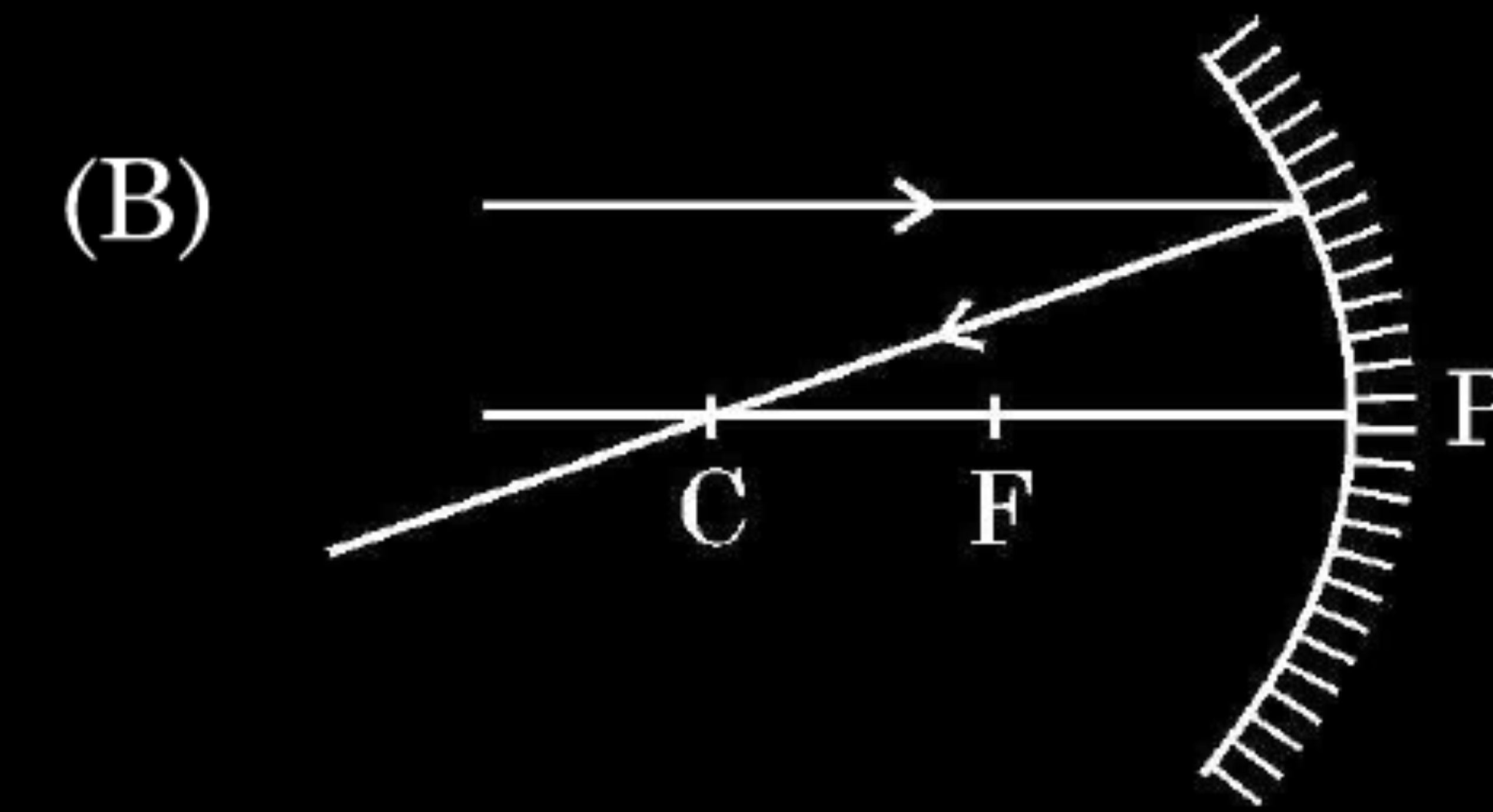
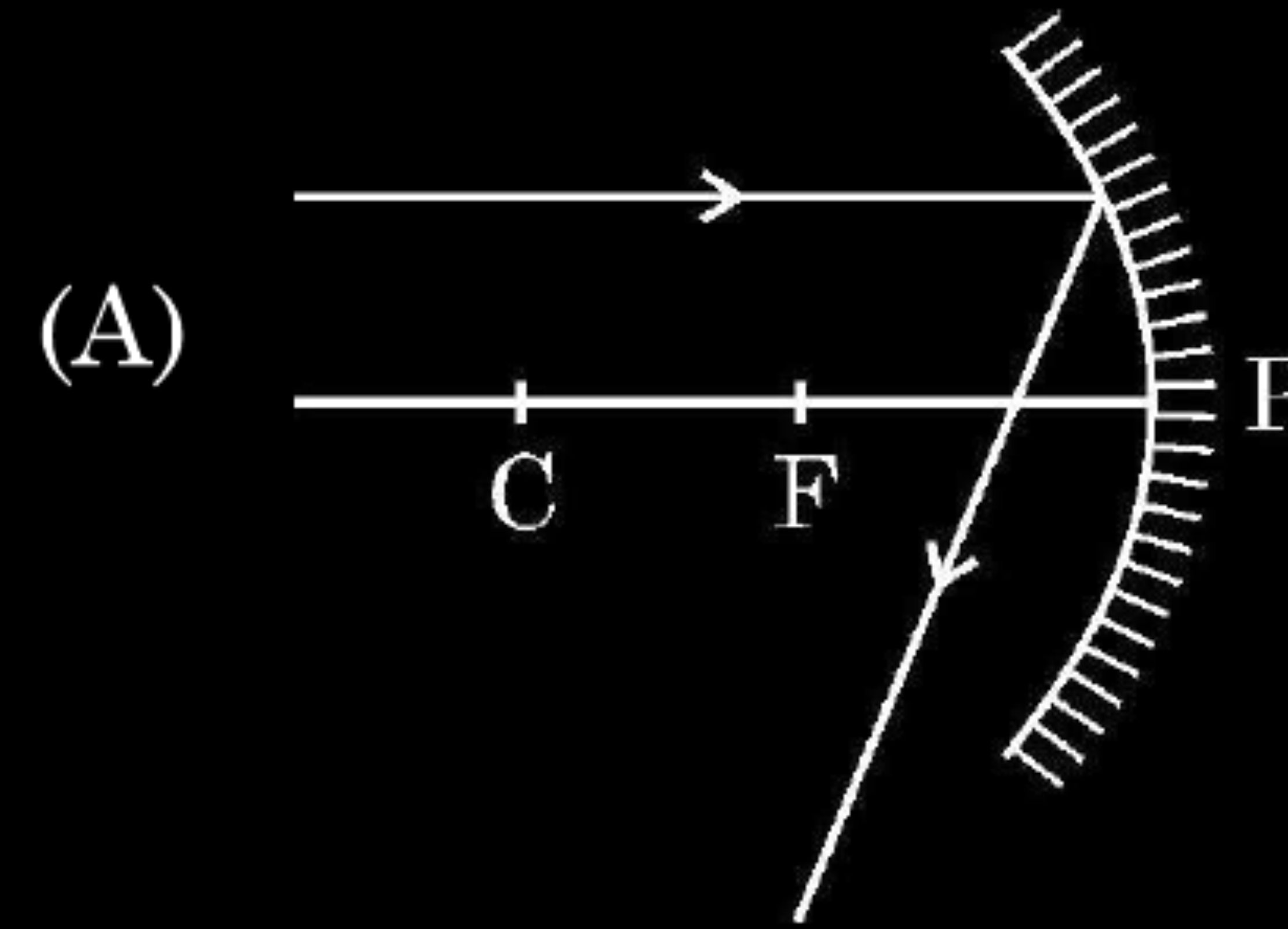
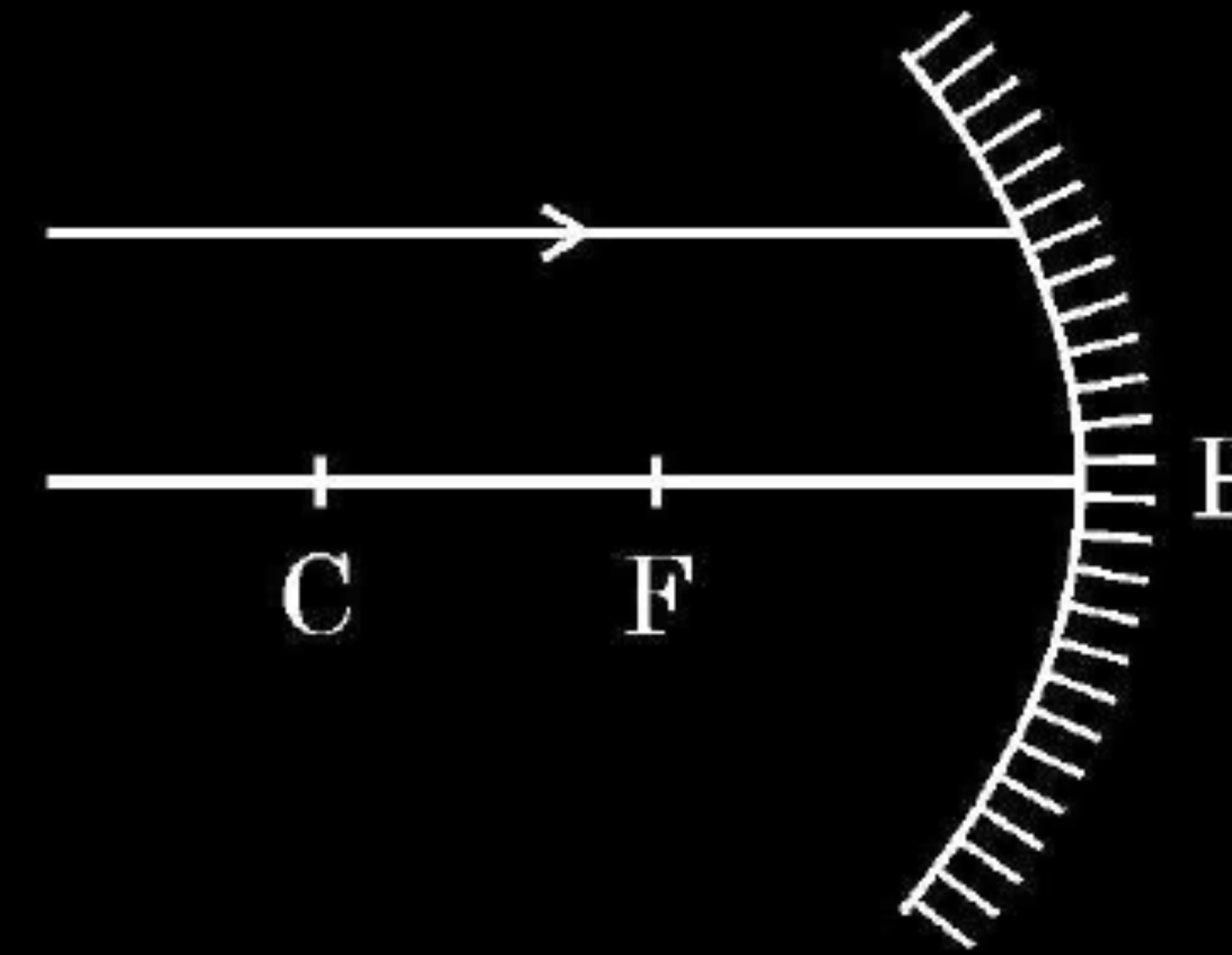
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14. Identify from the following the ray diagram which shows the correct path of the reflected ray for the ray incident on a concave mirror as shown :

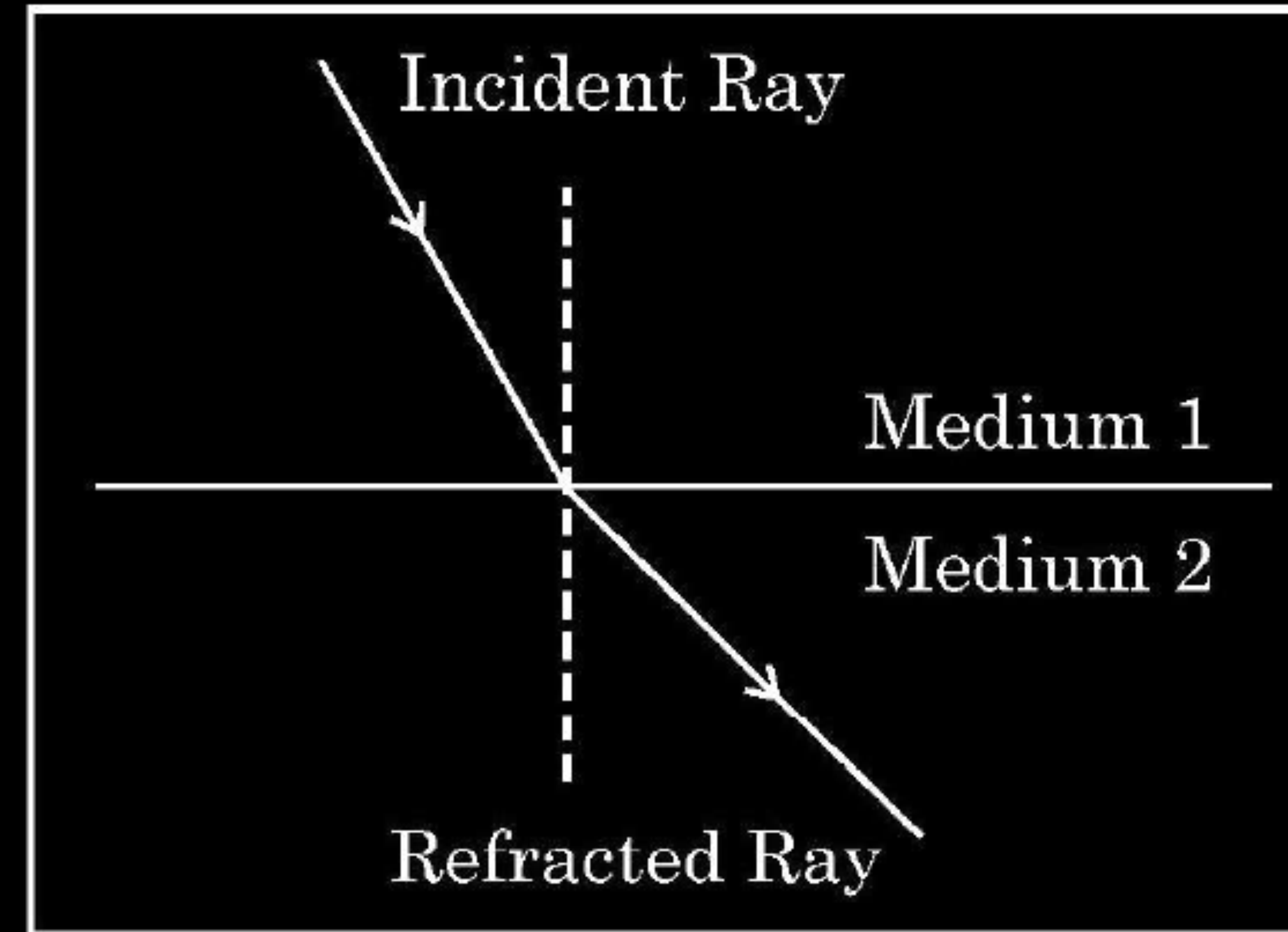
1

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(C)

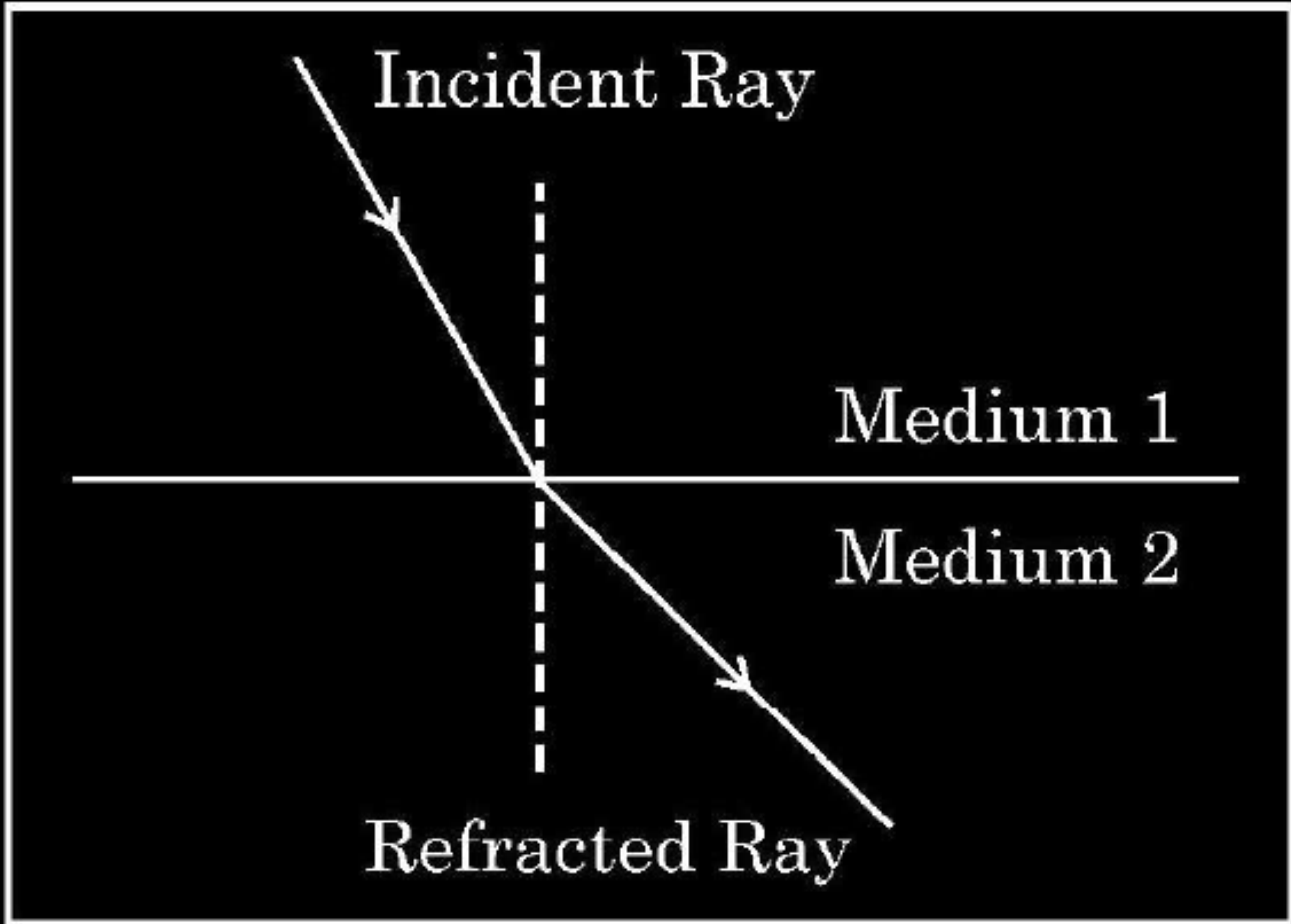
22. Study the figure in which the path of a ray of light going from Medium 1 to Medium 2 is shown.



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- (a) Out of the two Media – Medium 1 and Medium 2, in which is the speed of light more ?
- (b) State reason of bending of the refracted ray away from the normal.
- (c) Express refractive index of Medium 2 with respect to Medium 1 in terms of speed of light in two media.

22. Study the figure in which the path of a ray of light going from Medium 1 to Medium 2 is shown.



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- (a) Out of the two Media – Medium 1 and Medium 2, in which is the speed of light more ?
- (b) State reason of bending of the refracted ray away from the normal.
- (c) Express refractive index of Medium 2 with respect to Medium 1 in terms of speed of light in two media.

2

(a) Medium 2	$\frac{1}{2}$
(b) Ray bends away from the normal because it is travelling from optically denser medium to rarer medium / speed of light in medium 2 increases.	$\frac{1}{2}$
(c) $n_{21} = \frac{\text{Speed of light in medium 1}}{\text{Speed of light in medium 2}}$	1

29. A student placed a candle flame at different distances from a convex lens and focused its image on a screen. He recorded his observation in tabular form as given below :

S.No.	Distance of flame from the lens (cm)	Distance of the image from the lens
1	– 90	+ 18
2	– 60	+ 20
3	– 40	+ 24
4	– 30	+ 30
5	– 24	+ 40
6	– 20	+ 60
7	– 18	+ 90
8	– 12	+ 120

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Analyse the observation table and on the basis of your analysis only, answer the following questions (without doing any calculations) :

(a) What is the focal length of the convex lens used ? Give reason to justify your answer.

(b) Which one of the sets of observations is not correct and why ?

(c) Draw ray diagram to show image formation for any correct set of observation.

(a) Focal length = + 15 cm (Ignore units)

Reason: when object distance and image distance are same i.e. object is at $2F$, image is also formed at $2F$ on the other side

$$\therefore 2f = 30 \text{ cm}$$

(b) Observation No. 8

Reason: here the object is between optical centre and principal focus of the lens hence image is formed on the same side as the object and v is not equal to + 120 cm. (it should be -120 cm)

(c) Ray diagram

– Any one ray diagram

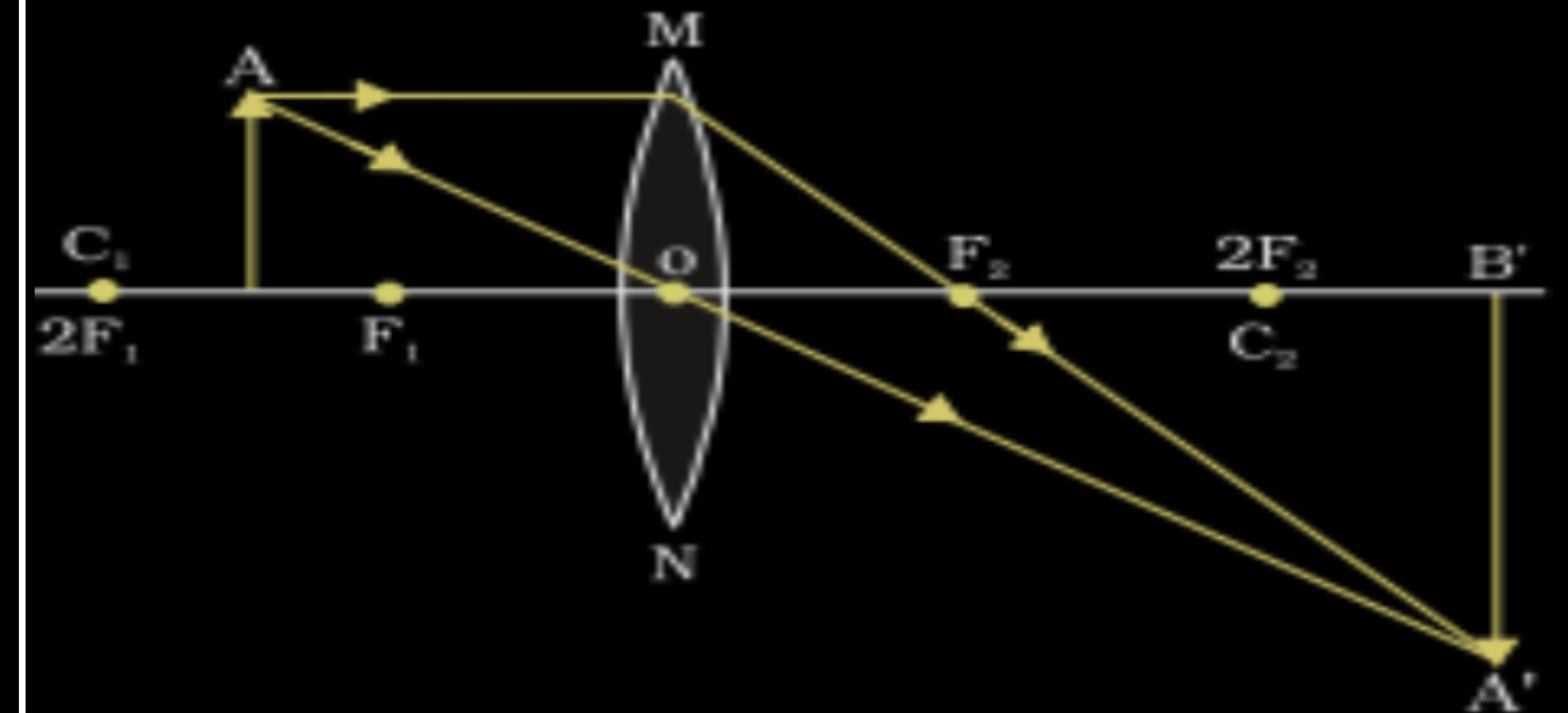
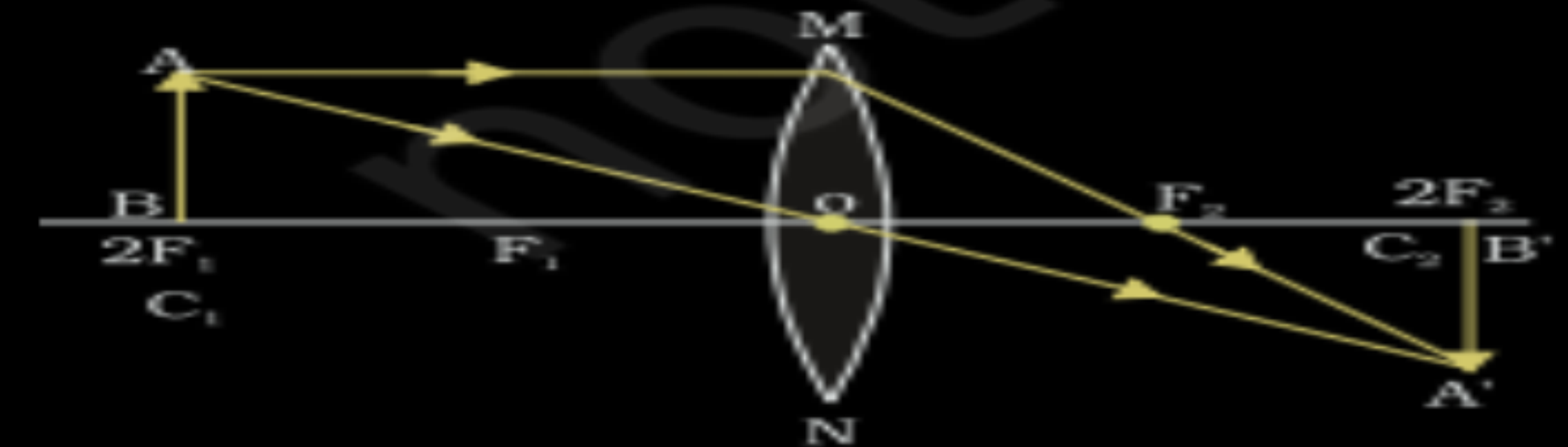
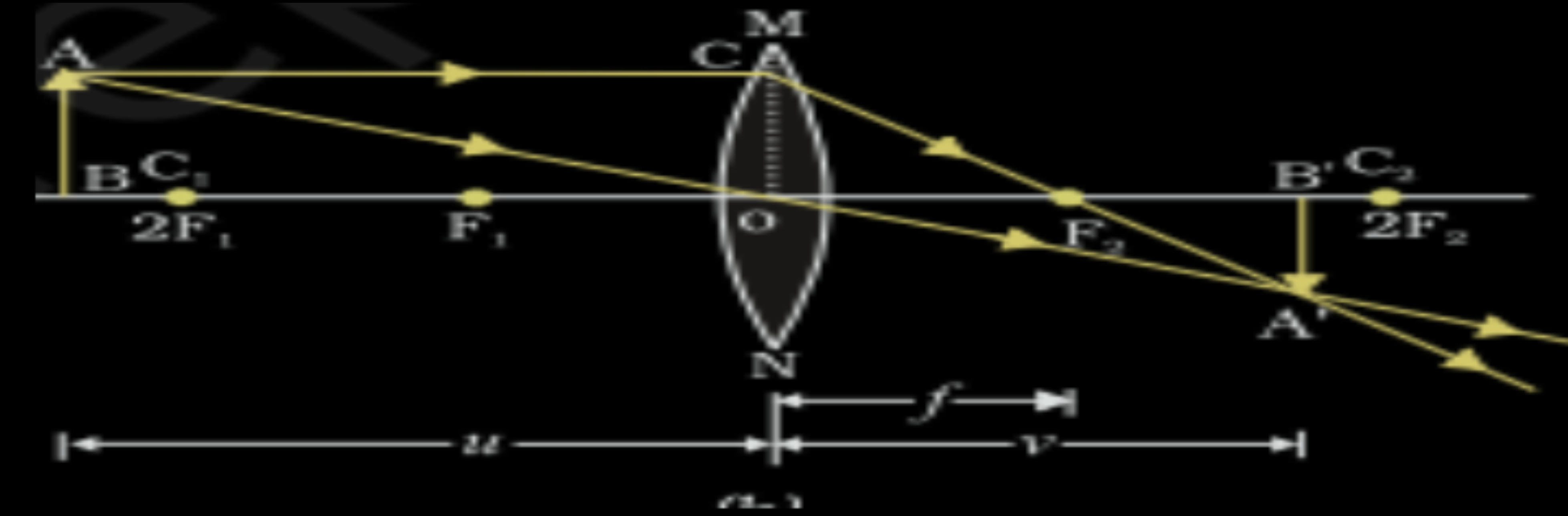
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1



29. An object of size 5 cm is placed at a distance of 30 cm from the optical centre of a converging lens of focal length 20 cm. Use lens formula to determine the position and size of the image formed.

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3

29. An object of size 5 cm is placed at a distance of 30 cm from the optical centre of a converging lens of focal length 20 cm. Use lens formula to determine the position and size of the image formed.

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3

Position: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{+20}$$

$$\frac{1}{v} + \frac{1}{30} = \frac{1}{20} \Rightarrow v = +60 \text{ cm}$$

position is 60 cm from the lens

Size: $m = \frac{h_i}{h_o} = \frac{v}{u}$

$$= \frac{60}{-30} = -2$$

$$h_i = -2 \times 5 \text{ cm} = -10 \text{ cm}$$

size of image is 10 cm(Enlarged)

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

12. An object is placed at a distance of 30 cm from the reflecting surface of a concave mirror of radius of curvature 40 cm. The image formed is
- (A) Virtual and magnified (B) Virtual and diminished
(C) Real and magnified (D) Real and diminished

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12. An object is placed at a distance of 30 cm from the reflecting surface of a concave mirror of radius of curvature 40 cm. The image formed is

- | | |
|---------------------------|----------------------------|
| (A) Virtual and magnified | (B) Virtual and diminished |
| (C) Real and magnified | (D) Real and diminished |

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(C)/ Real and magnified

25. (a) List four properties of the images formed by plane mirrors. 2

OR

(b) List four properties of the images formed by convex mirrors when objects are placed anywhere in front of their reflecting surfaces.

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25. (a) List four properties of the images formed by plane mirrors. 2

OR

(b) List four properties of the images formed by convex mirrors when objects are placed anywhere in front of their reflecting surfaces.

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(a) Properties of the images formed by plane mirror: • Virtual • Erect • Same size as the object • behind the mirror at the same distance • laterally inverted (Any Four)	$\frac{1}{2}$ x4
OR (b) • Virtual • Erect • Diminished • Formed behind the mirror	$\frac{1}{2}$ x4

31. An object of height 4.0 cm is placed at a distance of 30 cm from the optical centre of a convex lens of focal length 20 cm. Use lens formula to determine (i) the distance of the image from the optical centre and (ii) height of the image formed.

3

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(i) Lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $u = -30 \text{ cm}, f = +20 \text{ cm}, h_o = 4.0 \text{ cm}, v = ? ; h_i = ?$ $\therefore \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{20} + \frac{1}{-30}$ $\frac{1}{v} = \frac{3-2}{60}$ $\frac{1}{v} = \frac{1}{60}$ $v = 60 \text{ cm}$ (award marks, if solved by any other method)	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	(ii) Height of the image $h_i = \frac{v}{u} \times h_o$ $= \frac{+60 \text{ cm}}{-30 \text{ cm}} \times 4.0 \text{ cm}$ $= -8.0 \text{ cm}$
---	---	---

16. In order to obtain large images of the teeth of patients, the dentist holds the concave mirror in such a manner that the teeth are positioned
- (A) at the focus of mirror.
 - (B) between pole and focus of the mirror.
 - (C) between focus and centre of curvature of the mirror.
 - (D) at the centre of curvature of the mirror.

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16. In order to obtain large images of the teeth of patients, the dentist holds the concave mirror in such a manner that the teeth are positioned
- (A) at the focus of mirror.
 - (B) between pole and focus of the mirror.
 - (C) between focus and centre of curvature of the mirror.
 - (D) at the centre of curvature of the mirror.

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(B) / Between pole and focus of the mirror

21. During an experiment, a student observes that the given mirror is forming an erect and diminished image for all positions of the object placed in front of it.

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Based on the this information, answer the following questions :

- (a) Identify the mirror.
- (b) Using new Cartesian sign convention, write the magnitude of the magnification in this case.
- (c) What will be the radius of curvature of the mirror if its focal length is 20 cm ?

21. During an experiment, a student observes that the given mirror is forming an erect and diminished image for all positions of the object placed in front of it.

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Based on the this information, answer the following questions :

- (a) Identify the mirror.
- (b) Using new Cartesian sign convention, write the magnitude of the magnification in this case.
- (c) What will be the radius of curvature of the mirror if its focal length is 20 cm ?

(a) Convex mirror

$\frac{1}{2}$

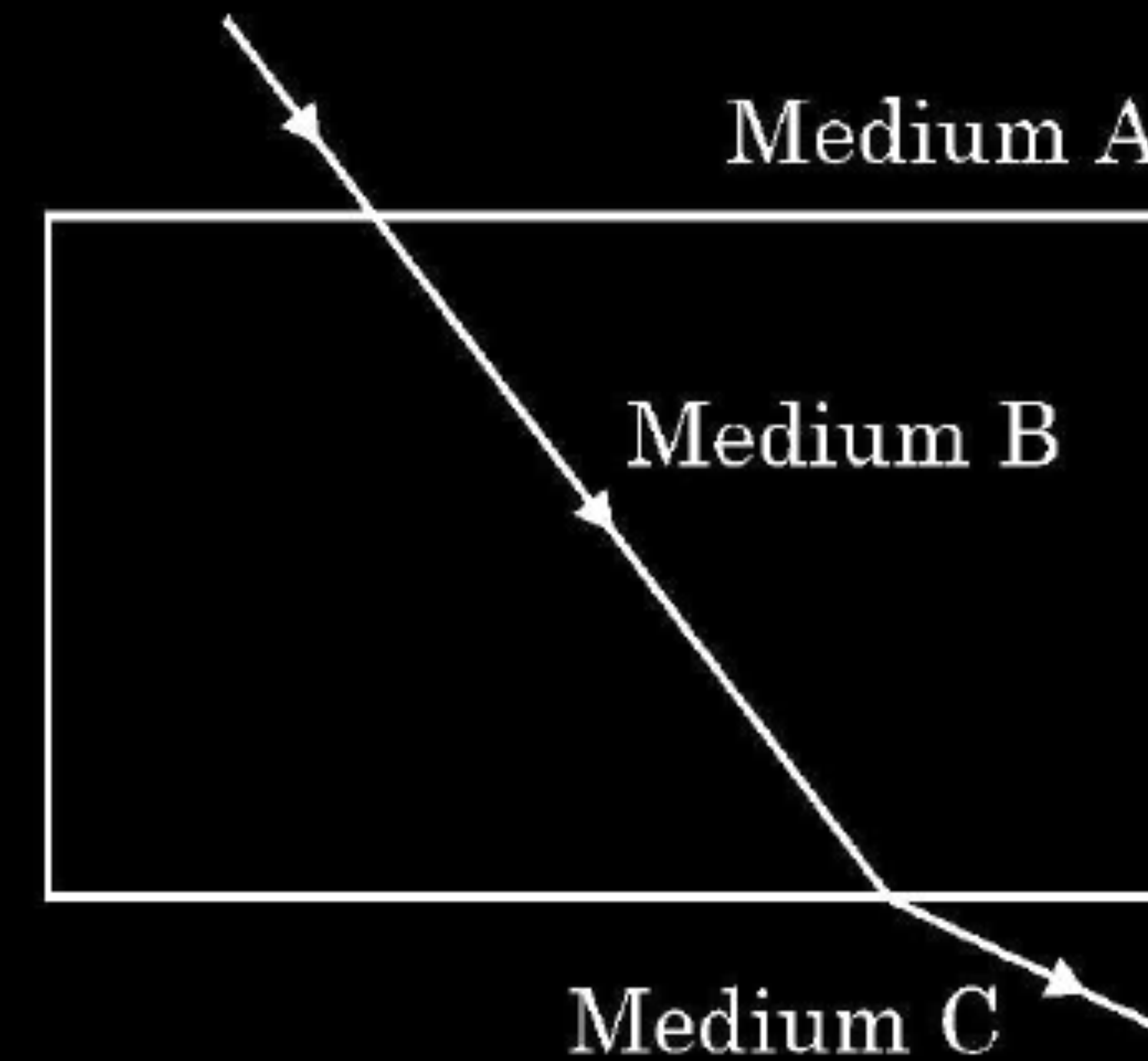
(b) Positive and less than 1

$\frac{1}{2}$ + $\frac{1}{2}$

(c) + 40 cm or 40 cm

$\frac{1}{2}$

36. (A) (a) Observe the following diagram and compare (i) speed of light and (ii) optical densities of the three media A, B and C. Also give justification for your answer of any one of the two cases in terms of refractive indices of A, B and C.



- (b) Redraw the path of a ray of light through the three media, if the ray of light starting from medium A falls on the medium B
- (i) Obliquely and the optical density of medium B is made more than that of A and C.
 - (ii) The ray falls normally from medium A to medium B.

(A) (a) (i) Speed of light in A and B is same whereas the speed of light in C is greater than that of A and B.

$\frac{1}{2} + \frac{1}{2}$

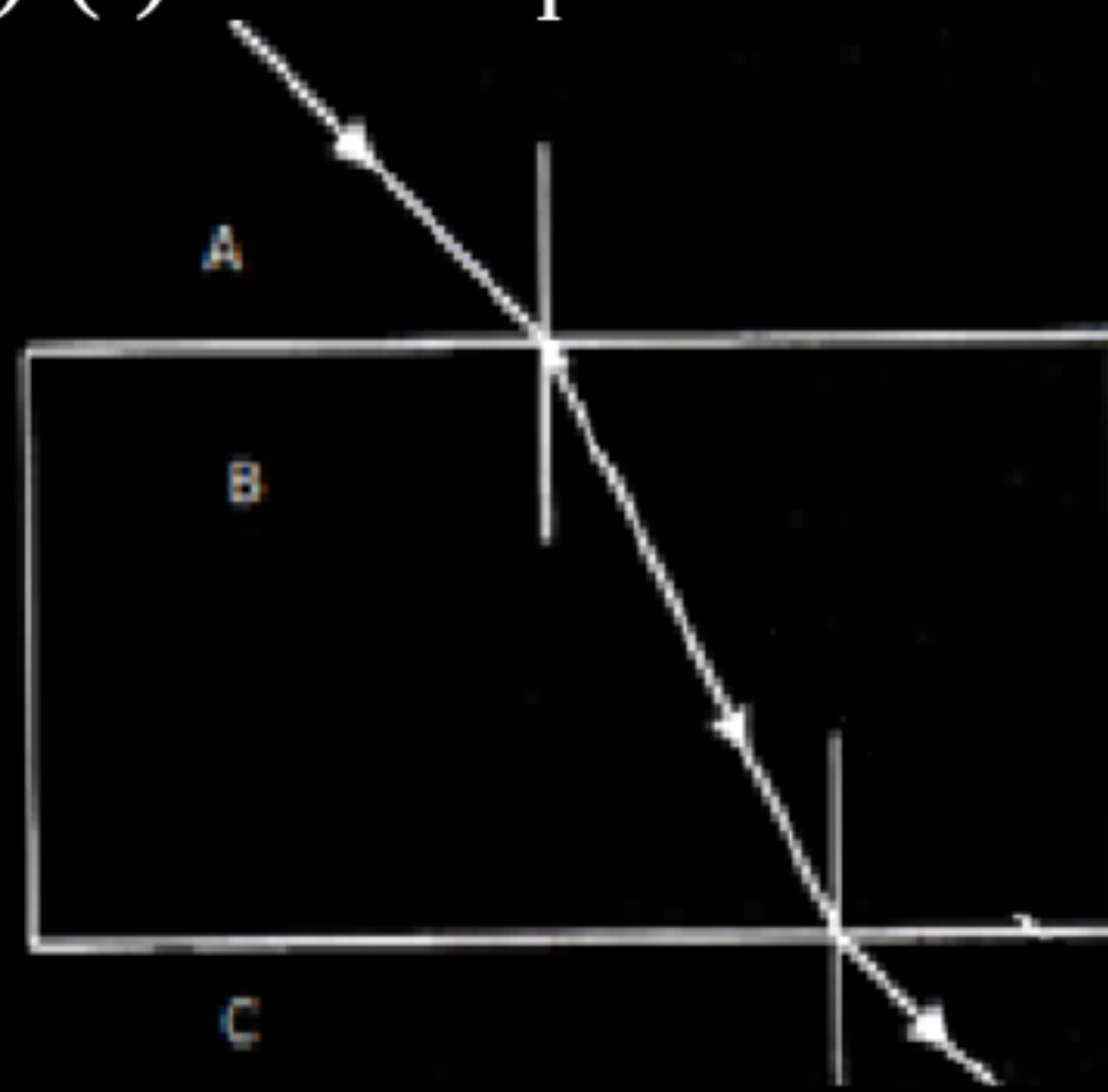
(ii) Optical density of A and B is same whereas optical density of C is less than that of A and B.

$\frac{1}{2} + \frac{1}{2}$

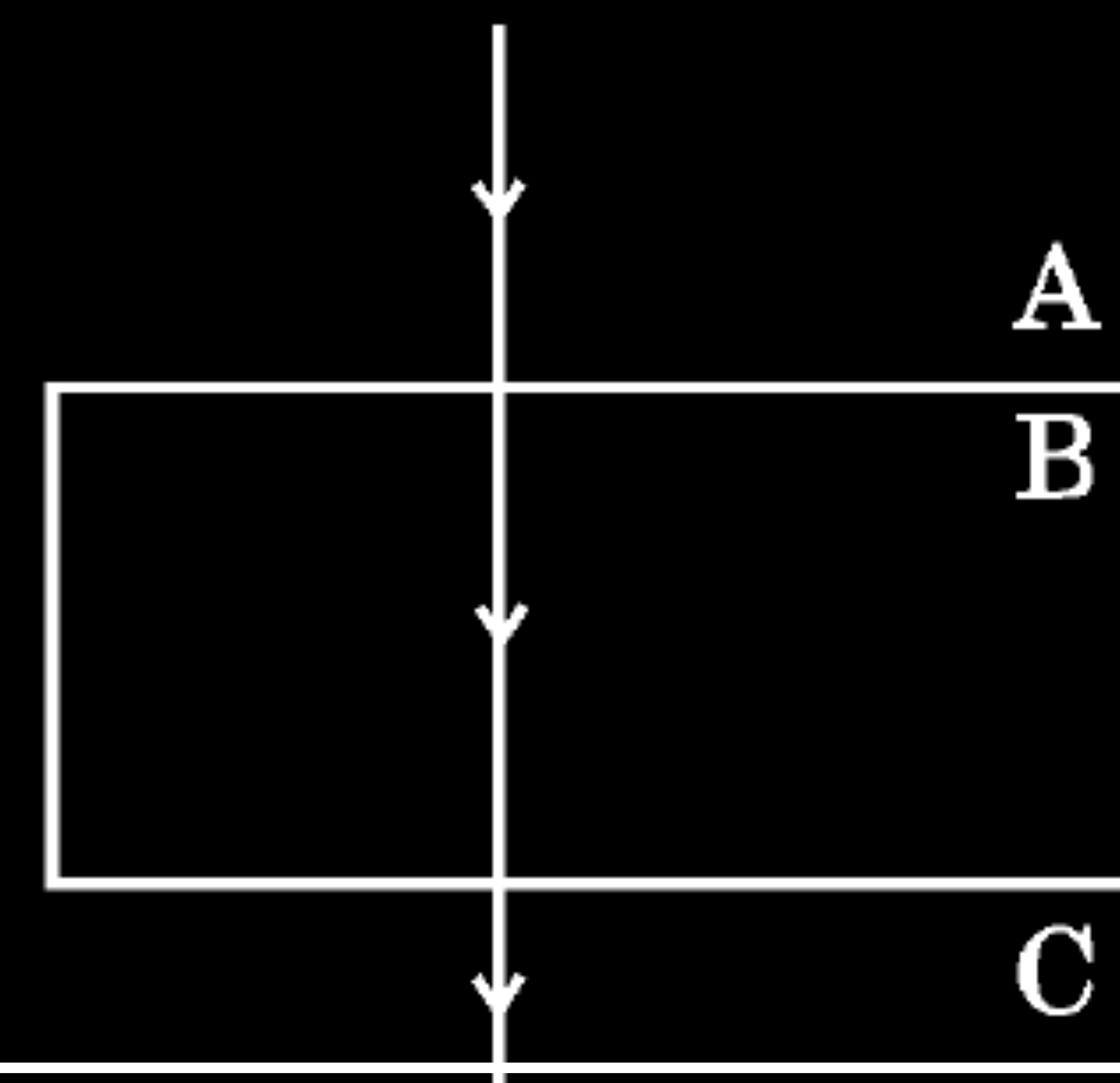
Since the ray of light does not bend while passing from A to B the refractive indices of A and B are same and since it bends away from the normal while passing from B to C the refractive index of C is less than that of A and B. / Refractive index of a medium is inversely proportional to the speed of light in that medium.

1

(b) (i) Oblique Incidence



(ii) Normal Incidence



1

1

- (B) Analyse the following observation table showing variation of image distance (v) with object distance (u) in case of a convex lens and answer the questions that follow without doing any calculations :

Observation Number	Object distance (u) in cm	Image distance (v) in cm
1	-15	-60
2	-25	+100
3	-30	+60
4	-40	+40
5	-60	+30
6	-100	+25

- (a) Determine the focal length of the lens. Give reason for your answer.
- (b) Find magnification of the image formed in Observation No. 3.
- (c) The numerical value of magnifications in cases of observation 1 and 2 is same. List two differences in the images formed in these two cases.

(B) (a) $2f = 40 \text{ cm}$ $\Rightarrow f = 20 \text{ cm}$	1
Reason: When an object is placed at $2f$ (-40 cm) of a convex lens its real image formed at $2f$ ($+40 \text{ cm}$) on the other side of the lens.	1
(b) $m = \frac{v}{u} = \frac{+60 \text{ cm}}{-30 \text{ cm}} = -2$	1
(c) Observation No.1- image is virtual and erect	1
Observation No.2 - image is real and inverted	1

23. When an incident ray of light enters a medium 'X' from air, it bends towards the normal.

Infer about the following with respect to air :

- (a) Optical density of medium 'X'.
- (b) Speed of light through medium 'X'

Give reason for your answer in each case.

23. When an incident ray of light enters a medium 'X' from air, it bends towards the normal.

Infer about the following with respect to air :

- (a) Optical density of medium 'X'.
- (b) Speed of light through medium 'X'

Give reason for your answer in each case.

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(a) Optical density of X is more than optical density of air because the ray coming from air bends towards the normal as it enters the medium X.	$\frac{1}{2} + \frac{1}{2}$
(b) Speed of light through medium X is less than the speed of light through air because X is optically denser than air.	$\frac{1}{2} + \frac{1}{2}$

24. The values of absolute refractive indices of kerosene and water are 1.44 and 1.33 respectively. Compare the two media on the basis of their

- (a) optical density
- (b) mass density
- (c) relative speed of propagation of light.

What do you infer on the basis of above comparisons ?

24. The values of absolute refractive indices of kerosene and water are 1.44 and 1.33 respectively. Compare the two media on the basis of their

- (a) optical density
- (b) mass density
- (c) relative speed of propagation of light.

What do you infer on the basis of above comparisons ?

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Kerosene (1.44)	Water (1.33)	
(a) Higher optical density	Lower optical density	$\frac{1}{2}$
(b) Lower mass density	Higher mass density	$\frac{1}{2}$
(c) Lower speed of light	Higher speed of light	$\frac{1}{2}$
•Inference: Although, kerosene is optically denser than water but its mass density is less than water/ An optically denser medium may not possess greater mass density.		$\frac{1}{2}$

13. The image formed by a concave mirror of focal length 50 cm is real and of magnification -1 . In this case the distance between the object from its own image is :

(A) 50 cm

(B) 100 cm

(C) 200 cm

(D) Zero

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13. The image formed by a concave mirror of focal length 50 cm is real and of magnification -1 . In this case the distance between the object from its own image is :

(A) 50 cm

(B) 100 cm

(C) 200 cm

(D) Zero

CBSE 2024 C

- 22.** Draw a labelled ray diagram to show the formation of image due to a concave mirror when an object is placed between its pole and principal focus.

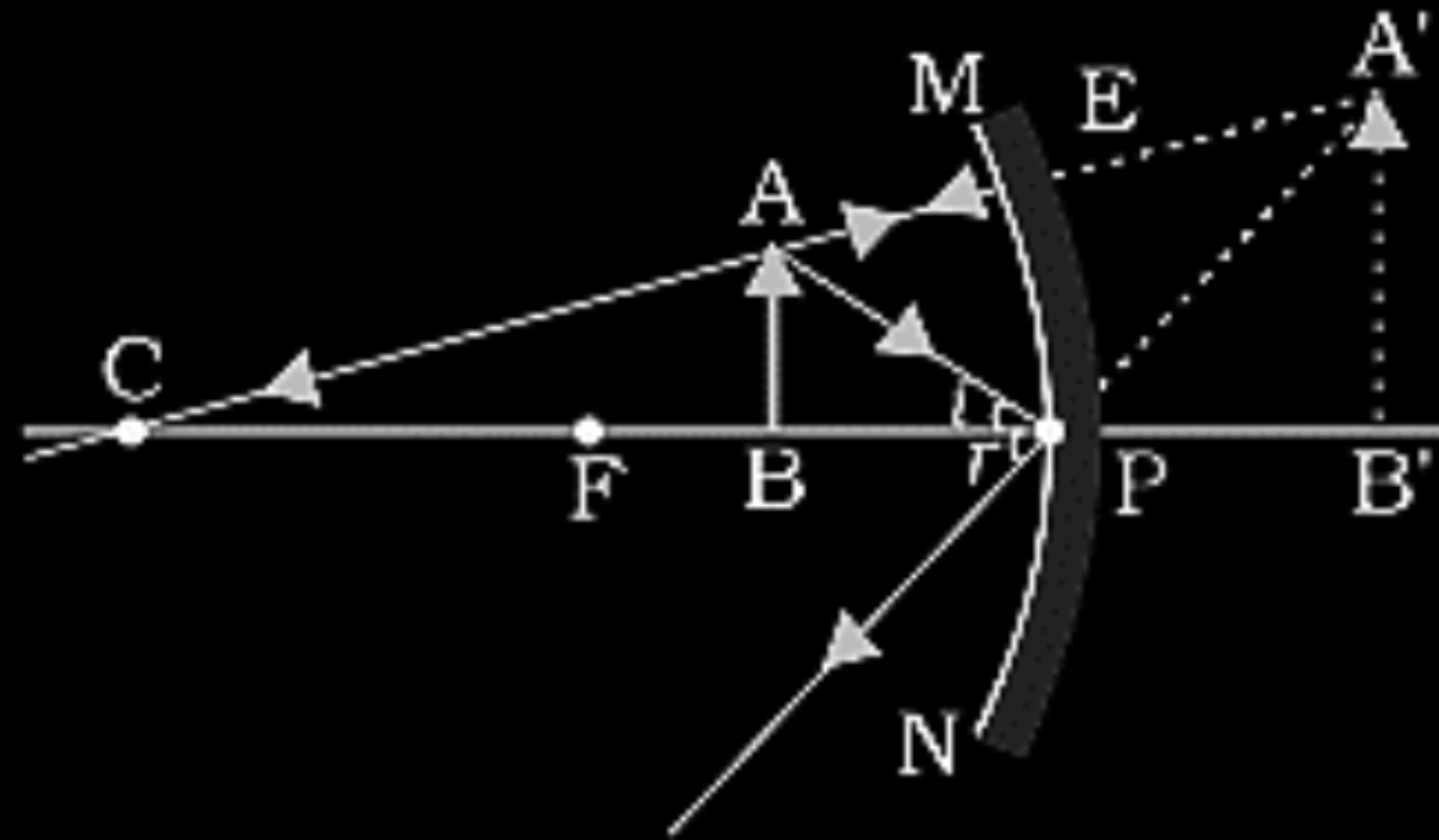
2

CBSE 2024 C

22. Draw a labelled ray diagram to show the formation of image due to a concave mirror when an object is placed between its pole and principal focus.

2

CBSE 2024 C



[P : Pole, F : Principal Focus, AB : Object, A'B' : Image]

[Diagram from NCERT fig 9.7(f) page 140]

Ray diagram

1

Labelling of pole and principal focus

$\frac{1}{2}$

Direction of rays

$\frac{1}{2}$

39. In our laboratories we use compound microscopes to see the magnified image of a microscopic object. A compound microscope is made up of two lenses. The lens nearest to the object to be viewed, called objective lens, forms real, inverted and magnified image of the object. This image serves as an object for the second lens called eyepiece. The eyepiece forms virtual, erect and magnified image of its object. Thus, the resultant image formed by a microscope is virtual, inverted and magnified with respect to the microscopic object viewed.

(a) The image of an object formed by a convex lens of focal length 2 cm is real, inverted and magnified. What is the range of object distance in this case ?

1

(b) The image of an object formed by a convex lens of focal length 6 cm is virtual, erect and magnified. What is the range of object distance in this case ?

1

39. In our laboratories we use compound microscopes to see the magnified image of a microscopic object. A compound microscope is made up of two lenses. The lens nearest to the object to be viewed, called objective lens, forms real, inverted and magnified image of the object. This image serves as an object for the second lens called eyepiece. The eyepiece forms virtual, erect and magnified image of its object. Thus, the resultant image formed by a microscope is virtual, inverted and magnified with respect to the microscopic object viewed.

(a) The image of an object formed by a convex lens of focal length 2 cm is real, inverted and magnified. What is the range of object distance in this case ? 1

(b) The image of an object formed by a convex lens of focal length 6 cm is virtual, erect and magnified. What is the range of object distance in this case ? 1

CBSE 2024 C	
(a) More than 2cm and less than 4cm from the optical centre of the lens	1
(b) Less than 6cm from the optical centre of the lens	1

- (c) (i) An object is placed at a distance of 12 cm from the optical centre of a convex lens of focal length 18 cm. Draw a labelled ray diagram to show the formation of image in this case.

2

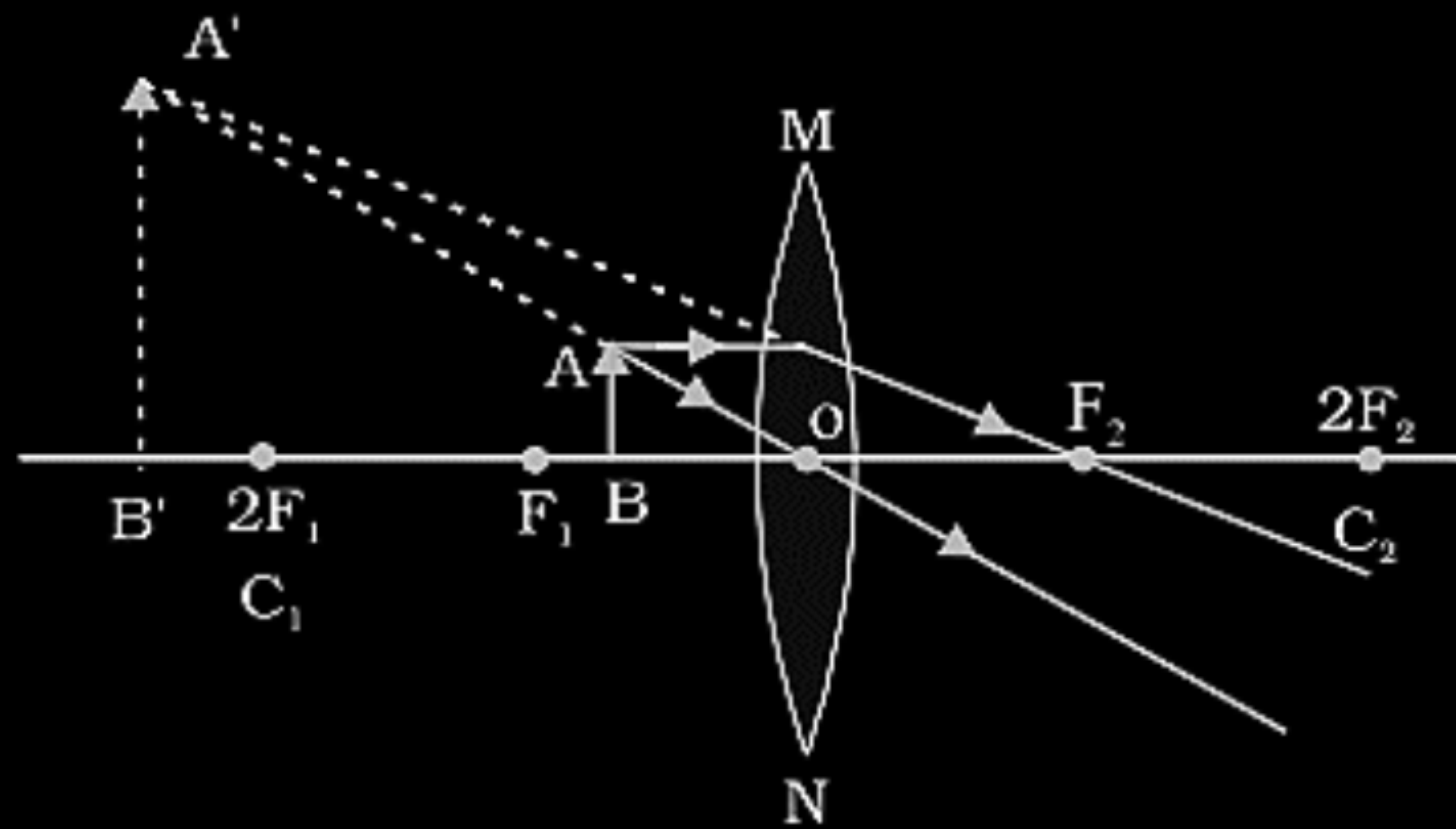
OR

- (c) (ii) The image formed by a convex lens is real, inverted and of the same size as the object. If the distance between the objects and the image is 60 cm, determine the focal length of the lens. Give justification for your answer.

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2

(c) (i)



OB = 12 cm
OF = 18 cm
A'B' : Image

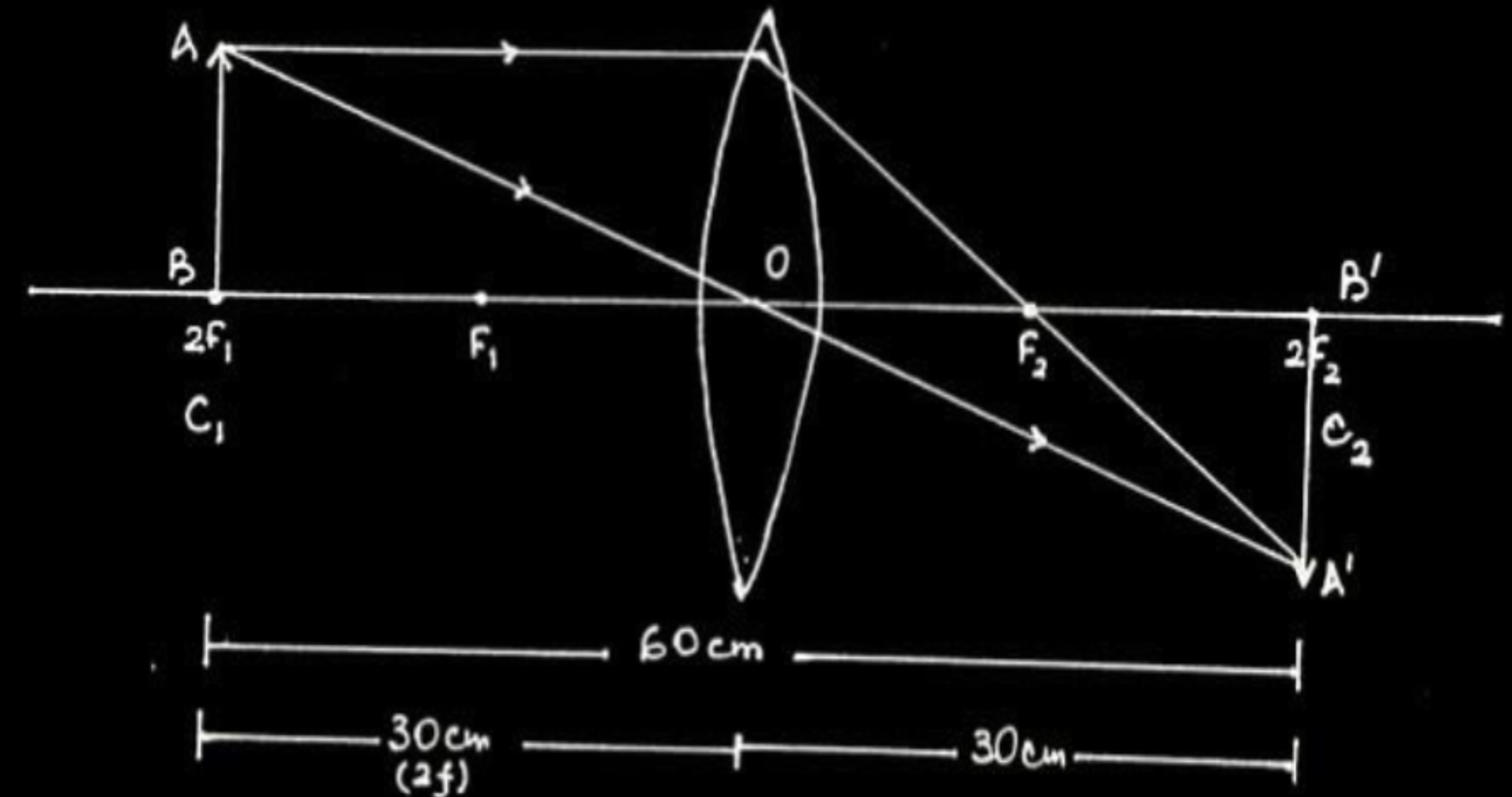
[Fig. 9.16 (f) page 155 NCERT]

(Deduct ½ mark if direction of rays is not shown)

2

(c) (ii)

As per the information given in the question, the lens lies at the centre of the object placed and image formed.



The position of the object is at 2F₁ therefore object distance = image distance = 2f

$$\therefore 2f + 2f = 60 \text{ cm}$$

$$4f = 60$$

$$f = 15 \text{ cm}$$

$\therefore$ The focal length of the lens = 15 cm

Note : Do not deduct marks if diagram is not drawn.

1

1

- 31.** (a) A student has focussed the image of an object of height 5 cm on a screen using a convex lens of focal length 20 cm. If the object distance is 30 cm find, using lens formula, the (i) image distance and (ii) height of the image produced. CBSE 2024 C 3

31. (a) A student has focussed the image of an object of height 5 cm on a screen using a convex lens of focal length 20 cm. If the object distance is 30 cm find, using lens formula, the (i) image distance and (ii) height of the image produced. CBSE 2024 C 3

(a)	
(i)	
Here $h = 5$ cm; $f = + 20$ cm; $u = - 30$ cm	$\frac{1}{2}$
Lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	$\frac{1}{2}$
$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$	
$\frac{1}{v} + \frac{1}{30} = \frac{1}{20}$	
$v = + 60$ cm, image distance is 60 cm from the lens.	1
(ii)	
$h' = \frac{v}{u} \times h$	$\frac{1}{2}$
$= \frac{+60 \text{ cm}}{-30 \text{ cm}} \times 5 \text{ cm} = -10 \text{ cm}$, height of the image is 10 cm.	$\frac{1}{2}$

- (b) Define power of a lens. The focal length of a lens is + 25 cm. Determine its nature (converging/diverging) and calculate its power. If an object is placed at a distance of 50 cm from the optical centre of this lens, list two properties of the image formed in this case.

- (b) Define power of a lens. The focal length of a lens is + 25 cm. Determine its nature (converging/diverging) and calculate its power. If an object is placed at a distance of 50 cm from the optical centre of this lens, list two properties of the image formed in this case.

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(b) • Converging or diverging ability of a lens / Reciprocal of focal length of a lens in metre / $P = \frac{1}{f(m)}$ • ➤ Converging ➤ $P = \frac{100}{+25}$ $= +4D$ • Same size • Real • Inverted (any two)	1 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$
--	--

39. Under the guidance of his teacher on a sunny day, a student took a concave mirror of large aperture in his hand and directed its reflecting surface towards the sun. After that he directed the light reflected by the concave mirror on a thick sheet of white paper held close to the mirror. He observed a bright circular spot of light on the paper. Then he moved the mirror back and forth gradually until he got the brightest and smallest spot of light on the paper. The teacher said this sharp spot of light is, in fact, the real and inverted image of the sun and the distance between the mirror and this image is the focal length of this concave mirror.

- (a) Define the term principal focus of a concave mirror. 1
- (b) If the distance between the pole and focus of a concave mirror is 12 cm, what is the radius of curvature of the mirror ? 1

39. Under the guidance of his teacher on a sunny day, a student took a concave mirror of large aperture in his hand and directed its reflecting surface towards the sun. After that he directed the light reflected by the concave mirror on a thick sheet of white paper held close to the mirror. He observed a bright circular spot of light on the paper. Then he moved the mirror back and forth gradually until he got the brightest and smallest spot of light on the paper. The teacher said this sharp spot of light is, in fact, the real and inverted image of the sun and the distance between the mirror and this image is the focal length of this concave mirror.

- (a) Define the term principal focus of a concave mirror. 1
- (b) If the distance between the pole and focus of a concave mirror is 12 cm, what is the radius of curvature of the mirror ? 1

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(a) It is a point on the principal axis of the mirror where rays incident, parallel to principal axis will converge after reflection.	1
(b) $R = 2f = 2 \times 12 \text{ cm} = 24 \text{ cm}$	1

- (c) (i) List two properties of the image formed due to a concave mirror of focal length 15 cm in each of the following cases :

2

(1) object distance 20 cm

(2) object distance 10 cm

OR

- (c) (ii) A 3 cm long candle flame is placed in front of a concave mirror. If the distance between the pole and the candle flame is 40 cm and the image is formed at the same place where the candle flame is located, find the

2

- (c) (i) List two properties of the image formed due to a concave mirror of focal length 15 cm in each of the following cases :

2

(1) object distance 20 cm

(2) object distance 10 cm

OR

- (c) (ii) A 3 cm long candle flame is placed in front of a concave mirror. If the distance between the pole and the candle flame is 40 cm and the image is formed at the same place where the candle flame is located, find the

2

CBSE 2024 C

(c) (i)

(1) Image is

- real
- inverted and
- magnified

(Any two)

(3) Image is

- virtual
- erect and
- magnified

(Any two)

(c)

(ii)

(1) In this case, the object is at C, therefore $R = 40$ cm

$$R = 2f$$

$$\therefore \text{Focal length } (f) = \frac{40}{2} \text{ cm} = 20 \text{ cm}$$

(2) Magnification(m) = -1 (Image of same size and inverted)

$\frac{1}{2}$

$\frac{1}{2}$

1

12. At what distance from a convex lens should an object be placed to get an image of the same size as that of the object on a screen ?

1

- (a) Beyond twice the focal length of the lens.
- (b) At the principal focus of the lens.
- (c) At twice the focal length of the lens.
- (d) Between the optical centre of the lens and its principal focus.

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12. At what distance from a convex lens should an object be placed to get an image of the same size as that of the object on a screen ? 1
- (a) Beyond twice the focal length of the lens.
 - (b) At the principal focus of the lens.
 - (c) At twice the focal length of the lens.
 - (d) Between the optical centre of the lens and its principal focus.

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(c) /At twice the focal length of the lens

24. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror.

2

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24. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position of the image formed by the mirror.

2

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$$u = -10 \text{ cm}; f = +15 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{15} = \frac{1}{v} + \frac{1}{-10 \text{ cm}}$$

$$\frac{1}{v} = \frac{1}{15 \text{ cm}} + \frac{1}{10 \text{ cm}}$$

$$v = +6 \text{ cm}$$

Image is formed behind the mirror.

$\frac{1}{2}$

$\frac{1}{2}$

1

39. Study the data given below showing the focal length of three concave mirrors A, B and C and the respective distances of objects placed in front of the mirrors :

Case	Mirror	Focal Length (cm)	Object Distance (cm)
1	A	20	45
2	B	15	30
3	C	30	20

- (i) In which one of the above cases the mirror will form a diminished image of the object ? Justify your answer. 1
- (ii) List two properties of the image formed in case 2. 1
- (iii) (A) What is the nature and size of the image formed by mirror C ?
Draw ray diagram to justify your answer. 2

OR

- (iii) (B) An object is placed at a distance of 18 cm from the pole of a concave mirror of focal length 12 cm. Find the position of the image formed in this case. 2

(i)

- Mirror A.
- as the object is placed beyond the centre of curvature of the mirror.

$\frac{1}{2}$

$\frac{1}{2}$

(ii) Same size/ Real / Inverted

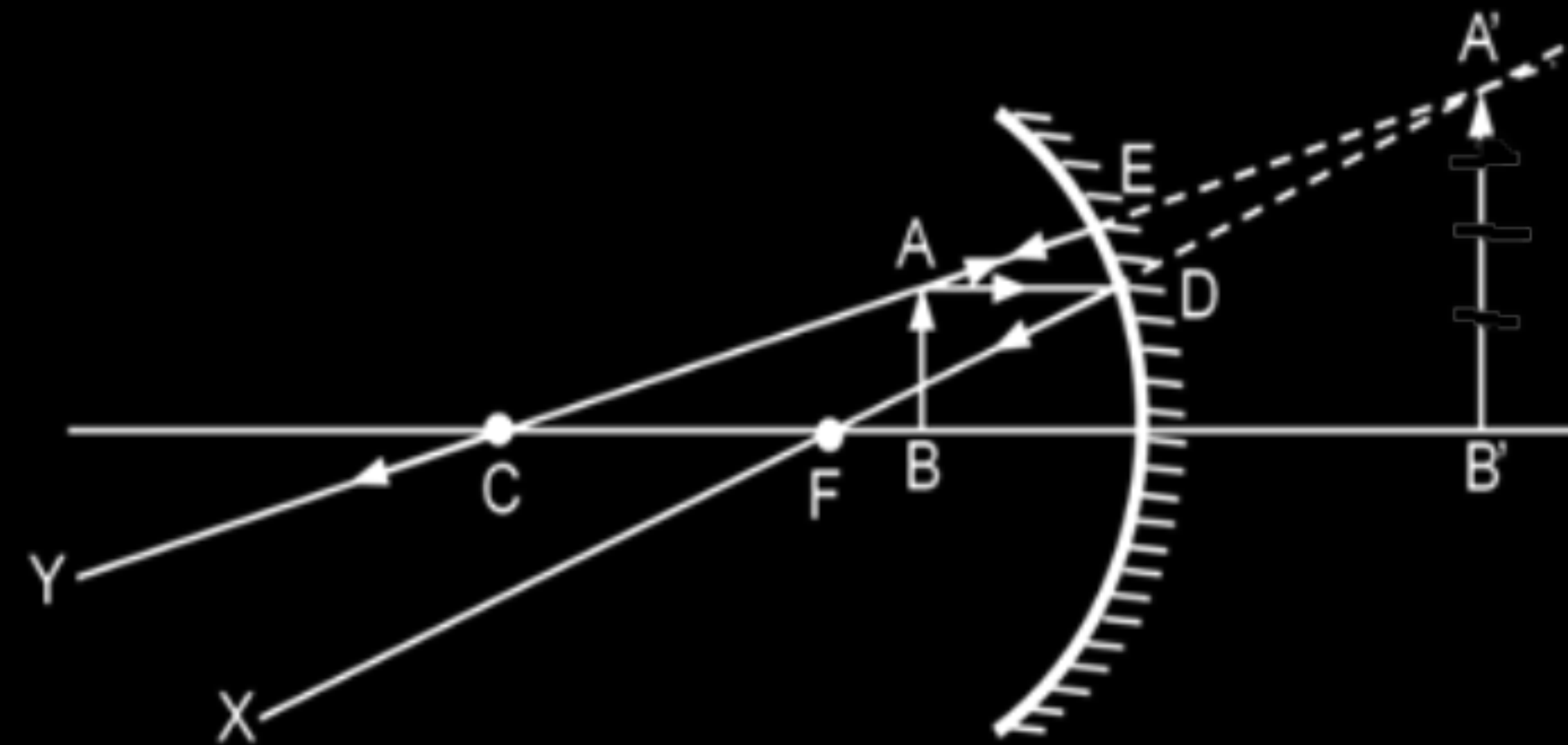
$\frac{1}{2} + \frac{1}{2}$

(Any two)

(iii) (A) Nature-Virtual and erect
Size-magnified

$\frac{1}{2}$

$\frac{1}{2}$



1

(Deduct $\frac{1}{2}$ mark if direction of rays are not marked)

$\frac{1}{2}$

(iii) (B) Here $f = -12$ cm, $u = -18$ cm, $v = ?$

Mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ or $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$$\frac{1}{v} = \frac{1}{-12} - \frac{1}{-18}$$

$\frac{1}{2}$

1

$$v = -36\text{cm}$$

In front of the mirror at a distance of 36 cm from the pole of the mirror.

- 23.** A ray of light falls making an angle of incidence θ on the surface of a glass slab. Draw a labelled ray diagram to show its path. Also mark lateral displacement on it.

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23. A ray of light falls making an angle of incidence θ on the surface of a glass slab. Draw a labelled ray diagram to show its path. Also mark lateral displacement on it.

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2

	1½
Lateral displacement labelling	½

15. The speed of light in vacuum is 3×10^8 m/s. If the speed of light in a medium is 2.25×10^8 m/s, the absolute refractive index of the medium is : **1**

(A) $\frac{7}{6}$

(B) $\frac{5}{4}$

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

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15. The speed of light in vacuum is 3×10^8 m/s. If the speed of light in a medium is 2.25×10^8 m/s, the absolute refractive index of the medium is : **1**

(A) $\frac{7}{6}$

(B) $\frac{5}{4}$

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

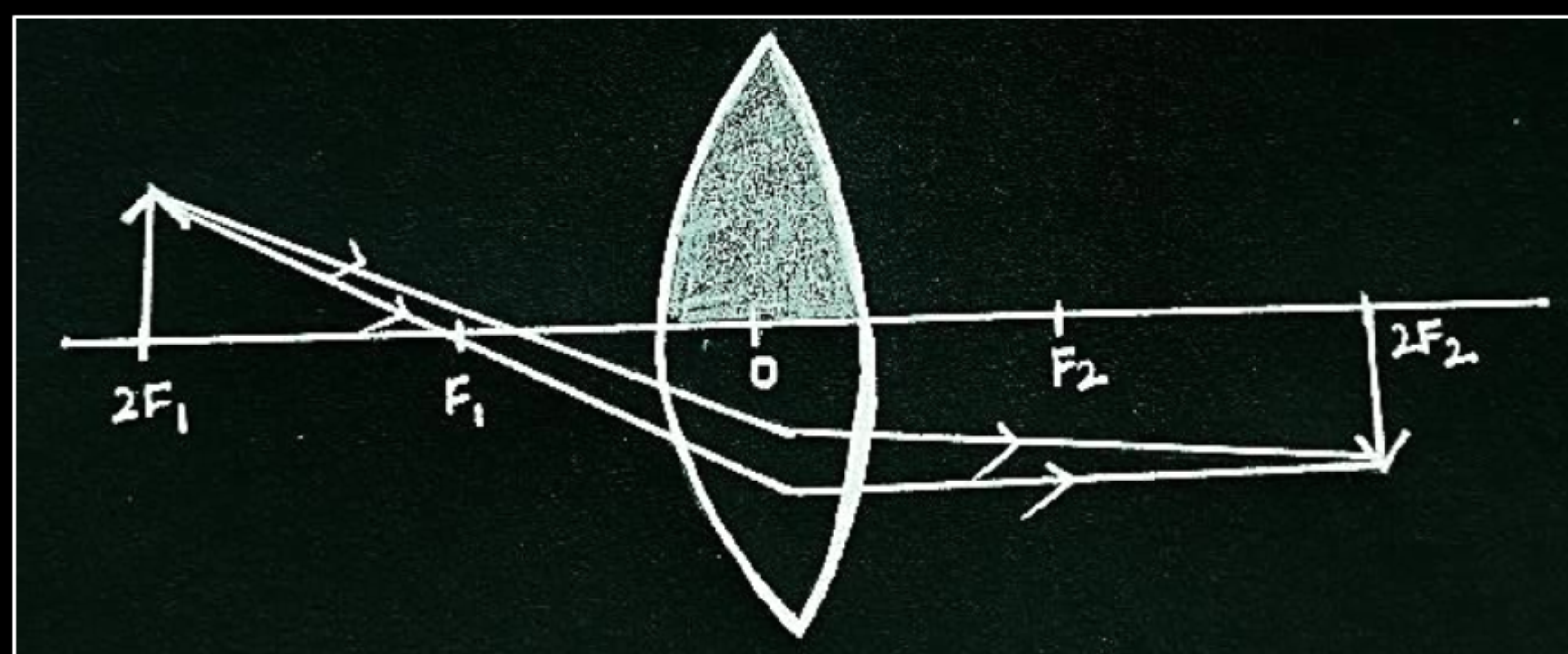
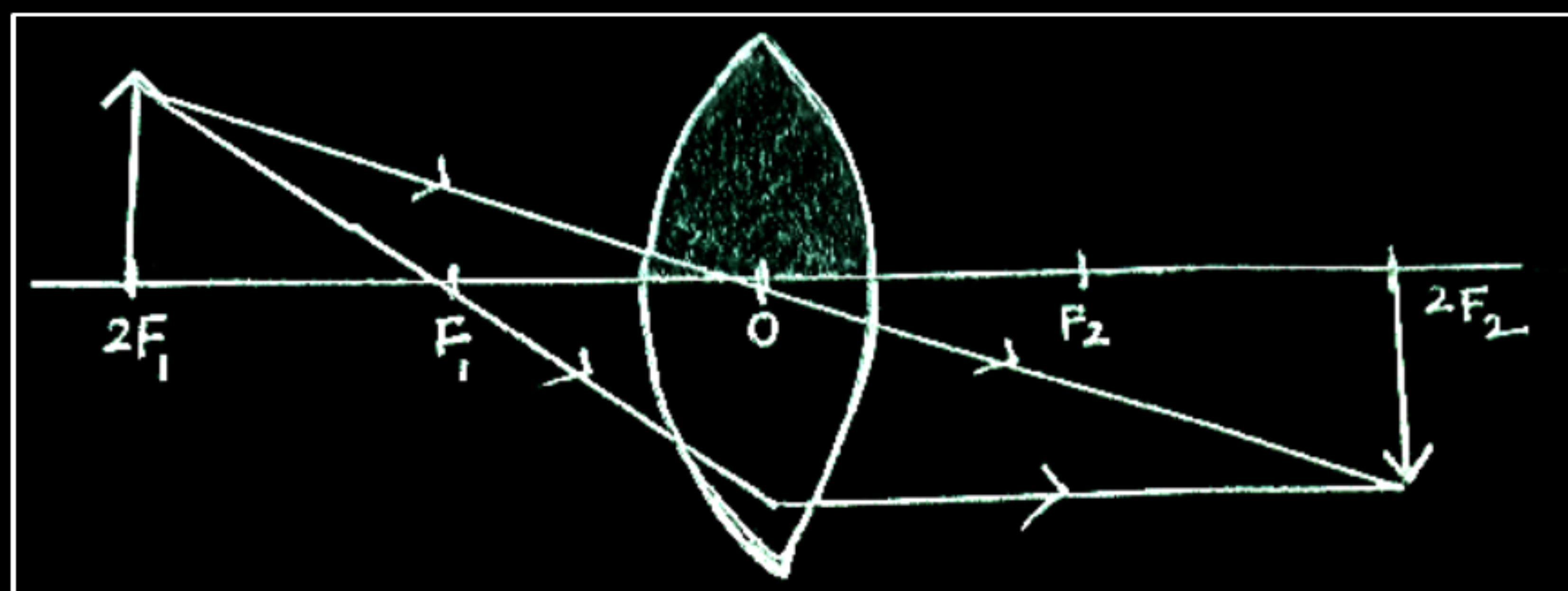
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(C)/ 4/3

35. (a) Upper half of a convex lens is covered with a black paper. Draw a ray diagram to show the formation of image of an object placed at a distance of $2F$ from such a lens. Mention the position and nature of the image formed. State the observable difference in the image obtained if the lens is uncovered. Give reason to justify your answer. 5
- (b) An object is placed at a distance of 30 cm from the optical centre of a concave lens of focal length 15 cm. Use lens formula to determine the distance of the image from the optical centre of the lens.

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(a)



1

Note: Any one of the above drawn ray diagrams should be marked.

When the upper half of lens is covered:

- Position of image: at $2F$ on the other side of the lens
- Nature of image: Real and inverted
- **Observable difference in the image, if the lens is uncovered**
The brightness of the image will increase
- **Reason:** More number of rays will pass through the lens to form the image.

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

(b)

Here $u = -30$ cm, $f = -15$ cm, $v = ?$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$= \frac{1}{-15} + \frac{1}{-30}$$

$$v = -10 \text{ cm}$$

 $\frac{1}{2}$ $\frac{1}{2}$

1

24. (a) State two laws of refraction of light.

2

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24. (a) State two laws of refraction of light.

2

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(a) Laws of Refraction of light :

(i) The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane.

1

(ii) The ratio of the sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for the given pair of media.

1

Note:

If a student writes $\frac{\sin i}{\sin r} = \text{constant}$ instead of statement, award ½ mark only)

- (b) Define the term absolute refractive index of a medium. A ray of light enters from vacuum to glass of absolute refractive index 1.5. Find the speed of light in glass. The speed of light in vacuum is 3×10^8 m/s. **2**

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- (b) Define the term absolute refractive index of a medium. A ray of light enters from vacuum to glass of absolute refractive index 1.5. Find the speed of light in glass. The speed of light in vacuum is 3×10^8 m/s.

2

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(b) Absolute refractive index of a medium is the ratio of speed of light in vacuum to the speed of light in the medium.

Given:

$$c = 3 \times 10^8 \text{ m/s}; n_m = 1.5; v_m = ?$$

Absolute refractive index of a medium (n_m)

$$= \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}} = \frac{c}{v_m}$$

$$v_m = \frac{c}{n_m} = 2 \times 10^8 \text{ m/s}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

31. (a) An object is placed in front of a concave mirror of focal length 12 cm. If distance of the object from the pole of the mirror is 8 cm, then use mirror formula to determine the position of the image formed. Draw a labelled ray diagram to justify your answer in this case.

3

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31. (a) An object is placed in front of a concave mirror of focal length 12 cm. If distance of the object from the pole of the mirror is 8 cm, then use mirror formula to determine the position of the image formed. Draw a labelled ray diagram to justify your answer in this case.

3

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(a)

Here $f = -12$ cm, $u = -8$ cm, $v = ?$

Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

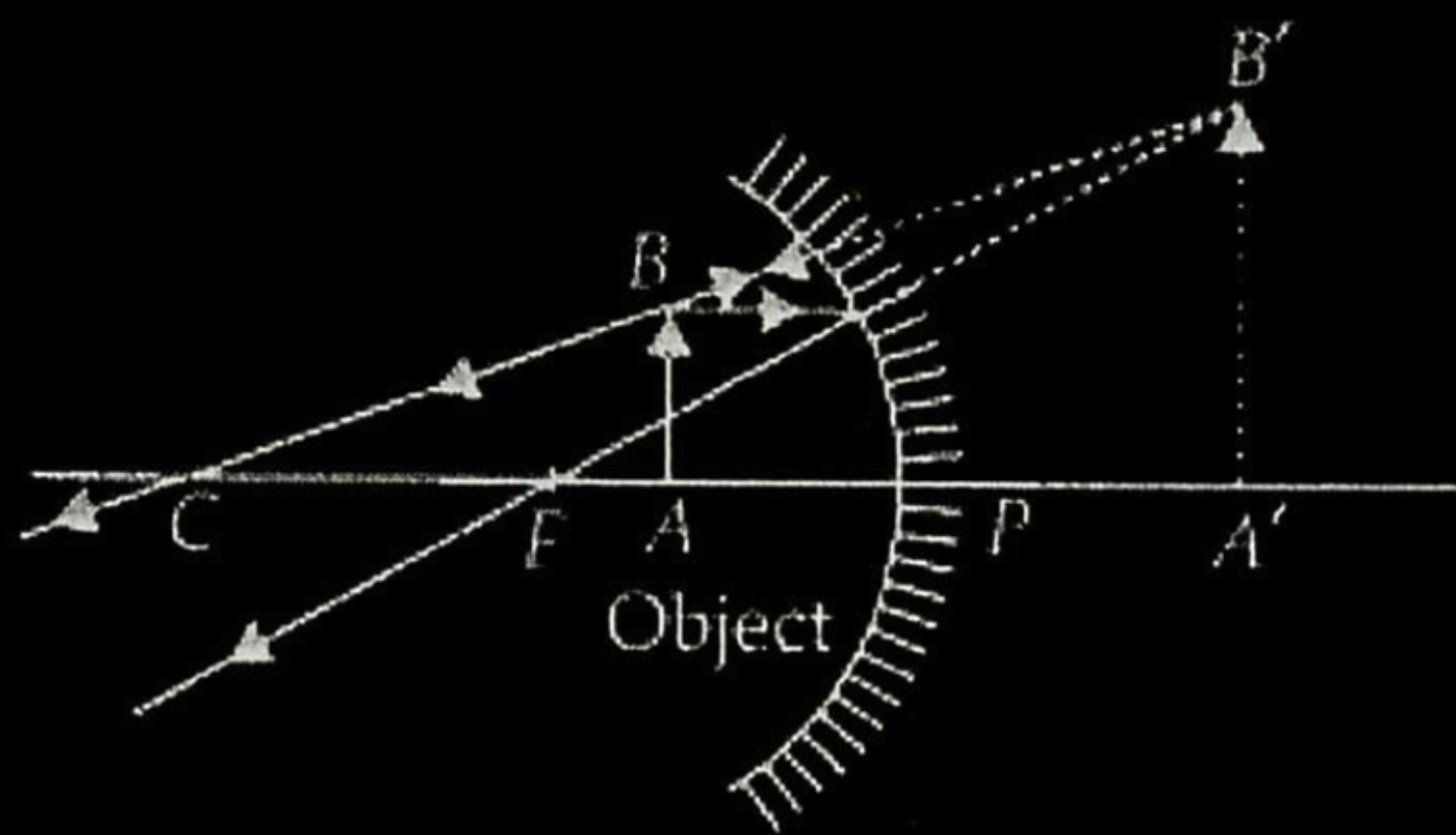
$$\therefore \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-12 \text{ cm}} - \frac{1}{-8 \text{ cm}}$$

$$= + \frac{1}{24 \text{ cm}}$$

$$v = +24 \text{ cm}$$

Image is formed at 24 cm behind the mirror.

1



Note: Deduct $\frac{1}{2}$ mark if direction of rays is not shown.

31. (b) (i) The image of an object formed by a mirror is real, inverted and is of magnification-1. If the image is at a distance of 30 cm from the mirror, where is the object placed ? Give reason to justify your answer.
- (ii) Where would the image be if the object is moved 15 cm towards the mirror ? Draw ray diagram for the new position of the object to justify your answer.

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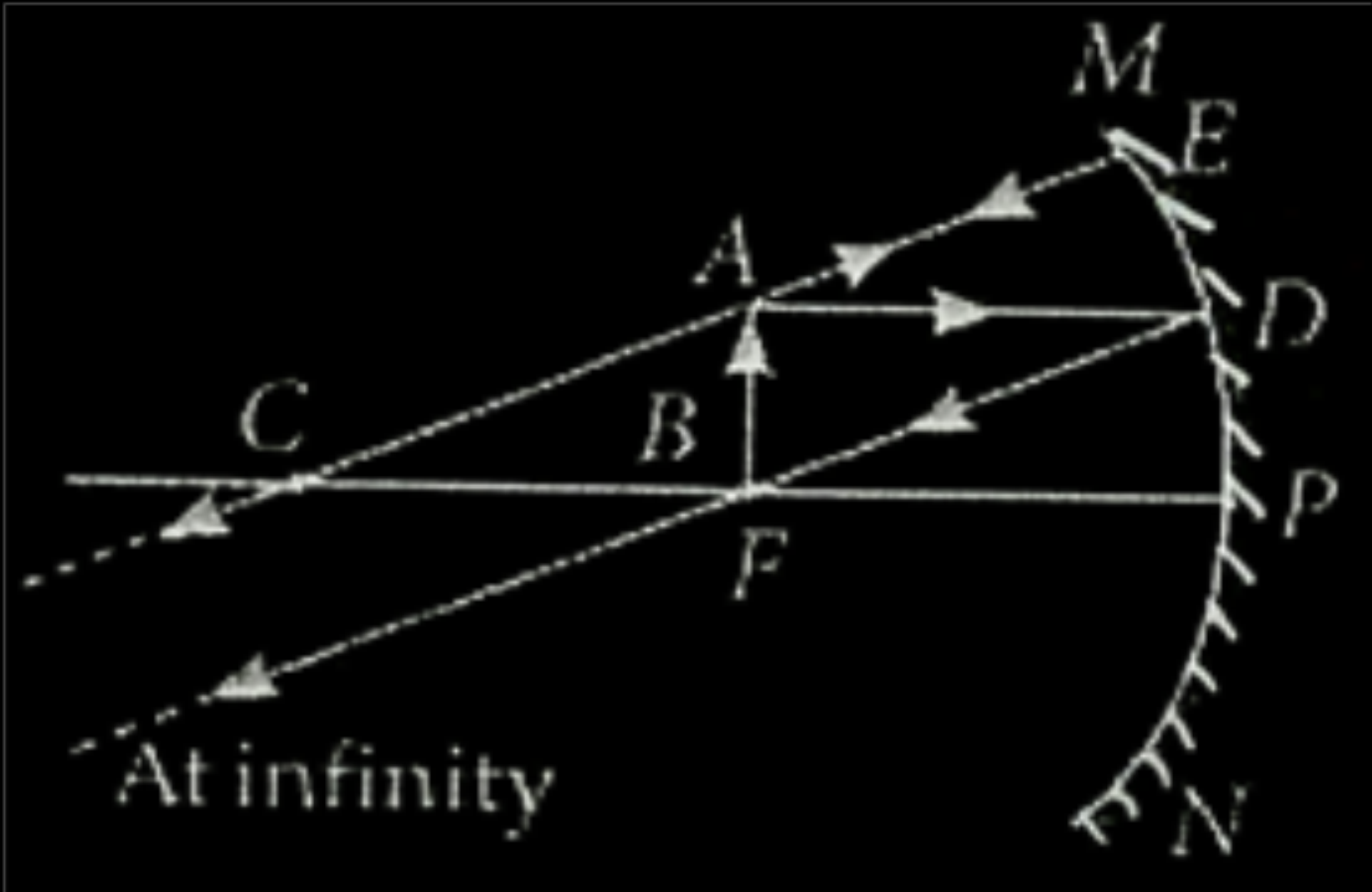
31. (b) (i) The image of an object formed by a mirror is real, inverted and is of magnification-1. If the image is at a distance of 30 cm from the mirror, where is the object placed ? Give reason to justify your answer.
- (ii) Where would the image be if the object is moved 15 cm towards the mirror ? Draw ray diagram for the new position of the object to justify your answer.

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(b)(i) Object is placed at 30 cm from the mirror.
Note: Award marks if expressed in the form of calculations.

Reason: Since magnification is -1, Image distance(v)= object distance(u)

(ii) Image is formed at infinity.



Note: Deduct ½ mark if direction of rays is not shown

36. (a) (i) Draw a ray diagram to show the path of the refracted ray in each of the following cases :

A ray of light incident on a concave lens

- (1) parallel to its principal axis, and
- (2) is directed towards its principal focus.

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- (ii) A 4 cm tall object is placed perpendicular to the principal axis of convex lens of focal length 24 cm. The distance of object from the lens is 16 cm. Find the position and size of image formed.

(1)

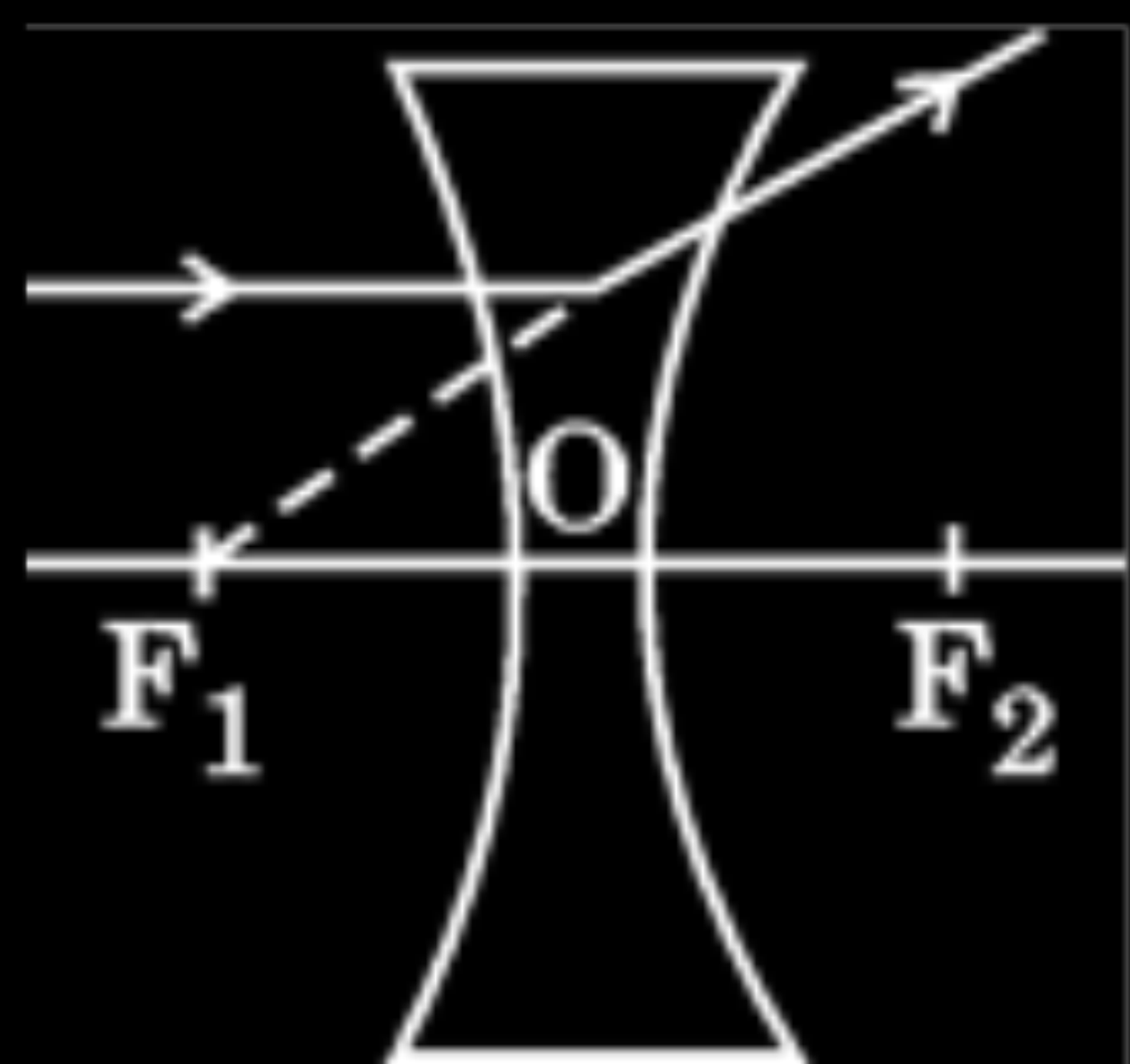


Fig.9.13(b)-Page-153, NCERT.

(2)

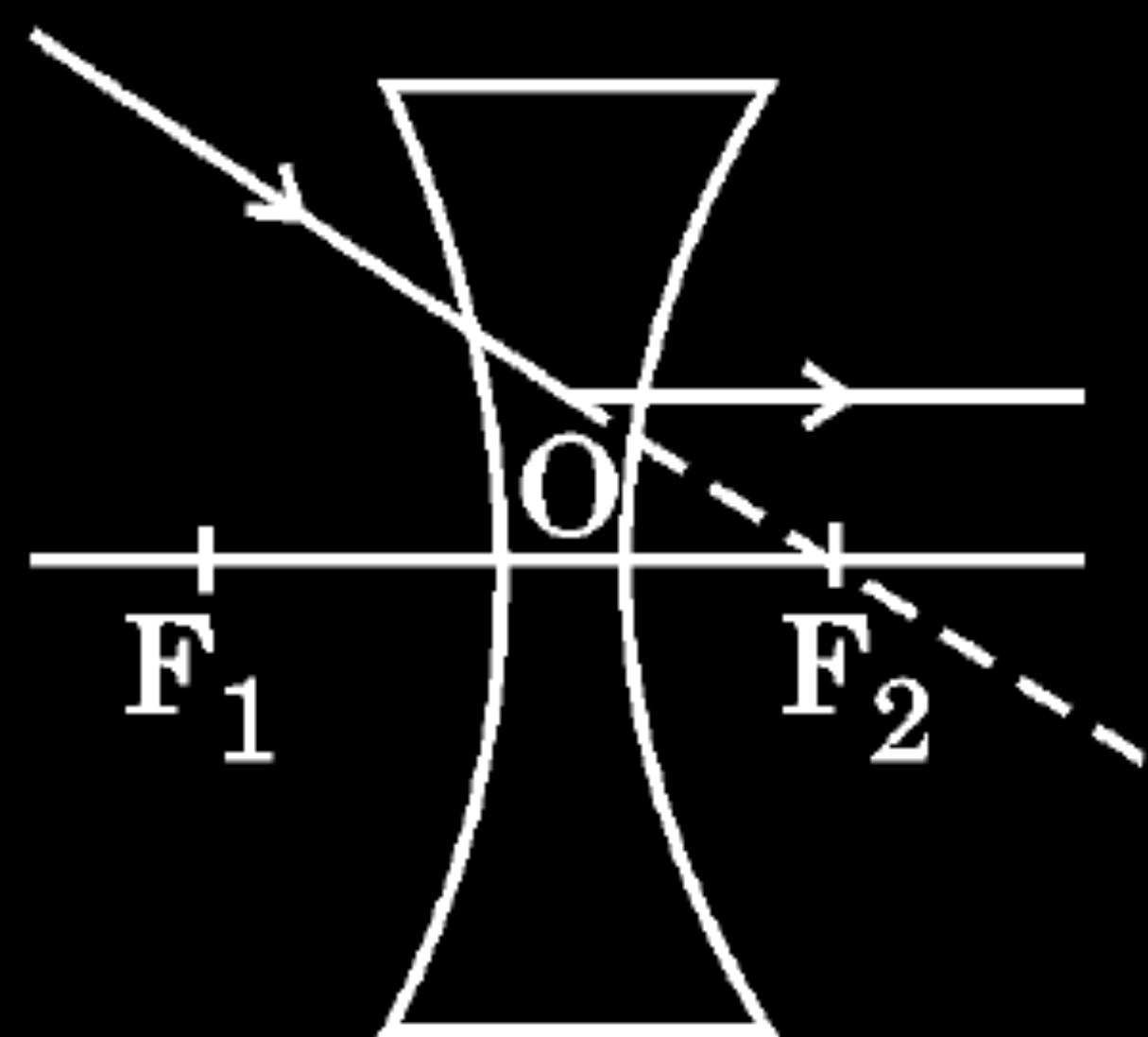


Fig.9.14(b)-Page-154, NCERT.

(Note:- Deduct half mark if directions of rays are not shown)

1

1

(ii) Given $u = -16$ cm, $f = +24$ cm, $h = 4$ cmFormula used $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\therefore \frac{1}{v} - \frac{1}{(-16)} = \frac{1}{+24}$$

$$\frac{1}{v} = \frac{-1}{48}$$

$$v = -48 \text{ cm}$$

Image is formed on the same side as the object

$$m = \frac{h'}{h} = \frac{v}{u}$$

$$\frac{h'}{4} = \frac{-48}{-16}$$

$$h' = 12 \text{ cm}$$

 $\frac{1}{2}$
 $\frac{1}{2}$

1

 $\frac{1}{2}$ $\frac{1}{2}$

36. (b) (i) Draw a ray diagram to show the path of the reflected ray in each of the following cases :

A ray of light incident on a convex mirror

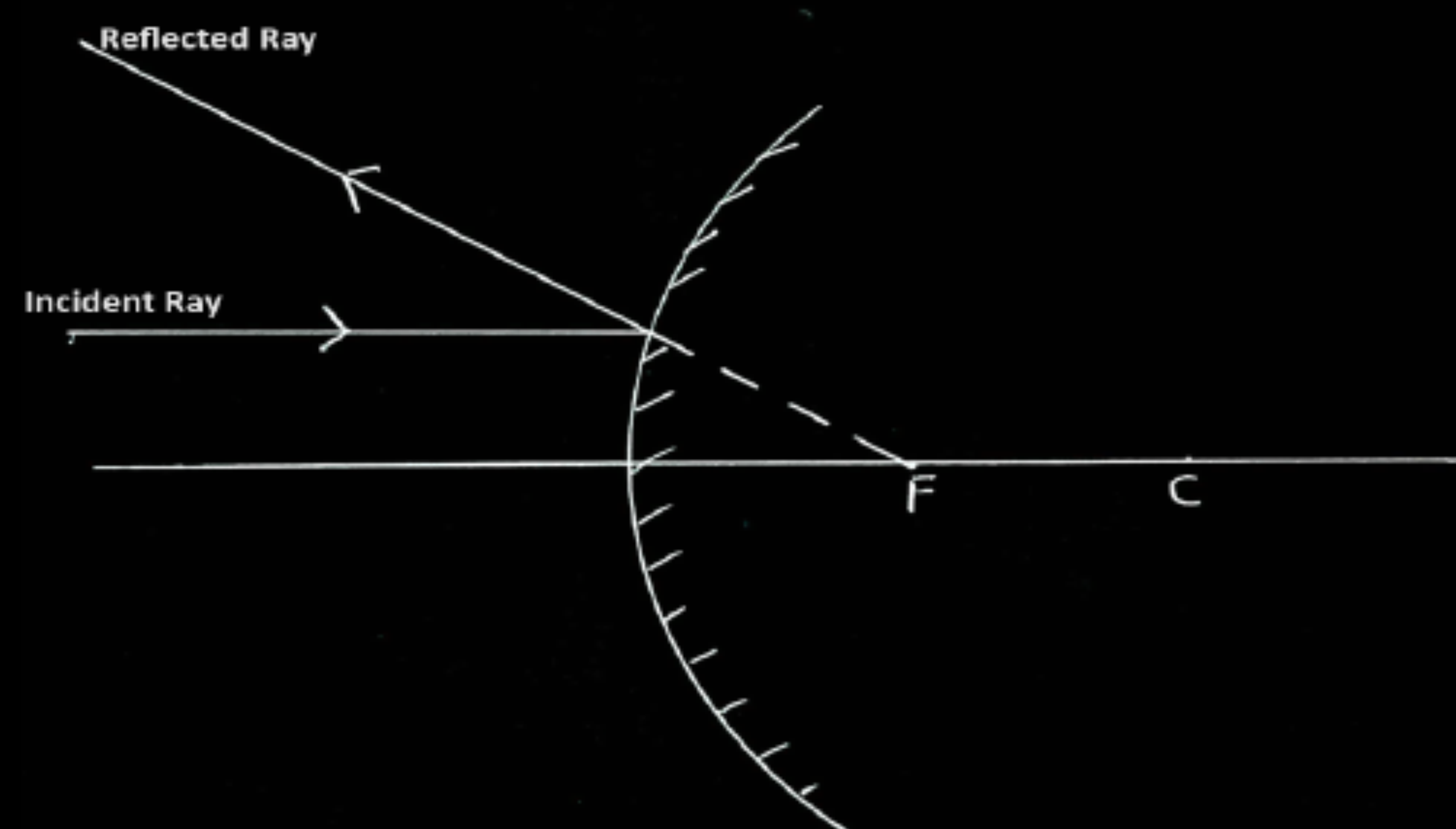
- (1) parallel to its principal axis, and
- (2) is directed towards its principal focus

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- (ii) A 1.5 cm tall candle flame is placed perpendicular to the principal axis of a concave mirror of focal length 12 cm. If the distance of the flame from the pole of the mirror is 18 cm, use mirror formula to determine the position and size of the image formed.

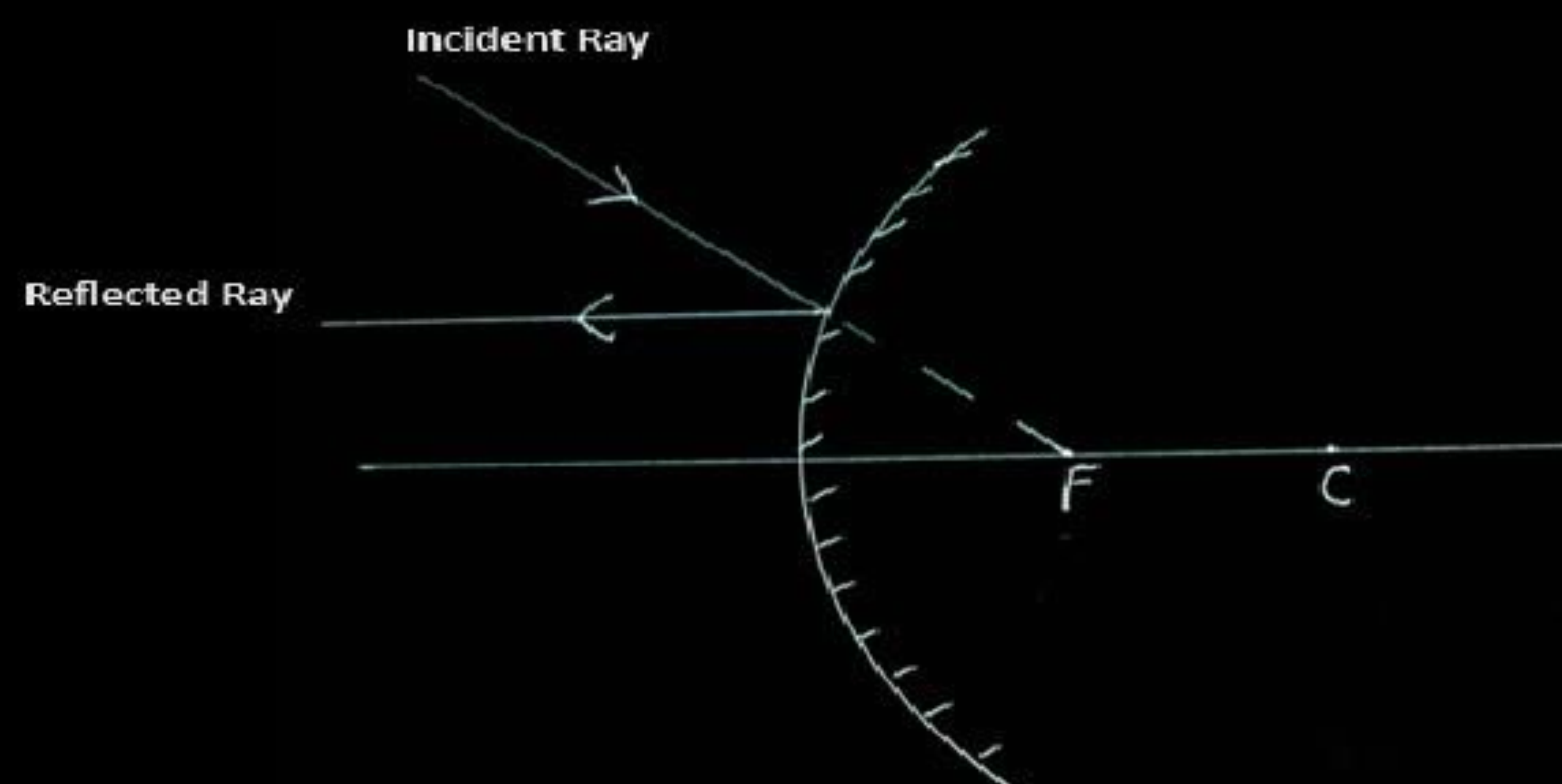
(b) (i)

(1)



1

(2)



1

(Note:- Deduct half mark if directions of rays are not shown)

(ii) Here $f = -12$ cm, $u = -18$ cm, $v = ?$, $h = 1.5$ cm, $h' = ?$

Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\therefore \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= \frac{1}{-12 \text{ cm}} - \frac{1}{-18 \text{ cm}}$$

$$= \frac{-1}{36}$$

$$\therefore v = -36 \text{ cm}$$

$$m = \frac{h'}{h} = -\frac{v}{u}$$

$$\frac{h'}{1.5} = -\frac{(-36)}{(-18)}$$

$$h' = -3.0 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

- 11.** How will the image formed by a convex lens be affected, if the upper half of the lens is wrapped with a black paper ?
- (A) The size of the image formed will be one-half of the size of the image due to complete lens.
 - (B) The image of upper half of the object will not be formed.
 - (C) The brightness of the image will reduce.
 - (D) The lower half of the inverted image will not be formed.

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11. How will the image formed by a convex lens be affected, if the upper half of the lens is wrapped with a black paper ?
- (A) The size of the image formed will be one-half of the size of the image due to complete lens.
 - (B) The image of upper half of the object will not be formed.
 - (C) The brightness of the image will reduce.
 - (D) The lower half of the inverted image will not be formed.

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(C) / The brightness of the image will reduce

39. A highly polished surface such as a mirror reflects most of the light falling on it. In our daily life we use two types of mirrors — plane and spherical. The reflecting surface of a spherical mirrors may be curved inwards or outwards. In concave mirrors, reflection takes place from the inner surface, while in convex mirrors reflection takes place from the outer surface.

- (a) Define the principal axis of a concave mirror. 1
- (b) A ray of light is incident on a concave mirror, parallel to its principal axis. If this ray after reflection from the mirror passes through the principal axis from a point at a distance of 10 cm from the pole of the mirror, find the radius of curvature of the mirror. 1
- (c) (i) An object is placed at a distance of 10 cm from the pole of a convex mirror of focal length 15 cm. Find the position of the image. 2

OR

- (c) (ii) A mirror forms a virtual, erect and diminished image of an object. Identify the type of this mirror. Draw a ray diagram to show the image formation in this case.

(a) It is straight line passing through the pole and centre of curvature of a concave mirror.

(b) Radius of curvature ,R= 20 cm

(c)

(i) $u = -10$ cm, $f = +15$ cm

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

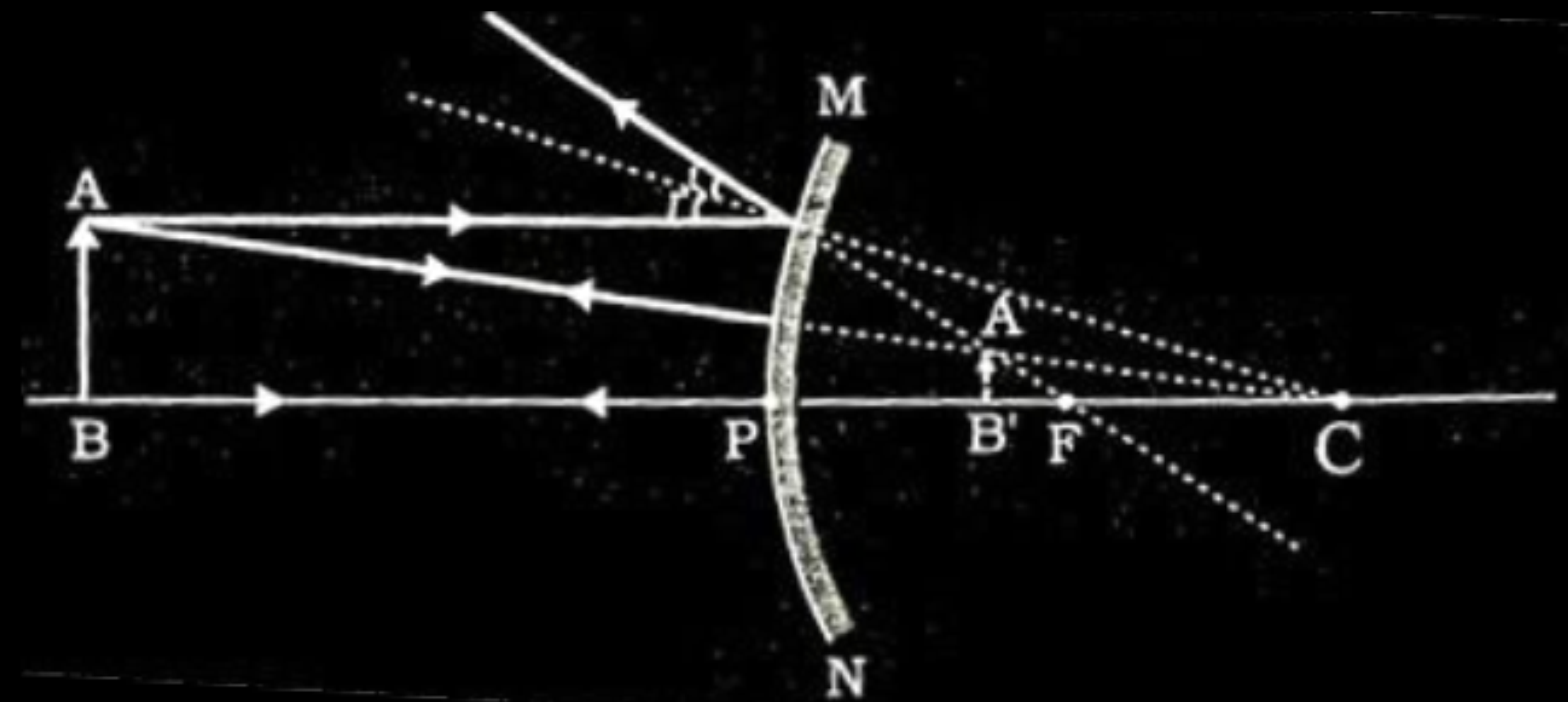
$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} - \frac{1}{-10}$$

$$\frac{1}{v} = \frac{1}{6}$$

$$\Rightarrow v = +6 \text{ cm}$$

OR

(c) (ii) Convex mirror / Diverging mirror



[Note: Deduct ½ mark if direction of rays is not shown]

25. Define power of a lens. Find power of a lens whose focal length is 50 cm. 2

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25. Define power of a lens. Find power of a lens whose focal length is 50 cm. 2

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- Power of a lens is the reciprocal of focal length in metre./ It is the degree of convergence or divergence of light rays achieved by a lens.

1

- $$P = \frac{1}{f} = \frac{100}{50} = 2 \text{ D}$$

1

25. “The linear magnification produced by a spherical mirror is $+3$.” Based on this statement answer the following questions :

2

- (a) What is the type of mirror ?
- (b) Where is the object located ?
- (c) List two properties of the image formed (other than the size/magnification).

25. “The linear magnification produced by a spherical mirror is + 3.” Based on this statement answer the following questions :

2

- (a) What is the type of mirror ?
- (b) Where is the object located ?
- (c) List two properties of the image formed (other than the size/magnification).

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(a) Concave mirror/ Converging mirror	$\frac{1}{2}$
(b) Between pole and focus	$\frac{1}{2}$
(c) • Virtual • Erect • Behind the mirror (Any two)	$\frac{1}{2} + \frac{1}{2}$

11. Absolute refractive index of glass and water is $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is :

(A) $\frac{9}{4} \times 10^8$ m/s

(B) $\frac{5}{2} \times 10^8$ m/s

(C) $\frac{7}{3} \times 10^8$ m/s

(D) $\frac{16}{9} \times 10^8$ m/s

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11. Absolute refractive index of glass and water is $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If the speed of light in glass is 2×10^8 m/s, the speed of light in water is :

(A) $\frac{9}{4} \times 10^8$ m/s

(B) $\frac{5}{2} \times 10^8$ m/s

(C) $\frac{7}{3} \times 10^8$ m/s

(D) $\frac{16}{9} \times 10^8$ m/s

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(A)/9/4 x 10⁸ m/s

- 36.** (a) The variation of image distance (v) with object distance (u) for a convex lens is given in the following observation table. Analyse it and answer the questions that follow :

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S. No.	Object distance (u) cm	Image distance (v) cm
1	-150	$+30$
2	-75	$+37.5$
3	-50	$+50$
4	-37.5	$+75$
5	-30	$+150$
6	-15	$+37.5$

- (i) Without calculation, find the focal length of the convex lens. Justify your answer.
- (ii) Which observation is not correct ? Why ? Draw ray diagram to find the position of the image formed for this position of the object.
- (iii) Find the approximate value of magnification for $u = -30$ cm.

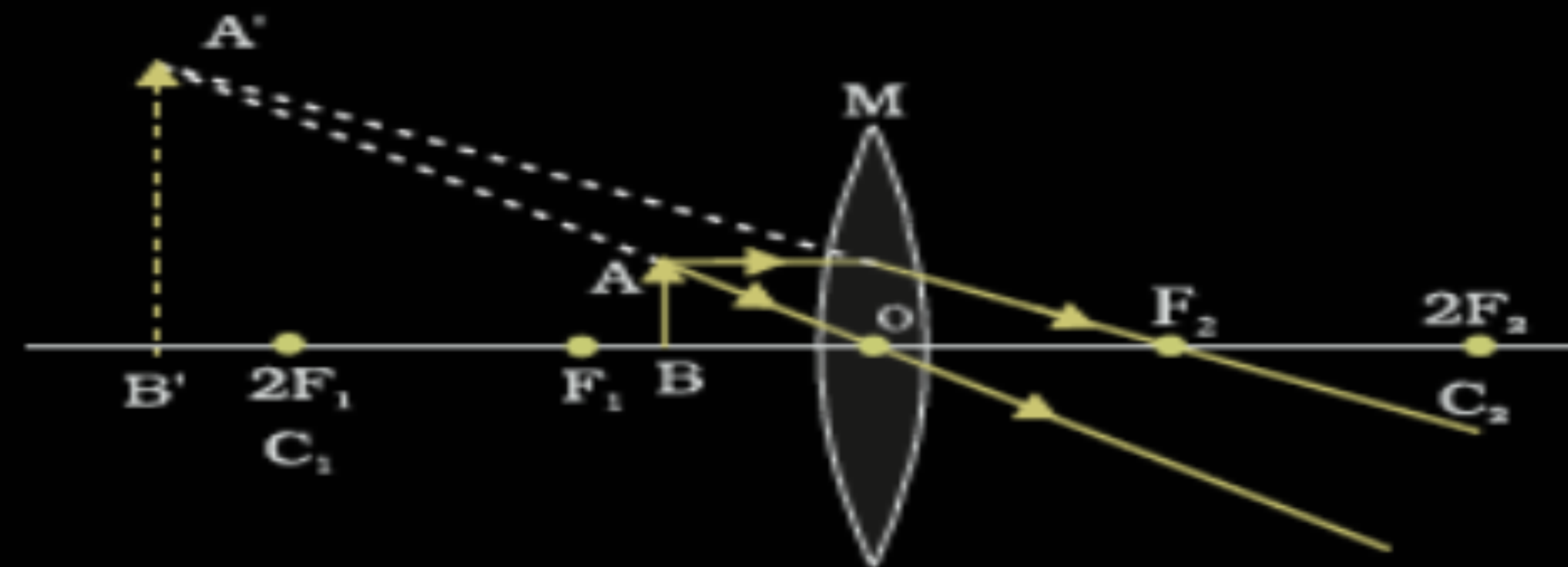
(a)

(i) S. No. 3, $2f$ is 50 cm. $\therefore 2f = 50$ cm, or $f = 25$ cm.

Justification: Object distance(u) and image distance (v) are same so it implies that object is placed at $2F$.

(ii) S. No. 6, is not correct.

Reason: For $u = -15$ cm, sign of v **must** be $-ve$ (as the image is formed on the same side of the lens as the object)



(deduct $\frac{1}{2}$ mark if the direction of the rays are not shown)

(iii) Magnification : $m = \frac{v}{u}$

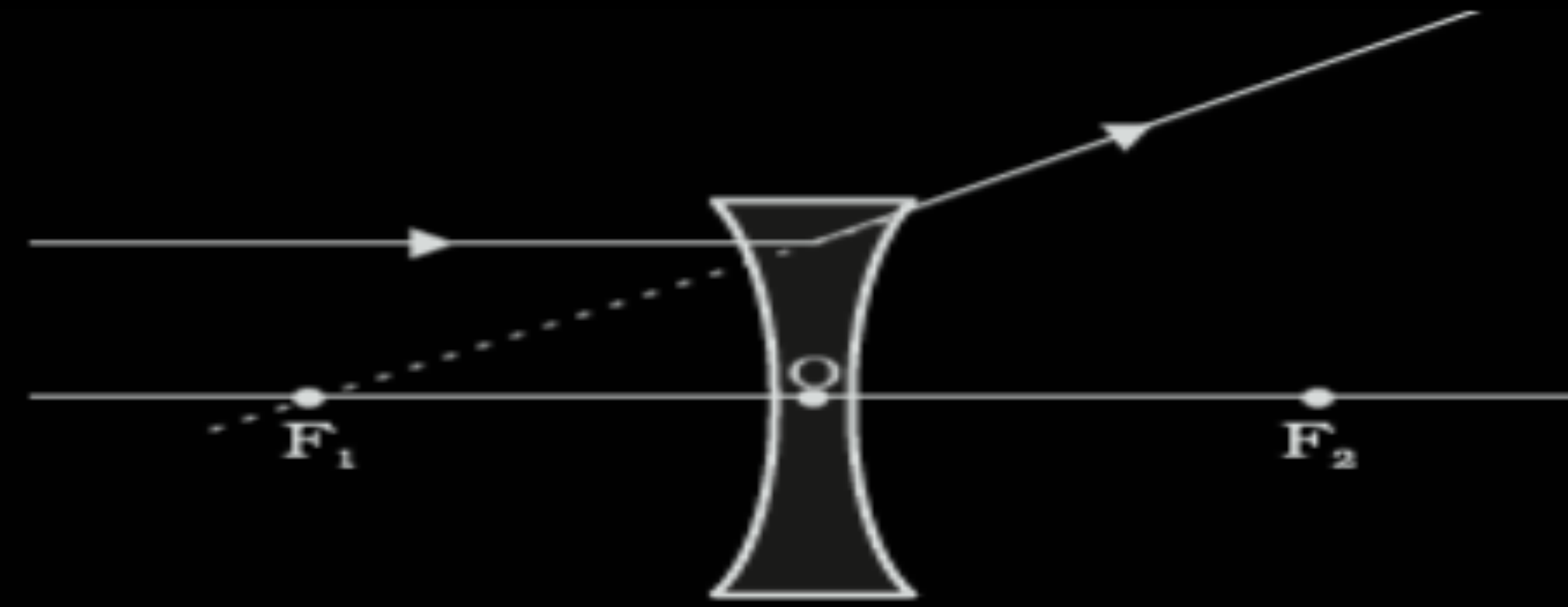
$$= \frac{+150 \text{ cm}}{-30 \text{ cm}} = -5 \text{ cm}$$

- (b) (i) Define principal axis of a lens. Draw a ray diagram to show what happens when a ray of light parallel to the principal axis of a concave lens passes through it.
- (ii) The focal length of a concave lens is 20 cm. At what distance from the lens should a 5 cm tall object be placed so that its image is formed at a distance of 15 cm from the lens ? Also calculate the size of the image formed.

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(b)

(i) Principal axis : It is an imaginary line passing through the two centres of curvatures of a lens.



(ii) $f = -20$ cm; $h = 5$ cm; $v = -15$ cm

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{or}$$

$$\begin{aligned} \frac{1}{u} &= \frac{1}{v} - \frac{1}{f} = \frac{1}{(-15)} - \frac{1}{(-20)} \\ &= \frac{-1}{60 \text{ cm}} \end{aligned}$$

or $u = -60$ cm object is at a distance of 60 cm from the lens

- Size of the image(magnification): $m = \frac{h'}{h} = \frac{v}{u}$

$$h' = \frac{v}{u} \times h = \frac{(-15)}{(-60)} \times 5 = 1.25 \text{ cm}$$

- 36.** (a) (i) State laws of reflection of light.
- (ii) An object of height 5.0 cm is placed at 15 cm in front of a concave mirror of focal length 10 cm. At what distance from the mirror should a screen be placed, so that a focussed image is obtained on it ? Find the height of the image.

36. (a) (i) State laws of reflection of light.

(ii) An object of height 5.0 cm is placed at 15 cm in front of a concave mirror of focal length 10 cm. At what distance from the mirror should a screen be placed, so that a focussed image is obtained on it ? Find the height of the image.

5

(a) (i)

- The angle of incidence is equal to the angle of reflection.
- The incident ray, the normal to the mirror at the point of incidence and the reflected ray, all lie in the same plane.

(ii) $u = -15$ cm, $f = -10$ cm (concave mirror)
 $h = 5.0$ cm

Mirror formula $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{v} = \frac{-1}{10 \text{ cm}} + \frac{1}{15 \text{ cm}} = \frac{-1}{30 \text{ cm}}$$

or $v = -30$ cm. The screen must be placed at a distance of 30 cm from the mirror in front of it

$$(m) = \frac{h'}{h} = -\frac{v}{u}$$

$$h = \frac{-v}{u} \times h = -\frac{-30 \text{ cm}}{-15 \text{ cm}} \times 5 \text{ cm} = -10 \text{ cm}$$

1+1

 $\frac{1}{2}$ $\frac{1}{2}$

1

1

(b) (i) State laws of refraction of light.

(ii) Draw a ray diagram to show refraction of a ray of light through a rectangular glass slab. How is the emergent ray related to incident ray ? Mark lateral displacement in the diagram.

(b) (i) State laws of refraction of light.

(ii) Draw a ray diagram to show refraction of a ray of light through a rectangular glass slab. How is the emergent ray related to incident ray ? Mark lateral displacement in the diagram.

5

(b)(i) -The incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane. - The ratio of sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for the given pair of media. / $\frac{\sin i}{\sin r} = \text{constant}$	1 1
(ii) • The emergent ray is parallel to the incident ray. • Labelling of lateral displacement (If labelling is not done deduct ½ marks)	2 ½ ½

2. For an object, a convex mirror always forms an image which is :

- (A) Virtual, erect and magnified
- (B) Virtual, erect and diminished
- (C) Virtual, inverted and diminished
- (D) Real, erect and diminished

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2. For an object, a convex mirror always forms an image which is :

- (A) Virtual, erect and magnified
- (B) Virtual, erect and diminished
- (C) Virtual, inverted and diminished
- (D) Real, erect and diminished

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(B)/ Virtual, erect and diminished

21. (a) List four properties of the images formed by a plane mirror.

2

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21. (a) List four properties of the images formed by a plane mirror.

2

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(a) Four properties:

- Virtual
- Erect
- Size of the image = Size of the object
- Image distance from the mirror = Object distance from the mirror
- Laterally inverted

(Any four)

$\frac{1}{2} \times 4$

(b) Write the definitions of the following terms in the context of a concave mirror :

2

- (i) Pole
- (ii) Centre of curvature
- (iii) Radius of curvature
- (iv) Principal axis

(b) Write the definitions of the following terms in the context of a concave mirror :

2

- (i) Pole
- (ii) Centre of curvature
- (iii) Radius of curvature
- (iv) Principal axis

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(b)		
(i)	Pole: Central point of the reflecting surface of a spherical mirror.	
(ii)	Centre of curvature: Centre of the sphere of which the reflecting surface of the spherical mirror forms a part.	
(iii)	Radius of curvature: Radius of the sphere of which the reflecting surface of the spherical mirror forms a part.	
(iv)	Principal axis: Straight line joining the pole and the centre of curvature of a spherical mirror	$\frac{1}{2} \times 4$

- 37.** A student has three convex lenses A, B and C of different focal lengths. He wants to observe the images formed by these lenses on a screen by placing a candle flame at different distances as given in following table :

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Case No.	Lens	Focal Length	Object Distance
1	A	50 cm	25 cm
2	B	20 cm	60 cm
3	C	15 cm	30 cm

- (i) List two properties of the image formed by the lens B in case no. 2. 1
- (ii) In which one of the three cases given above will the lens form a real image of the same size as the object and why ? 1
- (iii) (a) Name the type of lens presented by opticians for the correction of hypermetropia. State the role of such lenses in correcting the vision of the person suffering from this defect. 2

OR

- (iii) (b) Use Lens formula to find the distance of the image formed from the optical centre of the lens A in case no. 1. 2

(i)	Real; inverted; diminished	(Any two)	$\frac{1}{2} + \frac{1}{2}$
(ii)	Case 3: because object distance is twice the focal length		$\frac{1}{2} + \frac{1}{2}$
(iii) (a)	• Convex lens/ converging lens • It provides the additional focussing power required for forming the image on the retina. / These lenses decrease the focal length of the eye lens, thereby forming the image on the retina.		1
			1
	OR		
(iii)			
(b)	In case 1: $u = -25$ cm; $f = +50$ cm		
	Lens formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \therefore \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$		$\frac{1}{2}$
	$\therefore \frac{1}{v} = \frac{1}{50} + \frac{1}{-25}$		$\frac{1}{2}$
	$v = -50$ cm		1

12. When an object is placed beyond $2F$ of a convex lens, the nature of the image formed is :

- (a) Real, inverted and diminished
- (b) Real, erect and magnified
- (c) Virtual, erect and magnified
- (d) Real, inverted and magnified

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12. When an object is placed beyond $2F$ of a convex lens, the nature of the image formed is :

- (a) Real, inverted and diminished
- (b) Real, erect and magnified
- (c) Virtual, erect and magnified
- (d) Real, inverted and magnified

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- 24.** (a) Find the magnification of the image formed by a spherical mirror from the following data :
 $u = -20 \text{ cm}$, $f = -15 \text{ cm}$.

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24. (a) Find the magnification of the image formed by a spherical mirror from the following data :
 $u = -20$ cm, $f = -15$ cm.

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2

(a)	$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ $= \frac{1}{-15} - \frac{1}{-20} = \frac{-1}{60}$ $v = -60 \text{ cm}$ $m = \frac{-v}{u}$ $= -\frac{-60}{-20} = -3$
-----	---

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) Draw a labelled ray diagram for the image formation by a concave mirror when an object is placed between its centre of curvature and focus.

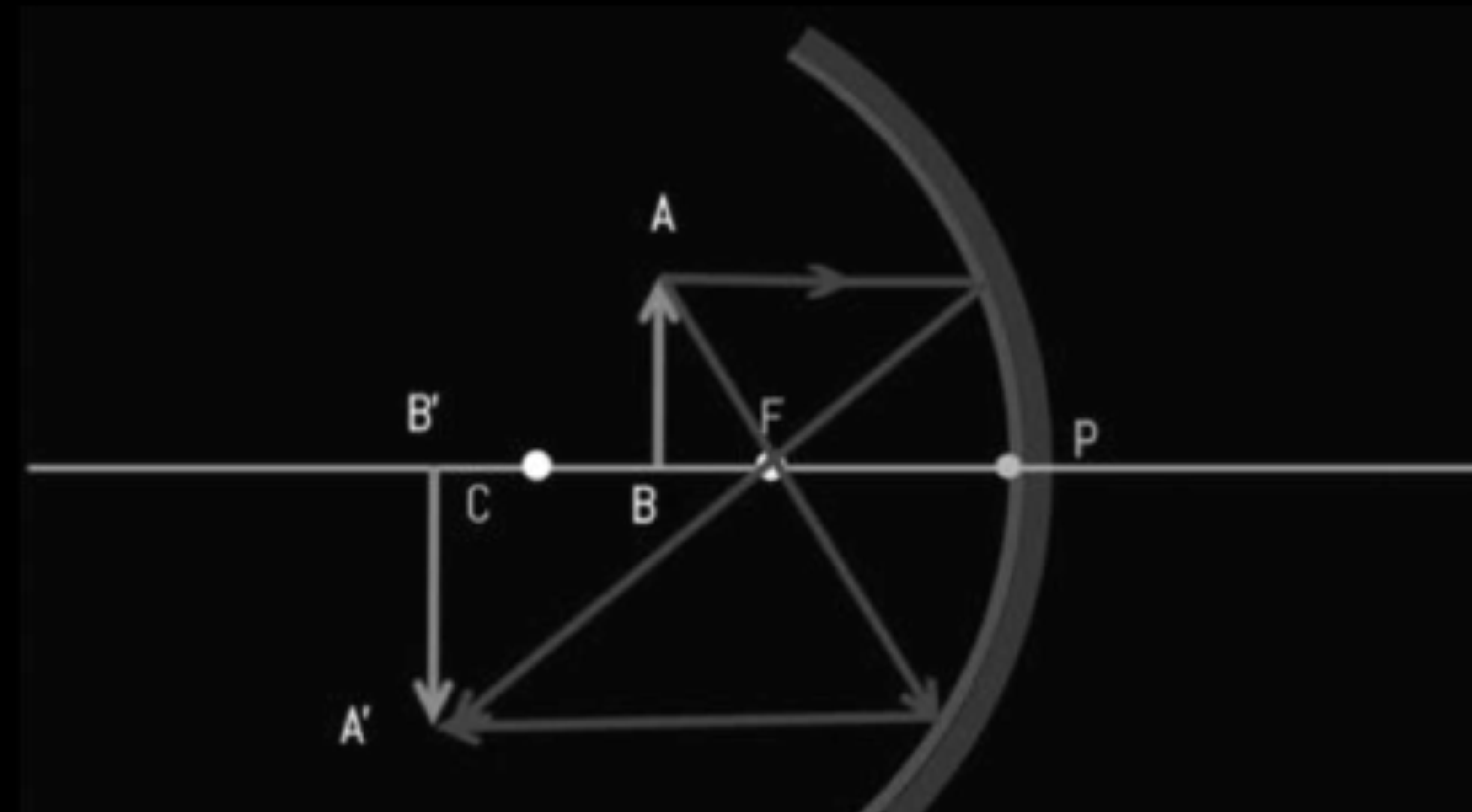
CBSE 2023 C

- (b) Draw a labelled ray diagram for the image formation by a concave mirror when an object is placed between its centre of curvature and focus.

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2

(b)



Diagram

1½

Direction of the rays

½

39. When a ray of light moving in a medium enters obliquely into another medium, it bends from its path. This phenomenon is called refraction of light. The ability of a medium to refract light is also expressed in terms of optical density. It is not the same as mass density. We use the terms ‘rarer medium’ and ‘denser medium’ which actually means ‘optically rarer medium’ and ‘optically denser medium’ respectively. When we say that a medium ‘A’ is optically denser than the other medium ‘B’, we mean that the refractive index of medium A is more than the refractive index of medium ‘B’. The speed of light is higher in a rarer medium than a denser medium. Thus a ray of light travelling from a rarer medium to a denser medium slows down and bends towards the normal.

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- (a) Define the term absolute refractive index of a medium. 1
- (b) Absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively.
- (i) In which one of the two media is the speed of light more ?
- (ii) If a ray of light enters obliquely from glass to water, will it bend towards the normal or away from the normal ? 1

- (c) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, find the speed of light in (i) water and (ii) vacuum. 2

OR

- (c) “A ray of light incident on a rectangular glass slab immersed in any medium emerges parallel to itself.” Draw a labelled ray diagram to justify this statement. CBSE 2023 C 2

(a) The refractive index of a medium with respect to air or vacuum
 / Absolute refractive index of a medium = $\frac{\text{speed of light in air(vacuum)}}{\text{speed of light in medium}}$

1

(b) (i) speed of light is more in water
 (ii) bends away from normal

$\frac{1}{2}$
 $\frac{1}{2}$

(c) Absolute refractive index of a medium = $\frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$

$$\frac{3}{2} = \frac{\text{speed of light in vacuum}}{2 \times 10^8}$$

$\frac{1}{2}$

Speed of light in vacuum = 3×10^8 m/s

$\frac{1}{2}$

Speed of light in water = $\frac{3}{4} \times 3 \times 10^8$ m/s = $\frac{9}{4} \times 10^8$ m/s

1

OR

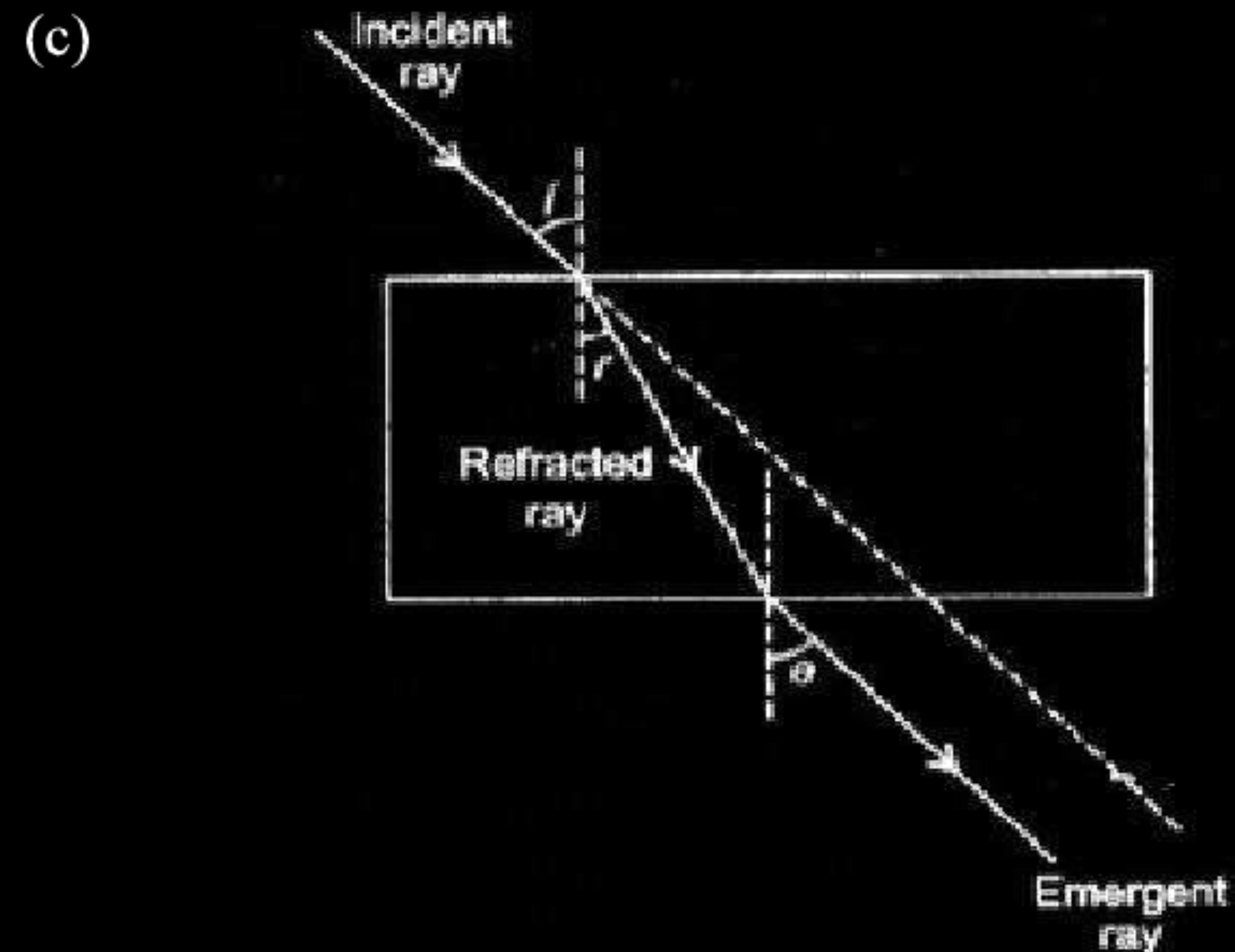


Diagram
 Labelling

1
 1

30. A candle flame is placed at a distance of 18 cm from the optical centre of a convex lens. If its image is formed on a screen placed at a distance of 36 cm from the lens, answer the following questions :

- (a) What is the magnification of the image formed ?
- (b) What is the focal length of the lens ? Use the “Lens Formula” to determine it.

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30. A candle flame is placed at a distance of 18 cm from the optical centre of a convex lens. If its image is formed on a screen placed at a distance of 36 cm from the lens, answer the following questions :

- (a) What is the magnification of the image formed ?
- (b) What is the focal length of the lens ? Use the “Lens Formula” to determine it.

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3

(a) magnification = $\frac{v}{u}$ $= \frac{+36}{-18}$ $m = -2$	$\frac{1}{2}$
(b) Lens formula : $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ $f = \frac{-18 \text{ cm} \times 36 \text{ cm}}{-18 \text{ cm} - 36 \text{ cm}} = \frac{18 \times 36 \text{ cm}}{-18 \text{ cm} - 36 \text{ cm}} = \frac{18 \times 36}{54} \text{ cm}$ $f = 12 \text{ cm}$	$\frac{1}{2}$
	1

31. (a) Define the following terms in the context of spherical mirrors :

(i) Pole

(ii) Centre of curvature

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(iii) Principal axis

(iv) Principal focus

(b) An object is placed at a distance of 60 cm from the pole of a concave mirror. If its real and inverted image is formed on a screen placed at a distance of 60 cm in front of the mirror, what is the focal length of the mirror ? Give justification for your answer.

31. (a) Define the following terms in the context of spherical mirrors :

- (i) Pole
- (ii) Centre of curvature
- (iii) Principal axis
- (iv) Principal focus

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(b) An object is placed at a distance of 60 cm from the pole of a concave mirror. If its real and inverted image is formed on a screen placed at a distance of 60 cm in front of the mirror, what is the focal length of the mirror ? Give justification for your answer.

3

(a)

- (i) Centre of the reflecting surface of a spherical mirror
- (ii) Centre of the sphere of which mirror was a part
- (iii) Straight line passing through pole and centre of curvature of spherical mirror
- (iv) Point on principal axis where the reflected rays meet / appear to meet.

(b) Focal length = $\frac{-60 \text{ cm}}{2} = -30 \text{ cm}$

Because the object is placed at the centre of curvature.

4 x 1/2

1/2

1/2

37. When a ray of light moving in a medium enters obliquely in another medium, it bends from its path. This phenomenon is called refraction of light. The ability of a medium to refract light is expressed in terms of optical density. The term optical density is not the same as mass density. If we compare two optical media, the one with the larger absolute refractive index is an optically denser medium than the other. Also, the speed of light through a given medium is inversely proportional to its optical density.

- (a) Define the term 'absolute refractive' index of a medium. 1
- (b) Refractive indices of glass, water and carbon disulphide are 1.50, 1.33 and 1.62 respectively. In which one of these media is the speed of light (i) highest, and (ii) lowest ? 1
- (c) (i) The absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If the speed of light in water is 2.25×10^8 m/s, find the speed of light in (I) vacuum, and (II) glass. 2

OR

- (ii) The absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. Find the refractive index of water with respect to glass. 2

(a) The refractive index of a medium with respect to air or vacuum	/
Absolute refractive index of a medium = $\frac{\text{speed of light in air(vacuum)}}{\text{speed of light in medium}}$	1
(b) (i) water (ii) carbon disulphide	$\frac{1}{2} + \frac{1}{2}$
(c) (i)	
$n_g = \frac{3}{2}$; $n_w = \frac{4}{3}$; $v_w = 2.25 \times 10^8 \text{ m/s}$; $c = ?$; $v_g = ?$	
(I) $n_w = \frac{c}{v_w}$	$\frac{1}{2}$
$\frac{4}{3} = \frac{c}{2.25 \times 10^8 \text{ m/s}}$	
$c = 3 \times 10^8 \text{ m/s}$	$\frac{1}{2}$
(II) $v_g = \frac{c}{n_g}$	$\frac{1}{2}$
$= \frac{3 \times 10^8 \text{ m/s}}{3/2} = 2 \times 10^8 \text{ m/s}$	$\frac{1}{2}$
OR	
(ii)	
Refractive index of water w.r.t. glass = $\frac{\text{speed of light in glass}}{\text{speed of light in water}}$	
Absolute refractive index of a medium = $\frac{\text{speed of light in vacuum}}{\text{speed of light in the medium}}$	$\frac{1}{2}$
$\therefore \text{ speed of light in glass} = \frac{c}{\text{Absolute refractive index of glass}} = \frac{c}{3/2} = \frac{2c}{3}$	$\frac{1}{2}$
$\text{speed of light in water} = \frac{c}{\text{Absolute refractive index of water}} = \frac{c}{4/3} = \frac{3c}{4}$	$\frac{1}{2}$
$\therefore \text{ Refractive index of water w. r. t. glass} = \frac{2c/3}{3c/4} = \frac{8}{9}$	$\frac{1}{2}$

29. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

- (i) Distance of the image from the mirror
- (ii) Height of the image

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29. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

- (i) Distance of the image from the mirror
- (ii) Height of the image

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3

(a) $h_1 = 3 \text{ cm}$	
$f = -12 \text{ cm}$	
$u = -18 \text{ cm}$	$\frac{1}{2}$

(i) Image Distance	
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	$\frac{1}{2}$

$$\frac{1}{-12} = \frac{1}{v} + \frac{1}{-18}$$

$$\frac{1}{-12} + \frac{1}{18} = \frac{1}{v}$$

$$\frac{1}{v} = \frac{-3+2}{36} = \frac{-1}{36}$$

$v = -36 \text{ cm}$	1
----------------------	---

(ii) Height of image

$m = \frac{h_2}{h_1} = \frac{-v}{u}$	$\frac{1}{2}$
--------------------------------------	---------------

$$\frac{h_2}{3} = \frac{-[-36]}{-18}$$

$$h_2 = \frac{-36}{18} \times 3$$

$h_2 = -6 \text{ cm}$	$\frac{1}{2}$
-----------------------	---------------

- (b) Define power of a lens. The focal length of a lens is -10 cm. Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from the optical centre of this lens, according to the New Cartesian Sign Convention, what will be the sign of magnification in this case ?

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- (b) Define power of a lens. The focal length of a lens is -10 cm. Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from the optical centre of this lens, according to the New Cartesian Sign Convention, what will be the sign of magnification in this case ?

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(b) Degree of convergence or divergence of light/ length of lens in metre.	1
• It is diverging/concave lens	$\frac{1}{2}$
• $P = \frac{1}{f(m)} = \frac{100}{f(cm)}$	$\frac{1}{2}$
$P = \frac{100}{-10 \text{ cm}} = -10 \text{ D}$	$\frac{1}{2}$
• Sign of magnification = + or positive	$\frac{1}{2}$

- 38.** Many optical instruments consist of a number of lenses. They are combined to increase the magnification and sharpness of the image. The net power (P) of the lenses placed in contact is given by the algebraic sum of the powers of the individual lenses $P_1, P_2, P_3 \dots$ as

$$P = P_1 + P_2 + P_3 \dots$$

This is also termed as the simple additive property of the power of lens, widely used to design lens systems of cameras, microscopes and telescopes. These lens systems can have a combination of convex lenses and also concave lenses.

- (a) What is the nature (convergent / divergent) of the combination of a convex lens of power + 4 D and a concave lens of power – 2 D ? 1
- (b) Calculate the focal length of a lens of power – 2.5 D. 1
- (c) Draw a ray diagram to show the nature and position of an image formed by a convex lens of power + 0.1 D, when an object is placed at a distance of 20 cm from its optical centre. 2

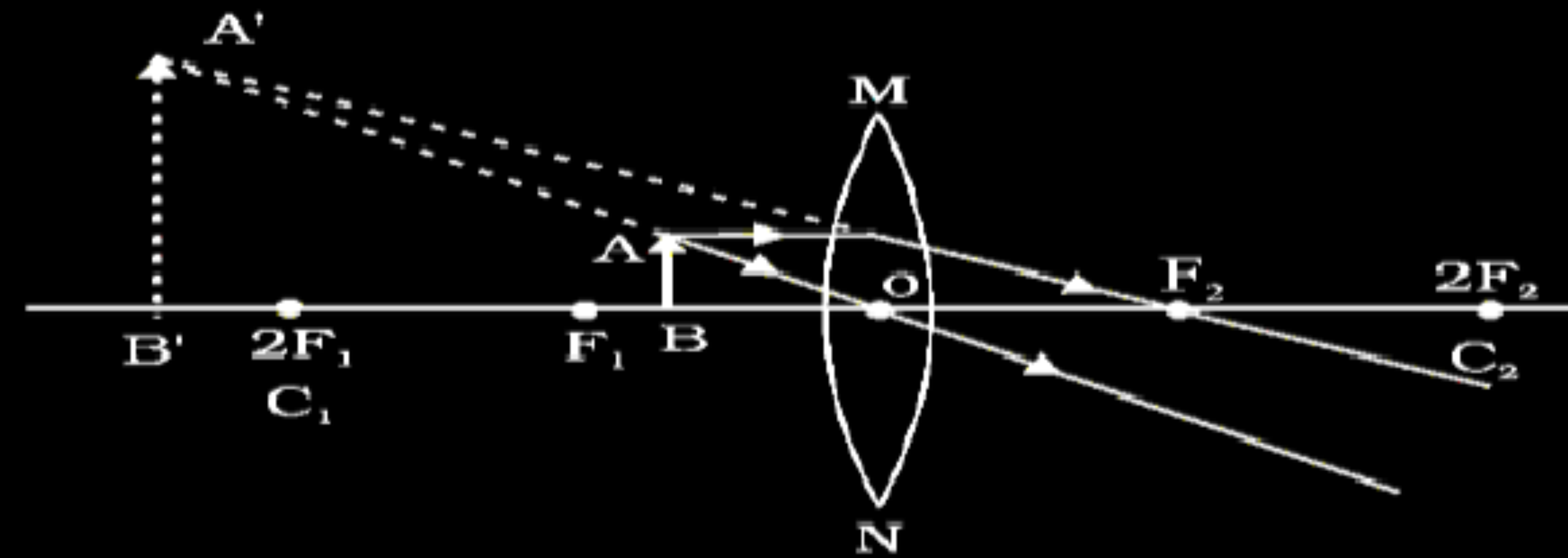
OR

- (c) How is a virtual image formed by a convex lens different from that formed by a concave lens ? Under what conditions do a convex and a concave lens form virtual images ?

(a) Convergent

(b) $f \text{ (m)} = \frac{1}{P} = \frac{1}{-2.5} = -0.4 \text{ m or } -40 \text{ cm}$

(c)



OR

- (c) • Convex lens – magnified
Concave lens – diminished
• Convex – object between O and F
Concave – object anywhere between optical centre and infinity.

1

$\frac{1}{2}$, $\frac{1}{2}$

2

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 30.** (a) (i) An object of 5 cm height is placed at a distance of 20 cm from the optical centre of a concave lens of focal length 18 cm. Calculate (1) image distance and (2) the magnification in this case.
- (ii) Compare the values of magnification obtained by a concave lens and a convex lens when both the lenses form virtual images.

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- 30.** (a) (i) An object of 5 cm height is placed at a distance of 20 cm from the optical centre of a concave lens of focal length 18 cm. Calculate (1) image distance and (2) the magnification in this case.
- (ii) Compare the values of magnification obtained by a concave lens and a convex lens when both the lenses form virtual images.

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3

(a) Given,

Height of object (h) = 5 cm

Object distance (u) = – 20 cm

Focal length (f) = – 18 cm

Image distance (v) = ?

(i)

$$(1) \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{1}{-20} + \frac{1}{-18} = - \left[\frac{18 + 20}{360} \right]$$

$$= \frac{-38}{360}$$

$$\Rightarrow v = \frac{-360}{38} = -9.47 \text{ cm}$$

$$(2) \quad m = \frac{v}{u} = \frac{-9.47}{-20} = 0.47$$

(ii) For convex lens : $m > 1$, for concave lens $m < 1$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

- (b) A convex lens can form a (i) real, inverted and magnified image as well as (ii) virtual, erect and magnified image of an object. If the focal length of the lens is 10 cm, what should be the range of the object distance in both cases ? Draw ray diagrams to justify your answer.

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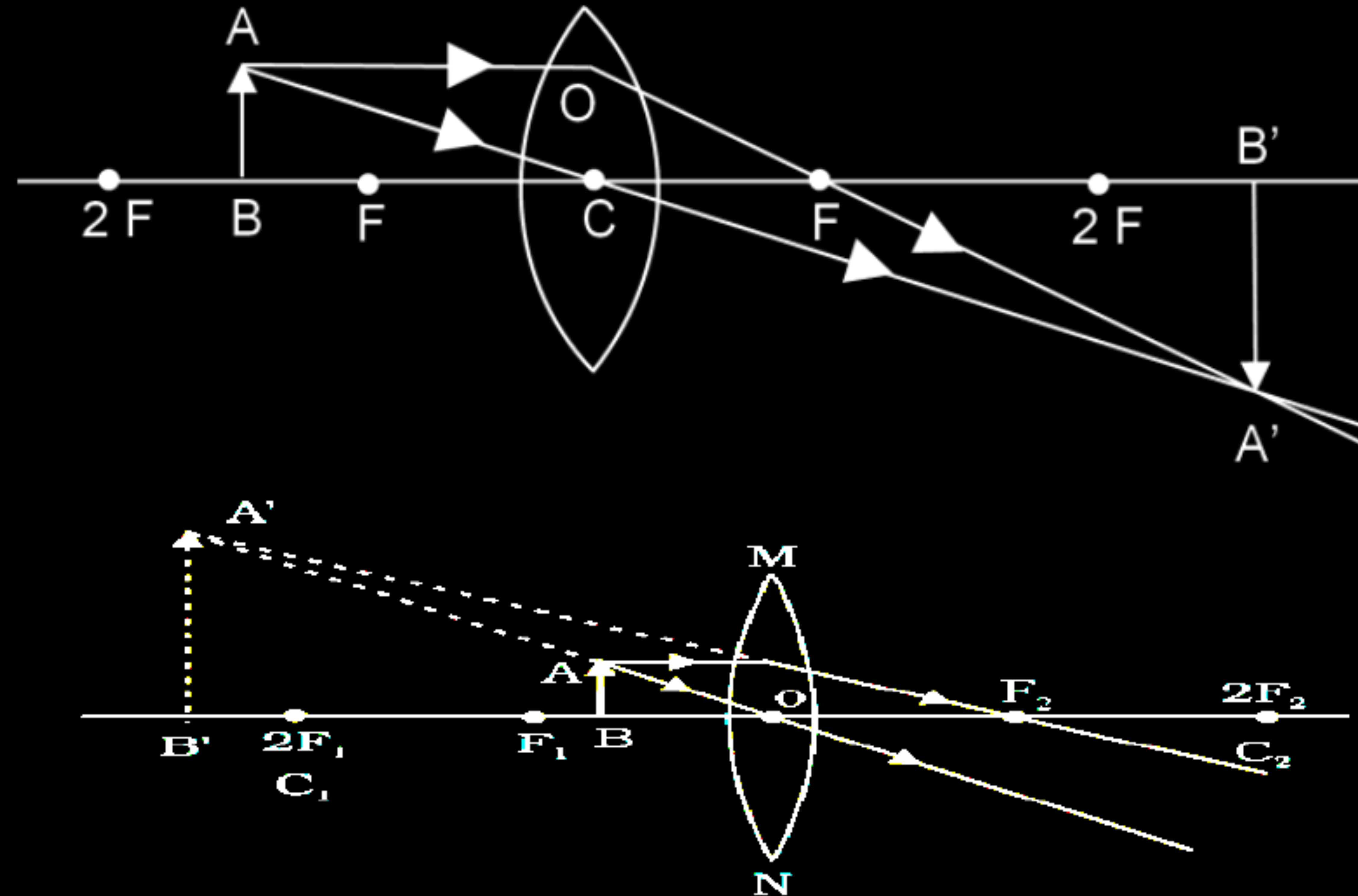
- (b) A convex lens can form a (i) real, inverted and magnified image as well as (ii) virtual, erect and magnified image of an object. If the focal length of the lens is 10 cm, what should be the range of the object distance in both cases ? Draw ray diagrams to justify your answer.

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3

- (b) (i) $20 \text{ cm} > u > 10 \text{ cm}$ / Between 10 cm and 20 cm
(ii) Object distance less than 10 cm / $10 > u > 0$

(i)



(Deduct $\frac{1}{2}$ mark for not marking the direction of ray)

13. An object is placed in front of a convex mirror at infinity. According to the New Cartesian Sign Convention, the sign of the focal length and the sign of the image distance in this case are respectively :

(a) $+$, $-$

(b) $-$, $+$

(c) $-$, $-$

(d) $+$, $+$

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13. An object is placed in front of a convex mirror at infinity. According to the New Cartesian Sign Convention, the sign of the focal length and the sign of the image distance in this case are respectively :

(a) $+$, $-$

(b) $-$, $+$

(c) $-$, $-$

(d) $+$, $+$

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(d)

31. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

(i) Distance of the image from the mirror

(ii) Height of the image

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31. (a) A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following :

(i) Distance of the image from the mirror

(ii) Height of the image

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3

(a) Here $h = 3$ cm; $f = -12$ cm, $u = -18$ cm,

(Award full marks if data not written but calculations are correct)

(i)

$$v = ?, h' = ?$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-12 \text{ cm}} - \frac{1}{-18 \text{ cm}}$$

$$\therefore v = -36 \text{ cm}$$

(ii)

$$h' = -\frac{v}{u} \times h$$

$$h' = (-) \frac{-36 \text{ cm}}{-18 \text{ cm}} \times 3 \text{ cm} = -6 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) Define power of a lens. The focal length of a lens is -10 cm. Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from the optical centre of this lens, according to the New Cartesian Sign Convention, what will be the sign of magnification in this case ?

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- (b) Define power of a lens. The focal length of a lens is -10 cm . Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from the optical centre of this lens, according to the New Cartesian Sign Convention, what will be the sign of magnification in this case ?

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3

(b)• Power of lens : Ability of a lens to converge or diverge light rays falling on it / Degree of convergence or divergence of light achieved by a lens /
Reciprocal of focal length of lens in metre. **(Any one)**

• It is diverging/concave lens

$$\bullet P = \frac{1}{f(m)} = \frac{100}{f(cm)}$$

$$P = \frac{100}{-10\text{ cm}} = -10\text{ D}$$

• Sign of magnification = + or positive

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

36. (a) An object is placed in front of a convex lens of focal length f . If the distance of the object from the lens is $2f$, draw a ray diagram to show the formation of the image. Write the value of magnification in this case.

(b) A student has focussed the image of a candle flame on a white screen using a convex lens. The situation is as given below :

Length of the flame = 2 cm

Focal length of the lens = 12 cm

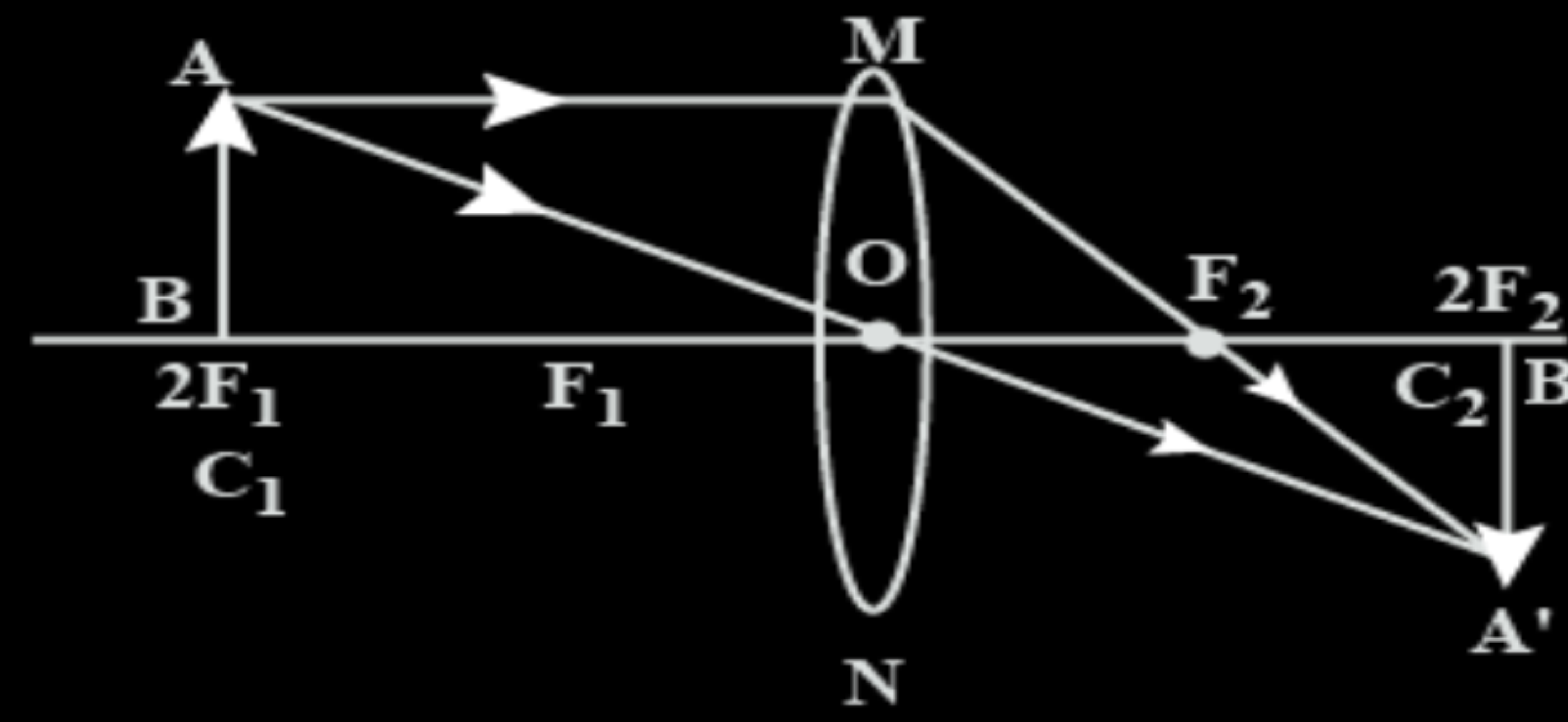
Distance of the flame from the lens = 16 cm

If the flame is perpendicular to the principal axis of the lens, calculate the values of the following :

(i) Distance of the image from the lens

(ii) Length of the image formed

(a) •



1½

(Deduct ½ mark for not marking arrows)

• Magnification = -1

½

(b) (i) $u = -16 \text{ cm}$; $f : +12 \text{ cm}$

½

Formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

½

Substitution Result $\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = \frac{1}{-16 \text{ cm}} + \frac{1}{12 \text{ cm}} + \frac{+1}{48 \text{ cm}}$

½

or $V = +48 \text{ cm}$ or $v = +48 \text{ cm}$

½

(Award full marks if data not written but calculations are correct)

(iv) $h_i = \frac{v}{u} \times h_o$

½

$$= \frac{+48 \text{ cm}}{-16 \text{ cm}} \times 2 \text{ cm}$$

$$= -6 \text{ cm}$$

Image is 6 cm in size.

½

- 36.** (a) Name the type of mirror that should be used to obtain the following types of images :
- (i) A magnified and virtual image of an object
 - (ii) A diminished and virtual image of an object
- (b) Draw labelled ray diagrams to justify your answers in each case mentioned above.
- (c) Which of these mirrors can also be used to obtain a magnified and real image of an object ? State the position of the object in this case.

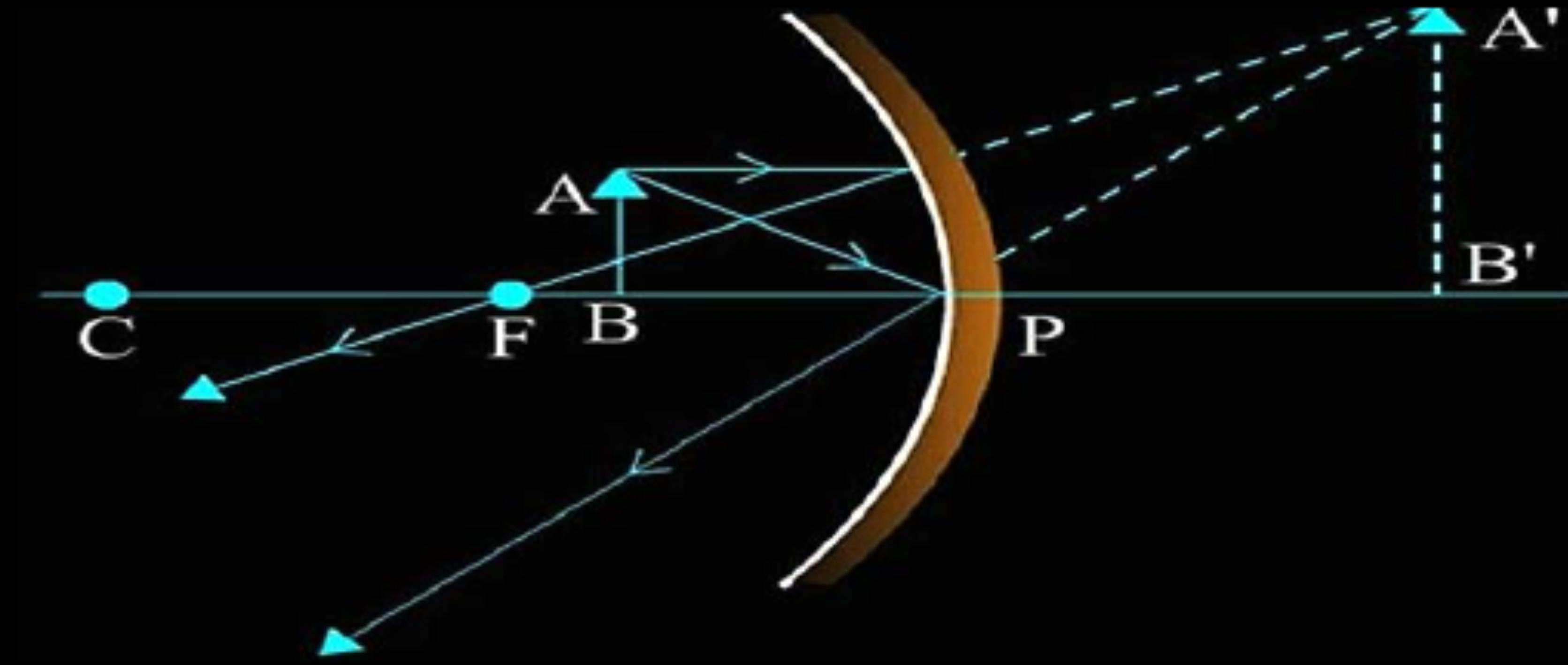
(a) (i) Concave mirror / Converging mirror

1

(ii) Convex mirror / Diverging mirror

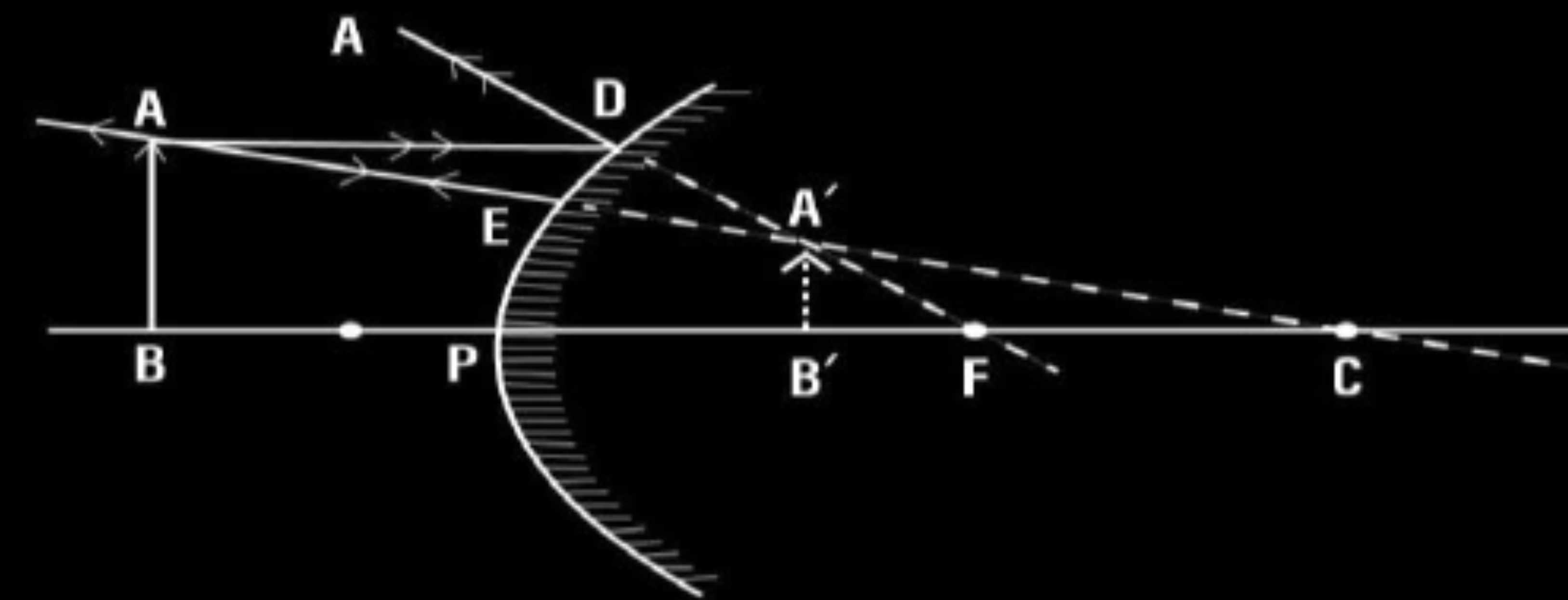
1

(b) (i)



1

(ii)



1

(Overall deduct $\frac{1}{2}$ mark for not drawing the arrows)

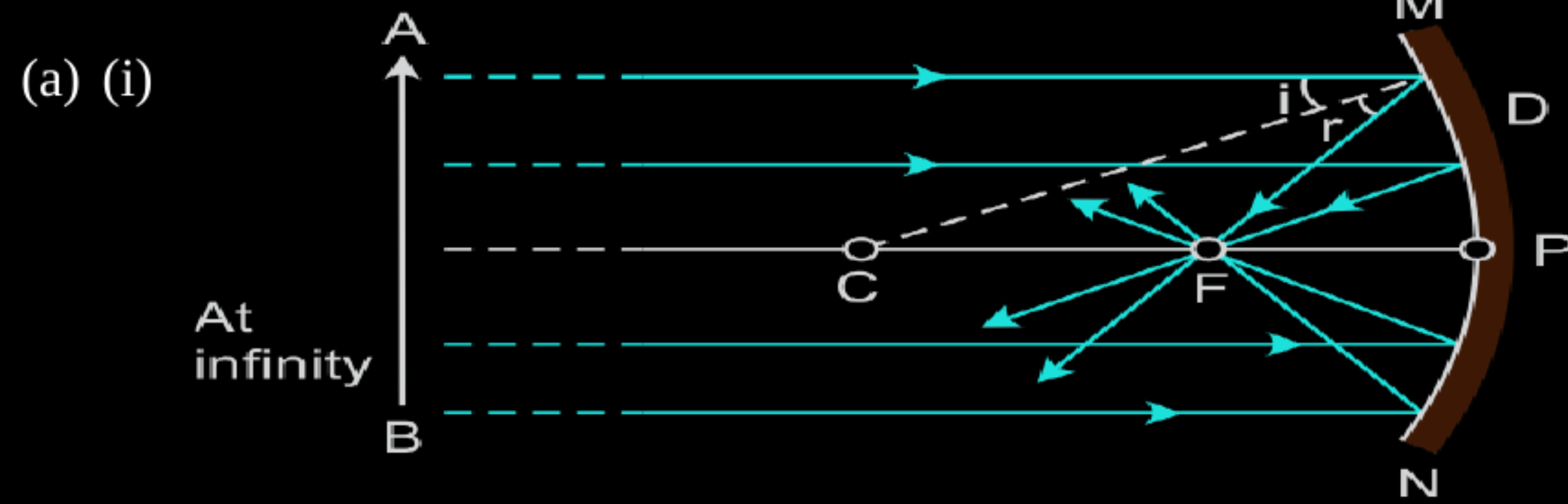
(c) • Concave mirror / Converging mirror

$\frac{1}{2}$

• Between focus and centre of curvature of the mirror / between F & C

$\frac{1}{2}$

- 36.** (a) Draw a ray diagram for showing the convergence of a parallel beam of light by a concave mirror and mark the positions of pole and the centre of curvature in it.
- (b) An object 4 cm in size is placed at 25 cm in front of a concave mirror of focal length 15 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image ? Find the nature and size of the image.
- (c) List any two uses of a concave mirror.



(P, F, C should be labelled correctly.)

(b) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-15} - \frac{1}{-25} = -\frac{1}{15} + \frac{1}{25} = \frac{-2}{75}$$

or $v = -37.5 \text{ cm}$

The screen should be placed at 37.5 cm from the mirror. The image is real and inverted.

Also, magnification, $m = \frac{h'}{h} = -\frac{v}{u}$

$$h' = \frac{vh}{u} = -\frac{(-37.5 \text{ cm})(+4.0 \text{ cm})}{(-25.0 \text{ cm})}$$

Height of the image = $h' = -6.0 \text{ cm}$. The image is inverted and enlarged.

(c) Torches, search lights

(or Any other)

1

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}, \frac{1}{2}$

30. A student wants to draw a ray diagram to show the formation of image due to a concave mirror when an object is placed in between its focus and centre of curvature.

- (a) Write the minimum number of incident rays required.
- (b) State the paths of the incident rays.
- (c) State the paths of the reflected rays.
- (d) Write the sign of magnification according to the New Cartesian Sign Convention.

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3

30. A student wants to draw a ray diagram to show the formation of image due to a concave mirror when an object is placed in between its focus and centre of curvature.

- (a) Write the minimum number of incident rays required.
- (b) State the paths of the incident rays.
- (c) State the paths of the reflected rays.
- (d) Write the sign of magnification according to the New Cartesian Sign Convention.

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3

(a)	Two / 2	$\frac{1}{2}$
(b)	• Parallel to the principal axis • Passing through the principal focus • Ray passing through centre of curvature • Ray incident obliquely to principal axis. (any two points)	$\frac{1}{2}, \frac{1}{2}$
(c)	• After reflection passes through the principal focus. • After reflection becomes parallel to the principal axis. • After reflection, reflected back along same path. • Reflected obliquely (any two points)	$\frac{1}{2}, \frac{1}{2}$
(d)	negative	$\frac{1}{2}$

31. (a) A candle flame and a screen are placed 1 m apart. When a lens is placed exactly in the middle of the two, a distinct image of the candle flame is observed on the screen.

(i) Name the type of the lens used.

(ii) How is focal length of the lens determined in this case ? State the method and determine its value.

(iii) Write the power of the lens used.

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3

31. (a) A candle flame and a screen are placed 1 m apart. When a lens is placed exactly in the middle of the two, a distinct image of the candle flame is observed on the screen.

(i) Name the type of the lens used.

(ii) How is focal length of the lens determined in this case ? State the method and determine its value.

(iii) Write the power of the lens used.

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3

(a)	
(i) Convex lens/Converging lens	$\frac{1}{2}$
(ii) Distance between the flame and screen = 1 m $\therefore$ Lens is placed exactly in the middle and a distinct image of the flame is formed on the screen. Object distance = Image distance = $2f$ $\therefore 2f + 2f = 4f = 1 \text{ m}$ $\therefore f = \frac{1}{4} \text{ m} = 25 \text{ cm} / 0.25 \text{ m}$	$\frac{1}{2}$
(iii) Power = $\frac{1}{f(\text{m})}$	1
= +4D	$\frac{1}{2}$

- (b) (i) Define absolute refractive index.
- (ii) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, find :
- (I) Speed of light in vacuum
- (II) Speed of light in water

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(b) (i) Define absolute refractive index.

(ii) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is 2×10^8 m/s, find :

(I) Speed of light in vacuum

(II) Speed of light in water

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3

(b)

(i) Absolute refractive index of a medium is the ratio of speed of light in vacuum (or air) to the speed of light in the medium.

/

$$\text{Absolute refractive index } n_m = \frac{\text{Speed of light in vacuum } (c)}{\text{Speed of light in the medium } v_m}$$

(ii) (I) $c = n_g \times v_g$

$$= \frac{3}{2} \times 2 \times 10^8 \text{ m/s} = 3 \times 10^8 \text{ m/s}$$

(II) $v_w = \frac{c}{n_w}$

$$= \frac{3 \times 10^8 \text{ m/s}}{\frac{4}{3}} = \frac{9}{4} \times 10^8 \text{ m/s or } 2.25 \times 10^8 \text{ m/s}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

39. A simple magnifier or microscope is a convex lens of small focal length. In order to use such a lens as a microscope, the lens is held near the object to get an erect, magnified and virtual image of the object. However, this microscope has a maximum limiting magnification. For much larger magnification we use a compound microscope in which two convex lenses are used, one compounding the effect of the other. The lens nearest the object, called the objective, forms a real, inverted and magnified image of the object. This image serves as the object for the second lens, the eyepiece which functions essentially like a simple microscope or magnifier and produces the final image.

(a) You have a compound microscope in which a convex lens of focal length 'f' is used as objective. What should be the distance of the object from the objective in terms of 'f' ?

1

(b) List two properties of the final image obtained from a compound microscope.

1

- (c) An object is placed in front of a convex lens at a **distance of two times the focal length of the lens**. List two properties of the image formed and write the value of magnification produced by the lens using the New Cartesian Sign Convention. 2

OR

- (c) An object is placed in front of a convex lens at a **distance more than two times the focal length of the lens**. List two properties of the image formed and write the value of magnification produced by the lens using the New Cartesian Sign Convention. 2

(a)	• Should be slightly more than f / between f and $2f$		1
(b)	• Virtual • Inverted with respect to object • Magnified	(any two)	$\frac{1}{2}$, $\frac{1}{2}$
(c)	• Real • Inverted • Image is of the same size as the object	(any two)	$\frac{1}{2}$, $\frac{1}{2}$
	$m = -1$ (Minus one)		1
OR			
(c)	• Real • inverted • diminished	(any two)	$\frac{1}{2}$, $\frac{1}{2}$
	$m < 1$ and negative		1

7. A student wants to obtain an erect image of an object using a concave mirror of 10 cm focal length. What will be the distance of the object from mirror ?

1

(a) Less than 10 cm

(b) 10 cm

(c) between 10 cm and 20 cm

(d) more than 20 cm

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7. A student wants to obtain an erect image of an object using a concave mirror of 10 cm focal length. What will be the distance of the object from mirror ?

1

(a) Less than 10 cm

(b) 10 cm

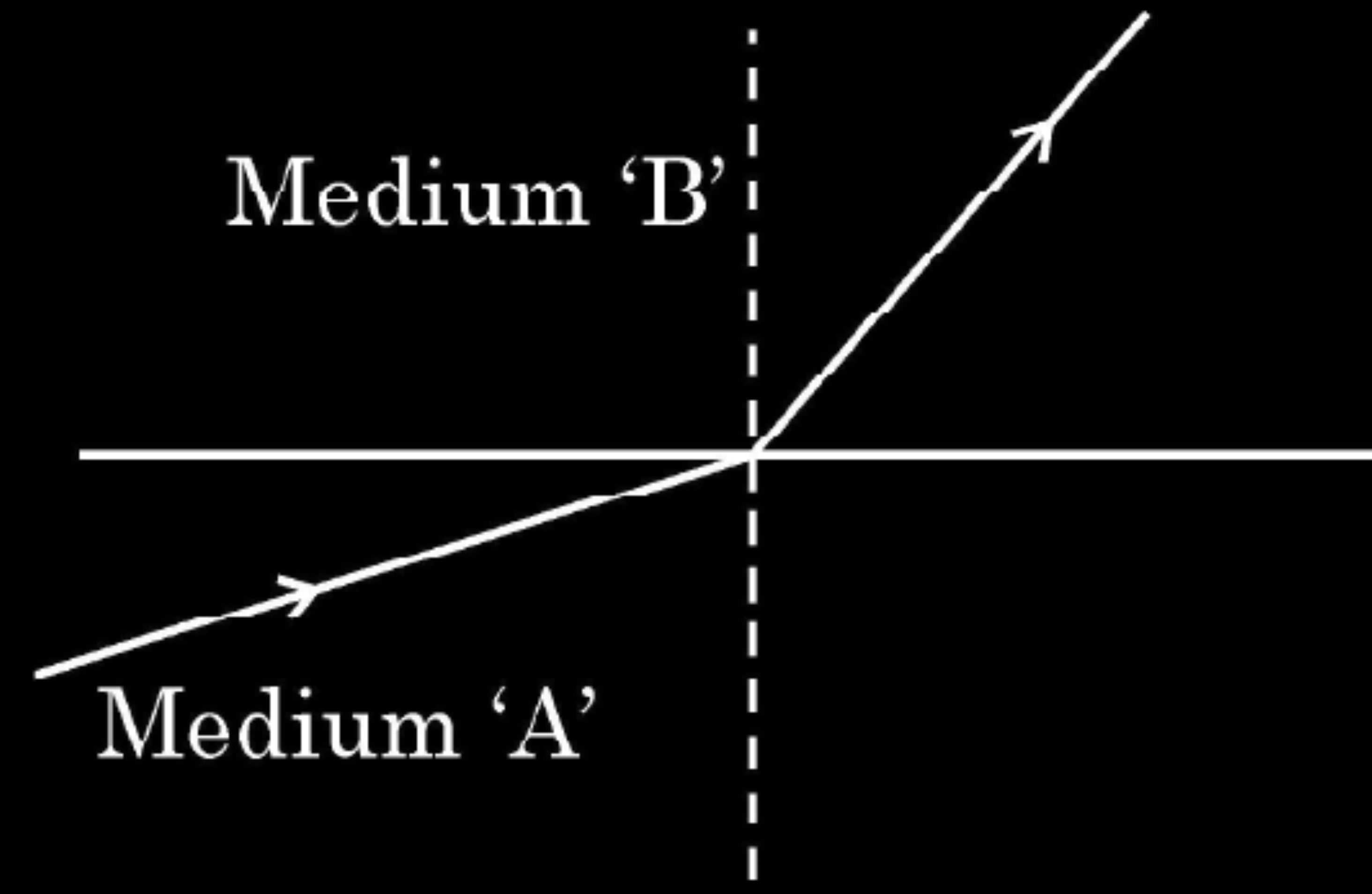
(c) between 10 cm and 20 cm

(d) more than 20 cm

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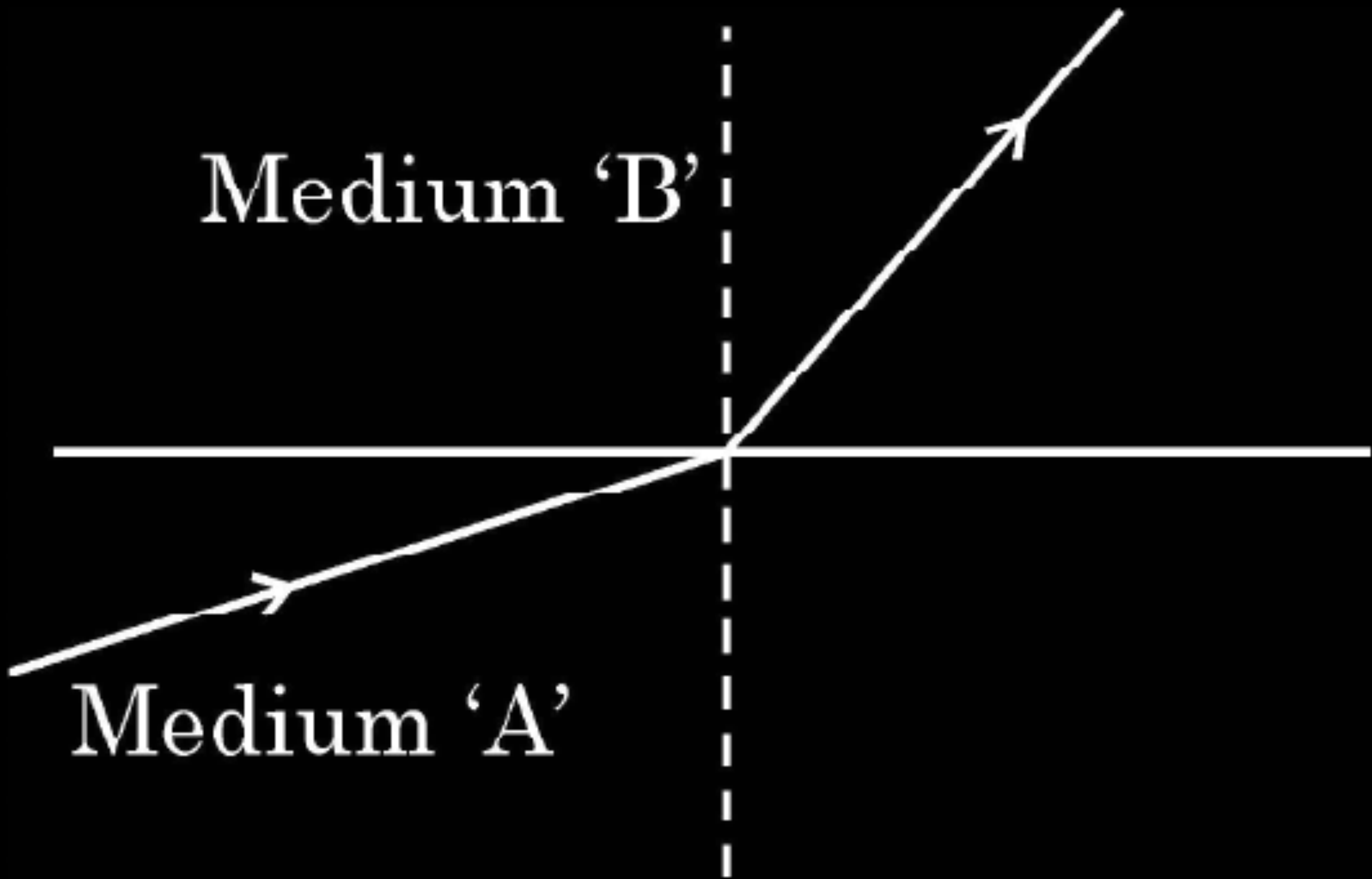
(a)

25. A light ray enters from medium A to medium B as shown in the figure.



- (a) Which one of the two media is denser w.r.t. other medium ? Justify your answer. 1
- (b) If the speed of light in medium A is v_a and in medium B is v_b , what is the refractive index of B with respect to A. 1

25. A light ray enters from medium A to medium B as shown in the figure.



- (a) Which one of the two media is denser w.r.t. other medium ? Justify your answer.

1
- (b) If the speed of light in medium A is v_a and in medium B is v_b , what is the refractive index of B with respect to A.

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1

(a) Medium B	$\frac{1}{2}$
In medium B ray of light bends towards normal / $\angle r < \angle i$	$\frac{1}{2}$
(b) Refractive index of Medium 'B' with respect of Medium 'A' is	
$n_{BA} = \frac{v_a}{v_b}$	1

- (a) A ray of light starting from diamond is incident on the interface separating diamond and water. Draw a labelled ray diagram to show the refraction of light in this case. 1
- (b) Absolute refractive indices of diamond and water are 2.42 and 1.33 respectively. Find the value of refractive index of water w.r.t. diamond. 1

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(a) A ray of light starting from diamond is incident on the interface separating diamond and water. Draw a labelled ray diagram to show the refraction of light in this case.

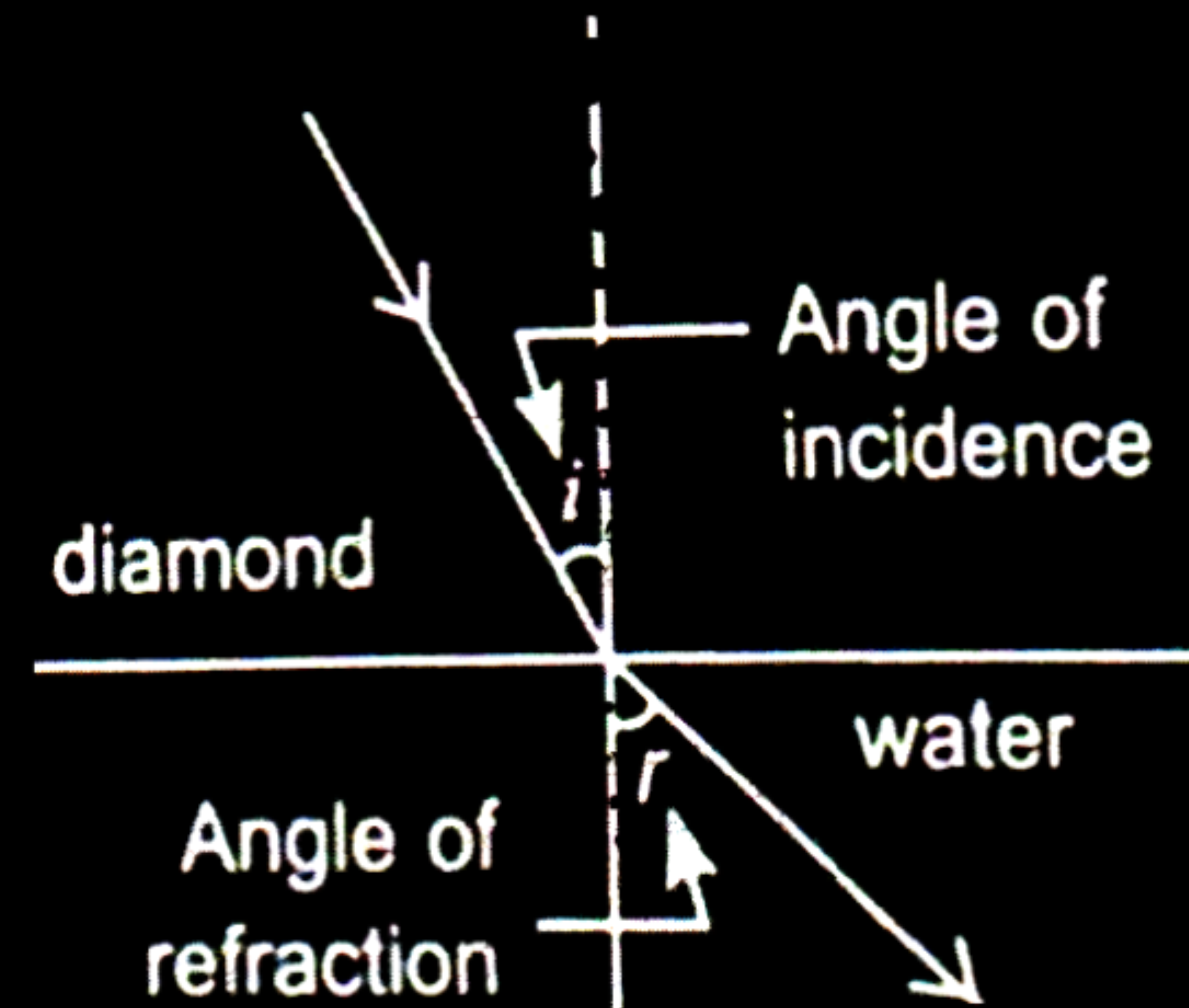
1

(b) Absolute refractive indices of diamond and water are 2.42 and 1.33 respectively. Find the value of refractive index of water w.r.t. diamond.

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1

(a)



(Credit marks for $\angle i$, $\angle r$ and arrows.)

1

(b)

$$n_{21} = \frac{n_{2a}}{n_{1a}}$$

$$\frac{1.33}{2.42} \quad \text{or} \quad 0.55$$

$\frac{1}{2}$

$\frac{1}{2}$

39. A student took three concave mirrors of different focal lengths and performed the experiment to see the image formation by placing an object at different distances with these mirrors as shown in the following table. 4

Case No.	Object-distance	Focal length
I	45 cm	20 cm
II	30 cm	15 cm
III	20 cm	30 cm

Now answer the following questions :

- (a) List two properties of the image formed in Case I.
- (b) In which one of the cases given in the table, the mirror will form real image of same size and why ?
- (c) Name the type of mirror used by dentists. Give reason why do they use such type of mirrors.

OR

- (c) Look at the table and identify the situation (object distance and focal length) which resembles the situation in which concave mirrors are used as shaving mirrors ? Draw a ray diagram to show the image formation in this case.

(a) Real, inverted, diminished (Any two)

$\frac{1}{2}$, $\frac{1}{2}$

(b) Case II

$\frac{1}{2}$

Because focal length of mirror is 15 cm, object distance is 30cm which means the object is placed at C.

$\frac{1}{2}$

(c) Dentists use concave mirrors

1

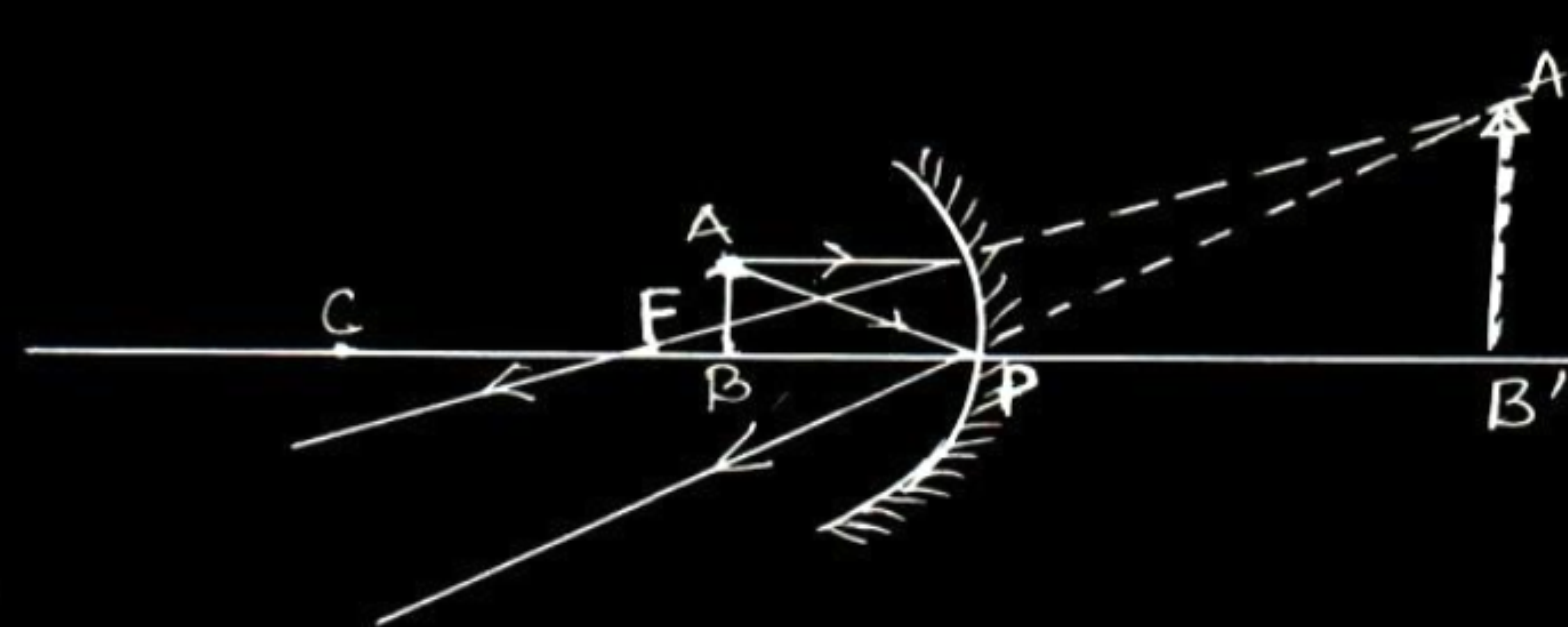
Because concave mirror forms erect and enlarged image when object is very close to the mirror.

1

OR

1

(c) Case III



1

(Deduct $\frac{1}{2}$ mark if direction of ray is not marked.)

7. To obtain a magnification of $+2$ with a concave mirror of radius of curvature 60 cm the object distance must be

1

(a) -90 cm

(b) -45 cm

(c) -30 cm

(d) -15 cm

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7. To obtain a magnification of + 2 with a concave mirror of radius of curvature 60 cm the object distance must be

1

(a) – 90 cm

(b) – 45 cm

(c) – 30 cm

(d) – 15 cm

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(d)

7. In torch lights and head lights of vehicles, the bulb is placed

1

- (a) between the pole and the focus of the reflector.
- (b) very near to the focus of the reflector.
- (c) between the focus and centre of curvature of the reflector.
- (d) at the centre of curvature of the reflector.

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7. In torch lights and head lights of vehicles, the bulb is placed

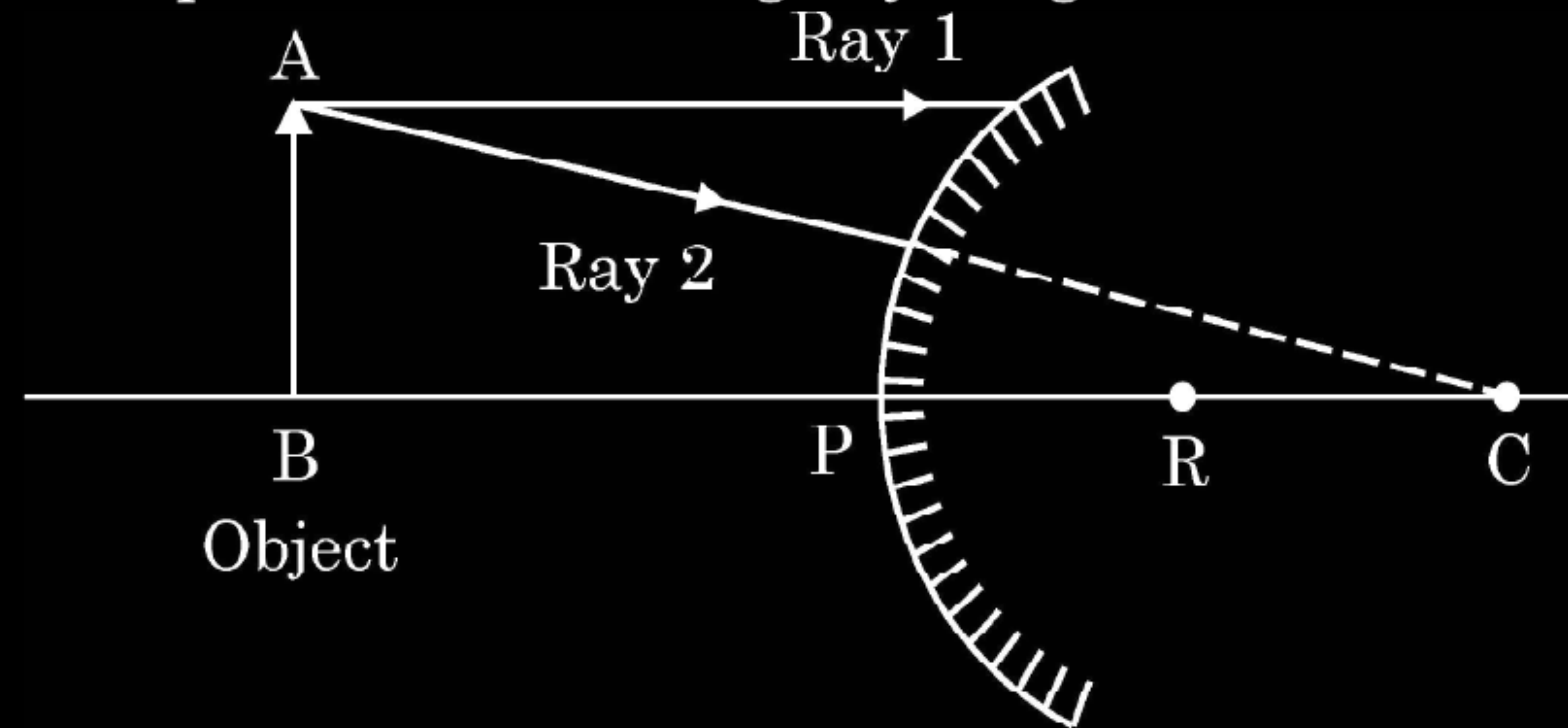
1

- (a) between the pole and the focus of the reflector.
- (b) very near to the focus of the reflector.
- (c) between the focus and centre of curvature of the reflector.
- (d) at the centre of curvature of the reflector.

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(b)

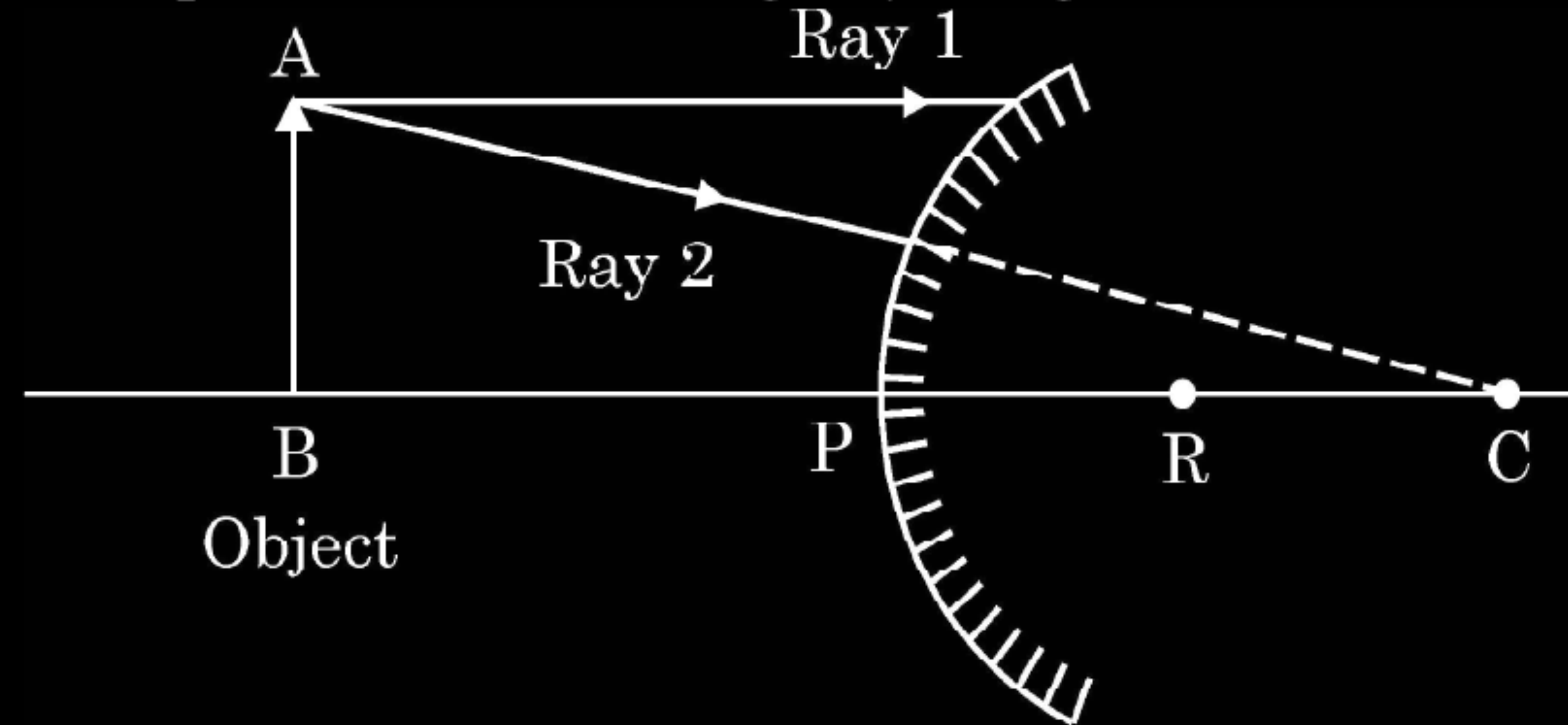
30. (a) Complete the following ray diagram to show the formation of image : 3



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- (b) Mention the nature, position and size of the image formed in this case.
- (c) State the sign of the image distance in this case using the Cartesian sign convention.

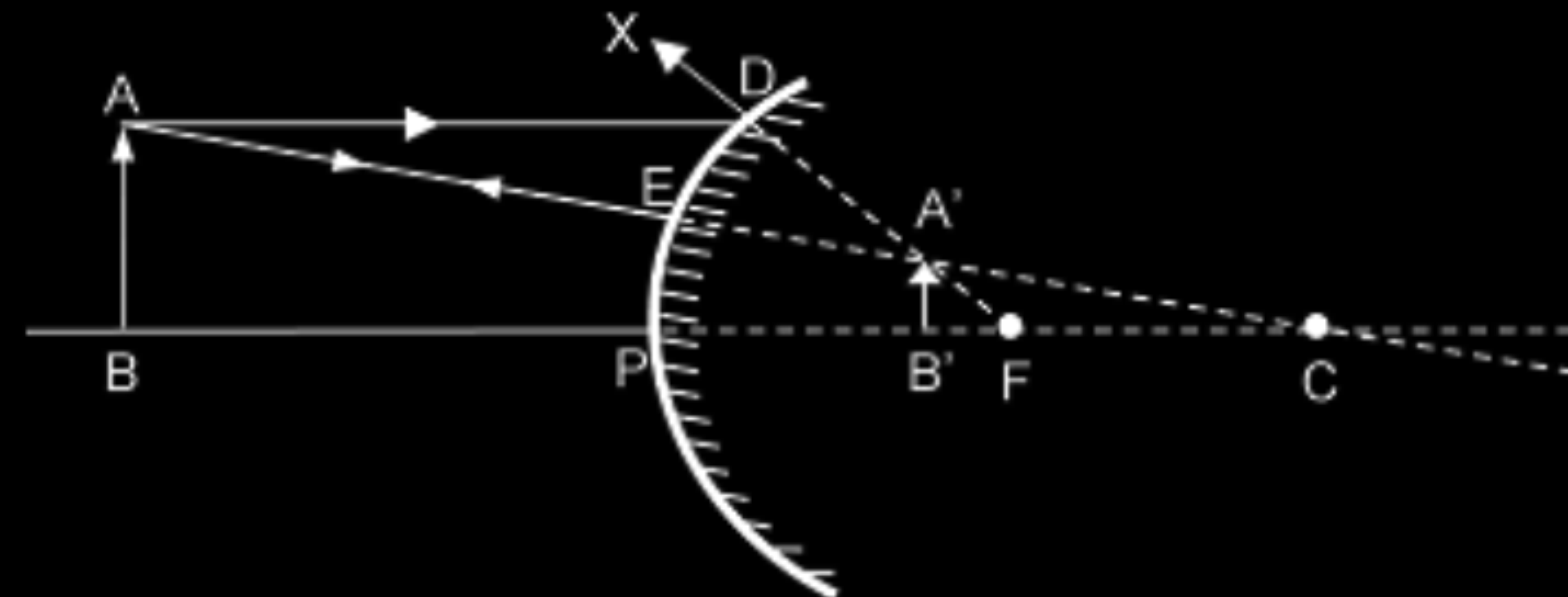
30. (a) Complete the following ray diagram to show the formation of image : 3



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- (b) Mention the nature, position and size of the image formed in this case.
- (c) State the sign of the image distance in this case using the Cartesian sign convention.

(a)



$A'B'$ is the image formed.

Credit full mark if attempted.

(b)

Nature: Virtual and erect

Position: Behind the mirror (between P and F)

1

$\frac{1}{2}$

$\frac{1}{2}$

Size: Diminished

$\frac{1}{2}$

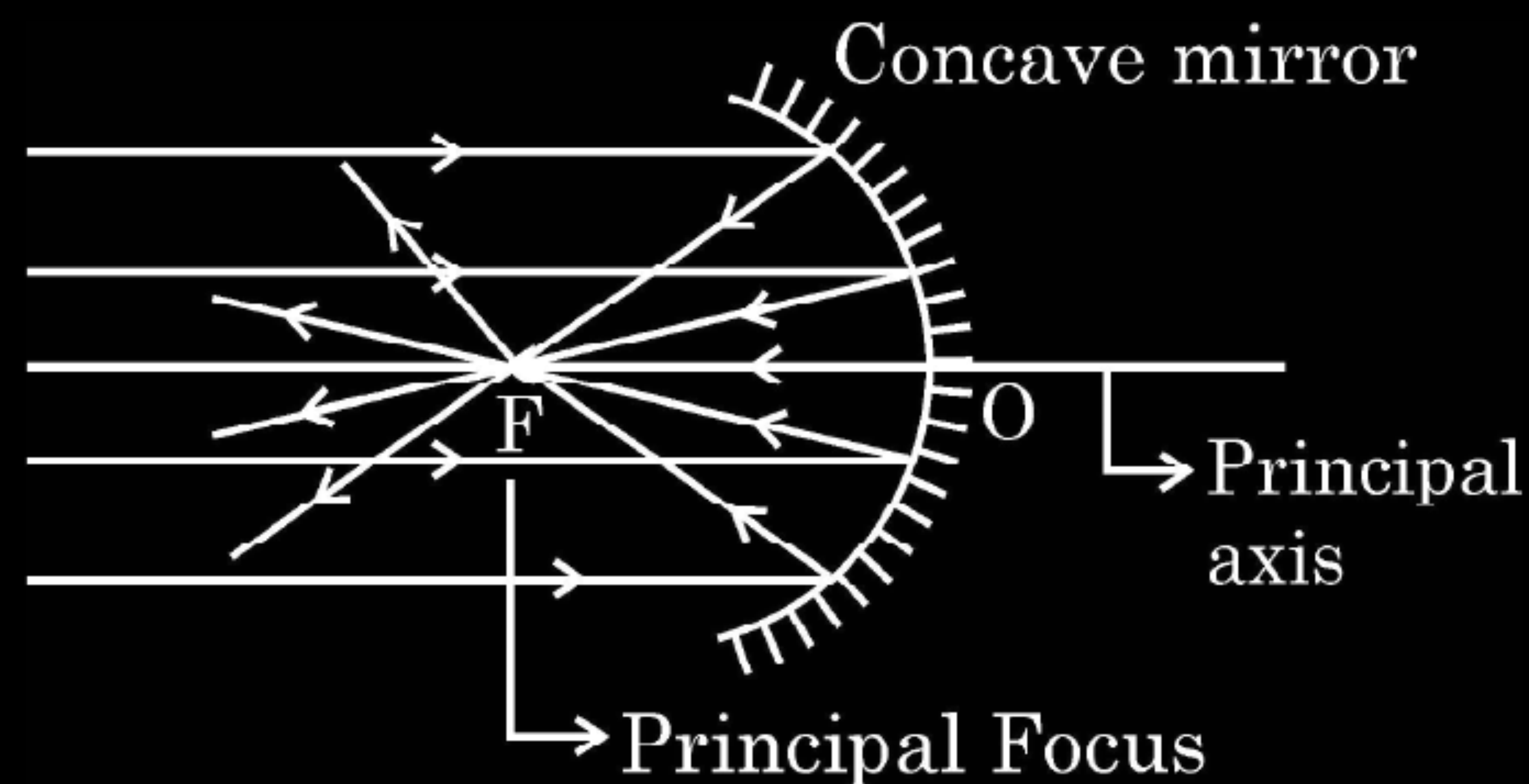
(c)

Positive

$\frac{1}{2}$

39. Hold a concave mirror in your hand and direct its reflecting surface towards the sun. Direct the light reflected by the mirror on to a white card-board held close to the mirror. Move the card-board back and forth gradually until you find a bright, sharp spot of light on the board. This spot of light is the image of the sun on the sheet of paper; which is also termed as “Principal Focus” of the concave mirror.

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- (a) List two applications of concave mirror. 1
- (b) If the distance between the mirror and the principal focus is 15 cm, find the radius of curvature of the mirror. 1

- (c) Draw a ray diagram to show the type of image formed when an object is placed between pole and focus of a concave mirror. **2**

OR

- (c) An object 10 cm in size is placed at 100 cm in front of a concave mirror. If its image is formed at the same point where the object is located, find : **2**
- (i) focal length of the mirror, and
 - (ii) magnification of the image formed with sign as per Cartesian sign convention.

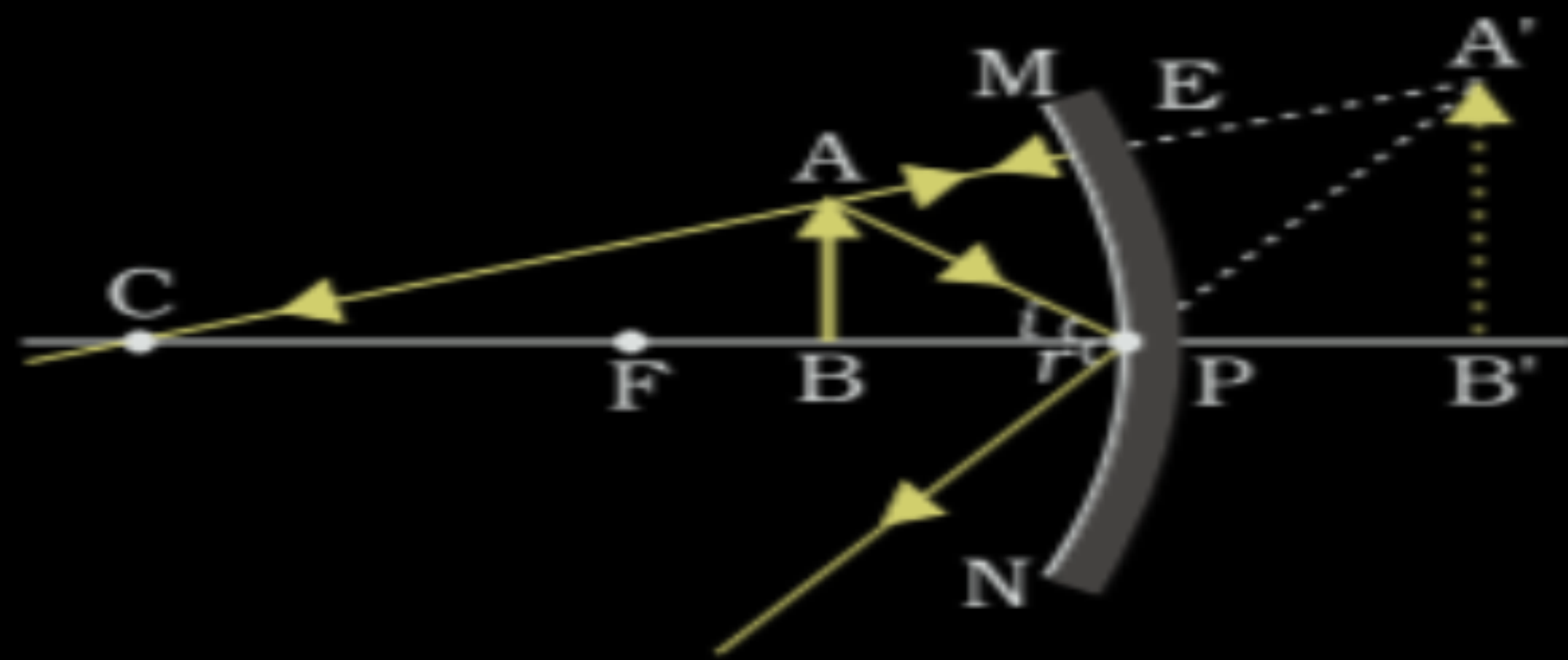
(a) Torches, search light, vehicles head lights, shaving mirrors,
dentist's mirror, Solar furnaces. (any two)

(b) $f = 15\text{cm}$

$$R = 2f$$

$$R = 2 \times 15\text{ cm} = 30\text{ cm}$$

(c)



(Note: $\frac{1}{2}$ mark to be deducted for not drawing the arrows.)

$\frac{1}{2}, \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

2

(c)

(i) $h = + 10\text{cm}$

$$u = - 100\text{ cm}$$

$$v = - 100\text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{100} - \frac{1}{100} = \frac{1}{f}$$

$$\frac{-2}{100} = \frac{1}{f}$$

$$f = -50\text{ cm}$$

Alternate answer for (i)

$$\text{Since } u = v$$

Therefore, object is placed at centre of curvature (C)

$$f = \frac{R}{2}$$

$$f = \frac{-100}{2}$$

$$f = -50\text{ cm}$$

$$(ii) \quad m = \frac{-v}{u} = \frac{-(-100)}{100} = -1$$

30. (a) The image of an object formed by a lens is of same size but inverted.

If the object distance is 30 cm, calculate

3

(i) The distance between the object and its image.

(ii) Focal length of the lens.

(b) Draw a ray diagram to show the image formed in above case.

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30. (a) The image of an object formed by a lens is of same size but inverted.
If the object distance is 30 cm, calculate
- The distance between the object and its image.
 - Focal length of the lens.
- (b) Draw a ray diagram to show the image formed in above case.

3

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(a)

(i) As the image is of same size

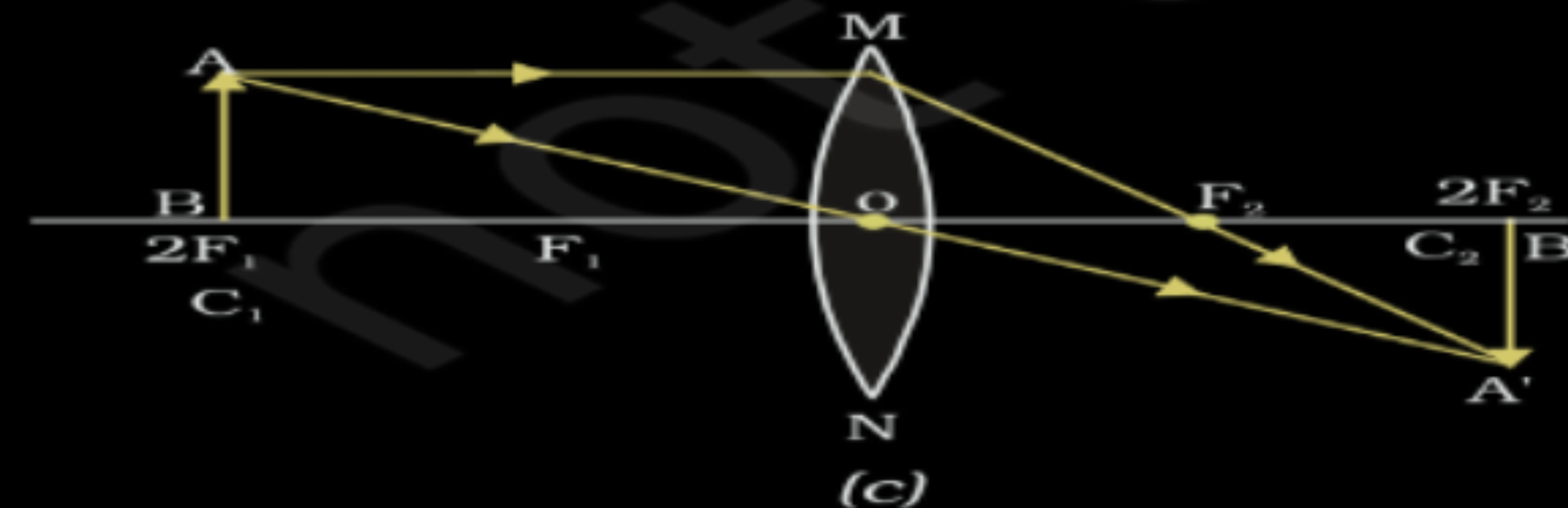
$\therefore$ Object distance = $2F = 30$ cm

$\therefore$ The image will be formed on the right side of lens at $2F' = 30$ cm

The distance between the object and its image = 60 cm

(ii) $f = 15$ cm

(iii)



$\frac{1}{2}$

$\frac{1}{2}$

1

1

29. A student wants to project the image of a candle flame on a screen 60 cm in front of a mirror by keeping the candle at a distance of 15 cm from its pole.

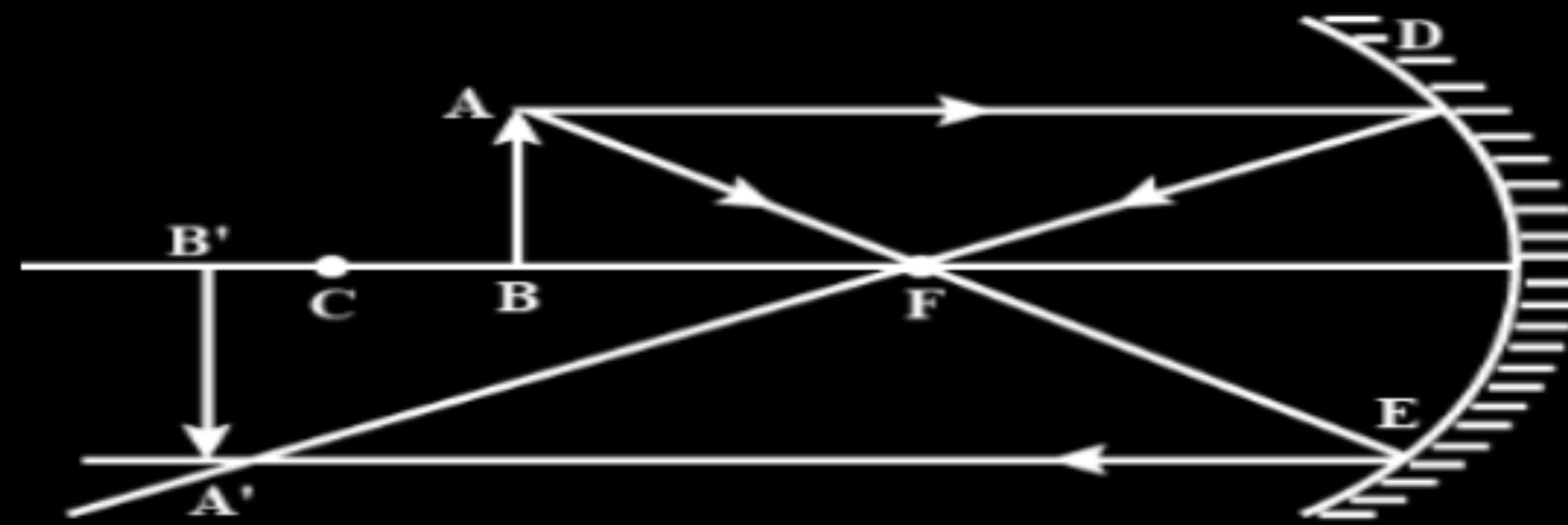
- (a) Name the type of mirror used.
- (b) Also calculate :
 - (i) Magnification of the image produced
 - (ii) Distance between object and its image
- (c) Draw a ray diagram to show the image formation.

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29. A student wants to project the image of a candle flame on a screen 60 cm in front of a mirror by keeping the candle at a distance of 15 cm from its pole.

- (a) Name the type of mirror used.
- (b) Also calculate :
 - (i) Magnification of the image produced
 - (ii) Distance between object and its image
- (c) Draw a ray diagram to show the image formation.

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(a) Concave Mirror / Converging Mirror	$\frac{1}{2}$
(b) (i). $m = \frac{-v}{u} = - \frac{(-60)}{(-15)} = -4$	$\frac{1}{2} + \frac{1}{2}$
(ii). 45 cm from the object	$\frac{1}{2}$
(c)  (Note: $\frac{1}{2}$ mark to be deducted for not drawing the arrows.)	1

30. (A) Define the following terms in the context of a diverging mirror :

3

(i) Principal focus

(ii) Focal length

Draw a labelled ray diagram to illustrate your answer.

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30. (A) Define the following terms in the context of a diverging mirror :

3

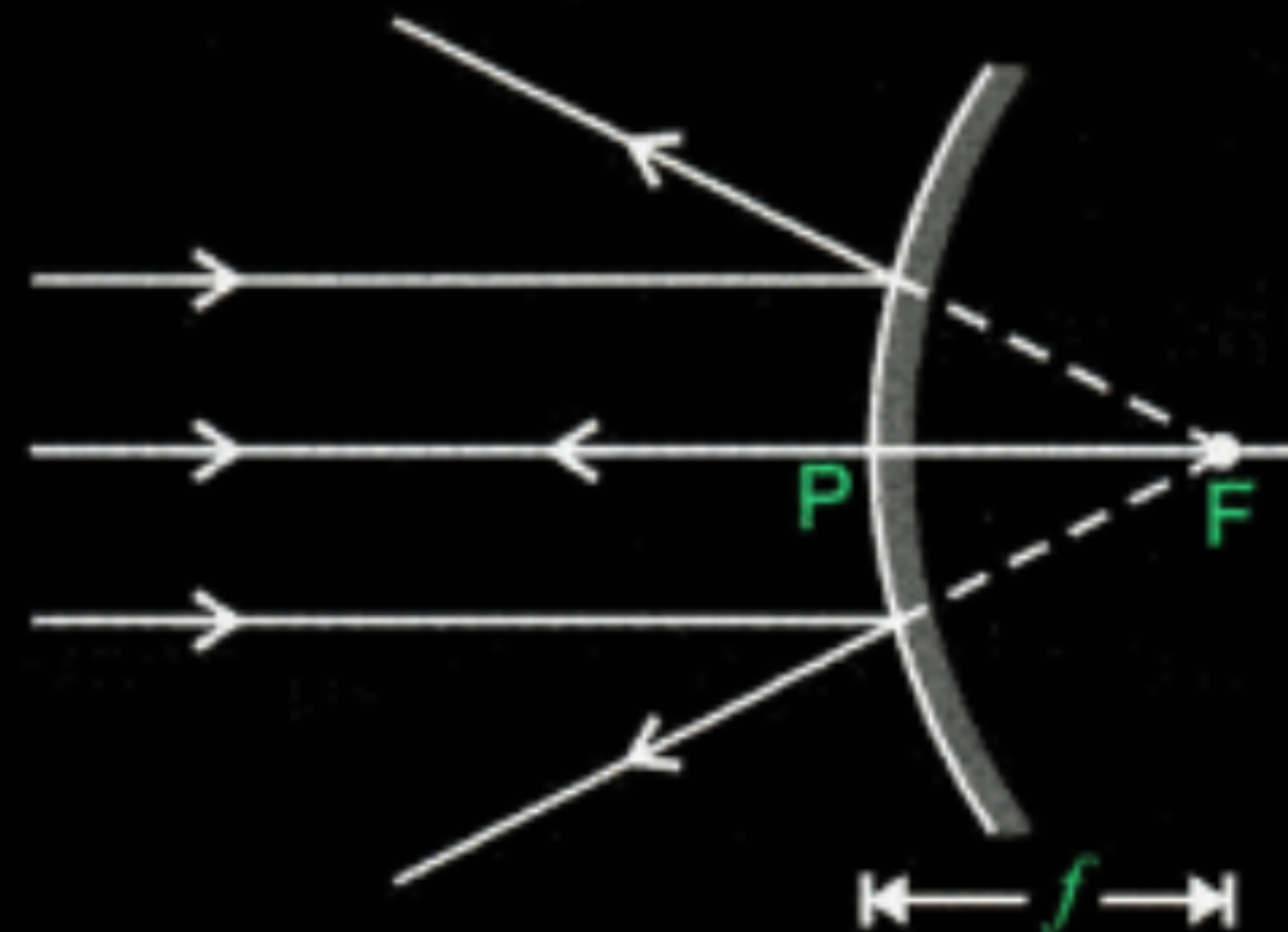
- (i) Principal focus
- (ii) Focal length

Draw a labelled ray diagram to illustrate your answer.

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(A)

- (i) It is a point on the principal axis of a diverging mirror from where the rays parallel to principal axis appear to diverge after reflection.
- (ii) The distance between the pole and the principal focus of a mirror.



1

1

1

(B) An object of height 10 cm is placed 25 cm away from the optical centre of a converging lens of focal length 15 cm. Calculate the image-distance and height of the image formed.

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3

(B) An object of height 10 cm is placed 25 cm away from the optical centre of a converging lens of focal length 15 cm. Calculate the image-distance and height of the image formed.

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3

$$(B) \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$f=15 \text{ cm}, u=-25 \text{ cm}, h=10 \text{ cm}$$

$$\frac{1}{v} = \frac{1}{15 \text{ cm}} + \frac{1}{-25 \text{ cm}} = \frac{2}{75} = + \frac{1}{37.5}$$

$$v = 37.5 \text{ cm}$$

$$\text{height of the image} = \frac{v}{u} \times \text{height of the object}$$

$$= \frac{37.5}{-25 \text{ cm}} \times 10 \text{ cm}$$

$$h' = -15 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

- 31.** The power of a lens is $+4\text{D}$. Find the focal length of this lens. An object is placed at a distance of 50 cm from the optical centre of this lens. State the nature and magnification of the image formed by the lens and also draw a ray diagram to justify your answer.

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31. The power of a lens is +4D. Find the focal length of this lens. An object is placed at a distance of 50 cm from the optical centre of this lens. State the nature and magnification of the image formed by the lens and also draw a ray diagram to justify your answer.

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3

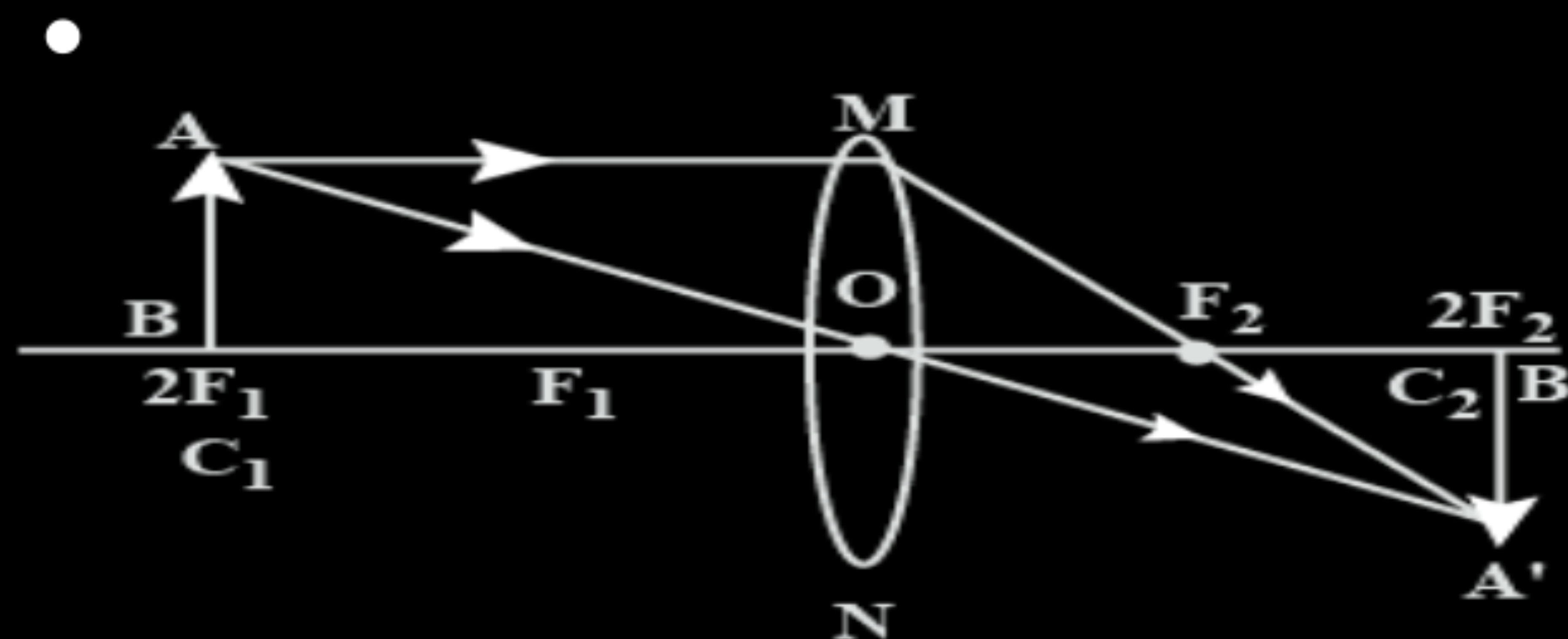
- Focal length of lens, $f(m) = \frac{1}{P}$

$$P = +4.0 \text{ D}$$

$$\Rightarrow f = \frac{1}{+4 \text{ D}} = 0.25 \text{ m} = 25 \text{ cm}$$

- Real and inverted

- Magnification = -1



[Deduct ½ mark if direction of rays is not marked]

½

½

½

½

1

39. The ability of a medium to refract light is expressed in terms of its optical density. Optical density has a definite connotation. It is not the same as mass density. On comparing two media, the one with the large refractive index is optically denser medium than the other. The other medium with a lower refractive index is optically rarer. Also the speed of light through a given medium is inversely proportional to its optical density. 4
- (i) Determine the speed of light in diamond if the refractive index of diamond with respect to vacuum is 2.42. Speed of light in vacuum is 3×10^8 m/s. 1
- (ii) Refractive indices of glass, water and carbon disulphide are 1.5, 1.33 and 1.62 respectively. If a ray of light is incident in these media at the same angle (say θ), then write the increasing order of the angle of refraction in these media. 1
- (iii) (A) The speed of light in glass is 2×10^8 m/s and in water is 2.25×10^8 m/s. 2
- (a) Which one of the two is optically denser and why ?
- (b) A ray of light is incident normally at the water-glass interface when it enters a thick glass container filled with water. What will happen to the path of the ray after entering the glass ? Give reason.

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OR

- (iii) (B) The absolute refractive indices of water and glass are $4/3$ and $3/2$ respectively. If the speed of light in glass is 2×10^8 m/s, find the speed of light in (i) vacuum and (ii) water. 2

(i) Refractive index of diamond = $\frac{\text{Speed of light in vacuum}}{\text{Speed of light in diamond}}$	$\frac{1}{2}$
Speed of light in diamond = $\frac{3 \times 10^8 \text{ m/s}}{2.42} = 1.23 \times 10^8 \text{ m/s}$	$\frac{1}{2}$
(ii) $\angle r$ in carbon disulphide < $\angle r$ in glass < $\angle r$ in water	1
(iii) (A)	$\frac{1}{2}$
(a) • Glass	
• The speed of light in water is more than the speed of light in glass. / Refractive index of glass is more than the refractive index of water.	$\frac{1}{2}$
(b) Light will enter from water to glass without bending (undeviated / straight) because in this case $\angle i = 0$; $\angle r = 0$.	1
OR	
(iii) (B)	
$n_{\text{glass}} = \frac{3}{2}$	
$n_{\text{water}} = \frac{4}{3}$	
$v_{\text{glass}} = 2 \times 10^8 \text{ m/s}$	
$n_{\text{glass}} = \frac{\text{speed of light in vacuum}(c)}{\text{speed of light in glass}(v_g)}$	
$c = n_{\text{glass}} \times v_{\text{glass}}$ $= \frac{3}{2} \times 2 \times 10^8 \text{ m/s}$	
$= 3 \times 10^8 \text{ m/s}$	1
$v_{\text{water}} = \frac{c}{n_{\text{water}}} = \frac{3 \times 10^8 \text{ m/s}}{\frac{4}{3}}$	
$= \frac{9}{4} \times 10^8 \text{ m/s}$ or $2.25 \times 10^8 \text{ m/s}$	1

The ability of a medium to refract light is expressed in terms of its optical density. Optical density has a definite connotation. It is not the same as mass density. On comparing two media, the one with the large refractive index is optically denser medium than the other. The other medium with a lower refractive index is optically rarer. Also the speed of light through a given medium is inversely proportional to its optical density.	4
(i) Determine the speed of light in diamond if the refractive index of diamond with respect to vacuum is 2.42. Speed of light in vacuum is $3 \times 10^8 \text{ m/s}$.	1
(ii) Refractive indices of glass, water and carbon disulphide are 1.5, 1.33 and 1.62 respectively. If a ray of light is incident in these media at the same angle (say θ), then write the increasing order of the angle of refraction in these media.	1
(iii) (A) The speed of light in glass is $2 \times 10^8 \text{ m/s}$ and in water is $2.25 \times 10^8 \text{ m/s}$.	2
(a) Which one of the two is optically denser and why ?	
(b) A ray of light is incident normally at the water-glass interface when it enters a thick glass container filled with water. What will happen to the path of the ray after entering the glass ? Give reason.	
OR	
(iii) (B) The absolute refractive indices of water and glass are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is $2 \times 10^8 \text{ m/s}$, find the speed of light in (i) vacuum and (ii) water.	2

30. (A) An object is kept at a distance of 1m from a lens of power +2D : **3**

(i) Identify the type of lens.

(ii) Calculate its focal length and distance of the image formed. **CBSE 2023**

30. (A) An object is kept at a distance of 1m from a lens of power +2D :

3

(i) Identify the type of lens.

(ii) Calculate its focal length and distance of the image formed.

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(A) (i) Convex lens / Converging lens

1

(ii) Power, $P = +2D$

$$f(m) = \frac{1}{P}$$

$\frac{1}{2}$

$$f = \frac{1}{2} = 0.5 \text{ m} = 50\text{cm}$$

$\frac{1}{2}$

$$u = -100\text{cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$\frac{1}{2}$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{50} - \frac{1}{100} = \frac{1}{100}$$

$$v = 100 \text{ cm} = 1 \text{ m}$$

$\frac{1}{2}$

Alternate answer only for image distance

Since object is at $2f$, image will be formed at $2f$. Therefore, image distance is 100 cm.

as $f = 50 \text{ cm}$, $u = 1\text{m}$ or $100 \text{ cm} = 2f$

$\therefore u = v = 1\text{m}$

(B) Define the following terms in the context of a diverging lens :

3

(i) Principal focus,

(ii) Focal length.

Draw a labelled ray diagram to illustrate your answer.

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(B) Define the following terms in the context of a diverging lens :

3

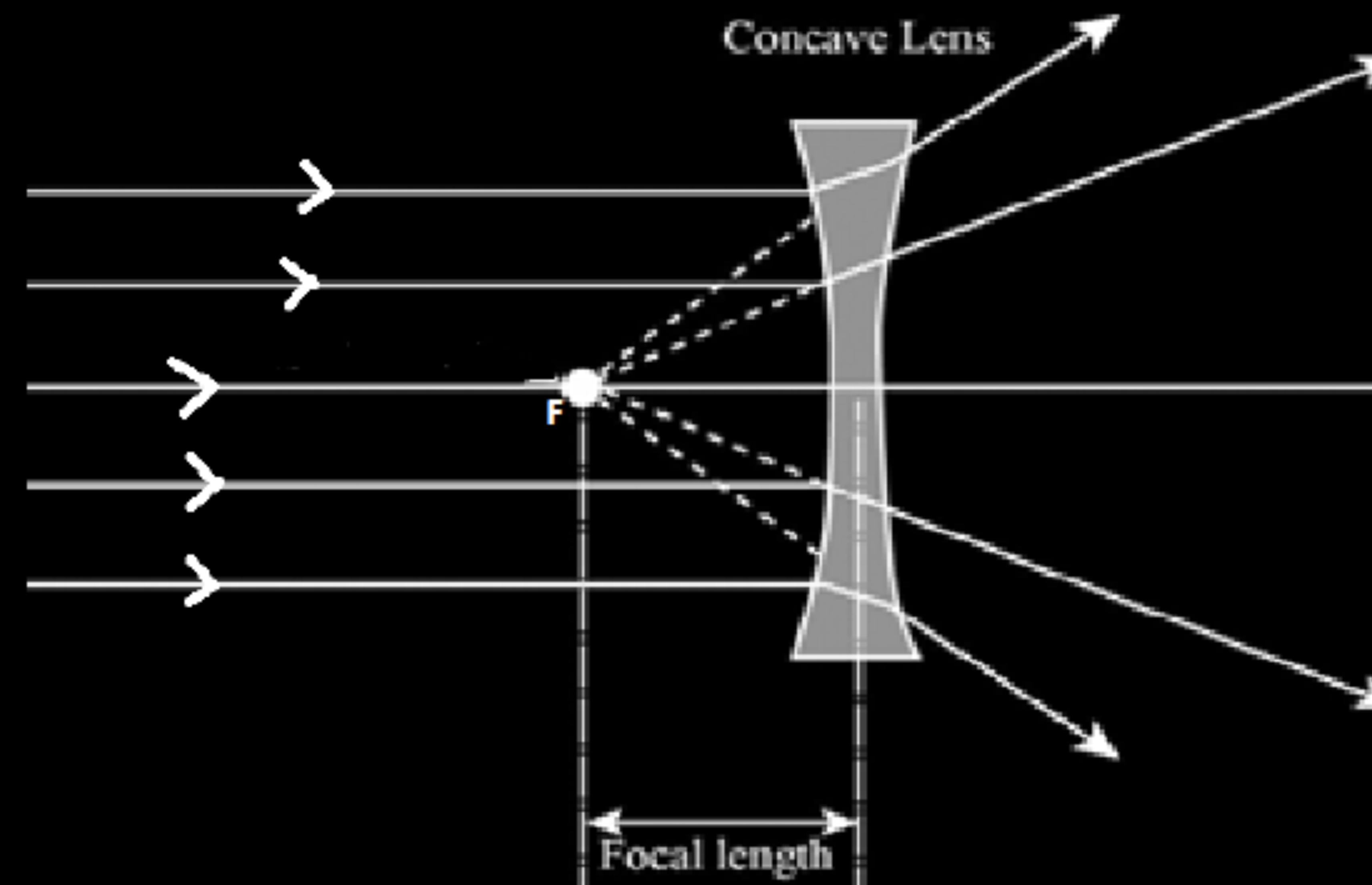
(i) Principal focus,

(ii) Focal length.

Draw a labelled ray diagram to illustrate your answer.

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- (B) (i) It is a point on the principal axis of a diverging lens from where the rays parallel to principal axis appear to diverge.
- (ii) The distance between the optical centre and the principal focus of the lens.



1

1

1

29. The magnification produced when an object is placed at a distance of 20 cm from a spherical mirror is $+1/2$. Where should the object be placed to reduce the magnification to $+1/3$.

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29. The magnification produced when an object is placed at a distance of 20 cm from a spherical mirror is $+1/2$. Where should the object be placed to reduce the magnification to $+1/3$.

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3

Given,

Distance of object, $u = -20$ cm

Magnification $m = \frac{1}{2}$

$$\therefore m = \frac{-v}{u} \Rightarrow v = -\frac{1}{2} \times -20 = 10 \text{ cm}$$

Using mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{10} - \frac{1}{20} = \frac{1}{20}$$

$$f = 20 \text{ cm}$$

If $m = \frac{1}{3}$ and v' and u' are the corresponding image distance and object

$$\text{distance. } \frac{1}{3} = \frac{-v'}{u'} \Rightarrow v' = \frac{-u'}{3}$$

$$\frac{1}{f} = \frac{1}{v'} + \frac{1}{u'}$$

$$\frac{1}{20} = \frac{-3}{u'} + \frac{1}{u'} \Rightarrow \frac{1}{20} = \frac{-2}{u'}$$

$$u' = -40 \text{ cm.}$$

$$\frac{1}{2}, \frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

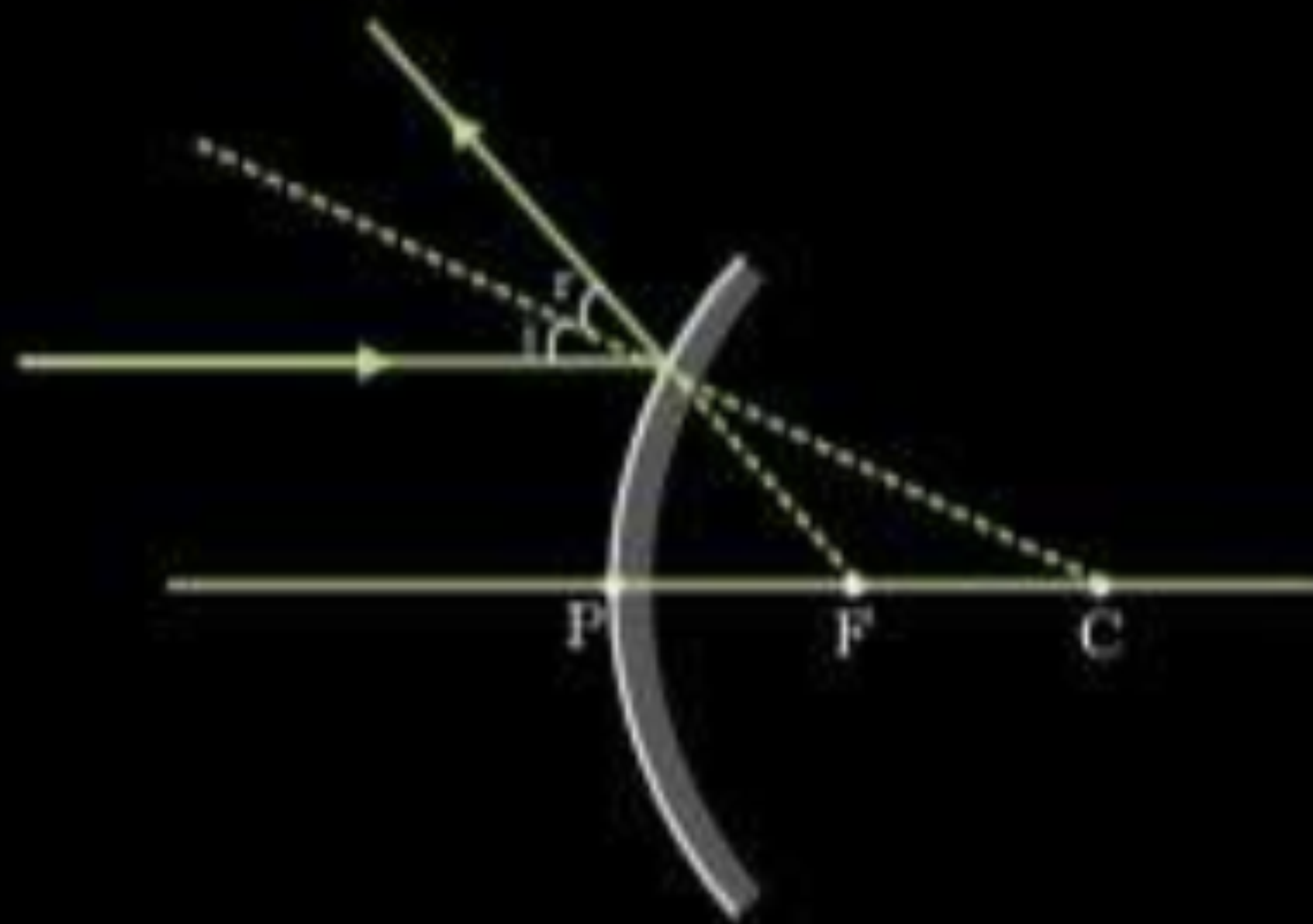
$$\frac{1}{2}$$

- 22.** Draw ray diagram in each of the following cases to show what happens after reflection to the incident ray when
- (a) it is parallel to the principal axis and falling on a convex mirror.
 - (b) it is falling on a concave mirror while passing through its principal focus.
 - (c) it is coming oblique to the principal axis and falling on the pole of a convex mirror.

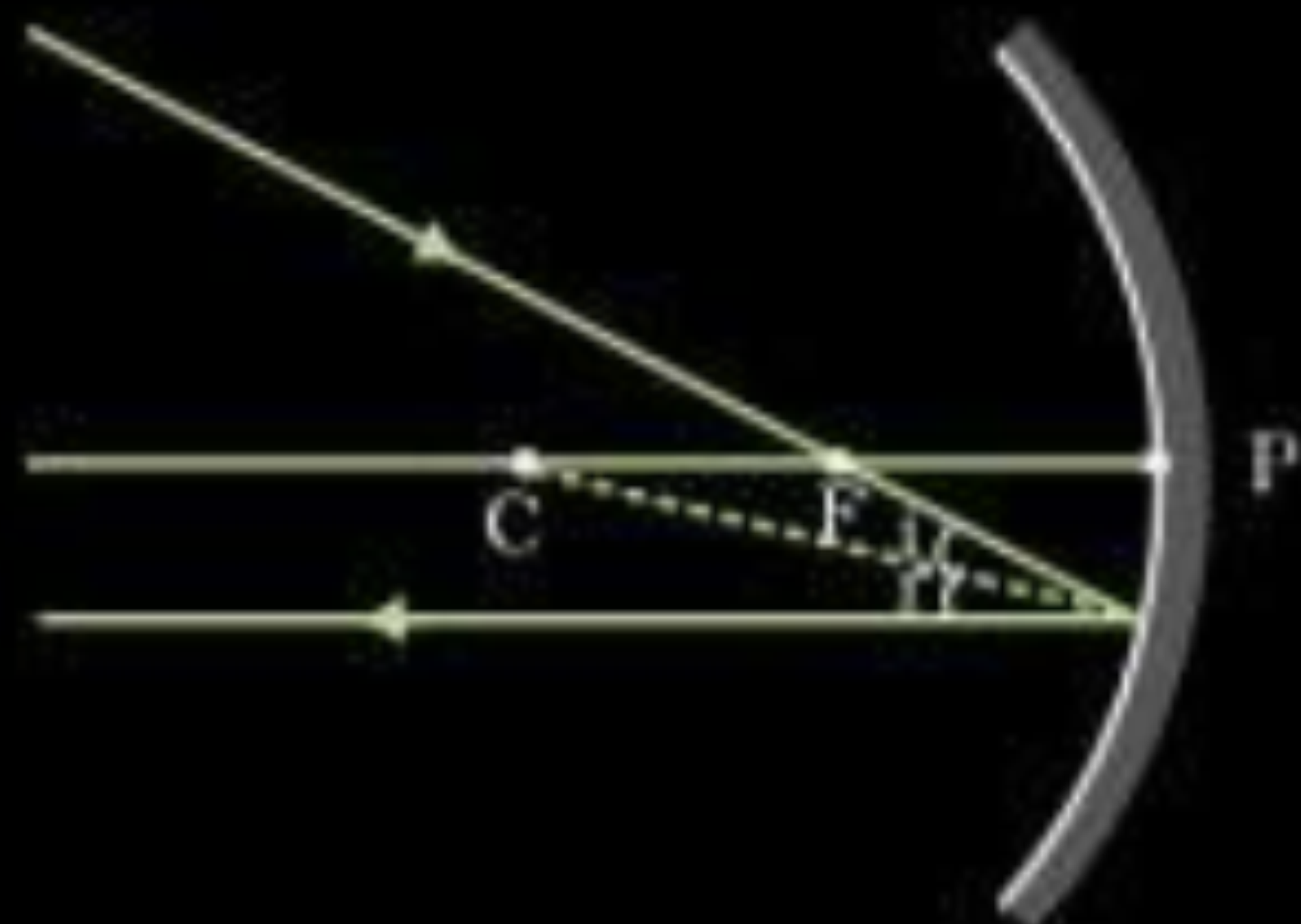
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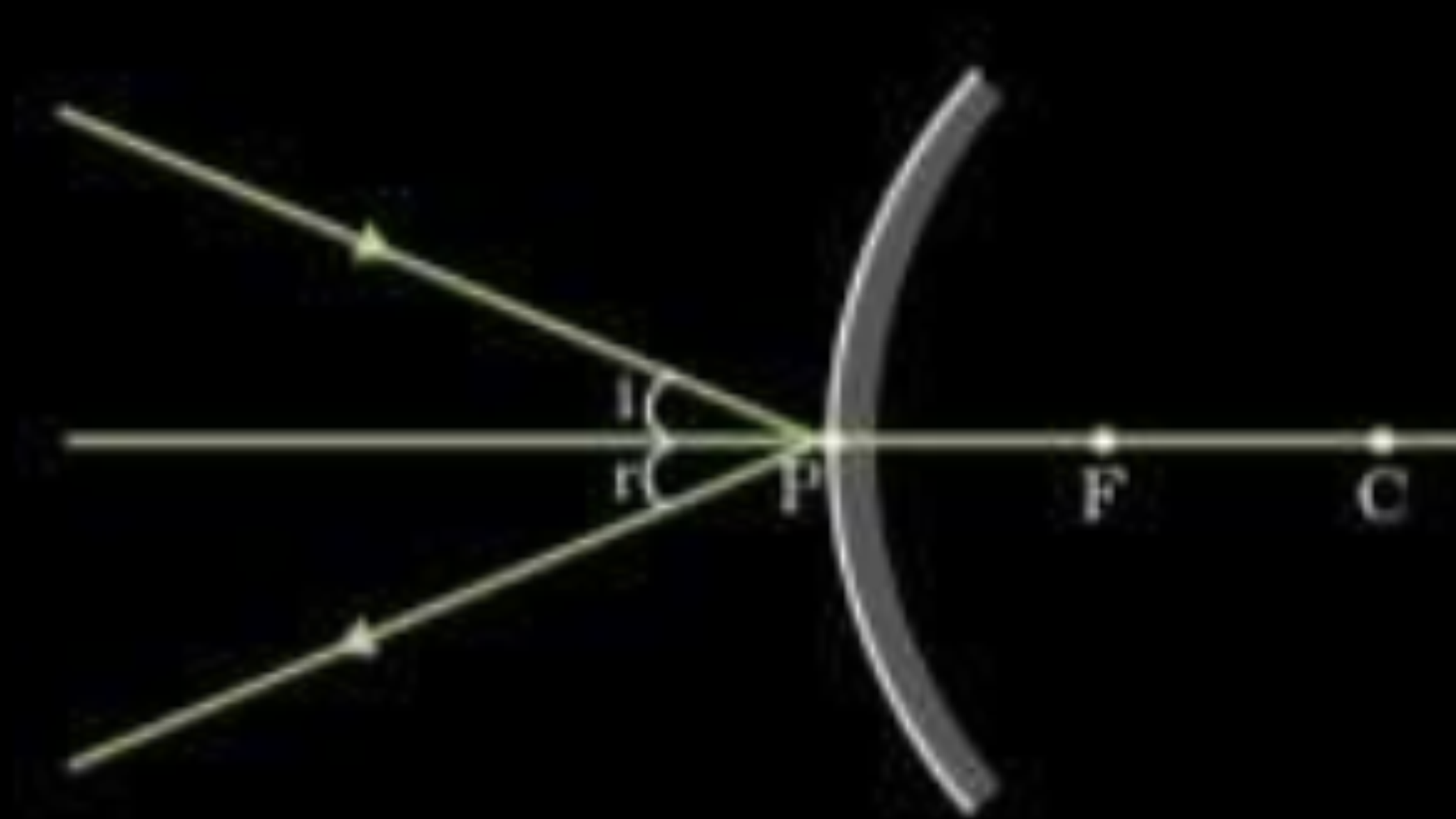
3



1



1



1

(Note : Deduct $\frac{1}{2}$ marks overall if no arrows are shown)

- 30.** (a) Draw a labelled ray diagram to show the path of a ray of light incident obliquely on one face of a glass slab.
- (b) Calculate the refractive index of the material of a glass slab. Given that the speed of light through the glass slab is 2×10^8 m/s and in air is 3×10^8 m/s.
- (c) Calculate the focal length of a lens, if its power is -2.5 D. CBSE 2020 5

- 30.** (a) Draw a labelled ray diagram to show the path of a ray of light incident obliquely on one face of a glass slab.
- (b) Calculate the refractive index of the material of a glass slab. Given that the speed of light through the glass slab is 2×10^8 m/s and in air is 3×10^8 m/s.
- (c) Calculate the focal length of a lens, if its power is -2.5 D. **CBSE 2020**

5

(a) .



(Note –Deduct $\frac{1}{2}$ mark if arrows are not shown)

$$(b) n_{ga} = \frac{\text{Speed of light in air}}{\text{Speed of light in glass}} = \frac{3 \times 10^8}{2 \times 10^8} = \frac{3}{2} = 1.5$$

$$(c) f (m) = \frac{1}{P (D)}$$

$$f = \frac{1}{P} = \frac{1}{-2.5D} = \frac{-10}{25D} = -0.4m$$

(Note –Deduct $\frac{1}{2}$ marks if unit is not mentioned)

2

$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

21. A concave mirror is used for image formation for different positions of an object. What inferences can be drawn about the following when an object is placed at a distance of 10 cm from the pole of a concave mirror of focal length 15 cm ?

- (a) Position of the image
- (b) Size of the image
- (c) Nature of the image

Draw a labelled ray diagram to justify your inferences.

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21. A concave mirror is used for image formation for different positions of an object. What inferences can be drawn about the following when an object is placed at a distance of 10 cm from the pole of a concave mirror of focal length 15 cm ?

- (a) Position of the image
- (b) Size of the image
- (c) Nature of the image

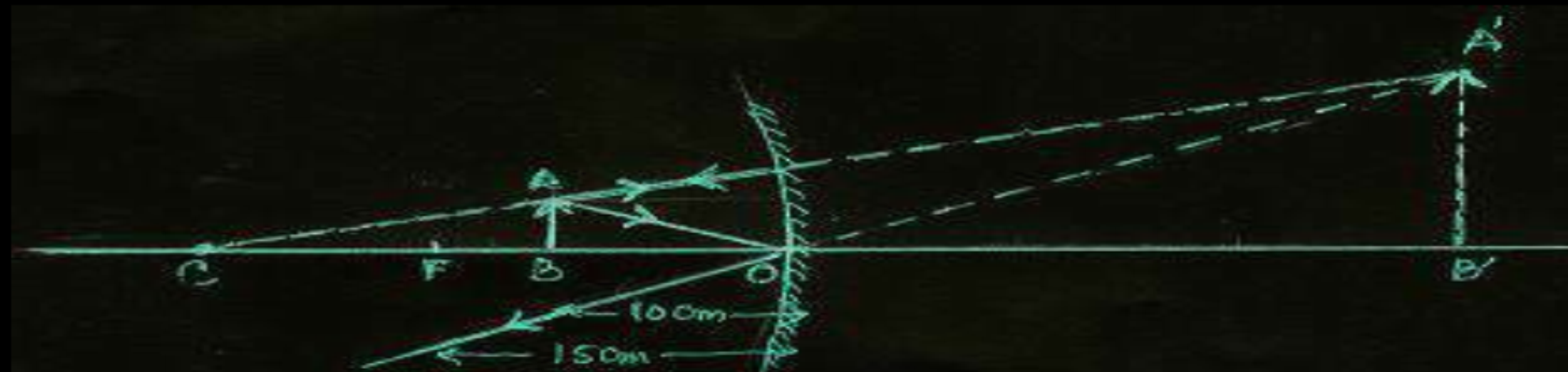
Draw a labelled ray diagram to justify your inferences.

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3

- (a) Behind the mirror
- (b) Magnified
- (c) Virtual and erect

Labelled ray diagram



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$1\frac{1}{2}$

- 22.** The refractive index of a medium 'x' with respect to a medium 'y' is $\frac{2}{3}$ and the refractive index of medium 'y' with respect to medium 'z' is $\frac{4}{3}$. Find the refractive index of medium 'z' with respect to medium 'x'. If the speed of light in medium 'x' is $3 \times 10^8 \text{ ms}^{-1}$, calculate the speed of light in medium 'y'.

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22. The refractive index of a medium 'x' with respect to a medium 'y' is $\frac{2}{3}$ and the refractive index of medium 'y' with respect to medium 'z' is $\frac{4}{3}$. Find the refractive index of medium 'z' with respect to medium 'x'. If the speed of light in medium 'x' is $3 \times 10^8 \text{ ms}^{-1}$, calculate the speed of light in medium 'y'.

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3

$$\bullet \quad n_{xy} = \frac{2}{3} \quad \therefore \quad n_{yx} = \frac{3}{2}$$

$$n_{yz} = \frac{4}{3} \quad \therefore \quad n_{zy} = \frac{3}{4}$$

$$n_{zx} = n_{zy} \times n_{yx}$$

$$\therefore \quad n_{zx} = \frac{3}{4} \times \frac{3}{2} = \frac{9}{8}$$

$$\bullet \quad n_{yx} = \frac{v_x}{v_y}$$

$$\frac{3}{2} = \frac{3 \times 10^8}{v_y}$$

$$v_y = \frac{3 \times 10^8 \times 2}{3} = 2 \times 10^8 \text{ m/s}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

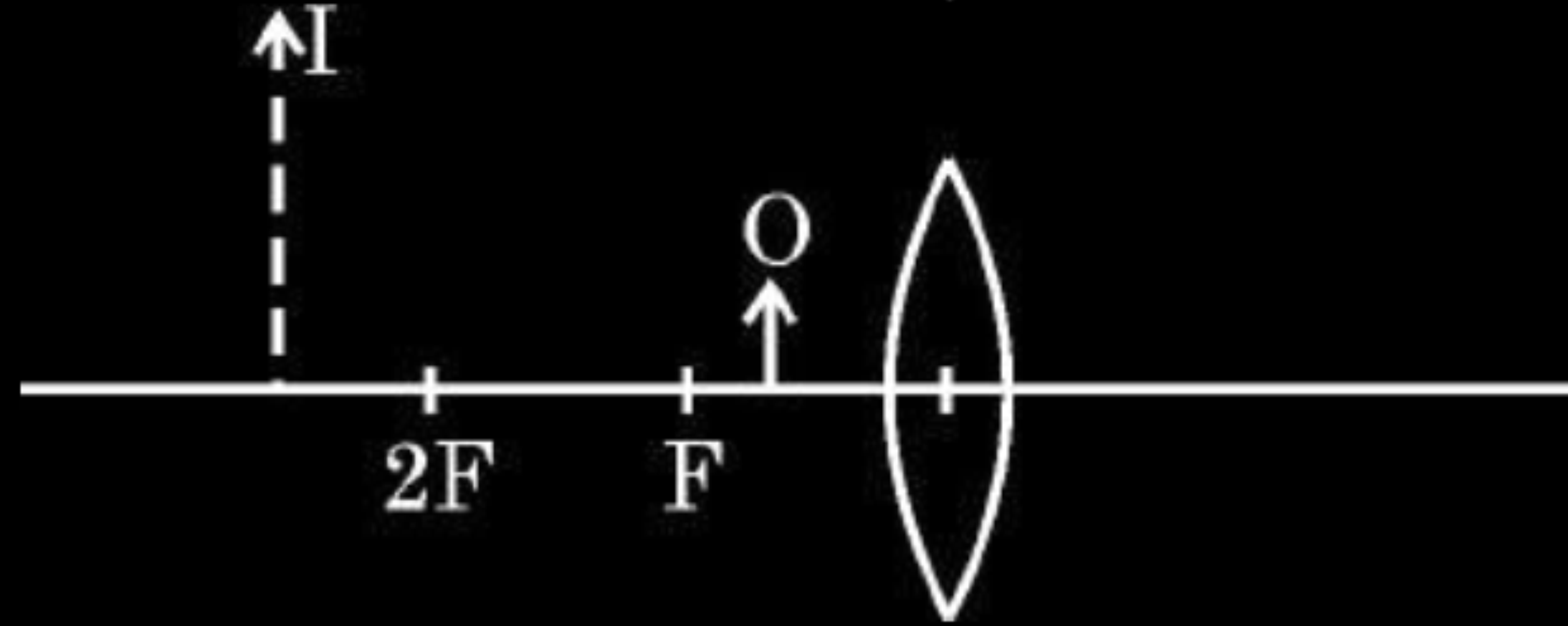
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

22. The diagram given below shows an object O and its image I.

CBSE 2020

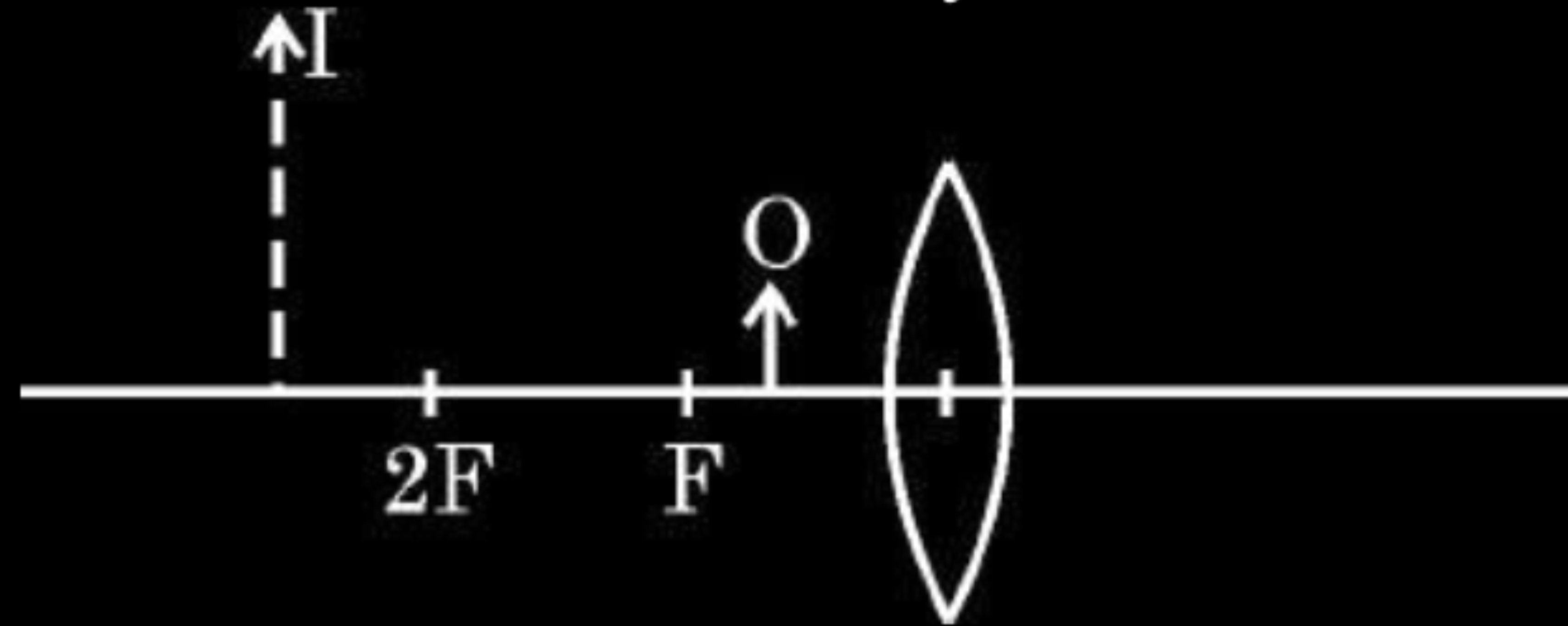


Without actually drawing the ray diagram, state the following :

- (i) Type of lens (Converging / Diverging)
- (ii) Name two optical instruments where such an image is obtained.
- (iii) List three characteristics of the image formed if this lens is replaced by a concave mirror of focal length ' f ' and an object is placed at a distance ' $f/2$ ' in front of the mirror.

22. The diagram given below shows an object O and its image I.

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Without actually drawing the ray diagram, state the following :

- (i) Type of lens (Converging / Diverging)
- (ii) Name two optical instruments where such an image is obtained.
- (iii) List three characteristics of the image formed if this lens is replaced by a concave mirror of focal length ' f ' and an object is placed at a distance ' $f/2$ ' in front of the mirror.

3

(i) Converging Lens	$\frac{1}{2}$
(ii) Magnifying Glass, Microscope	$\frac{1}{2}$, $\frac{1}{2}$
(iii) Three Characteristics of the image : (a) Virtual (b) Erect (c) Magnified	$\frac{1}{2} \times 3$

30. (a) A security mirror used in a big showroom has radius of curvature 5 m. If a customer is standing at a distance of 20 m from the cash counter, find the position, nature and size of the image formed in the security mirror.
- (b) Neha visited a dentist in his clinic. She observed that the dentist was holding an instrument fitted with a mirror. State the nature of this mirror and reason for its use in the instrument used by dentist.

5

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30. (a) A security mirror used in a big showroom has radius of curvature 5 m. If a customer is standing at a distance of 20 m from the cash counter, find the position, nature and size of the image formed in the security mirror.
- (b) Neha visited a dentist in his clinic. She observed that the dentist was holding an instrument fitted with a mirror. State the nature of this mirror and reason for its use in the instrument used by dentist.

5

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(a) It is a convex mirror. So focal length should be positive. Radius of curvature $R = +5 \text{ m}$ $\therefore$ focal length $f = \frac{5}{2} = +2.5 \text{ m}$ Object distance $u = -20 \text{ m}$ Mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} + \frac{1}{-20} = \frac{1}{2.5}$ $\frac{1}{v} = \frac{1}{20} + \frac{1}{2.5}$ $\frac{1}{v} = \frac{1}{20} + \frac{10}{25}$	$\frac{1}{v} = \frac{5+40}{100} = \frac{45}{100}$ $v = \frac{100}{45} = \frac{20}{9} = +2.2 \text{ m}$ - Nature of image = virtual and erect image - Size of image : diminished image (b) Concave Mirror Reason : to obtain erect and enlarged image of teeth
---	---

Rishi went to a palmist to show his palm. The palmist used a special lens for this purpose.

- (i) State the nature of the lens and reason for its use.
- (ii) Where should the palmist place/hold the lens so as to have a real and magnified image of an object ?
- (iii) If the focal length of this lens is 10 cm and the lens is held at a distance of 5 cm from the palm, use lens formula to find the position and size of the image.

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- (i) Convex lens to get a magnified image of the lines on the palm.
- (ii) Between F and 2F of the lens / or at F of the lens
- (iii) focal length $f = +10$ cm
object distance $u = -5$ cm

Lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} - \frac{1}{-5} = \frac{1}{10}$$

$$\frac{1}{v} + \frac{1}{5} = \frac{1}{10}$$

$$\frac{1}{v} = \frac{1}{10} - \frac{1}{5} = \frac{1-2}{10}$$

$$\frac{1}{v} = \frac{-1}{10}$$

$$v = -10 \text{ cm}$$

$$\bullet \quad m = \frac{h_{\text{image}}}{h_{\text{object}}} = \frac{v}{u}$$

$$= \frac{-10}{-5} = 2$$

Size of image is 2 times the size of object.

1

1

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

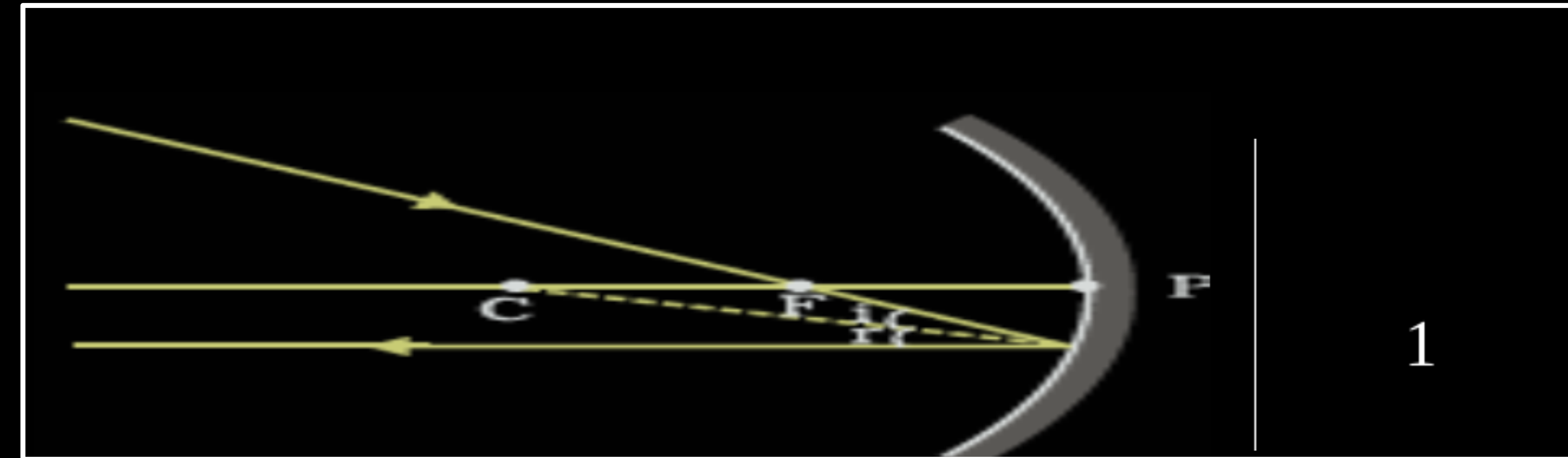
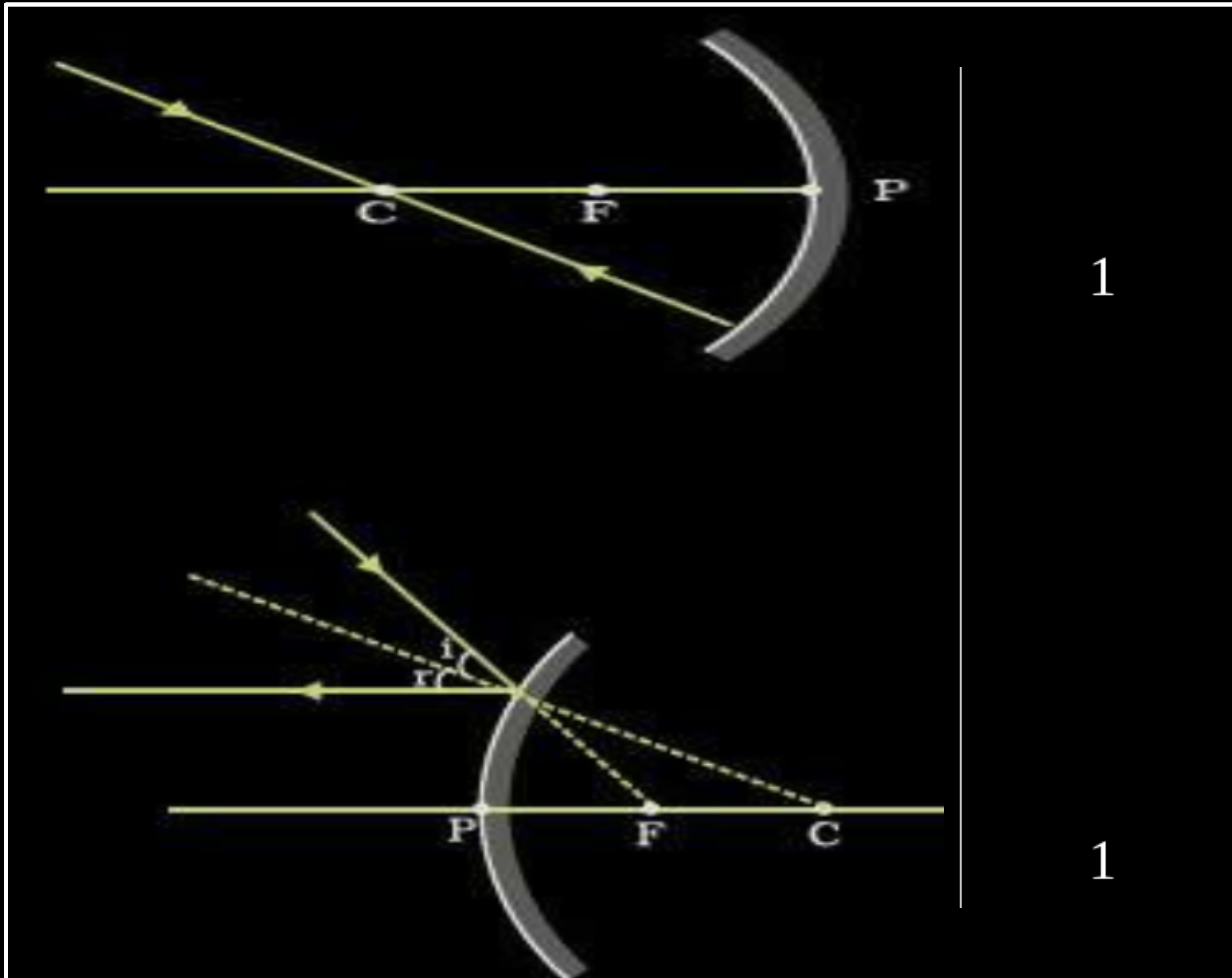
22. What happens after refraction, when :

- (i) a ray of light parallel to the principal axis passes through a concave lens ?
- (ii) a ray of light falls on a convex lens while passing through its principal focus ?
- (iii) a ray of light passes through the optical centre of a convex lens ?

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3



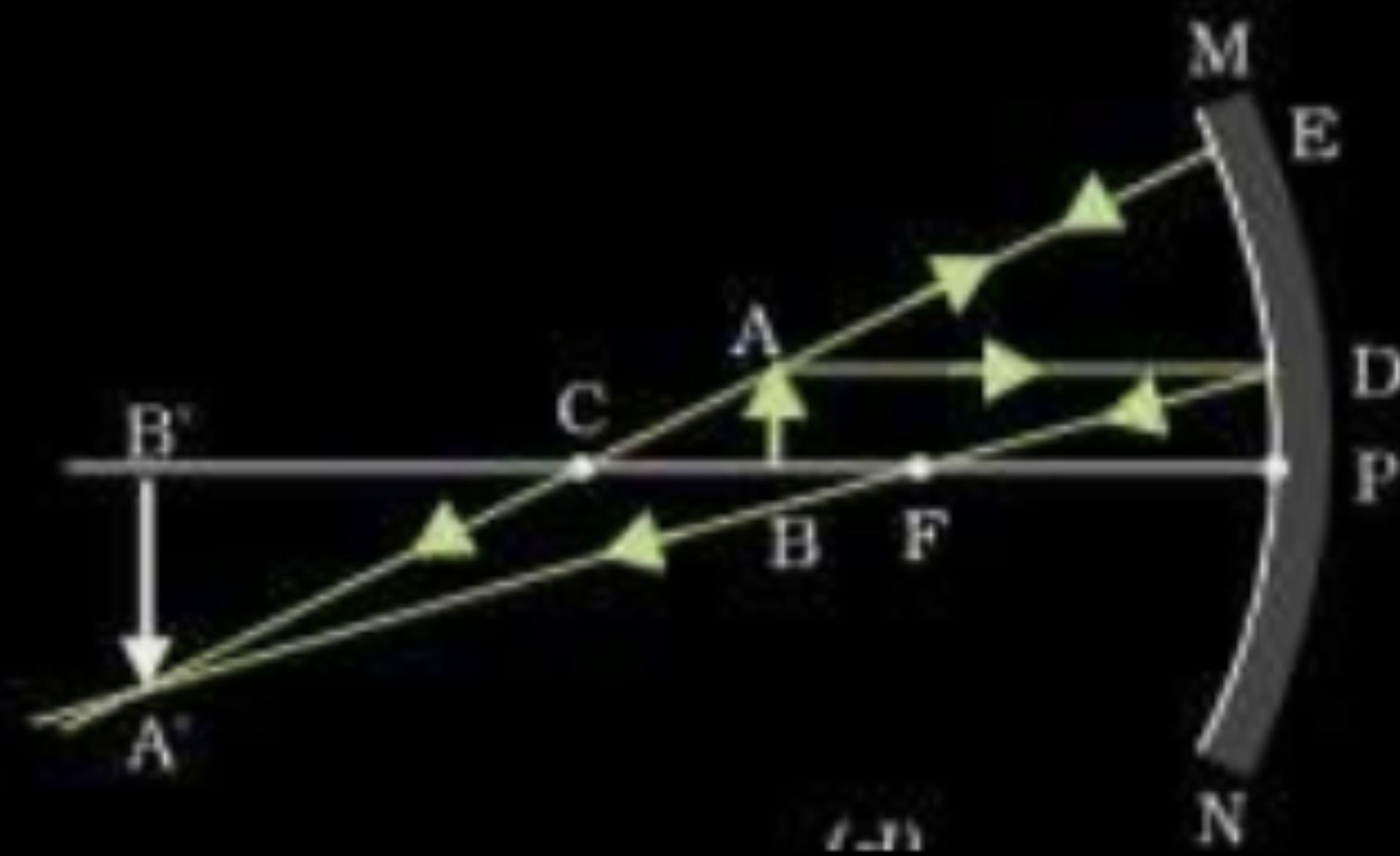
30. (a) A concave mirror of focal length 10 cm can produce a magnified real as well as virtual image of an object placed in front of it. Draw ray diagrams to justify this statement.

(b) An object is placed perpendicular to the principal axis of a convex mirror of focal length 10 cm. The distance of the object from the pole of the mirror is 10 cm. Find the position of the image formed.

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5

(a)
(i) Real and magnified



Object distance must be between 10 to 20 cm

(ii) Virtual and magnified



Object distance must be less than 10 cm

(b) $f = 10 \text{ cm}$; $u = -10 \text{ cm}$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{10} - \frac{1}{-10}$$

$$\frac{1}{v} = \frac{1}{10} + \frac{1}{10}$$

$$\frac{1}{v} = \frac{1}{5} \therefore v = 5 \text{ cm}$$

- (a) Define the following terms :
- (i) Power of a lens
 - (ii) Principal focus of a concave mirror
- (b) Write the relationship among the object distance (u), image distance (v) and the focal length (f) of a
- (i) Spherical lens
 - (ii) Spherical mirror
- (c) An object is placed at a distance of 10 cm from optical centre of a convex lens of focal length 15 cm. Draw a labelled ray diagram to show the formation of image in this case.

(a) (i) Ability of a lens to converge or diverge light rays/reciprocal of focal length of lens.

1

(ii) It is a point on principal axis at which light ray parallel to principal axis converges after reflection.

1

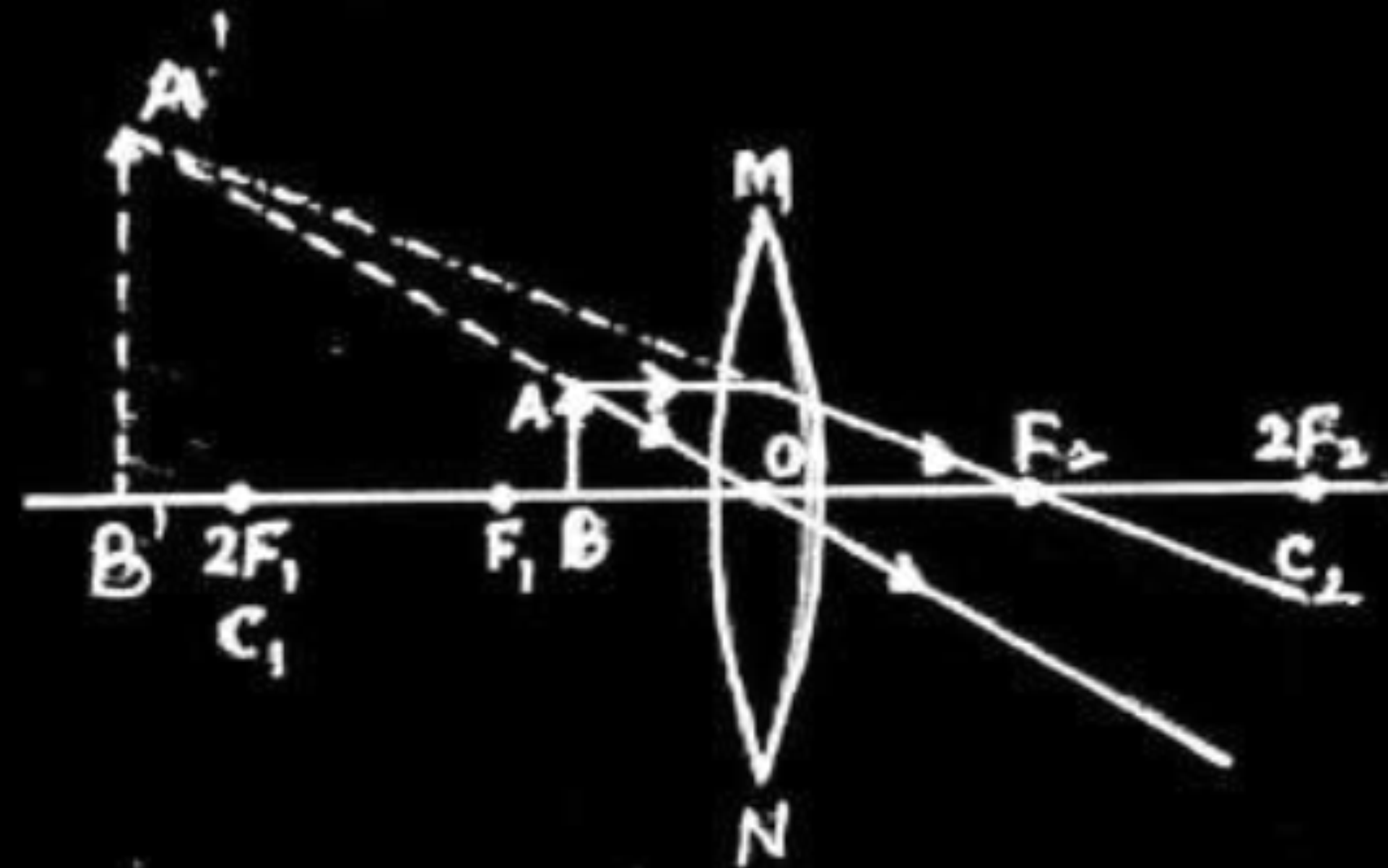
(b) (i) for spherical lens : $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{2}$

(ii) for spherical mirror : $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{2}$

(c)



Distance of object (BO) = 10 cm

Focal length (OF₁) = 15 cm

2

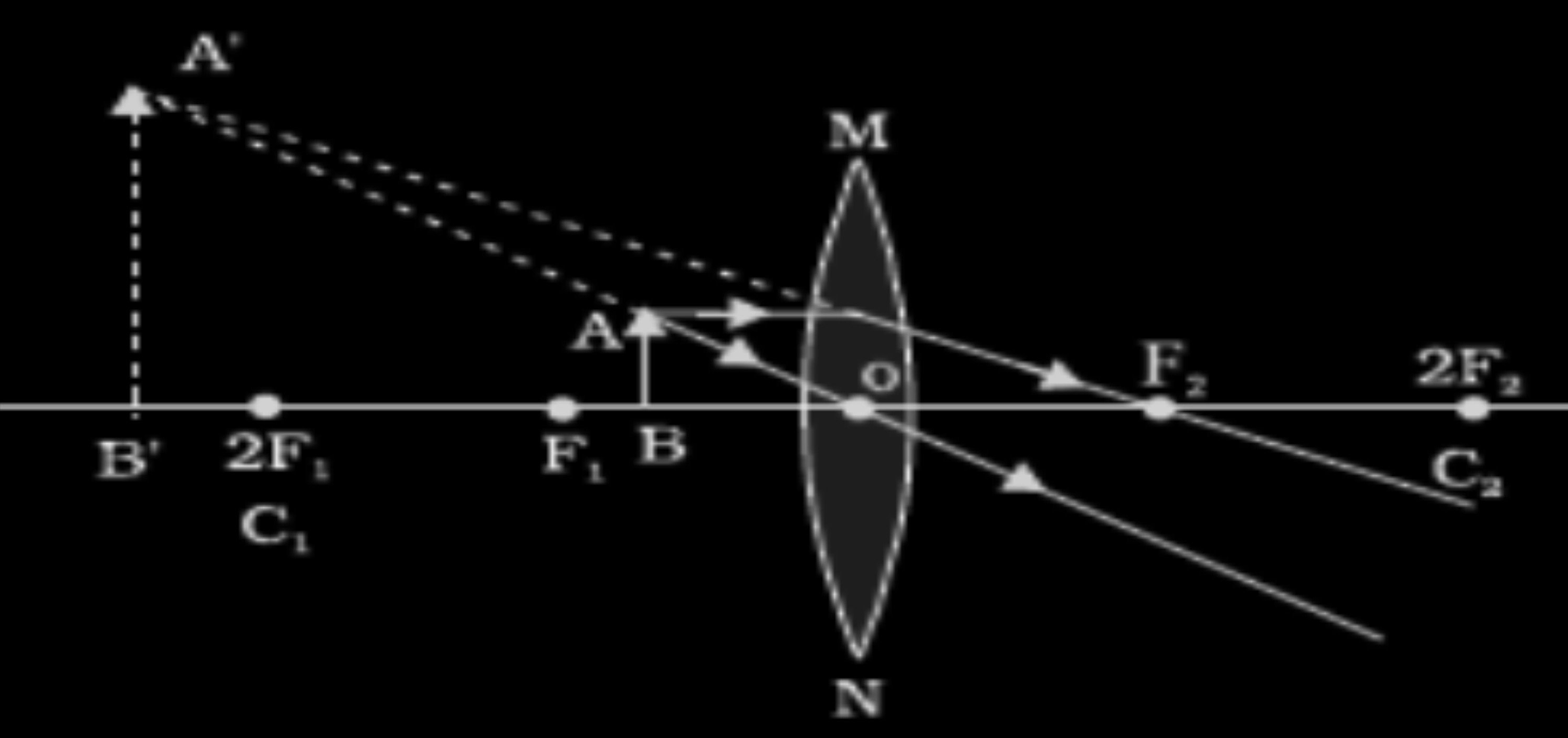
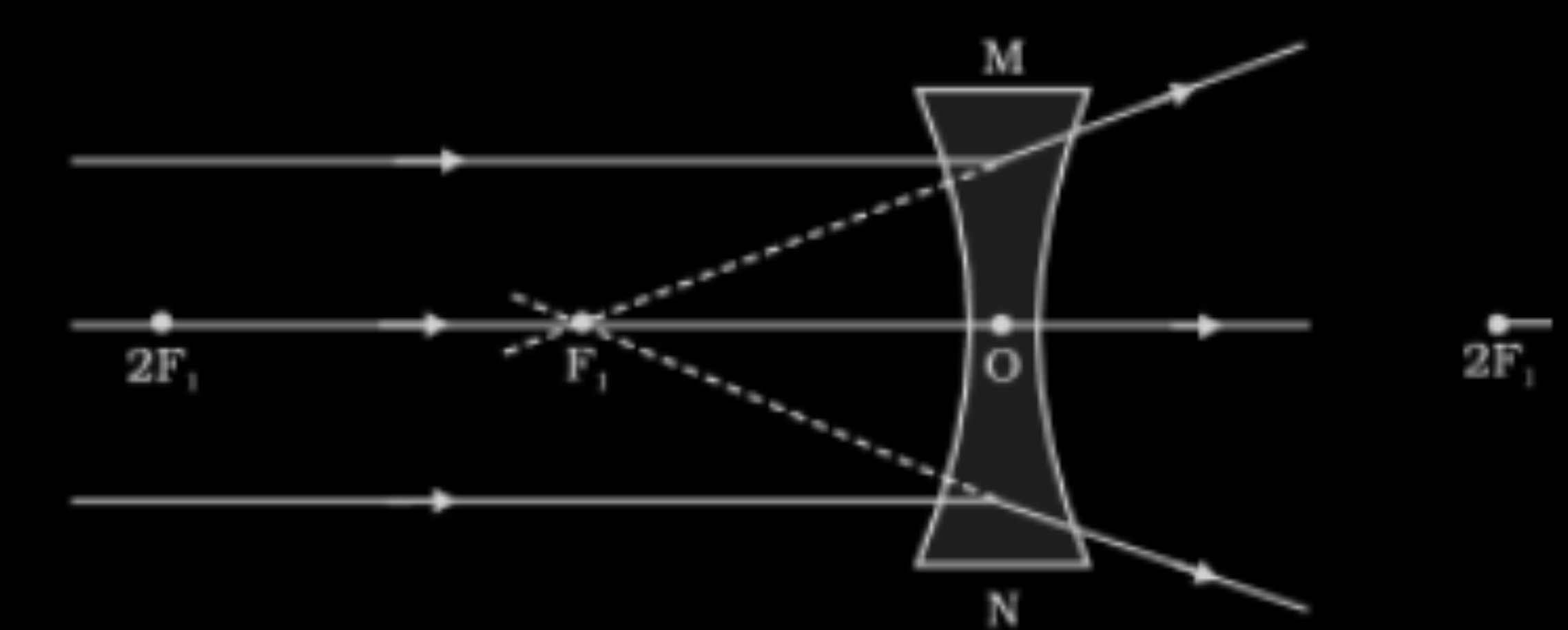
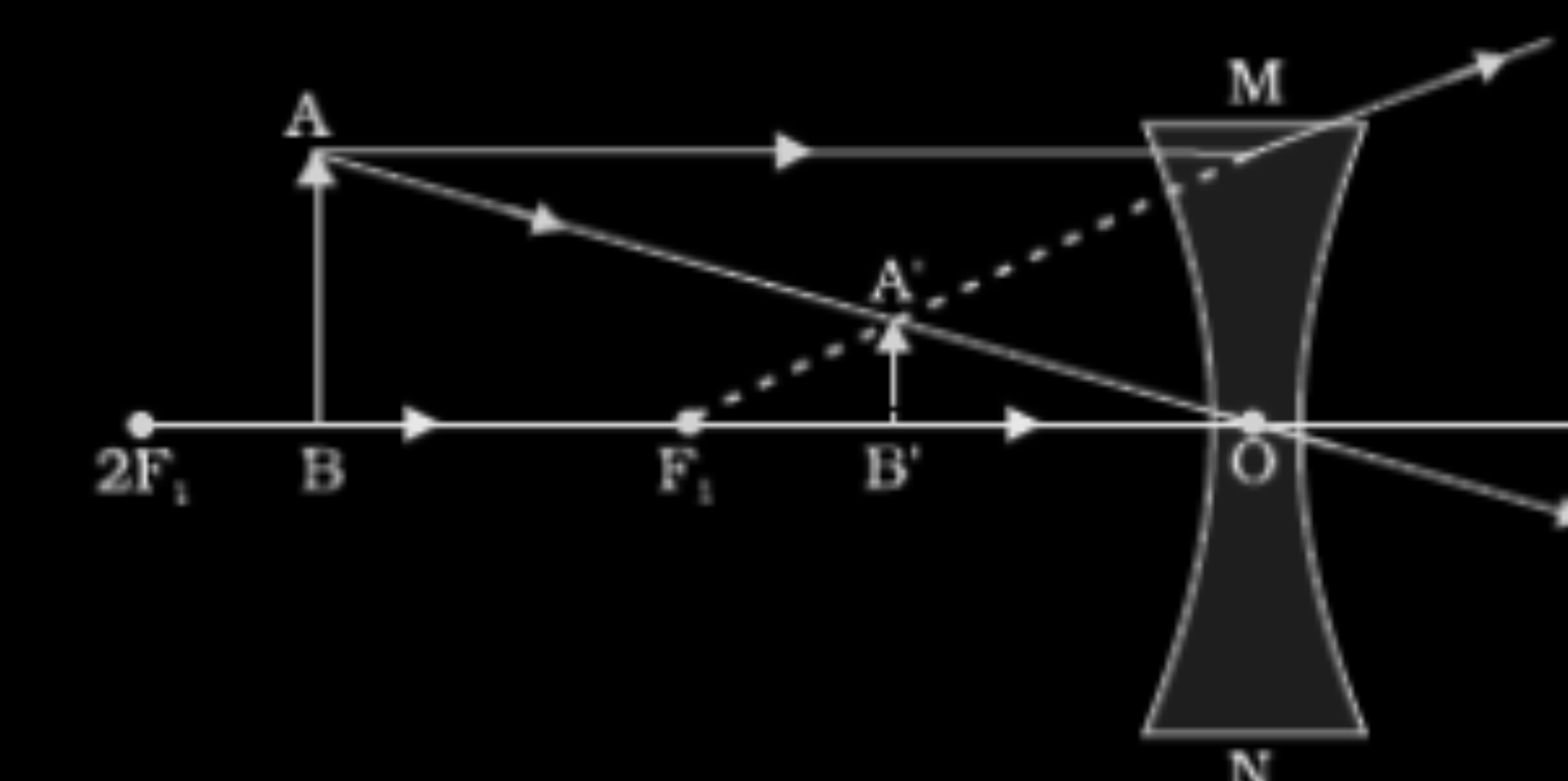
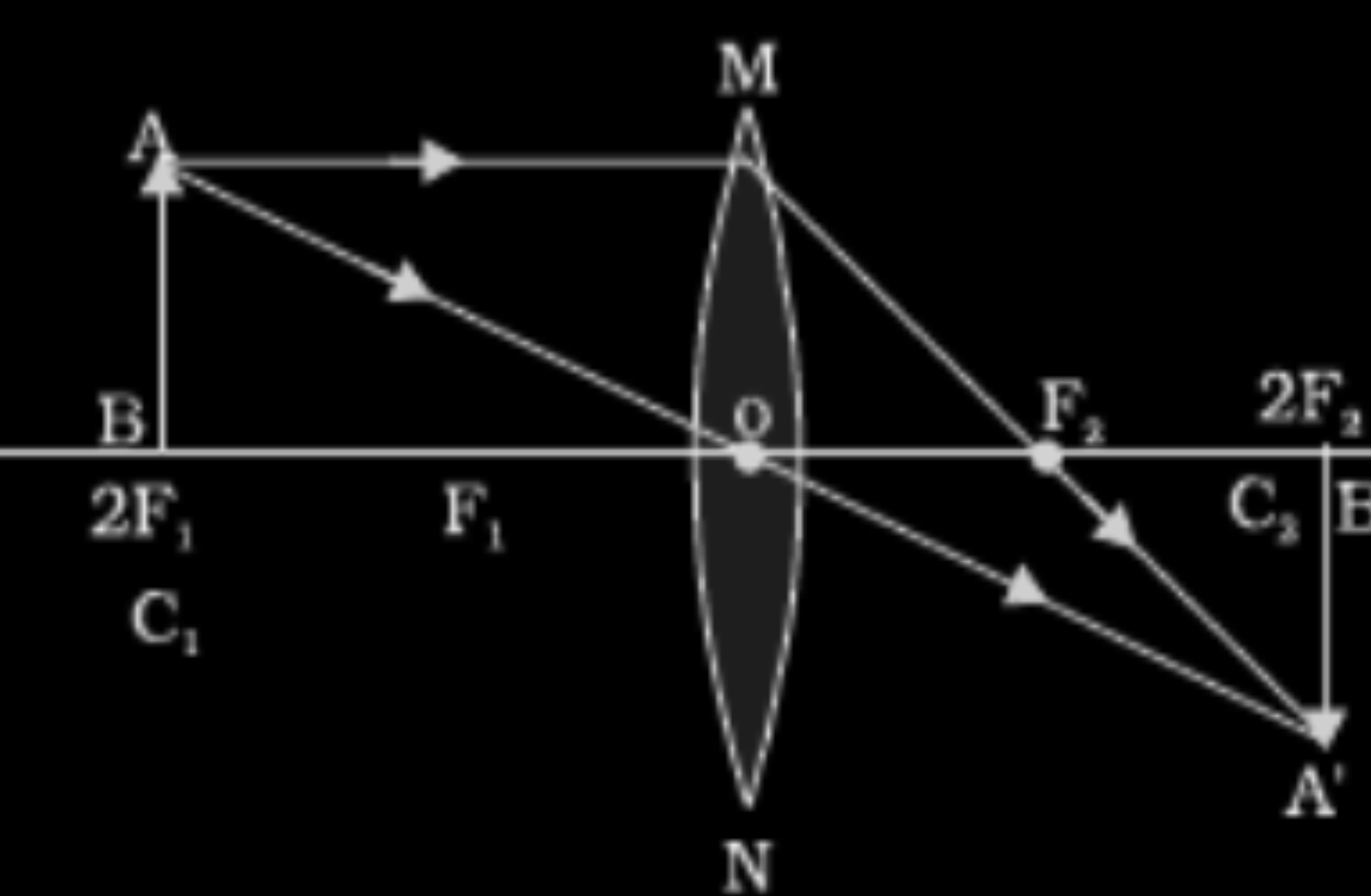
(If the distance in the diagram are not marked , deduct $\frac{1}{2}$ marks)

29. Draw a ray diagram in each of the following cases to show the formation of image, when the object is placed

- (i) between optical centre and principal focus of a convex lens.
- (ii) anywhere in front of a concave lens.
- (iii) at $2F$ of a convex lens.

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State the signs and values of magnifications in the above mentioned cases (i) and (ii).

i) 	1
ii)  (a) or 	1
iii)  In case (i) sign is positive and $m > 1$ (ii) sign is positive and $m < 1$	1 $\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$

An object 4.0 cm in size, is placed 25.0 cm in front of a concave mirror of focal length 15.0 cm.

- (i) At what distance from the mirror should a screen be placed in order to obtain a sharp image?
- (ii) Find the size of the image.
- (iii) Draw a ray diagram to show the formation of image in this case.

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5

Given $h = +4.0$ cm, $u = -25.0$ cm, $f = -15.0$ cm

$\frac{1}{2}$

i) image distance $v = ?$; mirror formula : $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$\frac{1}{2}$

$$\text{or } \frac{1}{v} = \frac{1}{f} - \frac{1}{u} ; = -\frac{1}{15} - \left(-\frac{1}{25}\right)$$

$$= \frac{-1}{15} + \frac{1}{25} = \frac{-5+3}{75}$$

$$= \frac{-2}{75}$$

$\frac{1}{2}$

$$v = -37.5 \text{ cm}$$

The screen should be placed 37.5 cm in front of the mirror.

ii) $m = \frac{h^1}{h} = -\frac{v}{u}$

$\frac{1}{2}$

$$: h^1 = -\frac{v}{u} \cdot h$$

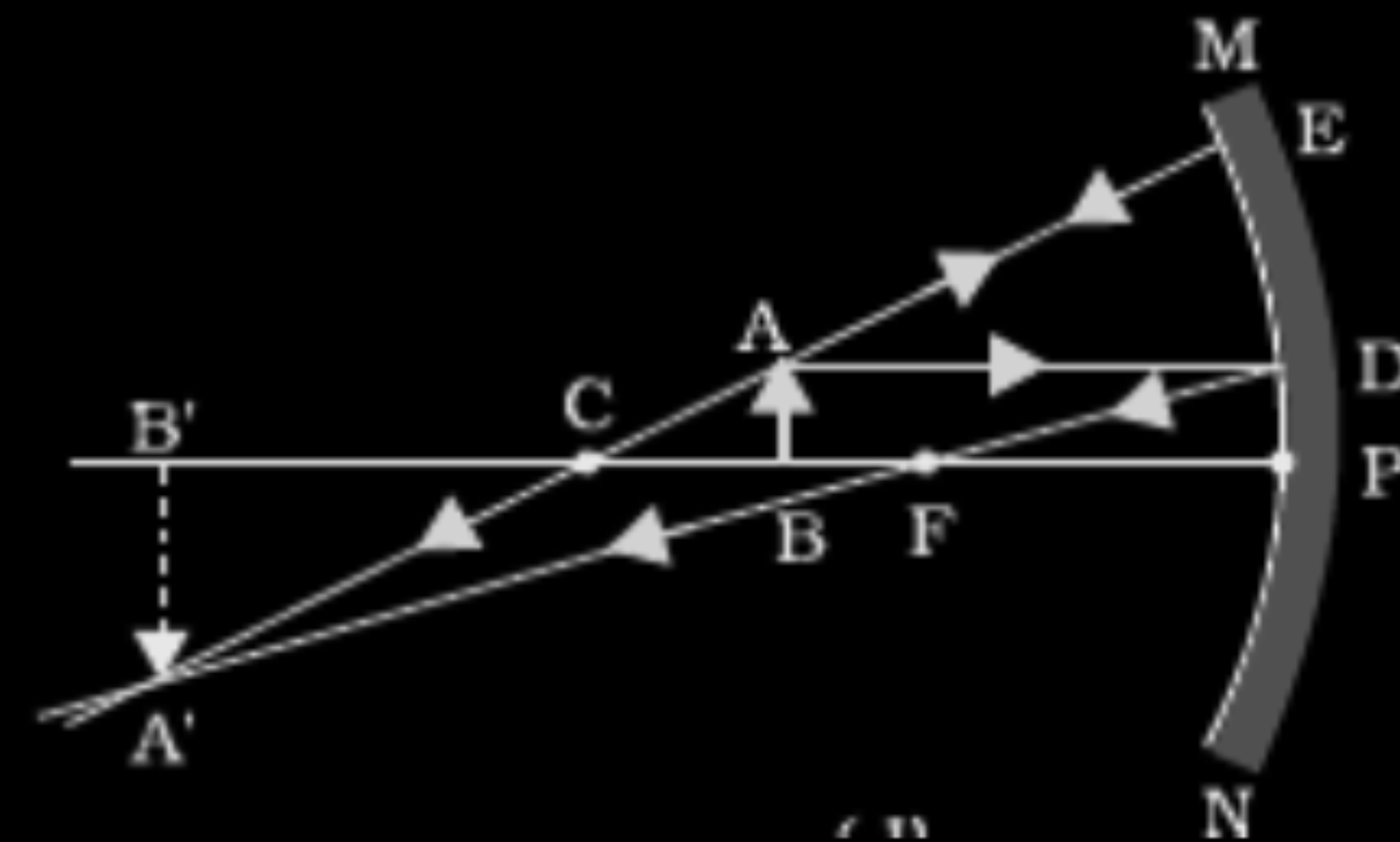
$\frac{1}{2}$

$$= -\frac{(-37.5 \times 4)}{-25}$$

$\frac{1}{2}$

$$h^1 = -6.0 \text{ cm (size of the image).}$$

iii)



1

Note: Deduct half mark for not showing arrows in ray diagrams.

The absolute refractive index of Ruby is 1.7. Find the speed of light in Ruby. The speed of light in vacuum is 3×10^8 m/s.

CBSE 2019

The absolute refractive index of Ruby is 1.7. Find the speed of light in Ruby. The speed of light in vacuum is 3×10^8 m/s.

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$\mu = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in Ruby}} = \frac{c}{v}$	$\frac{1}{2}$
$V = \frac{c}{\mu}$ c= velocity of light μ = refractive index	$\frac{1}{2}$
$v = \frac{3 \times 10^8}{1.7} = 1.76 \times 10^8 \text{ m/s}$	$\frac{1}{2} + \frac{1}{2}$

A student holding a mirror in his hand, directed the reflecting surface of the mirror towards the Sun. He then directed the reflected light on to a sheet of paper held close to the mirror.

- (a) What should he do to burn the paper ?
- (b) Which type of mirror does he have ?
- (c) Will he be able to determine the approximate value of focal length of this mirror from this activity ? Give reason and draw ray diagram to justify your answer in this case.

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A student holding a mirror in his hand, directed the reflecting surface of the mirror towards the Sun. He then directed the reflected light on to a sheet of paper held close to the mirror.

- (a) What should he do to burn the paper ?
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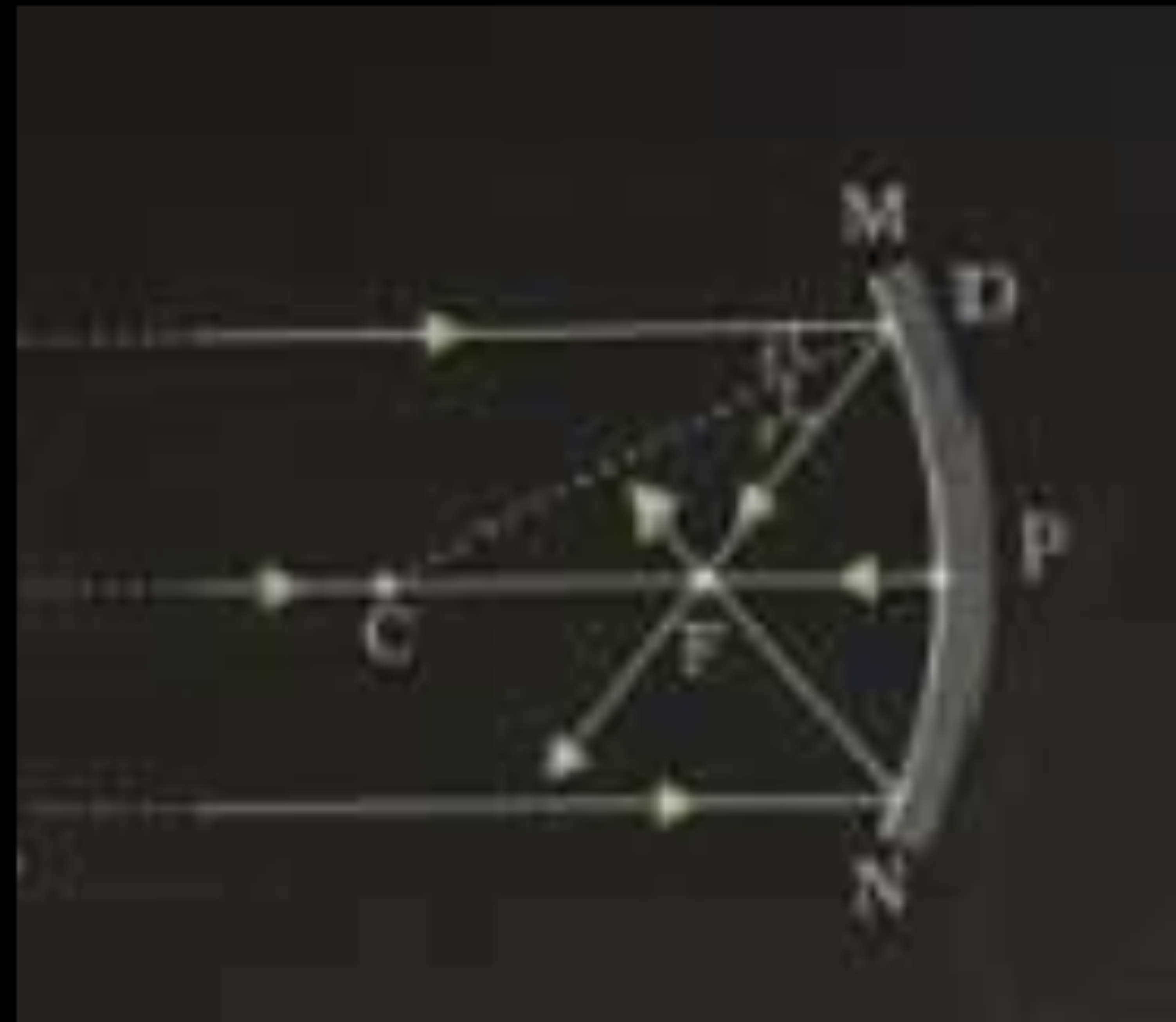
CBSE 2019

- (a) Move the mirror/paper to focus the rays at one point
- (b) Concave mirror
- (c) Yes, distance between mirror and focal point gives approximate focal length.

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$



$$\frac{1}{2} + \frac{1}{2}$$

A 10 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 12 cm. The distance of the object from the lens is 18 cm. Find the nature, position and size of the image formed.

CBSE 2019

A 10 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 12 cm. The distance of the object from the lens is 18 cm. Find the nature, position and size of the image formed.

CBSE 2019

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \therefore \quad \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\therefore \frac{1}{v} = \frac{1}{12} + \frac{1}{(-18)}$$

$$\therefore v = 36\text{cm}$$

$$m = \frac{v}{u} = \frac{h'}{h} \quad \therefore m =$$

$$\Rightarrow \frac{36}{-18} = \frac{h'}{10}$$

$$\Rightarrow h' = -20\text{cm (size of the image)}$$

Nature of image – Real and inverted

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

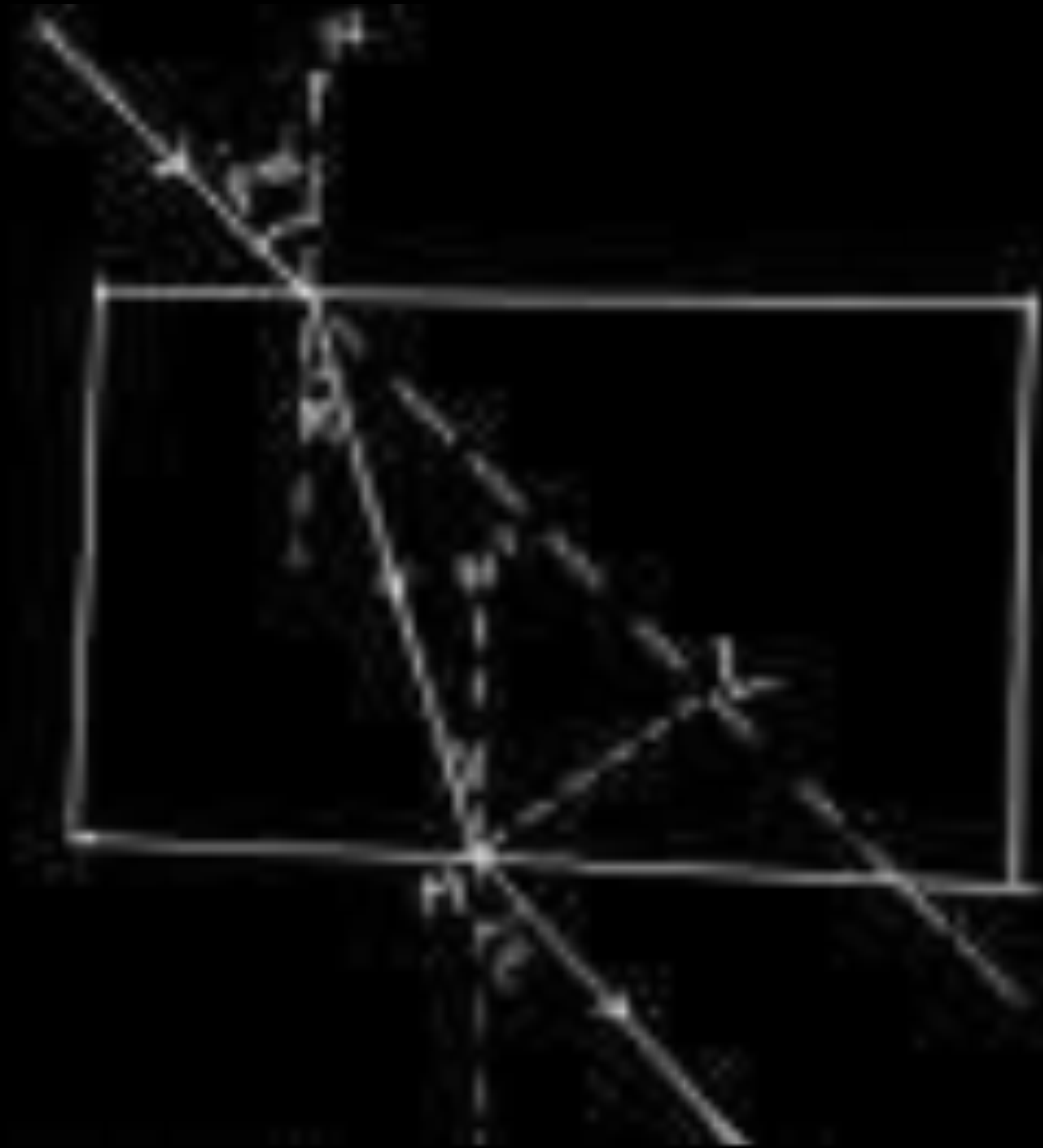
$\frac{1}{2}$

Draw the path of a ray of light when it enters one of the faces of a glass slab at an angle of nearly 45° . Label on it (i) angle of refraction, (ii) angle of emergence and (iii) lateral displacement.

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Draw the path of a ray of light when it enters one of the faces of a glass slab at an angle of nearly 45° . Label on it (i) angle of refraction, (ii) angle of emergence and (iii) lateral displacement.

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Labelling

- Angle of refraction (r_1)
- Angle of emergence (e)
- Lateral displacement (ML)

$\frac{1}{2}$

$\frac{1}{2} \times 3$

The power of a lens is +5 diopters. What is the nature and focal length of this lens ? At what distance from this lens should an object be placed so as to get its inverted image of the same size ?

CBSE 2019

The power of a lens is +5 diopters. What is the nature and focal length of this lens ? At what distance from this lens should an object be placed so as to get its inverted image of the same size ?

CBSE 2019

$P = +5D$	1
$f = \frac{1}{P} = \frac{100}{5} = 20\text{cm}$	
Nature of lens = convex (converging)	$\frac{1}{2}$
Distance is 40cm (at C)	$\frac{1}{2}$

A 6 cm tall object is placed perpendicular to the principal axis of a concave mirror of focal length 30 cm. The distance of the object from the mirror is 45 cm. Use mirror formula to determine the position, nature and size of the image formed. Also draw labelled ray diagram to show the image formation in this case.

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Given –

$$h = 6 \text{ cm}$$

$$f = -30 \text{ cm}$$

$$v = -45 \text{ cm}$$

mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$= -\frac{1}{30} - \frac{1}{(-45)}$$

$$= -\frac{1}{30} + \frac{1}{45} = -\frac{1}{90}$$

$f = -90 \text{ cm}$ from the pole of mirror

Size of the image

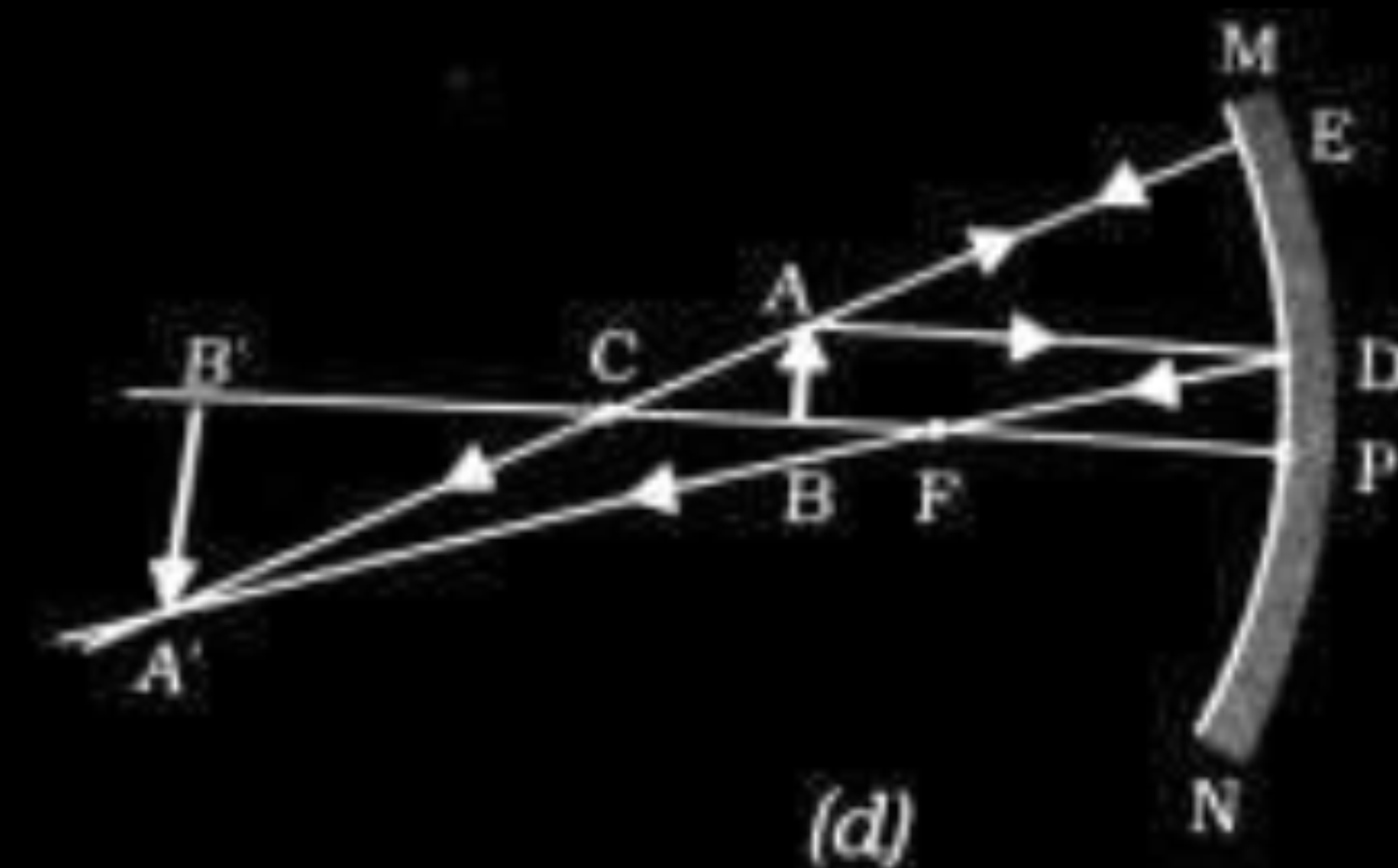
$$m = \frac{-v}{u}$$

$$= -\frac{90}{45} = -2$$

$$h^1 = -2 \times 6 \text{ cm}$$

$$= -12 \text{ cm}$$

Image formed will be real, inverted and enlarged.



Well labelled diagram

An object 6 cm in size is placed at 50 cm in front of a convex lens of focal length 30 cm. At what distance from the lens should a screen be placed in order to obtain a sharp image of the object ? Find the nature and size of the image. Also draw labelled ray diagram to show the image formation in this case.

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An object 6 cm in size is placed at 50 cm in front of a convex lens of focal length 30 cm. At what distance from the lens should a screen be placed in order to obtain a sharp image of the object ? Find the nature and size of the image. Also draw labelled ray diagram to show the image formation in this case.

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Given –

$$f = +30 \text{ cm}$$

$$u = -50 \text{ cm}$$

$$h = 6.0 \text{ cm}$$

lens formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$= \frac{1}{30} - \frac{1}{50}$$

$$\frac{5 - 3}{150} = \frac{2}{150} = \frac{1}{75}$$

$\therefore$

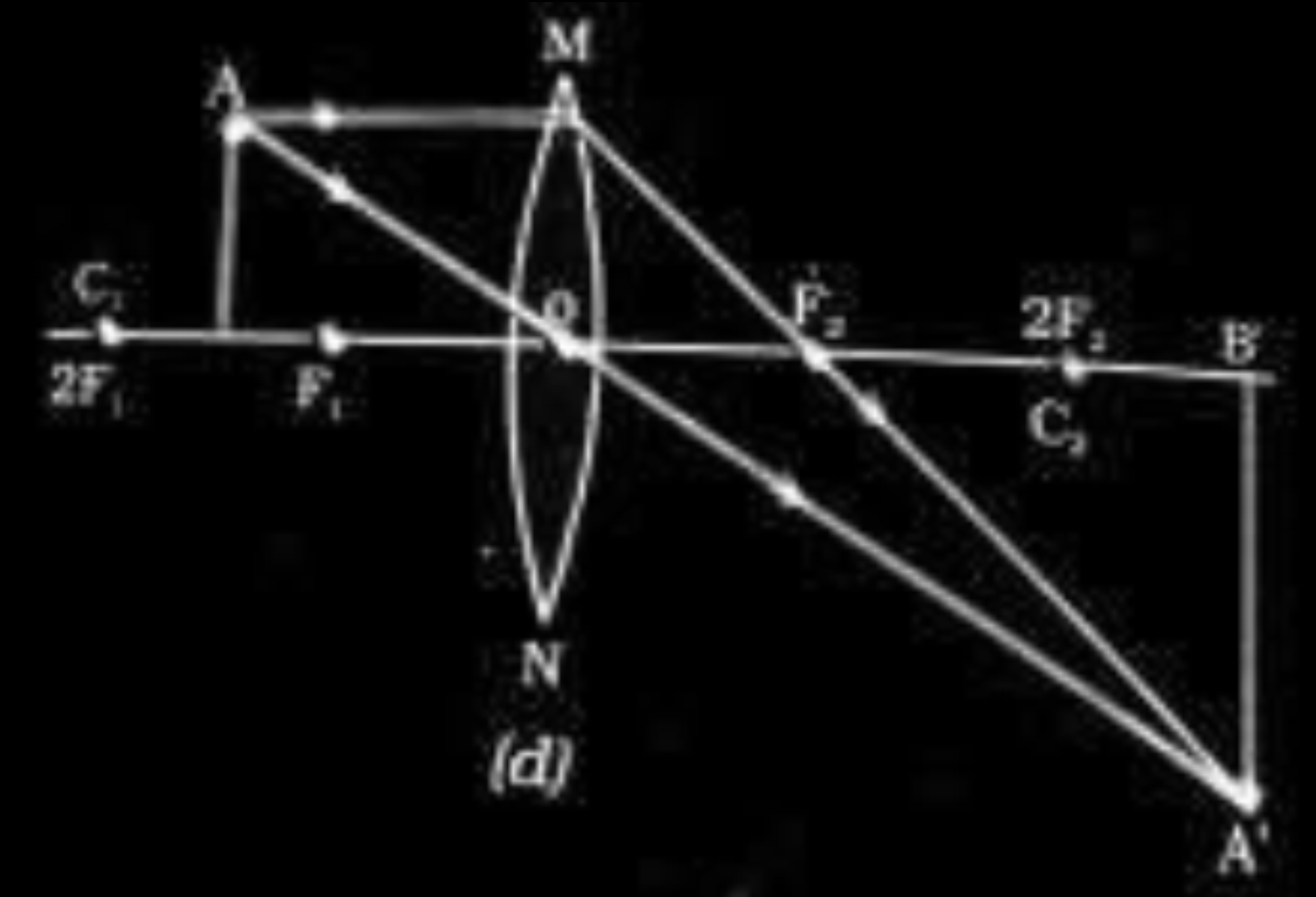
$$v = +75 \text{ cm}$$

$$m = \frac{v}{u} = \frac{h_1}{h}$$

$$= \frac{75}{-50} = \frac{h_1}{6}$$

$$h_1 = -9 \text{ cm}$$

Image formed is real, inverted and enlarged



Well labelled diagram

- (a) List four characteristics of the images formed by plane mirrors.
- (b) A 5 cm tall object is placed at a distance of 20 cm from a concave mirror of focal length 30 cm. Use mirror formula to determine the position and size of the image formed.

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- (a) List four characteristics of the images formed by plane mirrors.
- (b) A 5 cm tall object is placed at a distance of 20 cm from a concave mirror of focal length 30 cm. Use mirror formula to determine the position and size of the image formed.

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(a) Characteristics:

- i) The image is same size as the object.
- ii) The image is erect and virtual.
- iii) The image is laterally inverted.
- iv) The distance between the object and mirror is same as the distance between image and mirror.

$\frac{1}{2} \times 4$

(b)

$$h = 5\text{cm}$$

$$u = -20\text{cm}$$

$$f = -30$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$\frac{1}{2}$

$$\frac{1}{v} = \frac{1}{-30} + \frac{1}{-20}$$

1

$$v = -60\text{cm}$$

$\frac{1}{2}$

$$\frac{h'}{h} = \frac{v}{u}$$

$\frac{1}{2}$

$$h' = 15\text{cm}$$

$\frac{1}{2}$

Size - enlarged

List in proper sequence the steps of the experiment for determining the approximate focal length of a given concave mirror by obtaining the image of a distant object.

CBSE 2019

List in proper sequence the steps of the experiment for determining the approximate focal length of a given concave mirror by obtaining the image of a distant object.

CBSE 2019

- | | |
|---|------------------------|
| <ul style="list-style-type: none">i) Fix a concave mirror on a stand and place it near a source of bright lightii) Place a screen fitted on a stand in front of the mirroriii) Move the screen back and forth, until a sharp and clear image of a distant object like a tree is obtained on the screeniv) Mark the position of mirror and screen on the scale and note the distance between them | $\frac{1}{2} \times 4$ |
|---|------------------------|

A student has to trace the path of a ray of light passing through a rectangular glass slab for four different values of angle of incidence.

- (a) Write two important precautions for this experiment.
- (b) List two conclusions the student will draw based on his experiment.

A student has to trace the path of a ray of light passing through a rectangular glass slab for four different values of angle of incidence.

- (a) Write two important precautions for this experiment.
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CBSE 2019

The student should take the following precaution	
(a) Precaution -	
(i) See that the pins are in a straight line and atleast 3cm apart.	
(ii) Angle of incidence should be between 30^0 to 60^0 .	
(iii) Glass slab should always remain inside the boundary.	$\frac{1}{2}$
(any two)	$\frac{1}{2}$
(b) Conclusion -	
(i) The emergent ray is parallel to incident ray	
(ii) Lateral displacement takes place.	$\frac{1}{2}$
(iii) Angle of incidence = Angle of emergence	$\frac{1}{2}$
(any two)	

- (a) List four characteristics of the image formed by a convex lens when an object is placed between its optical centre and principal focus.
- (b) Size of the image of an object by a concave lens of focal length 20 cm is observed to be reduced to $\frac{1}{3}$ rd of its size. Find the distance of the object from the lens.

- (a) List four characteristics of the image formed by a convex lens when an object is placed between its optical centre and principal focus.
- (b) Size of the image of an object by a concave lens of focal length 20 cm is observed to be reduced to $\frac{1}{3}$ rd of its size. Find the distance of the object from the lens.

CBSE 2019	
a) Four characteristics: i. Image is formed on the same side of the lens as the object. ii. The image is enlarged / magnified, virtual and erect. b) $h' = \frac{h}{3}$ Focal length = -20cm As per the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $u = -40\text{cm}$	$\frac{1}{2}$ $\frac{1}{2} \times 3$ 1 1 1

- (a) List four characteristics of the image formed by a concave lens of focal length 20 cm when the object is placed at a distance of 40 cm from its optical centre.
- (b) The size of image of an object by a convex lens of focal length 20 cm is observed to be reduced to $\frac{1}{3}$ rd of its size. Find the distance of the object from the optical centre of the lens.

CBSE 2019

- (a) List four characteristics of the image formed by a concave lens of focal length 20 cm when the object is placed at a distance of 40 cm from its optical centre.
- (b) The size of image of an object by a convex lens of focal length 20 cm is observed to be reduced to $\frac{1}{3}$ rd of its size. Find the distance of the object from the optical centre of the lens.

CBSE 2019

(a)

- Virtual
- Erect
- Diminished
- On the same side of the object

$$\frac{1}{2} \times 4$$

(b) Focal Length = 20cm.

$$u = -x \text{ cm}$$

$$v = \frac{x}{3}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$x = 80 \text{ cm}$$

$$1$$

$$1$$

$$1$$

State laws of reflection of light.

CBSE 2019

OR

Define absolute refractive index and express it mathematically.

State laws of reflection of light.

CBSE 2019

OR

Define absolute refractive index and express it mathematically.

- **First law:** The incident ray, the reflected ray and the normal to the surface at the point of incidence all lie in the same plane.
 - **Second law:** Angle of incidence is equal to angle of reflection.
- OR
- Refractive index of a transparent medium with respect to vacuum or air is called absolute refractive index.
 - Mathematically

$$\text{Absolute refractive index of a medium} = \frac{\text{Speed of light in vacuum or air}}{\text{Speed of light in the medium}}$$
$$n = c/v$$

1

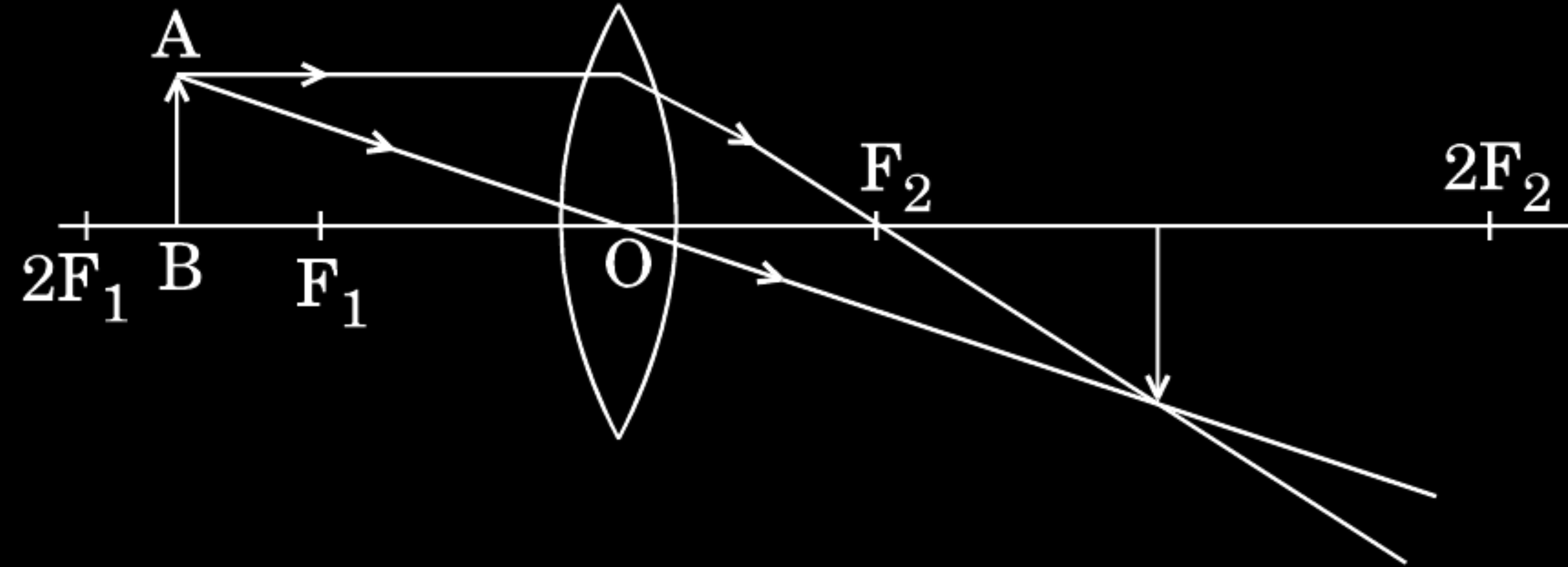
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1

1

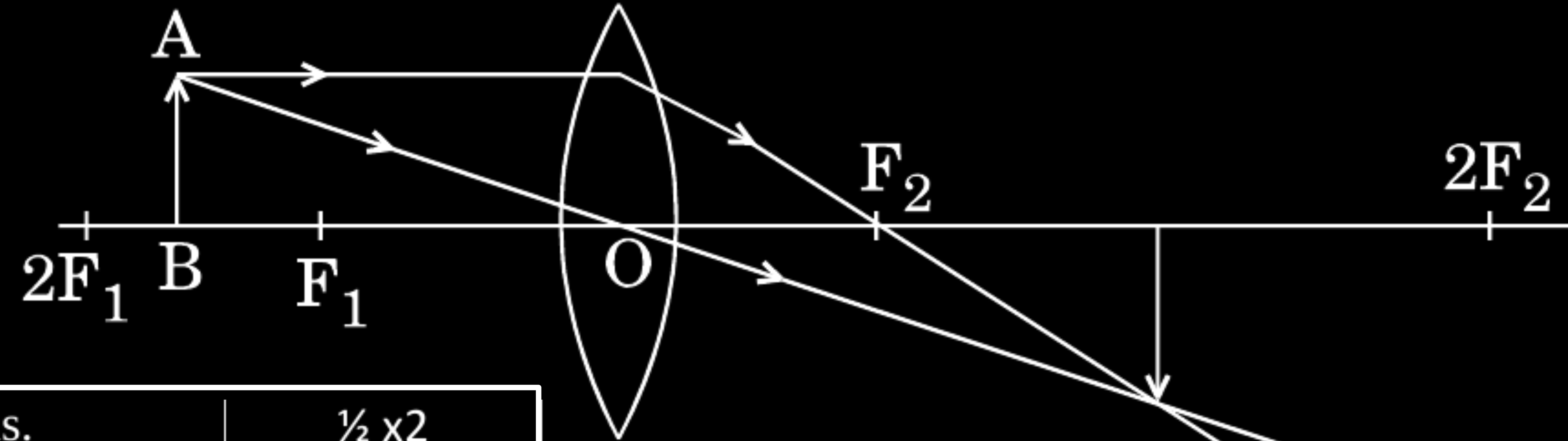
Study the following ray diagram and list two mistakes committed by the student while tracing it. Rectify these mistakes by drawing the correct ray diagram to show the real position and size of the image corresponding to the position of the object AB.

CBSE 2019



Study the following ray diagram and list two mistakes committed by the student while tracing it. Rectify these mistakes by drawing the correct ray diagram to show the real position and size of the image corresponding to the position of the object AB.

CBSE 2019



Mistakes : F_1 and F_2 are not equidistant from the optical center of the lens.

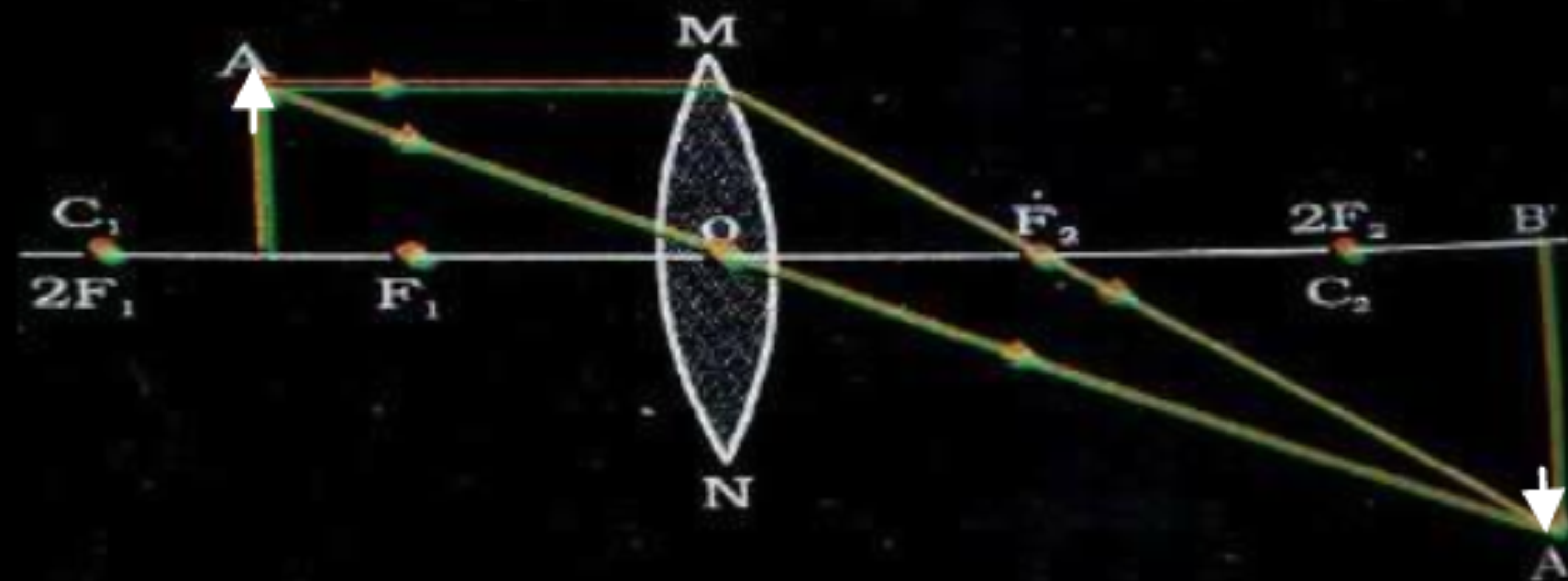
$OF_1 \neq OF_2$; $2OF_1 \neq 2OF_2$

Image should form beyond $2F_2$

Image should be magnified

$\frac{1}{2} \times 2$

(any two)



1

OR

- (i) Prism should be within the boundary all through the experiment .
- (ii) Pins should be fixed vertically and the feet of the pins should be observed.
- (iii) Protractor should be used correctly .
- (iv) Angle should be taken between 30° and 60° to observe the refraction clearly.
- (v) Separation between the pins should be kept at least 5cm .

$\frac{1}{2} \times 4$

(any four)

State laws of refraction of light.

CBSE 2019

OR

List four characteristics of the image formed by a concave mirror of focal length 40 cm when the object is placed in front of it at a distance of 20 cm from its pole.

State laws of refraction of light.

CBSE 2019

OR

List four characteristics of the image formed by a concave mirror of focal length 40 cm when the object is placed in front of it at a distance of 20 cm from its pole.

First law :The incident ray ,the refracted ray and the normal to the interface of two mediums at the point of incidence, all lie in the same plane.

1

Second law :The ratio of sine of angle of incidence to sine of angle of refraction is always constant for a given pair of medium .This constant is called refractive index.

1

OR

(i)Erect (ii) virtual (iii) magnified (iv) Behind the mirror

$\frac{1}{2} \times 4$

List four characteristics of the image formed by a convex lens of focal length 20 cm when the object is placed in front of it at a distance of 10 cm from its optical centre.

CBSE 2019

List four characteristics of the image formed by a convex lens of focal length 20 cm when the object is placed in front of it at a distance of 10 cm from its optical centre.

CBSE 2019

(i) Erect (ii) virtual (iii) magnified (iv) same side of the object

$\frac{1}{2} \times 4$

Define refractive index of a transparent medium. The speed of light in a medium of absolute refractive index 1.5 is $2 \times 10^8 \text{ ms}^{-1}$. What is the speed of light in vacuum ?

CBSE 2019

Define refractive index of a transparent medium. The speed of light in a medium of absolute refractive index 1.5 is $2 \times 10^8 \text{ ms}^{-1}$. What is the speed of light in vacuum ?

CBSE 2019

- Refractive index is the ratio of speed of light in air or vacuum to speed of light in medium.

Mathematically: Refractive index of a medium 1.5

Speed of light in medium = $2 \times 10^8 \text{ ms}^{-1}$

Speed of light in vacuum?

$\mu = \text{speed of light in vacuum} / \text{speed of light in medium}$

speed of light in vacuum = $1.5 \times 2 \times 10^8 \text{ ms}^{-1}$

$3 \times 10^8 \text{ ms}^{-1}$

1

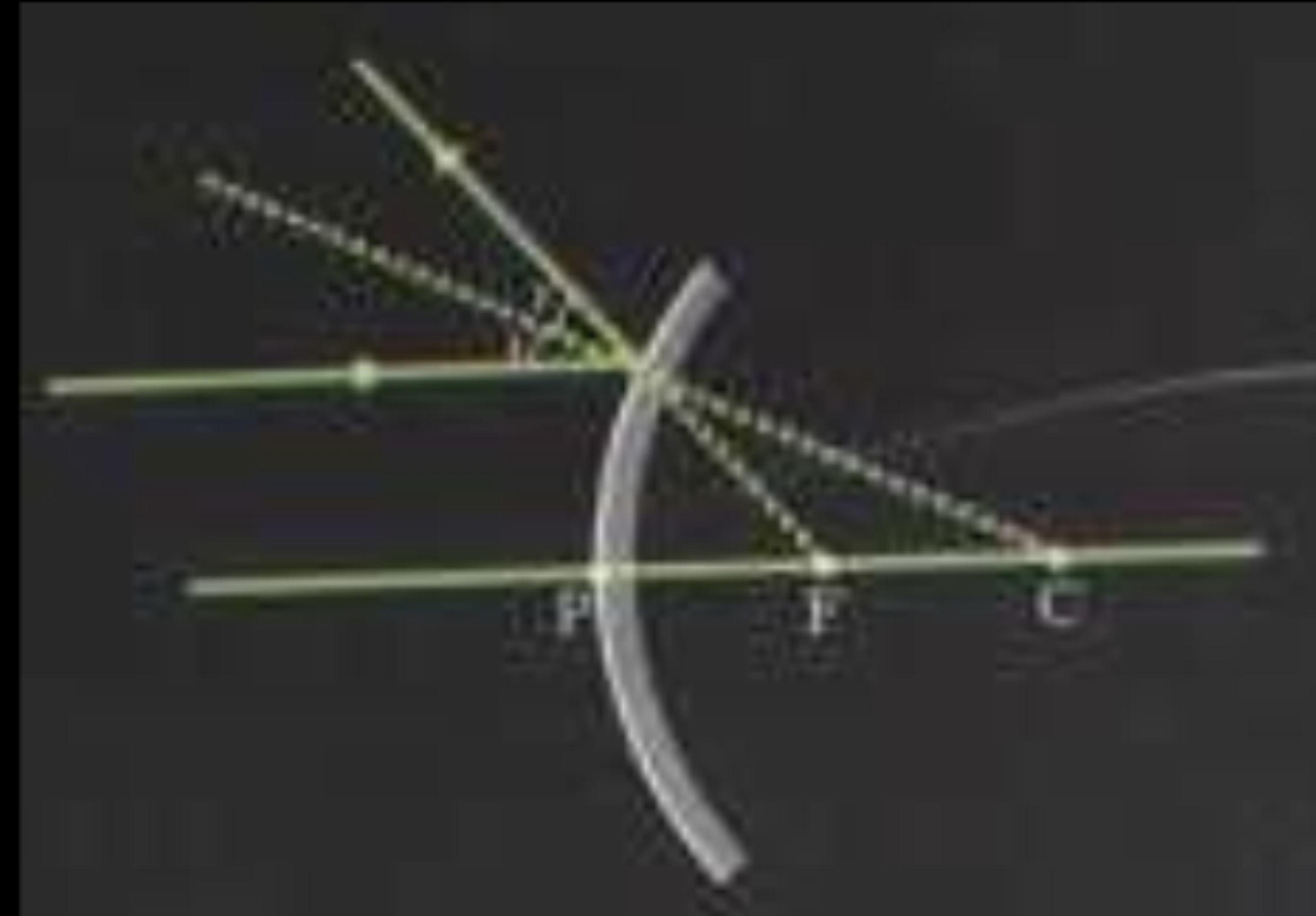
1

Draw a labelled ray diagram to show the path of the reflected ray corresponding to an incident ray of light parallel to the principal axis of a convex mirror. Mark the angle of incidence and angle of reflection on it.

CBSE 2019

Draw a labelled ray diagram to show the path of the reflected ray corresponding to an incident ray of light parallel to the principal axis of a convex mirror. Mark the angle of incidence and angle of reflection on it.

CBSE 2019



- (i) draw mirror.
- (ii) Complete ray diagram
- (iii) $\angle i$ and $\angle r$ islabelled
- (iv) Arrows marked

$\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$
 $\frac{1}{2}$

A concave mirror has a focal length of 20 cm. At what distance from the mirror should a 4 cm tall object be placed so that it forms an image at a distance of 30 cm from the mirror ? Also calculate the size of the image formed.

CBSE 2019

A concave mirror has a focal length of 20 cm. At what distance from the mirror should a 4 cm tall object be placed so that it forms an image at a distance of 30 cm from the mirror ? Also calculate the size of the image formed.

CBSE 2019

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

1/2

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{u} = \frac{1}{-20} + \frac{1}{30}$$

$$u = -60\text{cm (without unit (cm) deduct } \frac{1}{2} \text{ mark)}$$

1

$$m = \frac{-v}{u}$$

1/2

$$m = -\left[\frac{-30}{-60}\right]$$

$$m = -\frac{1}{2}$$

$$\frac{h_2}{h_1} = m$$

$$h_2 = h_1 \times m$$

$$h_2 = 4 \times \left[-\frac{1}{2}\right]$$

$$h_2 = -2 \text{ cm}$$

1

A real image $\frac{2}{3}$ rd of the size of an object is formed by a convex lens when the object is at a distance of 12 cm from it. Find the focal length of the lens.

CBSE 2019

A real image $\frac{2}{3}$ rd of the size of an object is formed by a convex lens when the object is at a distance of 12 cm from it. Find the focal length of the lens.

CBSE 2019

$$\left. \begin{aligned} m &= \frac{v}{u} \\ m &= -\frac{2}{3} \\ -\frac{2}{3} &= \frac{v}{-12} \\ v &= 8 \text{ cm} \end{aligned} \right\}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{8} - \frac{1}{-12}$$

$$\frac{1}{f} = \frac{1}{8} + \frac{1}{12}$$

$$\frac{1}{f} = \frac{5}{24}$$

$$f = \frac{24}{5}$$

$$f = 4.8 \text{ cm}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

- (a) A 5 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. The distance of the object from the lens is 30 cm. Find the position, nature and size of the image formed.
- (b) Draw a labelled ray diagram showing object distance, image distance and focal length in the above case.

CBSE 2019

- (a) A 5 cm tall object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. The distance of the object from the lens is 30 cm. Find the position, nature and size of the image formed.
- (b) Draw a labelled ray diagram showing object distance, image distance and focal length in the above case.

CBSE 2019

$$f = 20 \text{ cm}, u = -30 \text{ cm}$$

(a)

$$(i) \quad \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$\frac{1}{v} = \frac{1}{20} + \frac{1}{-30}$$

$$\frac{1}{v} = \frac{1}{60}$$

$$v = 60 \text{ cm}$$

(ii) Real, inverted and magnified

$$(iii) \quad m = \frac{v}{u}$$

$$m = \frac{60}{-30}$$

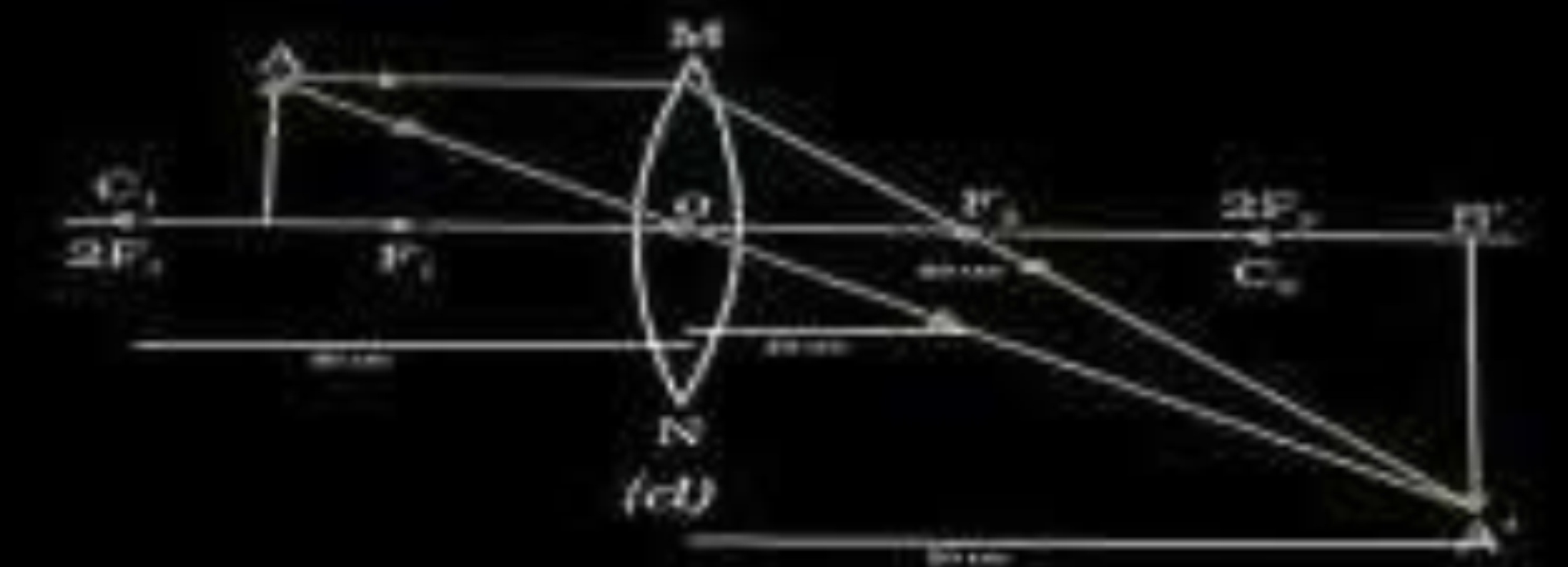
$$m = -2$$

$$h' = m \times h$$

$$h' = -2 \times 5$$

$$h' = -10 \text{ cm}$$

(b)



A teacher gives a convex lens and a concave mirror of focal length of 20 cm each to his student and asks him to find their focal lengths by obtaining the image of a distant object. The student uses a distant tree as the object and obtains its sharp image, one by one, on a screen. The distances d_1 and d_2 between the lens/mirror and the screen in the two cases and the nature of their respective sharp images are likely to be

- (a) (20 cm, 40 cm) and (erect and erect)
- (b) (20 cm, 40 cm) and (inverted and inverted)
- (c) (20 cm, 20 cm) and (inverted and inverted)
- (d) (20 cm, 40 cm) and (erect and inverted)

Give reason for your answer.

A teacher gives a convex lens and a concave mirror of focal length of 20 cm each to his student and asks him to find their focal lengths by obtaining the image of a distant object. The student uses a distant tree as the object and obtains its sharp image, one by one, on a screen. The distances d_1 and d_2 between the lens/mirror and the screen in the two cases and the nature of their respective sharp images are likely to be

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- (c) (20 cm, 20 cm) and (inverted and inverted)
- (d) (20 cm, 40 cm) and (erect and inverted)

Give reason for your answer.

CBSE 2019	
(c) (20 cm, 20 cm) and (inverted and inverted)	1
Reason: Only real and inverted image can be obtained on the screen and in both cases the image is formed at the principal focus.	1

A concave lens of focal length 60 cm is used to form an image of an object of length 9 cm kept at a distance of 30 cm from it. Use lens formula to determine the nature, position and length of the image formed. Also draw labelled ray diagram to show the image formation in the above case. CBSE 2019

A concave lens of focal length 60 cm is used to form an image of an object of length 9 cm kept at a distance of 30 cm from it. Use lens formula to determine the nature, position and length of the image formed. Also draw labelled ray diagram to show the image formation in the above case. **CBSE 2019**

$$f = -60 \text{ cm} \quad h = 9 \text{ cm} \quad u = -30 \text{ cm}$$

$$\text{Lens formula : } \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$= \frac{-1}{60} + \left[\frac{-1}{30} \right]$$

$$= \frac{-1}{60} - \frac{1}{30}$$

$$\frac{1}{v} = \frac{-1-2}{60}$$

$$v = -20 \text{ cm}$$

$$m = \frac{v}{u} = \frac{-20}{-30} = \frac{2}{3}$$

$$m = \frac{h'}{h} \Rightarrow h' = m \times h$$

$$= \frac{2}{3} \times 9$$

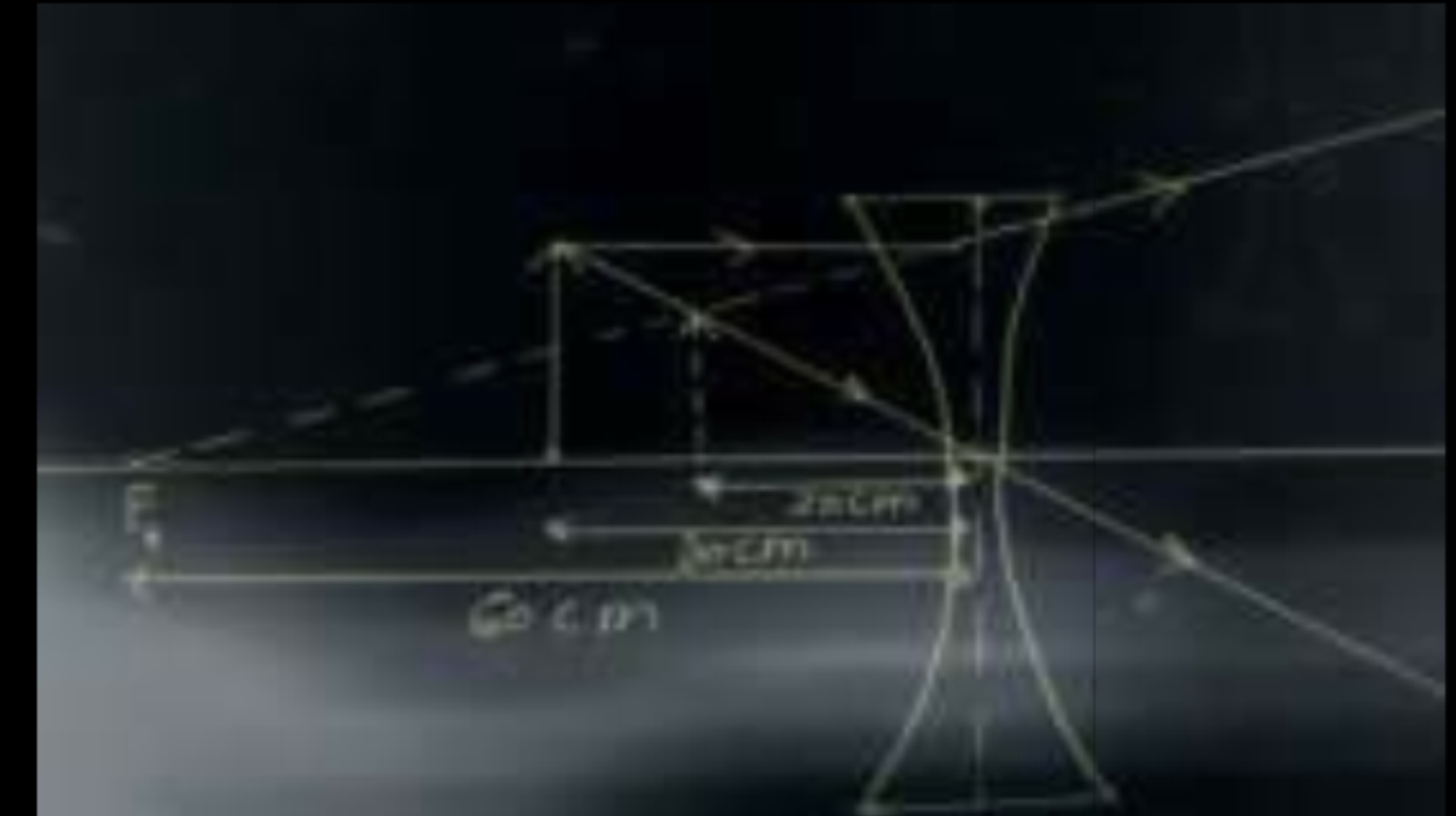
$$h' = 6 \text{ cm}$$

1

1

1

Image is : virtual, erect, smaller than object



1

1

An object is placed at a distance of 60 cm from a concave lens of focal length 30 cm.

- (i) Use lens formula to find the distance of the image from the lens.
- (ii) List four characteristics of the image (nature, position, size, erect/inverted) formed by the lens in this case.
- (iii) Draw ray diagram to justify your answer of part (ii).

CBSE 2019

i) $u = -60 \text{ cm}$ $f = -30 \text{ cm}$ $v = ?$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\therefore \frac{1}{v} = \frac{1}{f} + \frac{1}{u}$$

$$= \frac{1}{(-30 \text{ cm})} + \frac{1}{(-60 \text{ cm})} = \frac{-3}{60}$$

$$\therefore v = -20 \text{ cm}$$

$$m = v/u = \frac{-20 \text{ cm}}{-60 \text{ cm}} = \frac{1}{3}$$

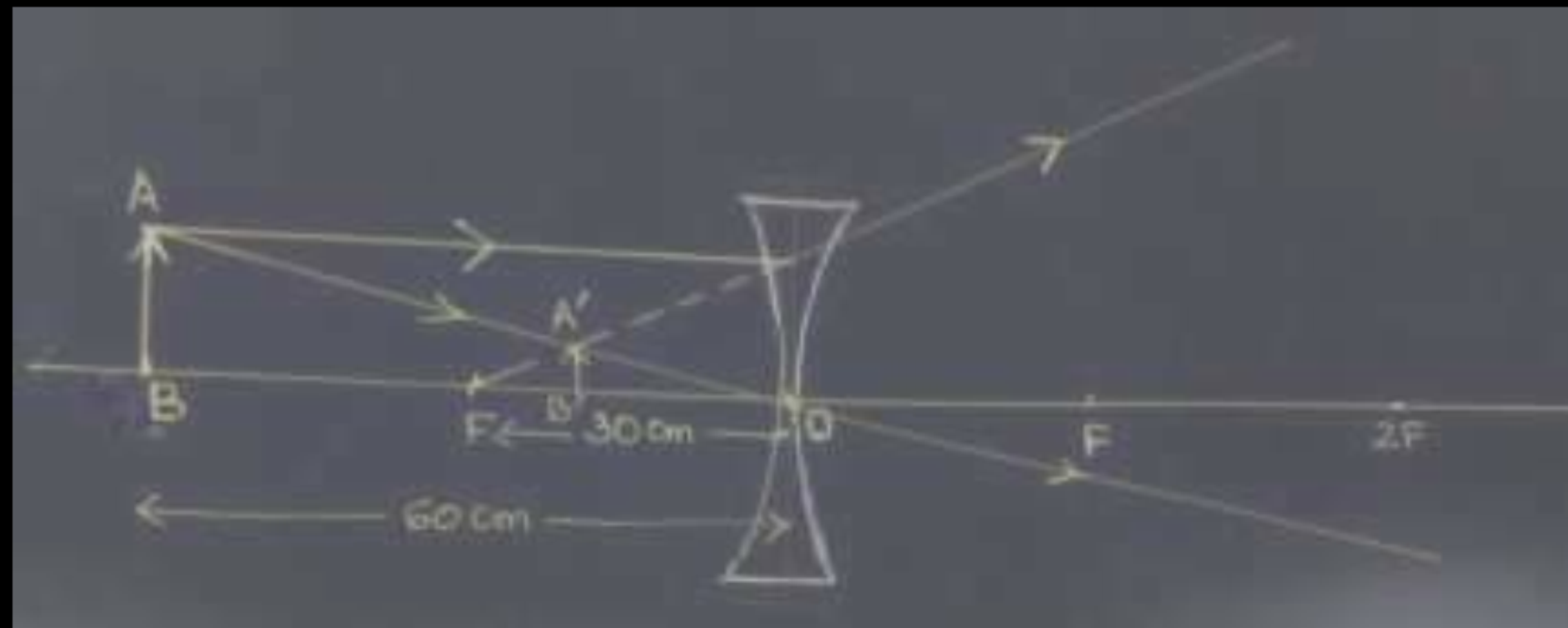
ii) Nature:- Virtual

Position:- 20 cm from lens on the same side as the object

Size:- Diminished

Erect/Inverted:- Erect

(iii)



$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

List four precautions which a student should observe while determining the focal length of a given convex lens by obtaining image of a distant object on a screen.

CBSE 2019

List four precautions which a student should observe while determining the focal length of a given convex lens by obtaining image of a distant object on a screen.

CBSE 2019

Precautions: 1) Lens should be held in vertical position with its faces parallel to the screen 2) Clear and sharpest image should be obtained by adjusting the position of lens 3) Three observations should be taken at least. 4) Base of lens, screen and measuring scale should be in straight line (or any other)	$\frac{1}{2} \times 4$
---	--

Draw a labelled ray diagram to show the path of the reflected ray corresponding to the ray which is incident obliquely to the principal axis, towards the pole of a convex mirror. Mark the angle of incidence and angle of reflection on it.

CBSE 2019

Draw a labelled ray diagram to show the path of the reflected ray corresponding to the ray which is incident obliquely to the principal axis, towards the pole of a convex mirror. Mark the angle of incidence and angle of reflection on it.

CBSE 2019

Section-B



- Correct ray diagram
- labelling of $\angle i$ and $\angle r$

1
 $\frac{1}{2} \times 2 = 1$

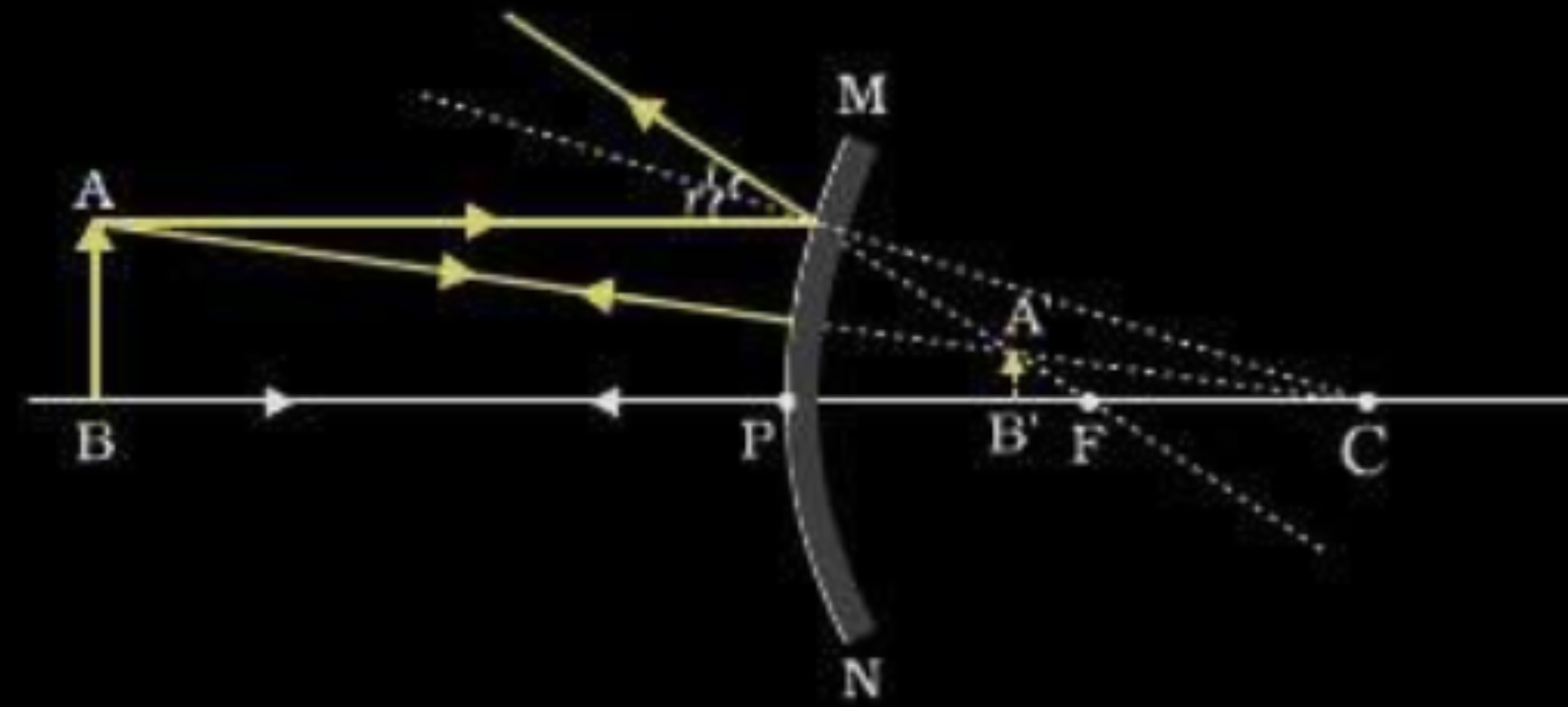
If the image formed by a spherical mirror for all positions of the object placed in front of it is always erect and diminished, what type of mirror is it ? Draw a labelled ray diagram to support your answer.

CBSE 2018

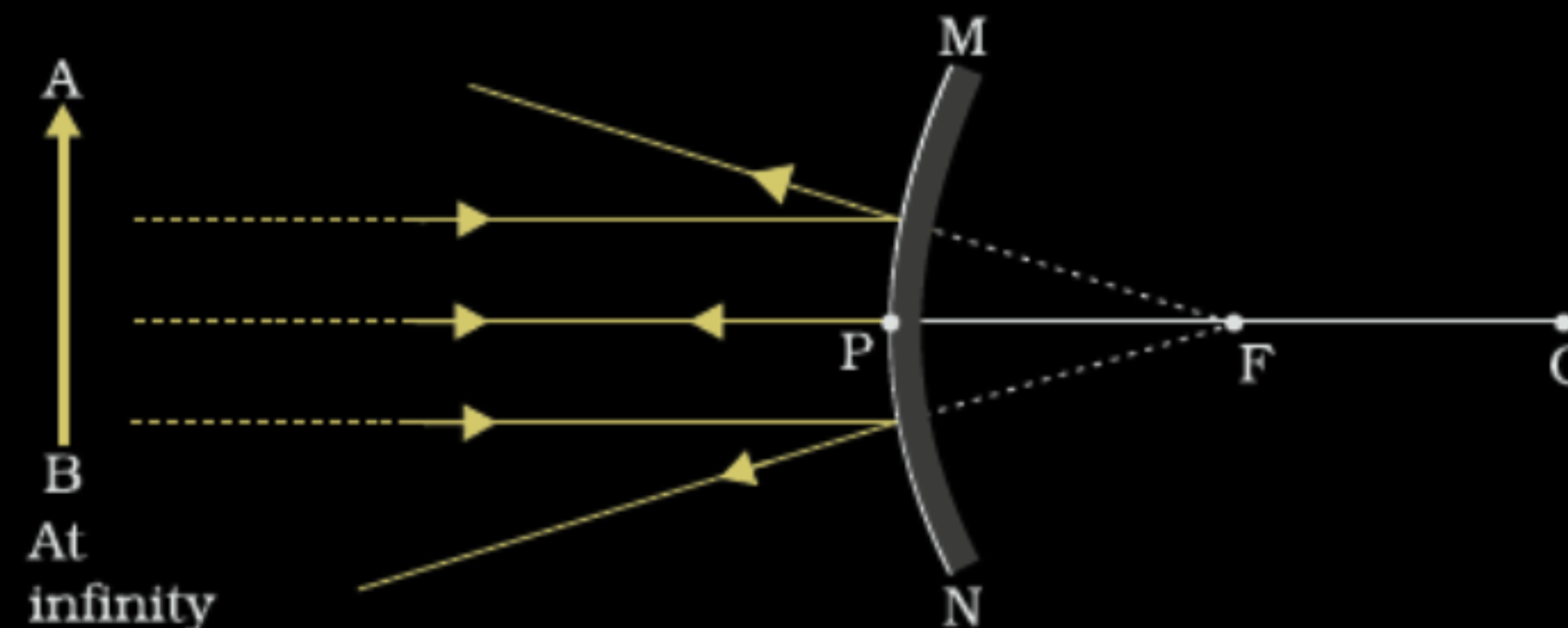
If the image formed by a spherical mirror for all positions of the object placed in front of it is always erect and diminished, what type of mirror is it ? Draw a labelled ray diagram to support your answer.

CBSE 2018

- Convex Mirror
- Labelled Ray diagram for any position of object



OR



Note: If arrows not marked, $\frac{1}{2}$ mark to be deducted.

$\frac{1}{2}$

$1 \frac{1}{2}$

State the laws of refraction of light. Explain the term ‘absolute refractive index of a medium’ and write an expression to relate it with the speed of light in vacuum.

CBSE 2018

State the laws of refraction of light. Explain the term ‘absolute refractive index of a medium’ and write an expression to relate it with the speed of light in vacuum.

CBSE 2018	
<u>1st law</u> : the incident ray, refracted ray and normal to the interface at the point of incidence lie in the same plane.	1
<u>2nd law</u> : The sine of angle of incidence bears a constant ratio with sine of angle of refraction for a given pair of media. Or $\frac{\sin i}{\sin r} = \text{constant}$	1
Absolute Refractive Index of a medium = $\frac{\text{Speed of light in air or vacuum}}{\text{Speed of light in the medium}}$ (Award full marks if the same thing is given in the form of statement)	1

What is meant by power of a lens ? Write its SI unit. A student uses a lens of focal length 40 cm and another of -20 cm. Write the nature and power of each lens.

CBSE 2018

What is meant by power of a lens ? Write its SI unit. A student uses a lens of focal length 40 cm and another of -20 cm. Write the nature and power of each lens.

CBSE 2018

Power of lens = Ability to converge/ diverge light rays passing through it/ reciprocal of the focal length in metres $/ \frac{1}{f}$ (*in meters*)

SI unit of power is Diopetre

$$\text{Power of 1}^{\text{st}} \text{ lens } P_1 = \frac{100}{f_1} = \frac{100}{40 \text{ cm}} = +2.5 \text{ D}$$

Nature: Converging lens / Convex lens

$$\text{Power of 2}^{\text{nd}} \text{ lens } P_2 = \frac{100}{f_2} = \frac{100}{-20 \text{ cm}} = -5 \text{ D}$$

Nature: Diverging lens / Concave lens

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

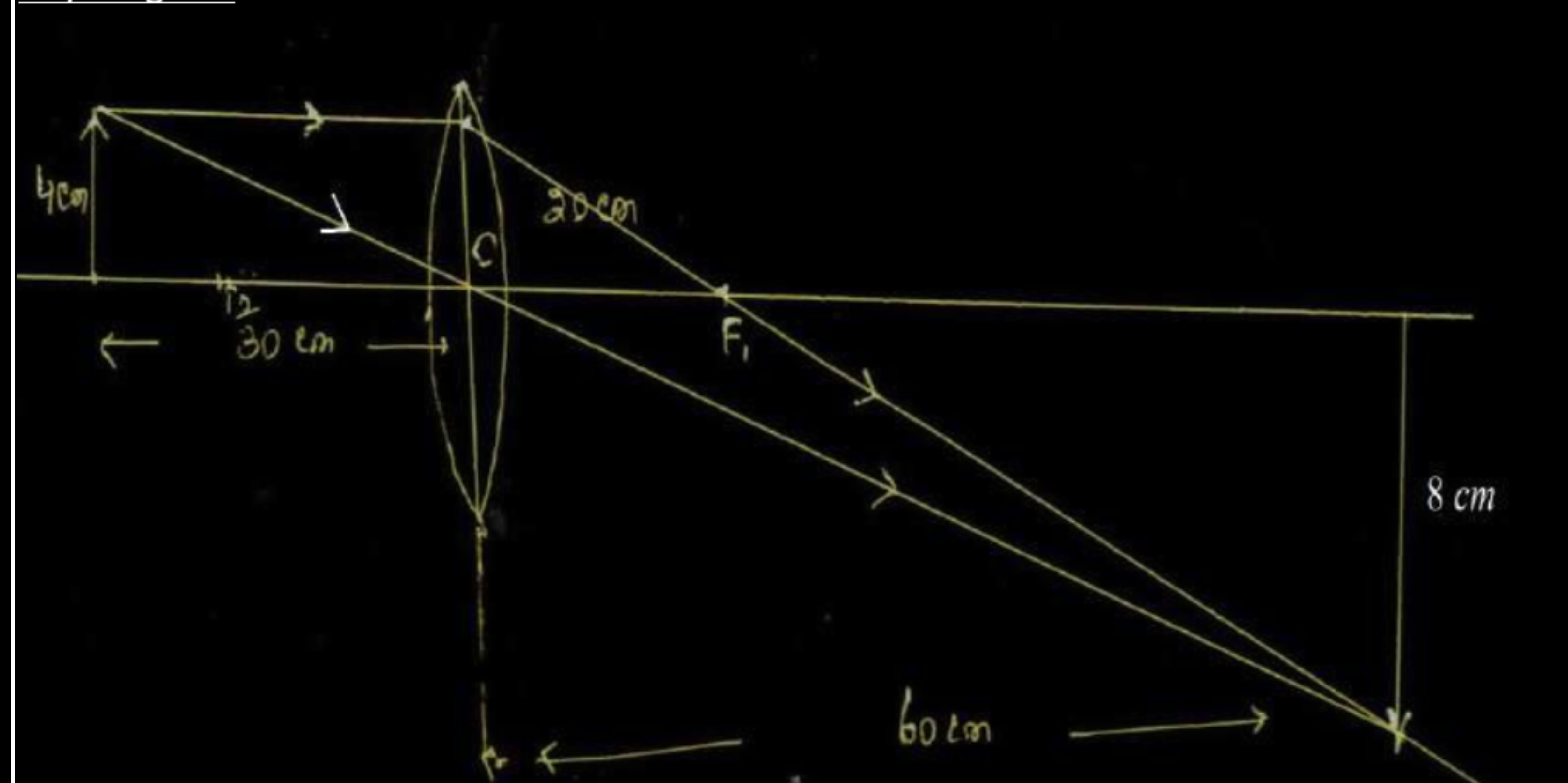
$\frac{1}{2}$

An object of height 4.0 cm is placed at a distance of 30 cm from the optical centre 'O' of a convex lens of focal length 20 cm. Draw a ray diagram to find the position and size of the image formed. Mark optical centre 'O' and principal focus 'F' on the diagram. Also find the approximate ratio of size of the image to the size of the object.

CBSE 2018

An object of height 4.0 cm is placed at a distance of 30 cm from the optical centre 'O' of a convex lens of focal length 20 cm. Draw a ray diagram to find the position and size of the image formed. Mark optical centre 'O' and principal focus 'F' on the diagram. Also find the approximate ratio of size of the image to the size of the object.

CBSE 2018

<u>Ray diagram</u>  Position of O and F Ratio=h_i/h_o approximately 2:1	1 $\frac{1}{2}$ $\frac{1}{2}$
---	---

An object is placed at a distance of 30 cm from a concave lens of focal length 15 cm. List four characteristics (nature, position, etc.) of the image formed by the lens.

CBSE 2017

An object is placed at a distance of 30 cm from a concave lens of focal length 15 cm. List four characteristics (nature, position, etc.) of the image formed by the lens.

CBSE 2017

- | | |
|---|------------------------|
| <ul style="list-style-type: none">• Virtual• Erect• Diminished• On the same side as the object / or any other characteristic | $\frac{1}{2} \times 4$ |
|---|------------------------|

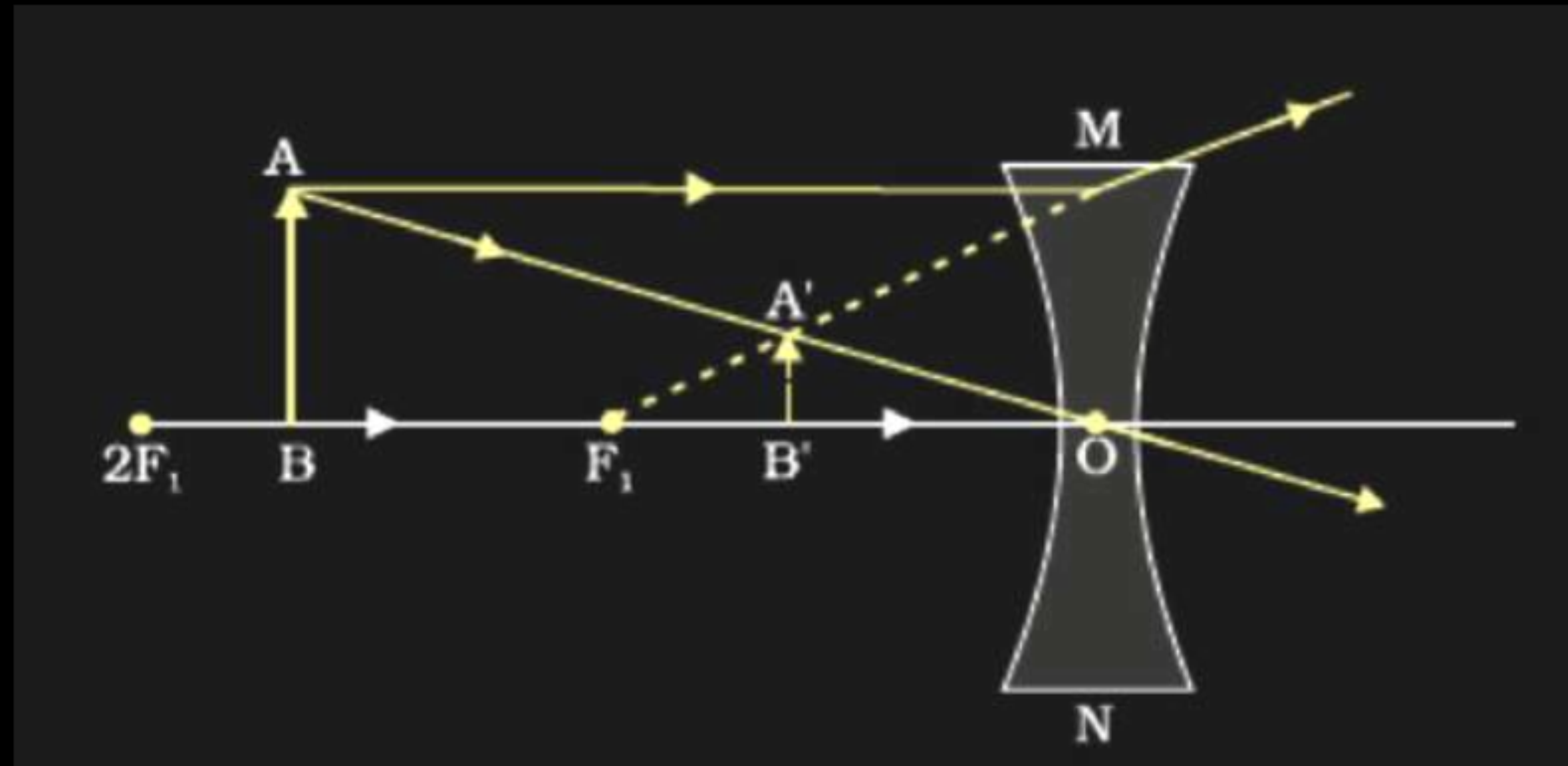
If the image formed by a lens for all positions of an object placed in front of it is always erect and diminished, what is the nature of this lens ? Draw a ray diagram to justify your answer. If the numerical value of the power of this lens is 10 D, what is its focal length in the Cartesian system ?

CBSE 2017

If the image formed by a lens for all positions of an object placed in front of it is always erect and diminished, what is the nature of this lens ? Draw a ray diagram to justify your answer. If the numerical value of the power of this lens is 10 D, what is its focal length in the Cartesian system ?

CBSE 2017

Concave / diverging lens.



Direction of rays

$$f = \frac{1}{P},$$

$$P = -10\text{D},$$

$$f = \frac{1}{-10\text{D}} = -0.1\text{ m} / -10\text{ cm}$$

1/2

1

1/2

1/2

1/2

Analyse the following observation table showing variation of image-distance (v) with object-distance (u) in case of a convex lens and answer the questions that follow without doing any calculations :

S.No.	Object-Distance u (cm)	Image-Distance v (cm)
1	– 100	+ 25
2	– 60	+ 30
3	– 40	+ 40
4	– 30	+ 60
5	– 25	+ 100
6	– 15	+ 120

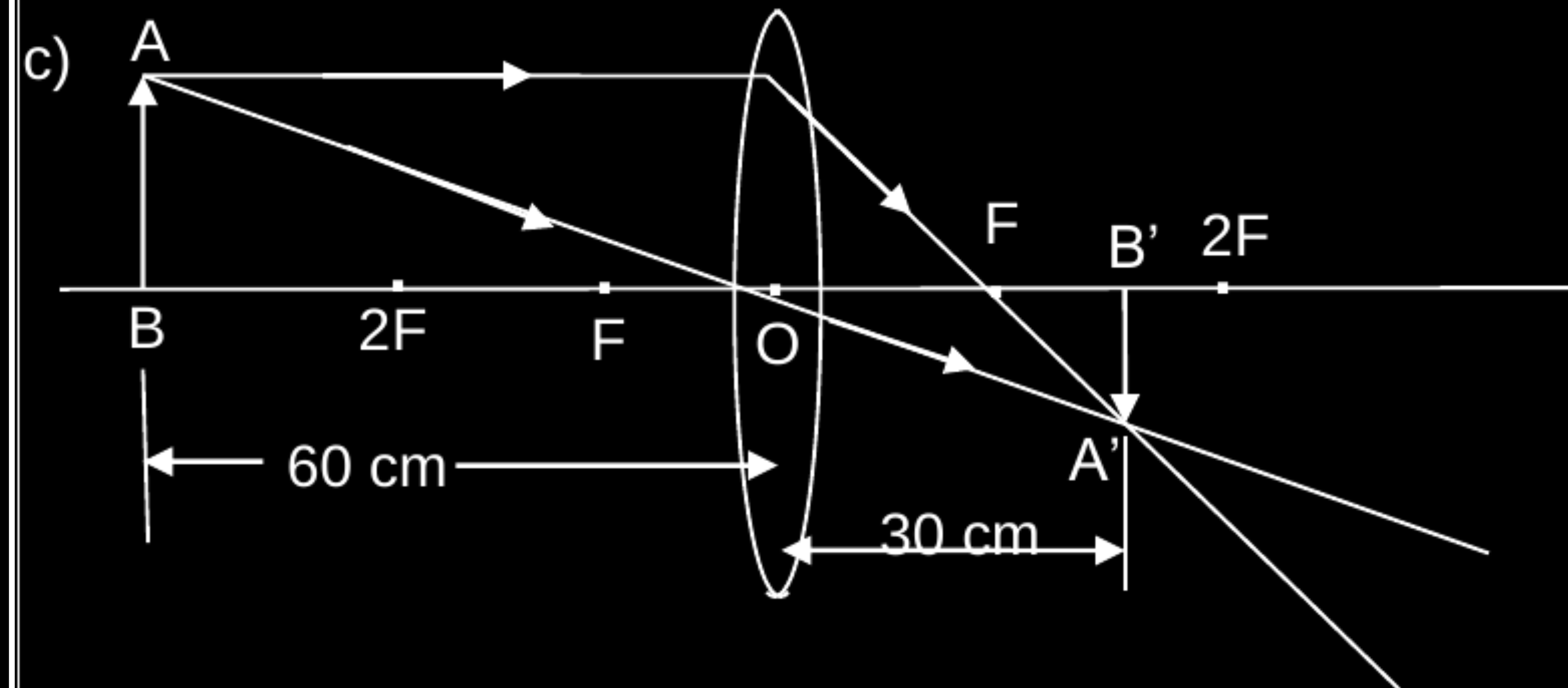
- (a) What is the focal length of the convex lens ? Give reason to justify your answer.
- (b) Write the serial number of the observation which is not correct. On what basis have you arrived at this conclusion ?
- (c) Select an appropriate scale and draw a ray diagram for the observation at S.No.2. Also find the approximate value of magnification.

a) $f = 20 \text{ cm}$

Sl. No. 3, Since $u = -40 \text{ cm}$ and $v = +40 \text{ cm}$, it may be concluded that object is at $2F$

b) Sl. No. 6

When $u = -15 \text{ cm}$, the object is between optical centre and principal focus. So image is virtual and it forms on the same side as the object. Hence, v should be $-ve$, but here it is $+ve$ ($+120 \text{ cm}$)



Direction of rays

$$\text{Magnification } m = \frac{v}{u} = \frac{30 \text{ cm}}{-60 \text{ cm}} = -0.5 / -\frac{1}{2}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

1

1

$\frac{1}{2}$

$\frac{1}{2}$

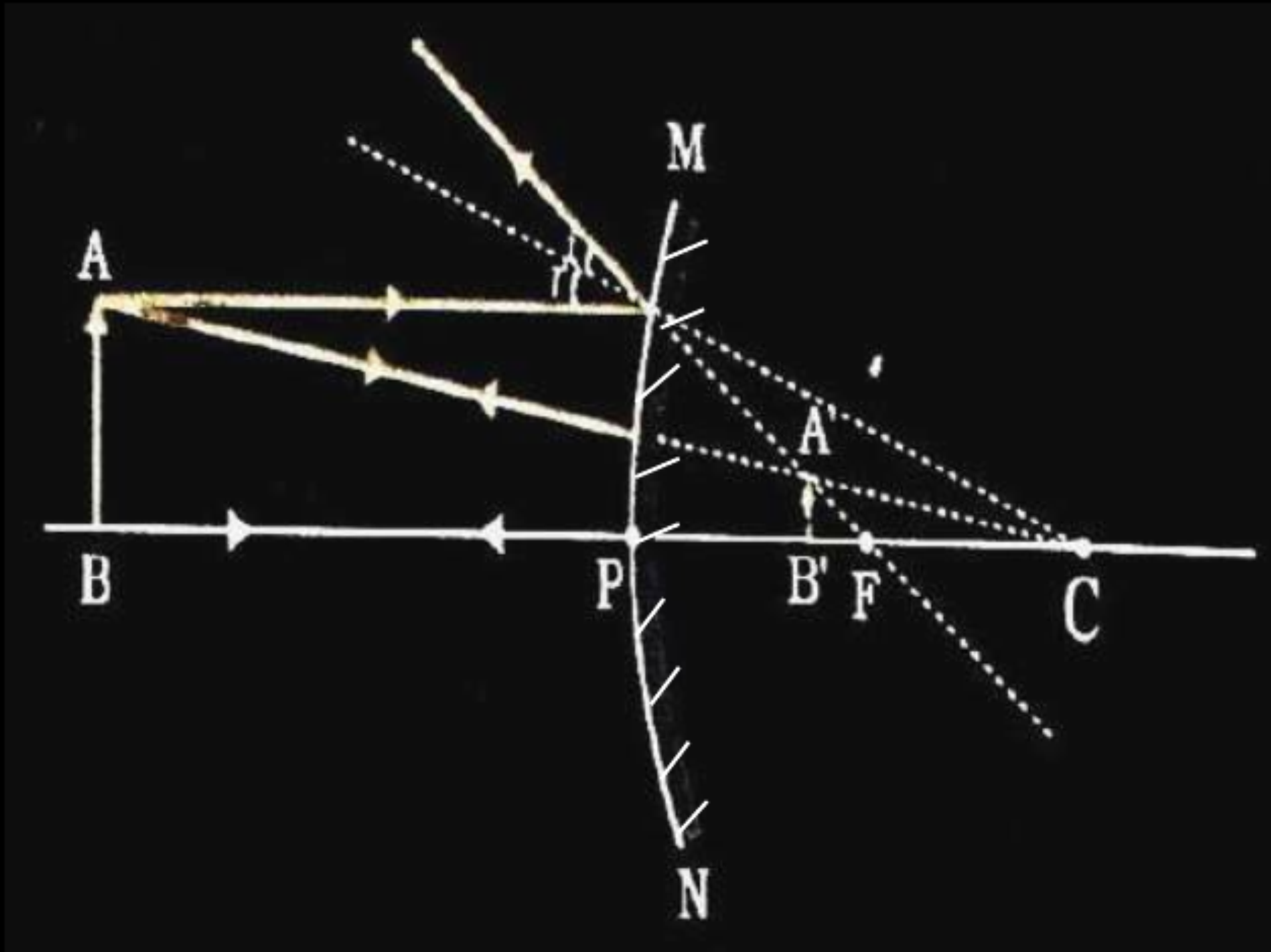
- (a) If the image formed by a mirror for all positions of the object placed in front of it is always diminished, erect and virtual, state the type of the mirror and also draw a ray diagram to justify your answer. Write one use such mirrors are put to and why.
- (b) Define the radius of curvature of spherical mirrors. Find the nature and focal length of a spherical mirror whose radius of curvature is + 24 cm.

CBSE 2017

- (a) If the image formed by a mirror for all positions of the object placed in front of it is always diminished, erect and virtual, state the type of the mirror and also draw a ray diagram to justify your answer. Write one use such mirrors are put to and why.
- (b) Define the radius of curvature of spherical mirrors. Find the nature and focal length of a spherical mirror whose radius of curvature is + 24 cm.

CBSE 2017

a) Convex / diverging mirror)



Direction of rays

Use:- As a rear view mirror / any other use

Reason :- Always give erect and diminished image / Large field of view

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b) The radius of the sphere of which the mirror forms a part / The distance between pole and center of curvature of a mirror.

Nature of the mirror – convex / diverging mirror

$$R = 2f = 24 \text{ cm}$$

$$\therefore f = +12 \text{ cm}$$

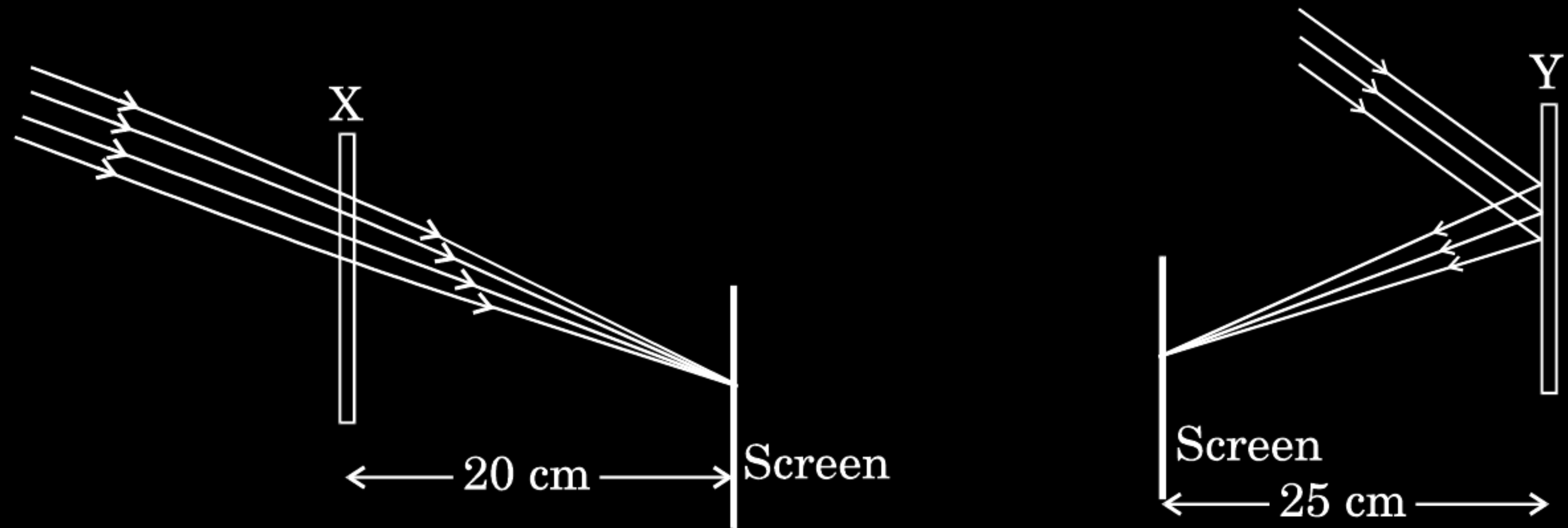
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

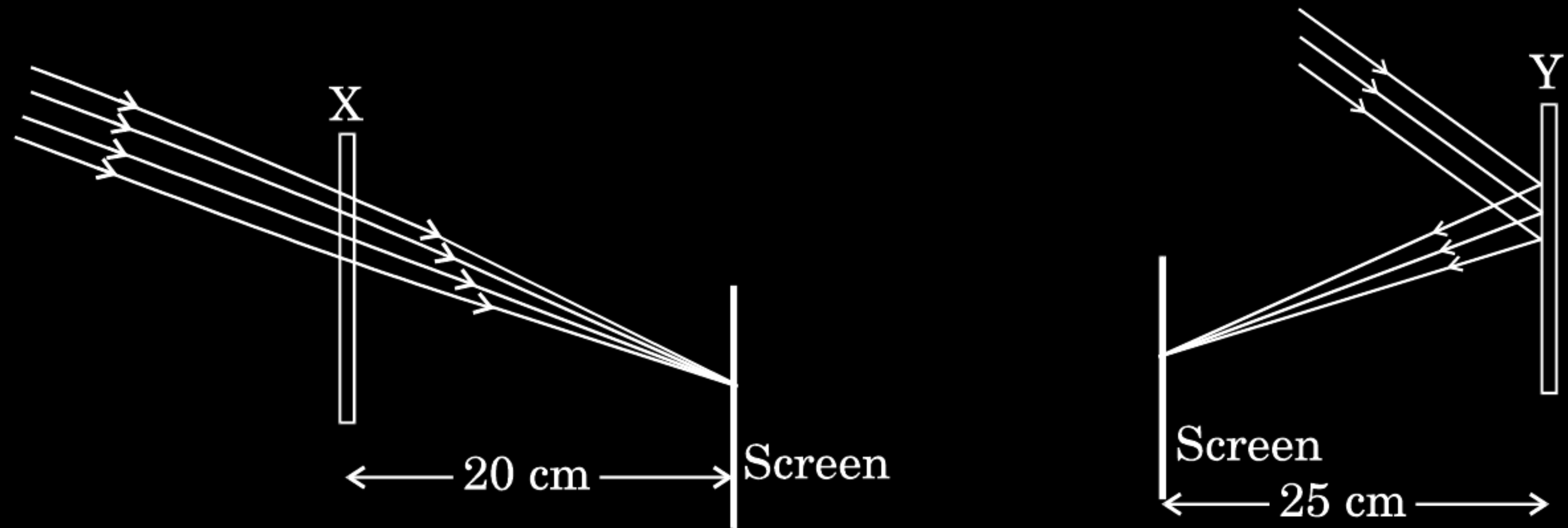
$\frac{1}{2}$

Study the given ray diagrams and select the correct statement from the following :



- (A) Device X is a concave mirror and device Y is a convex lens, whose focal lengths are 20 cm and 25 cm respectively.
- (B) Device X is a convex lens and device Y is a concave mirror, whose focal lengths are 10 cm and 25 cm respectively.
- (C) Device X is a concave lens and device Y is a convex mirror, whose focal lengths are 20 cm and 25 cm respectively.
- (D) Device X is a convex lens and device Y is a concave mirror, whose focal lengths are 20 cm and 25 cm respectively.

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- (D) Device X is a convex lens and device Y is a concave mirror, whose focal lengths are 20 cm and 25 cm respectively.

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A student obtains a blurred image of a distant object on a screen using a convex lens. To obtain a distinct image on the screen he should move the lens

- (A) away from the screen
- (B) towards the screen
- (C) to a position very far away from the screen
- (D) either towards or away from the screen depending upon the position of the object

CBSE 2017

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CBSE 2017

A student very cautiously traces the path of a ray through a glass slab for different values of the angle of incidence ($\angle i$). He then measures the corresponding values of the angle of refraction ($\angle r$) and the angle of emergence ($\angle e$) for every value of the angle of incidence. On analysing these measurements of angles, his conclusion would be

- (A) $\angle i > \angle r > \angle e$
- (B) $\angle i = \angle e > \angle r$
- (C) $\angle i < \angle r < \angle e$
- (D) $\angle i = \angle e < \angle r$

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CBSE 2017

A student focuses the image of a candle flame, placed at about 2 m from a convex lens of focal length 10 cm, on a screen. After that he moves gradually the flame towards the lens and each time focuses its image on the screen.

- (A) In which direction does he move the lens to focus the flame on the screen ?
- (B) What happens to the size of the image of the flame formed on the screen ?
- (C) What difference is seen in the intensity (brightness) of the image of the flame on the screen ?
- (D) What is seen on the screen when the flame is very close (at about 5 cm) to the lens ?

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a) Note: For part (a) $\frac{1}{2}$ mark to be awarded to every student	$\frac{1}{2}$
b) Size of the image increases	$\frac{1}{2}$
c) Intensity / brightness of the image decreases	$\frac{1}{2}$
d) No distinct image is formed. Only a patch of light is seen.	$\frac{1}{2}$

“The magnification produced by a spherical mirror is -3 ”. List four informations you obtain from this statement about the mirror/image.

CBSE 2016

“The magnification produced by a spherical mirror is -3 ”. List four informations you obtain from this statement about the mirror/image.

CBSE 2016

1) Inverted image; 2) magnified; 3) concave mirror ; 4) real image

4 x $\frac{1}{2}$

The image of an object formed by a lens is of magnification -1 . If the distance between the object and its image is 60 cm, what is the focal length of the lens ? If the object is moved 20 cm towards the lens, where would the image be formed ? State reason and also draw a ray diagram in support of your answer.

CBSE 2016

The image of an object formed by a lens is of magnification -1 . If the distance between the object and its image is 60 cm, what is the focal length of the lens ? If the object is moved 20 cm towards the lens, where would the image be formed ? State reason and also draw a ray diagram in support of your answer.

CBSE 2016

Image with magnification -1 means image is inverted and of the same size.

Therefore, object is at $2F$ and the image is also at $2F$ on the other side of the lens.

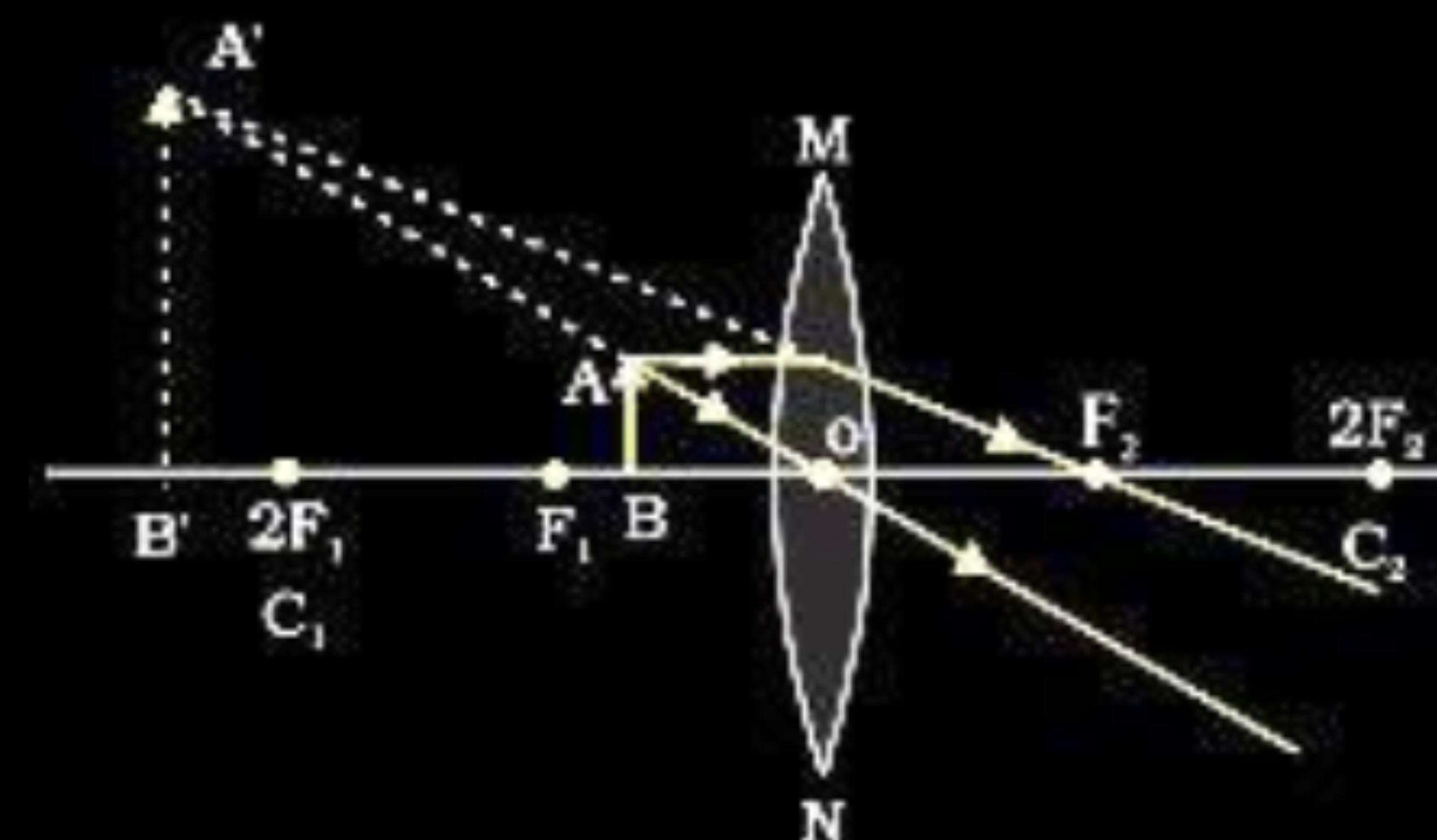
Therefore, distance between the object and its image is $4f = 60$ cm 1

$$\Rightarrow f = 15$$
 cm $\frac{1}{2}$

Object distance $2f = 30$ cm. if the object is shifted towards the lens by 20 cm, the new object distance = 30 cm $-$ 20 cm = 10 cm. $\frac{1}{2}$

This distance is less than the focal length , and the image formed in this case would be virtual, erect and will form on the same side as the object

1



- (a) Define focal length of a spherical lens.
- (b) A divergent lens has a focal length of 30 cm. At what distance should an object of height 5 cm from the optical centre of the lens be placed so that its image is formed 15 cm away from the lens ? Find the size of the image also.
- (c) Draw a ray diagram to show the formation of image in the above situation.

CBSE 2016

- (a) Define focal length of a spherical lens.
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CBSE 2016

a) Distance between optical centre and focus of the lens.

1

b) $f = -30$ cm; $u = ?$; $h_1 = 5$ cm; $h_2 = ?$; $v = -15$ cm

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \implies \frac{1}{u} = \frac{1}{v} - \frac{1}{f} \implies u = \frac{vf}{f-v}$$

$\frac{1}{2}$

$$\implies u = \frac{-15 \text{ cm} \times -30 \text{ cm}}{-30 \text{ cm} - (-15 \text{ cm})} = -30 \text{ cm}$$

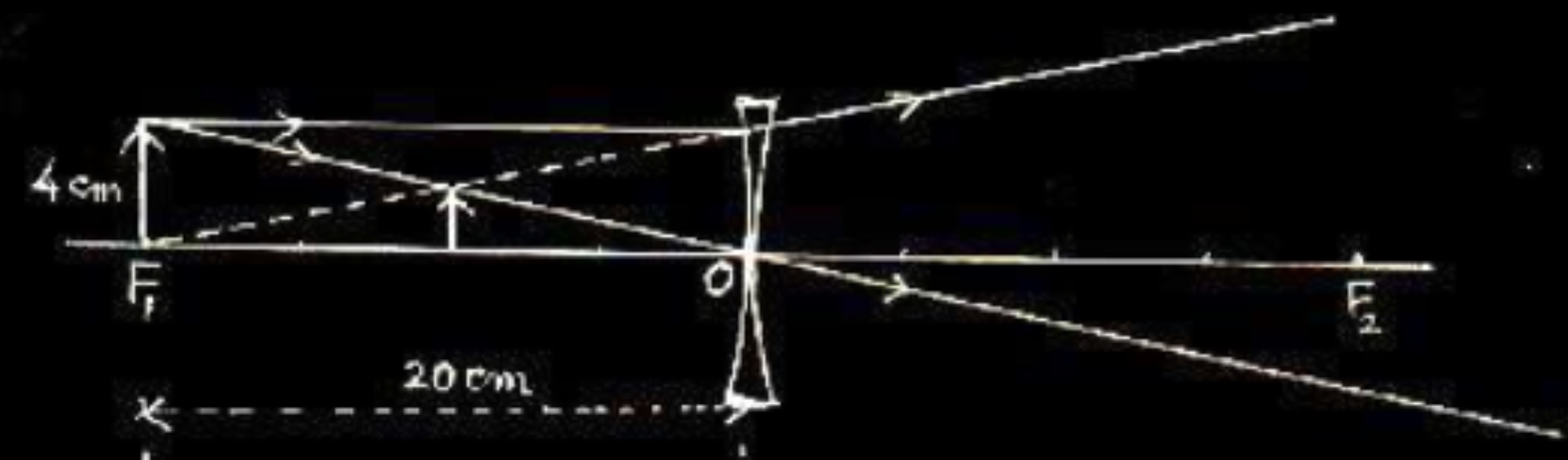
$1 \frac{1}{2}$

$$m = \frac{v}{u} = \frac{h_2}{h_1} \implies h_2 = \frac{v}{u} \times h_1$$

$$= \frac{-15 \text{ cm}}{-30 \text{ cm}} \times 5 \text{ cm} = 2.5 \text{ cm}$$

1

1



It is desired to obtain an erect image of an object, using concave mirror of focal length of 12 cm.

- (i) What should be the range of distance of an object placed in front of the mirror ?
- (ii) Will the image be smaller or larger than the object. Draw ray diagram to show the formation of image in this case.
- (iii) Where will the image of this object be, if it is placed 24 cm in front of the mirror ? Draw ray diagram for this situation also to justify your answer.

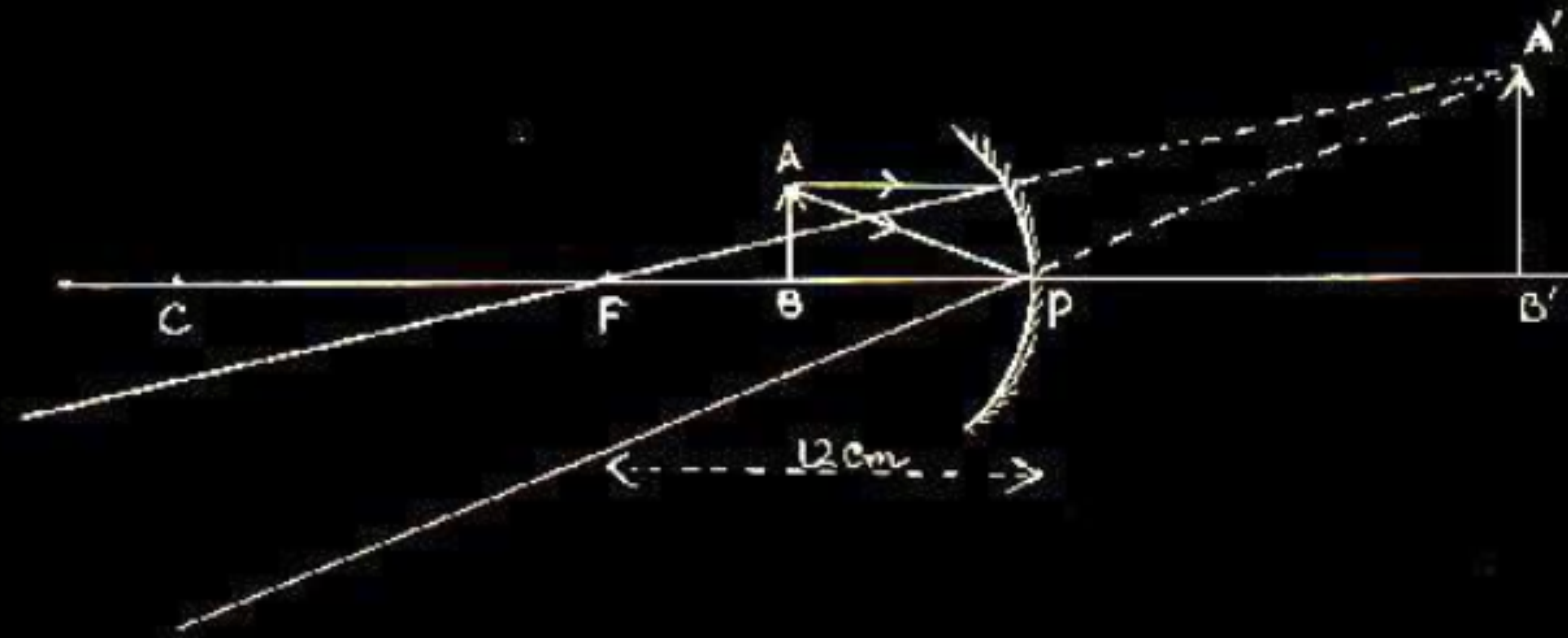
Show the positions of pole, principal focus and the centre of curvature in the above ray diagrams.

(i) Range of distance – between 0 cm - < 12 cm

1

ii) larger than the object

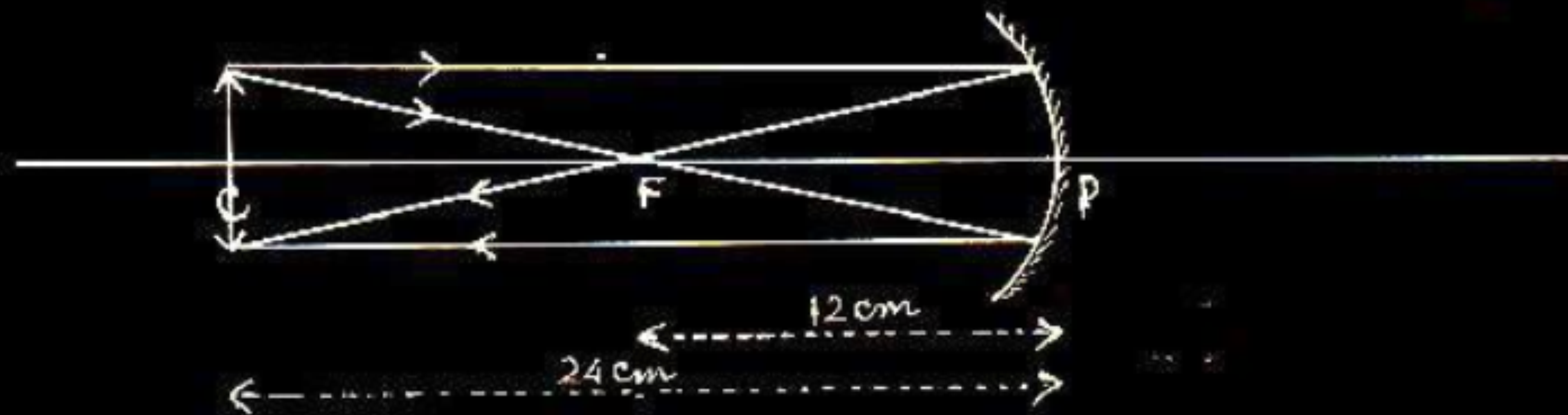
$\frac{1}{2}$



iii) Image also at 24 cm in front of the mirror

1 $\frac{1}{2}$

$\frac{1}{2}$



1 $\frac{1}{2}$

Suppose you have focused on a screen the image of candle flame placed at farthest end of the laboratory table using a convex lens. If your teacher suggests you to focus the parallel rays of the sun, reaching your laboratory table, on the same screen, what you are expected to do is to move the :

- (a) lens slightly towards the screen
- (b) lens slightly away from the screen
- (c) lens slightly towards the sun
- (d) lens and screen both towards the sun

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- (d) lens and screen both towards the sun

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(a)

To determine the approximate value of the focal length of a given concave mirror, you focus the image of a distant object formed by the mirror on a screen. The image obtained on the screen, as compared to the object is always :

- (a) Laterally inverted and diminished
- (b) Inverted and diminished
- (c) Erect and diminished
- (d) Erect and highly diminished

To determine the approximate value of the focal length of a given concave mirror, you focus the image of a distant object formed by the mirror on a screen. The image obtained on the screen, as compared to the object is always :

- (a) Laterally inverted and diminished
- (b) Inverted and diminished
- (c) Erect and diminished
- (d) Erect and highly diminished

CBSE 2016

(b)

In your laboratory you trace the path of light rays through a glass slab for different values of angle of incidence ($\angle i$) and in each case measure the values of the corresponding angle of refraction ($\angle r$) and angle of emergence ($\angle e$).

On the basis of your observations your correct conclusion is :

- (a) $\angle i$ is more than $\angle r$, but nearly equal to $\angle e$
- (b) $\angle i$ is less than $\angle r$, but nearly equal to $\angle e$
- (c) $\angle i$ is more than $\angle e$, but nearly equal to $\angle r$
- (d) $\angle i$ is less than $\angle e$, but nearly equal to $\angle r$

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In your laboratory you trace the path of light rays through a glass slab for different values of angle of incidence ($\angle i$) and in each case measure the values of the corresponding angle of refraction ($\angle r$) and angle of emergence ($\angle e$). On the basis of your observations your correct conclusion is :

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CBSE 2016

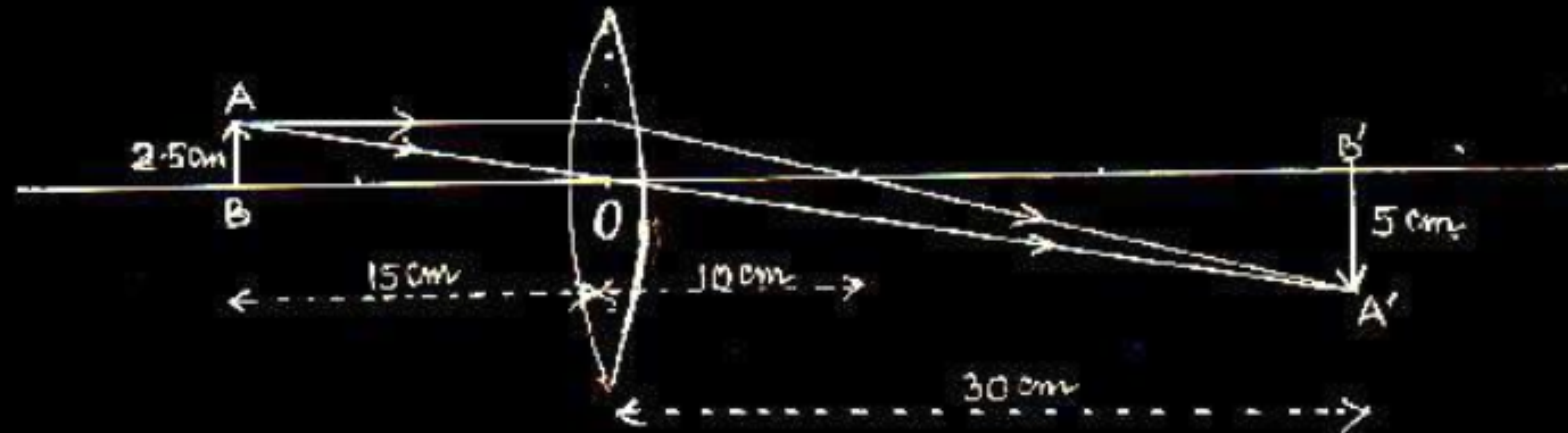
(a)

An object of height 2.5 cm is placed at a distance of 15 cm from the optical centre 'O' of a convex lens of focal length 10 cm. Draw a ray diagram to find the position and size of the image formed. Mark optical centre 'O', principal focus F and height of the image on the diagram.

CBSE 2016

An object of height 2.5 cm is placed at a distance of 15 cm from the optical centre 'O' of a convex lens of focal length 10 cm. Draw a ray diagram to find the position and size of the image formed. Mark optical centre 'O', principal focus F and height of the image on the diagram.

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Marking of O , F and size of the image

The refractive indices of glass and water with respect to air are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If speed of light in glass is 2×10^8 m/s, find the speed of light in water.

CBSE 2016

The refractive indices of glass and water with respect to air are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If speed of light in glass is 2×10^8 m/s, find the speed of light in water.

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$$\text{Refractive index of a medium} = \frac{\text{Speed of light in air}}{\text{Speed of light in the medium}}$$

$$\frac{3}{2} = \frac{\text{Speed of light in air}}{2 \times 10^8 \text{ m/s}}$$

1

$$\text{Speed of light in air} = 3 \times 10^8 \text{ m/s}$$

$$\text{Speed of light in water} = \frac{3 \times 10^8 \text{ m/s}}{4/3} = 2.25 \times 10^8 \text{ m/s}$$

1

The image formed by a spherical mirror is real, inverted and is of magnification -2 . If the image is at a distance of 30 cm from the mirror, where is the object placed ? Find the focal length of the mirror. List two characteristics of the image formed if the object is moved 10 cm towards the mirror.

CBSE 2016

The image formed by a spherical mirror is real, inverted and is of magnification -2 . If the image is at a distance of 30 cm from the mirror, where is the object placed ? Find the focal length of the mirror. List two characteristics of the image formed if the object is moved 10 cm towards the mirror.

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$$m = -2 \quad \frac{v}{u} = 2 \quad v = -30 \text{ cm}$$

$$u = -15$$

1

$$f = \frac{uv}{u+v} = \frac{-15 \text{ cm} \times -30 \text{ cm}}{-15 \text{ cm} + (-30) \text{ cm}} = \frac{450}{-45} = -10 \text{ cm}$$

1

If the object is shifted 10 cm towards the mirror $u = -5 \text{ cm}$

Therefore the object is between pole and focus and the image formed is

i) Virtual ii) Erect iii) magnified (any two) $\frac{1}{2}$, $\frac{1}{2}$

Name the type of mirrors used in the design of solar furnaces. Explain how high temperature is achieved by this device.

CBSE 2016

Name the type of mirrors used in the design of solar furnaces. Explain how high temperature is achieved by this device.

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- Concave Mirrors / Converging Mirrors 1/2
- When a solar furnace is placed at the focus of a large concave mirror/ reflector, it focuses a parallel beam of light on the furnace, consequently a high temperature is achieved after some time. 3 x 1/2

What is meant by power of a lens? What does its sign (+ve or –ve) indicate? State its S.I. unit. How is this unit related to focal length of a lens?

CBSE 2016

What is meant by power of a lens? What does its sign (+ve or –ve) indicate? State its S.I. unit. How is this unit related to focal length of a lens?

CBSE 2016

- | | |
|--|-----|
| • Ability of lens to converge or diverge the light rays. | 1/2 |
| • +ve sign $\square$ converging lens/ convex lens | |
| –ve sign $\square$ diverging lens/ concave lens | 1/2 |
| • S.I. unit – diopetre | 1/2 |
| 1 diopetre = 1/ focal length (m) | 1/2 |

- (a) Draw a ray diagram to show the refraction of light through a glass slab and mark angle of refraction and the lateral shift suffered by the ray of light while passing through the slab.
- (b) If the refractive index of glass for light going from air to glass is $\frac{3}{2}$, find the refractive index of air for light going from glass to air.

CBSE 2016

- (a) Draw a ray diagram to show the refraction of light through a glass slab and mark angle of refraction and the lateral shift suffered by the ray of light while passing through the slab.
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air.

a)

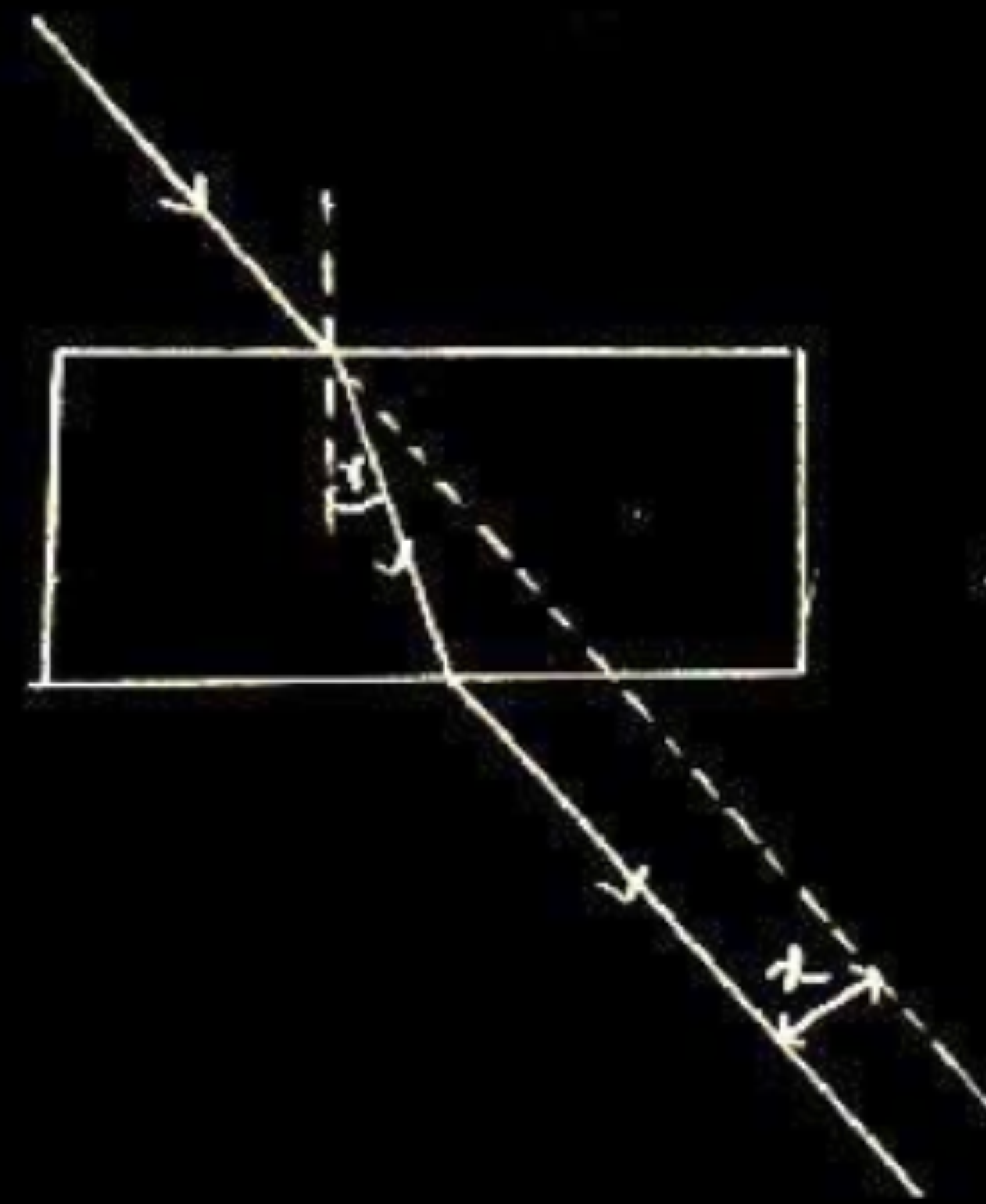


Diagram 1
Marking $\square$ r and x $\frac{1}{2}, \frac{1}{2}$

b)

$${}_a n_g = \frac{3}{2}$$

$$\square {}_g n_a = \frac{1}{{}_a n_g} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$$

Alternately,

$$\frac{c_{air}}{c_{glass}} = \frac{3}{2}$$

$$\square \frac{c_{glass}}{c_{air}} = \frac{2}{3}$$

(a) Define the following terms in the context of spherical mirrors :

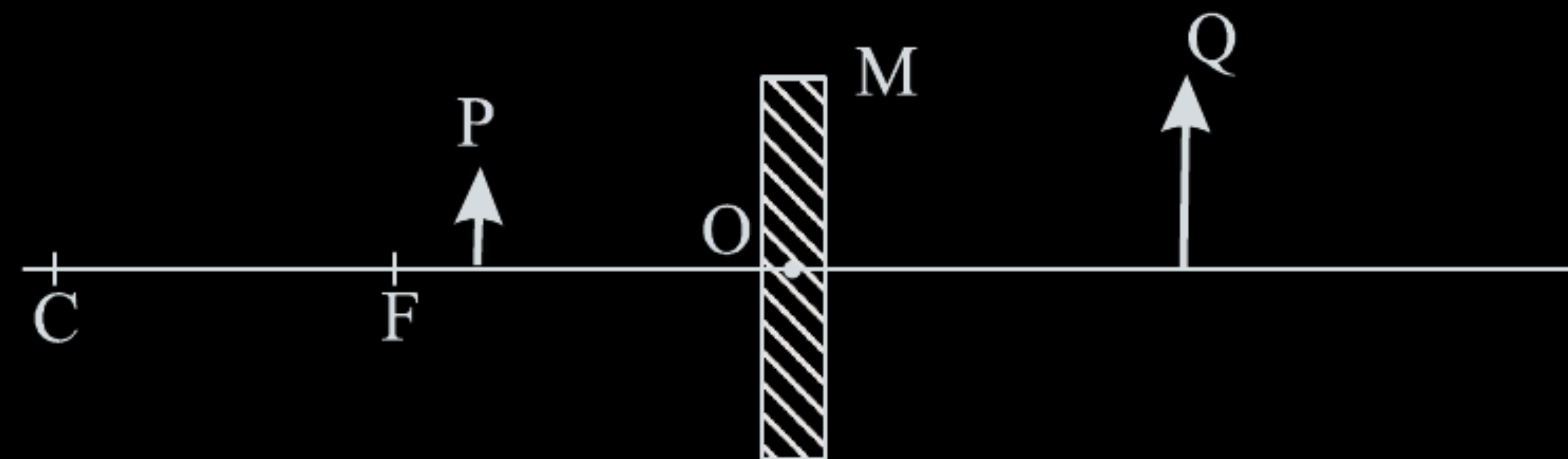
- (i) Pole
- (ii) Centre of curvature
- (iii) Principal axis
- (iv) Principal focus

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(b) Draw ray diagrams to show the principal focus of a

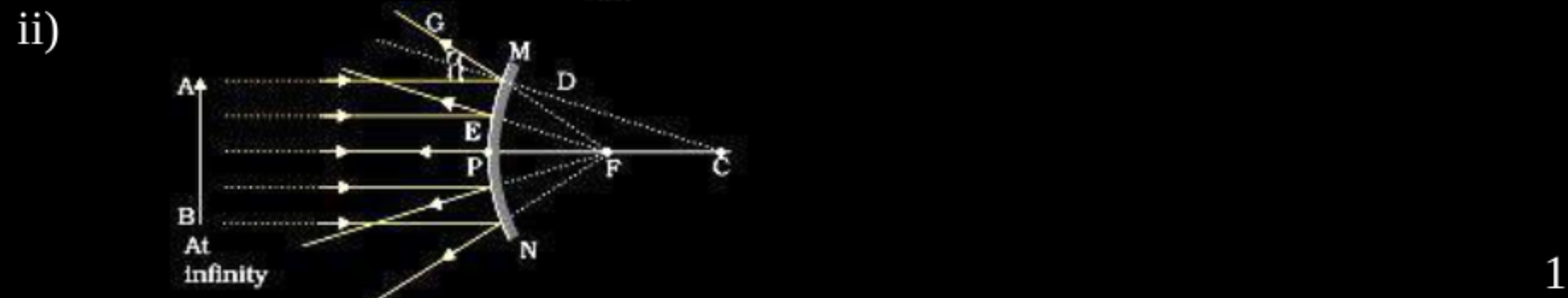
- (i) Concave mirror
- (ii) Convex mirror

(c) Consider the following diagram in which M is a mirror and P is an object and Q is its magnified image formed by the mirror.



State the type of the mirror M and one characteristic property of the image Q.

- a)
- i) Pole – Centre of the reflecting surface of the mirror.
 - ii) Centre of curvature – The centre of the hollow sphere of which the reflecting surface of mirror forms a part.
 - iii) Principal axis – Straight-line passing through the pole and the centre of curvature of a spherical mirror.
 - iv) Principal focus – Incident rays parallel to principal axis, after reflection, either converge to or appear to diverge from a fixed point on the principal axis called principal focus of the spherical mirror. 4x ½



- c) Concave mirror ½
 Image formed is virtual ½

- (a) Draw a ray diagram to show the formation of image by a convex lens when an object is placed in front of the lens between its optical centre and principal focus.
- (b) In the above ray diagram mark the object-distance (u) and the image-distance (v) with their proper signs (+ve or – ve as per the new Cartesian sign convention) and state how these distances are related to the focal length (f) of the convex lens in this case.
- (c) Find the power of a convex lens which forms a real, and inverted image of magnification -1 of an object placed at a distance of 20 cm from its optical centre.

a)

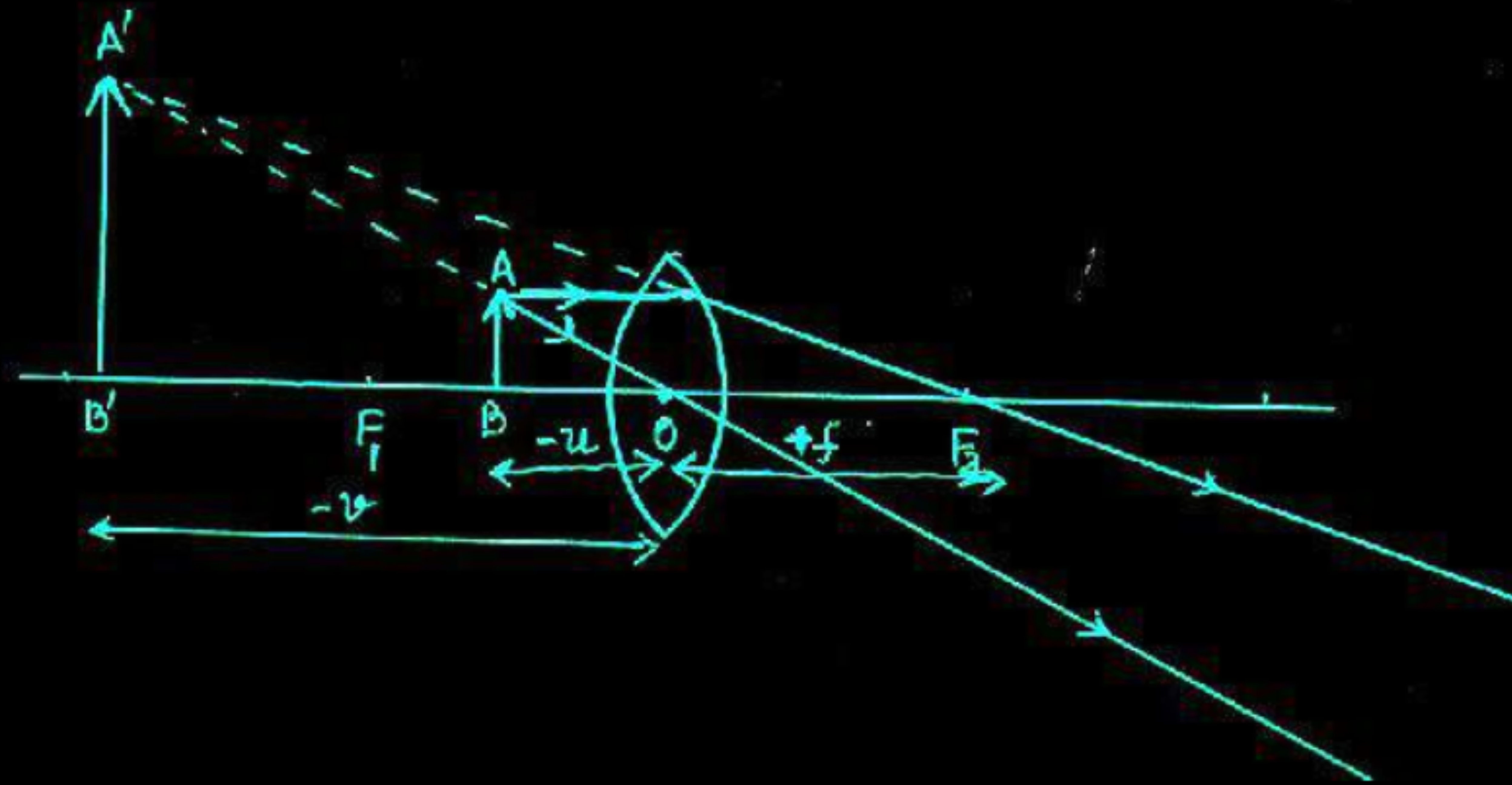


Diagram 1
Marking of u & v $\frac{1}{2}, \frac{1}{2}$

b)

Relation : $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$\frac{1}{2}$

c) $m = -1; u = -20\text{cm}; v = ? \quad f = ?$

$m = \frac{v}{u} \quad \square \quad v = +20\text{cm}$

$\frac{1}{2}$

Thus object is at $2F$

i.e. $2f = 20\text{cm}$

$\square \quad f = 10\text{cm} = 0.1\text{m}$

1

$P = \frac{1}{f} = \frac{1}{0.1} = +10\text{D}$

1

A student obtained a sharp image of a candle flame placed at the distant end of the laboratory table on a screen using a concave mirror to determine its focal length. The teacher suggested him to focus a distant building about 1 km far from the laboratory, for getting more correct value of the focal length. In order to focus the distant building on the same screen the student should slightly move the :

CBSE 2016

- (a) mirror away from the screen
- (b) screen away from the mirror
- (c) screen towards the mirror
- (d) screen towards the building

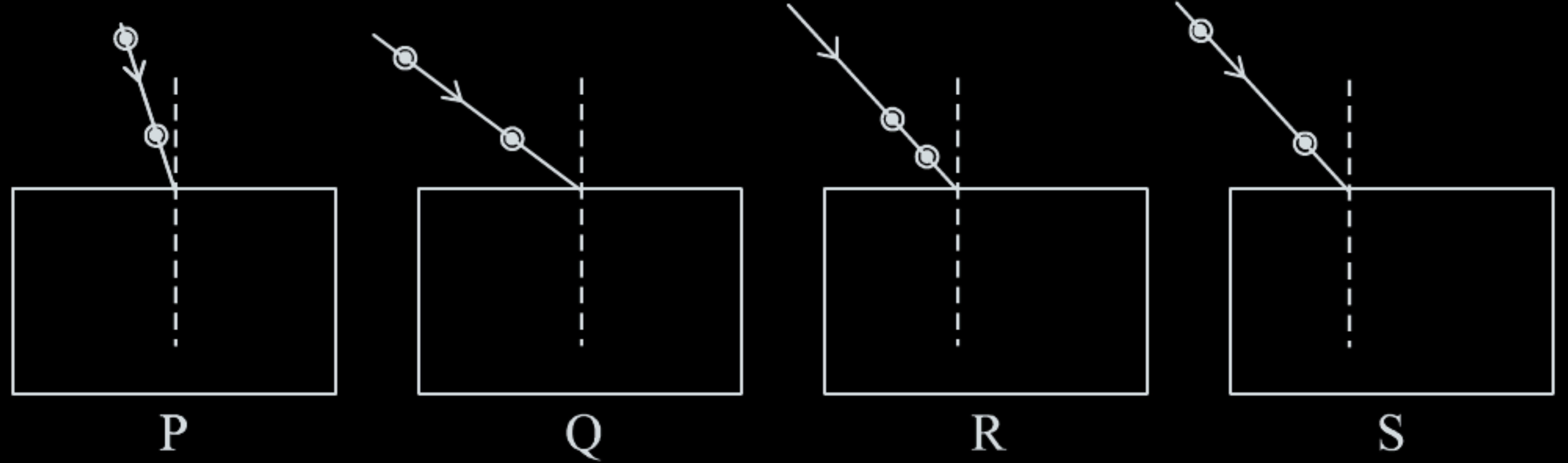
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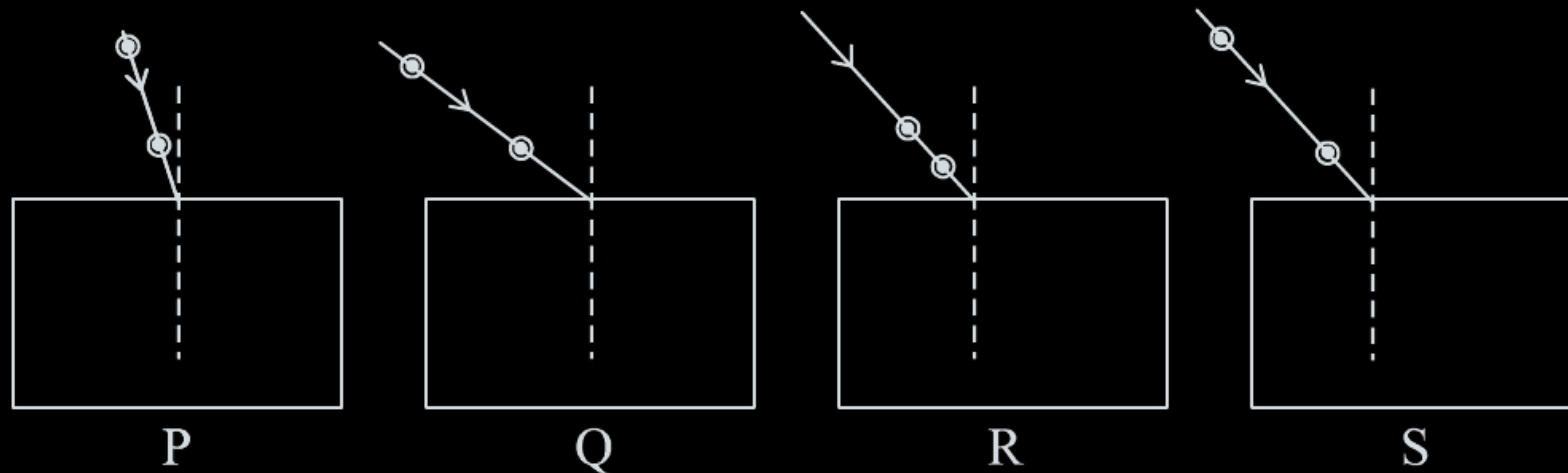
(c)

Select from the following the best experimental set-up for tracing the path of a ray of light passing through a rectangular glass slab :



- (a) P
- (b) Q
- (c) R
- (d) S

Select from the following the best experimental set-up for tracing the path of a ray of light passing through a rectangular glass slab :



- (a) P
- (b) Q
- (c) R
- (d) S

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(d)

To determine the approximate focal length of the given convex lens by focussing a distant object (say, a sign board), you try to focus the image of the object on a screen. The image you obtain on the screen is always :

- (a) erect and laterally inverted
- (b) erect and diminished
- (c) inverted and diminished
- (d) virtual, inverted and diminished

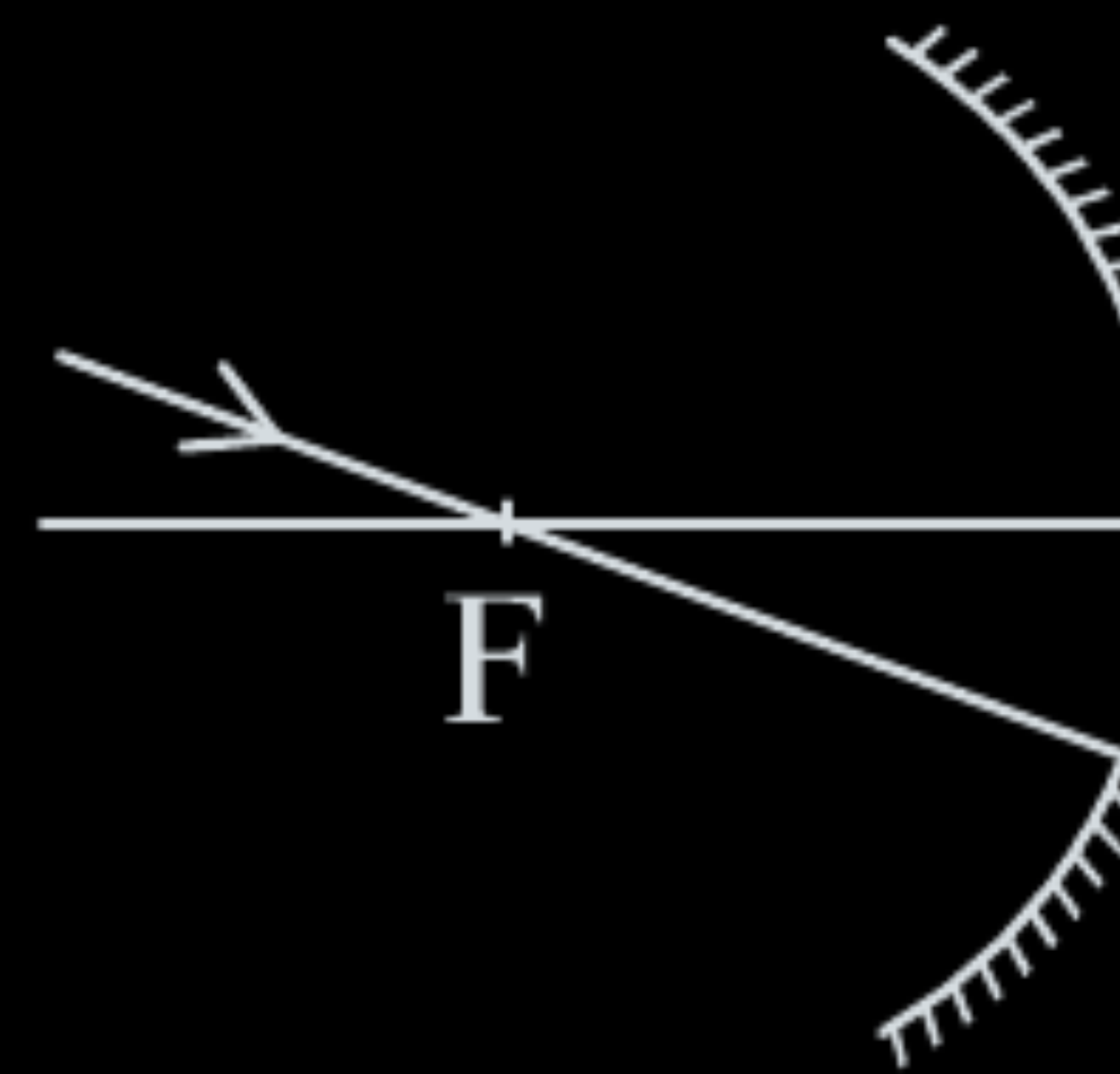
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To determine the approximate focal length of the given convex lens by focussing a distant object (say, a sign board), you try to focus the image of the object on a screen. The image you obtain on the screen is always :

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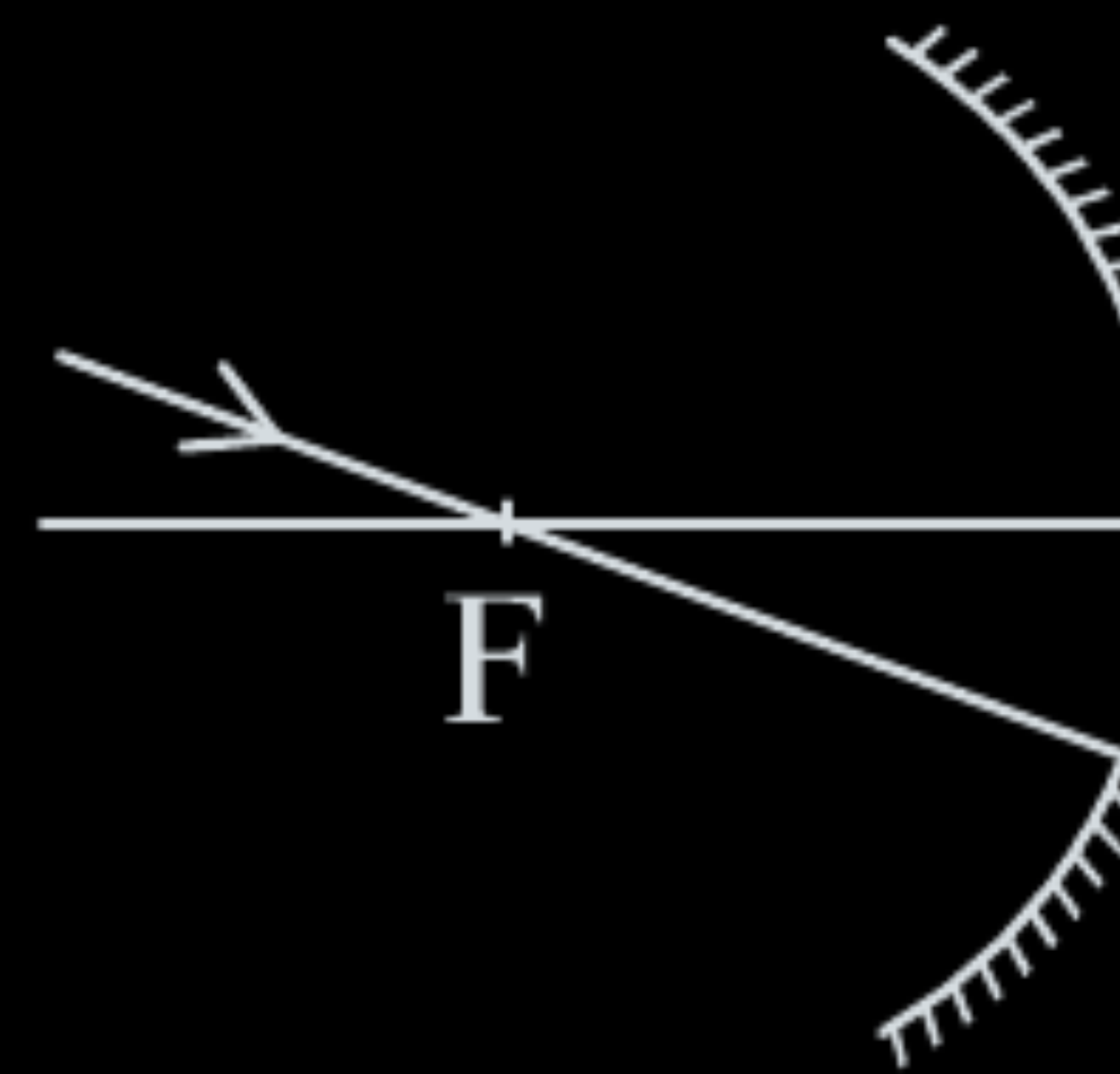
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Draw the following ray diagram. A student focuses the image of a well illuminated distant object on a screen using a convex lens. After that he gradually moves the object towards the lens and each time focuses its image on the screen by adjusting the lens.



- (i) In which direction towards the screen or away from the screen, does he move the lens?
- (ii) What happens to the size of the image does it decrease or increase?
- (iii) What happens to the image on the screen when he moves the object very close to the lens?

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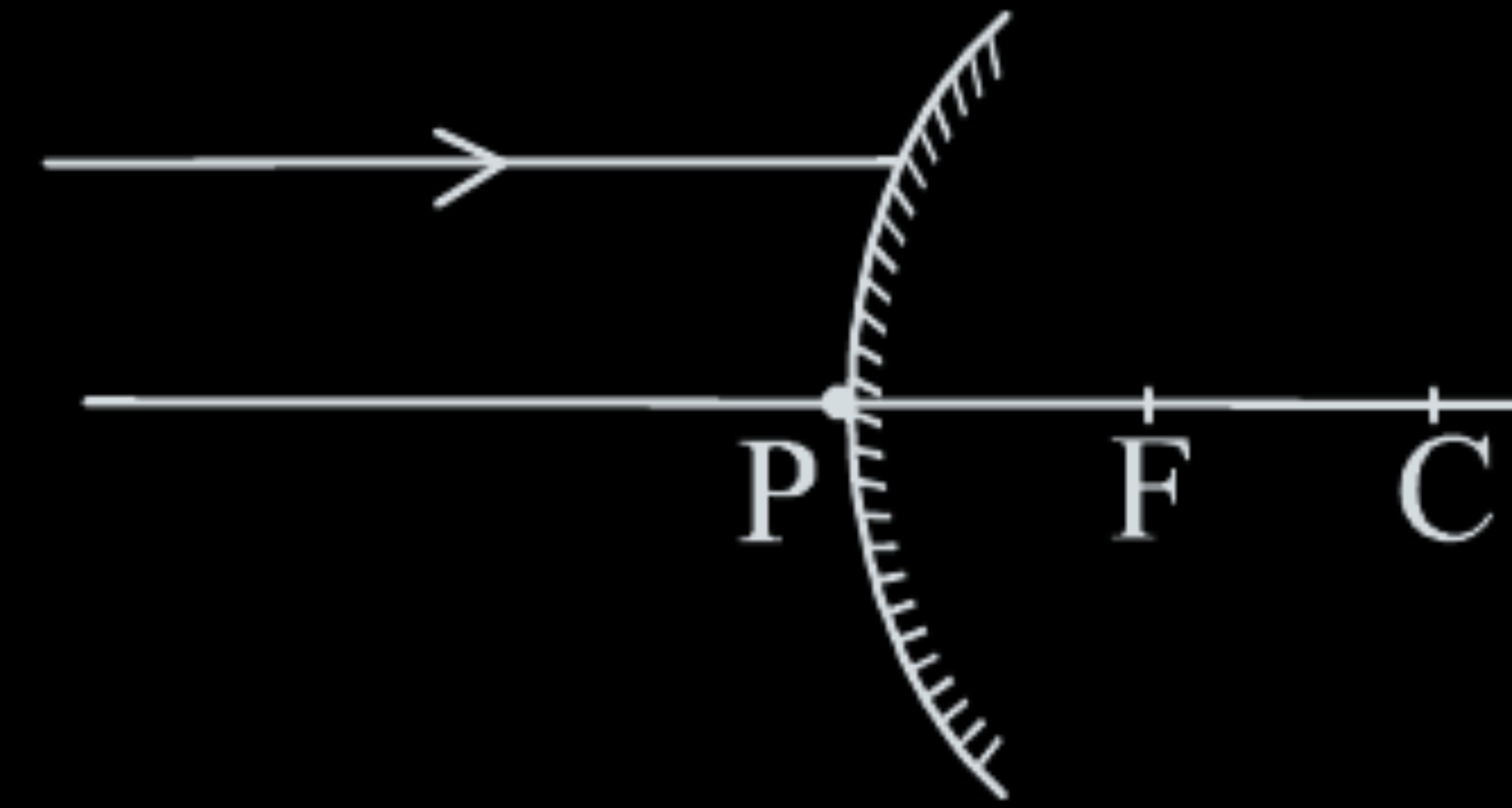


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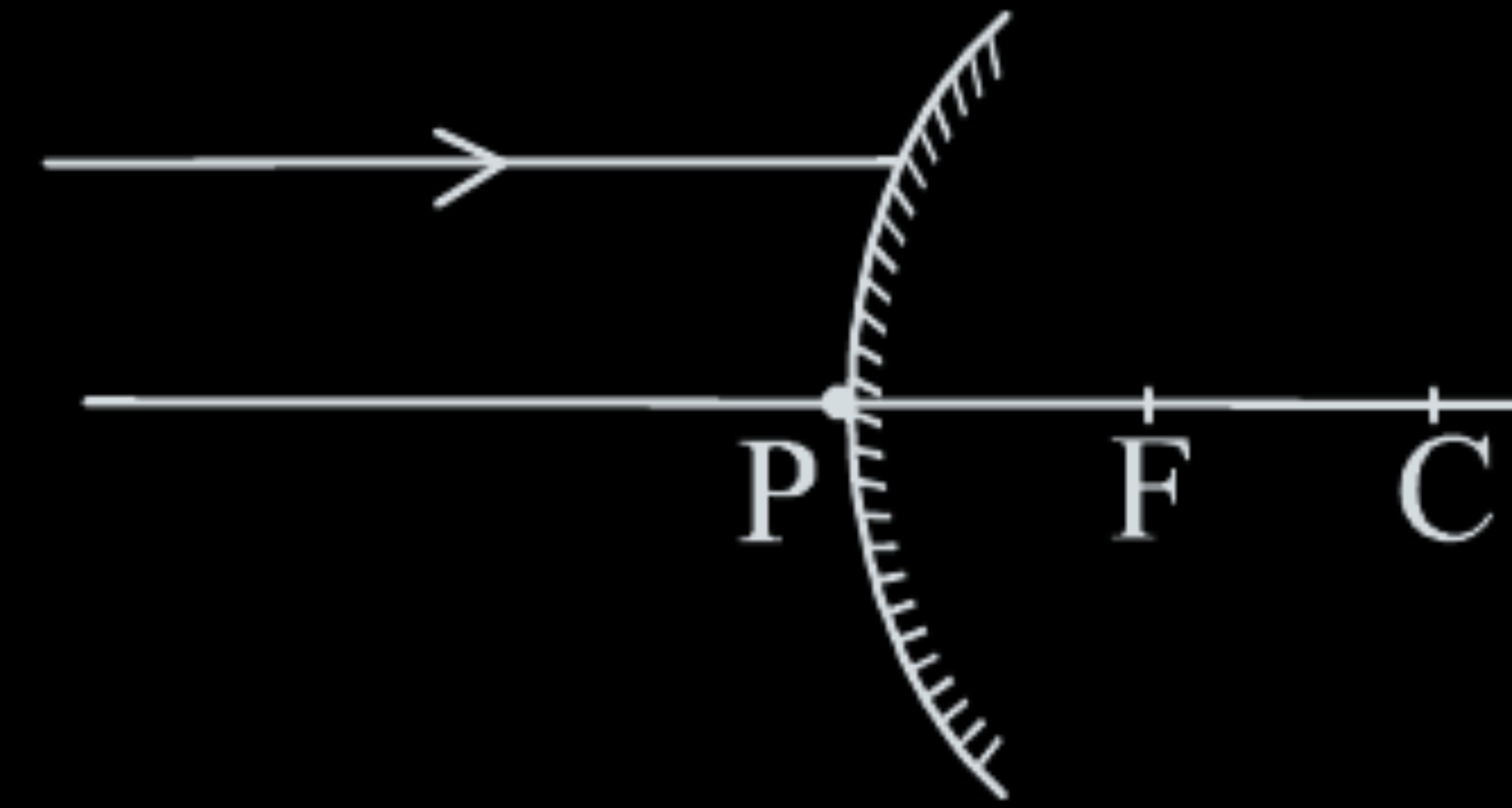
- | | | |
|-------|---|---------------|
| (i) | Lens towards the screen/ screen away from the lens
(Note: one mark to be awarded for any other answer) | 1 |
| (ii) | Increase | $\frac{1}{2}$ |
| (iii) | No image on the screen | $\frac{1}{2}$ |

A ray of light is incident on a convex mirror as shown. Redraw the diagram and complete the path of this ray after reflection from the mirror. Mark angle of incidence and angle of reflection on it.

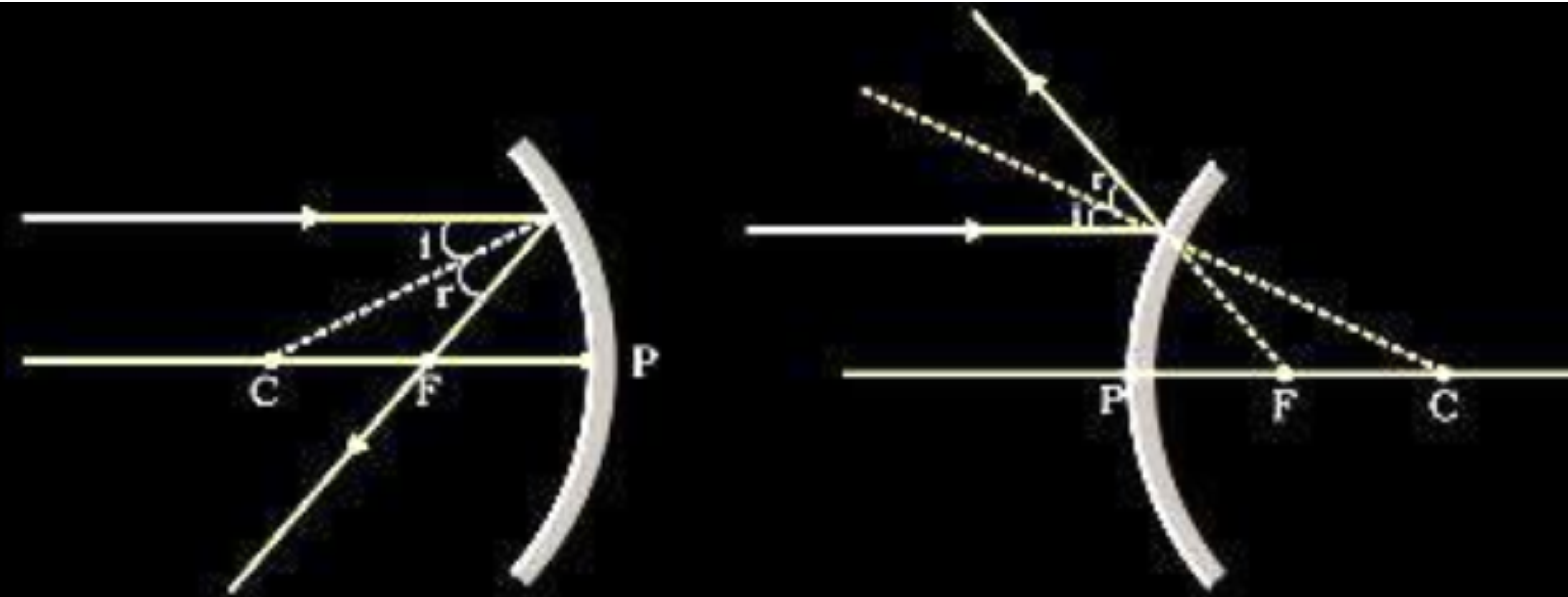


CBSE 2016

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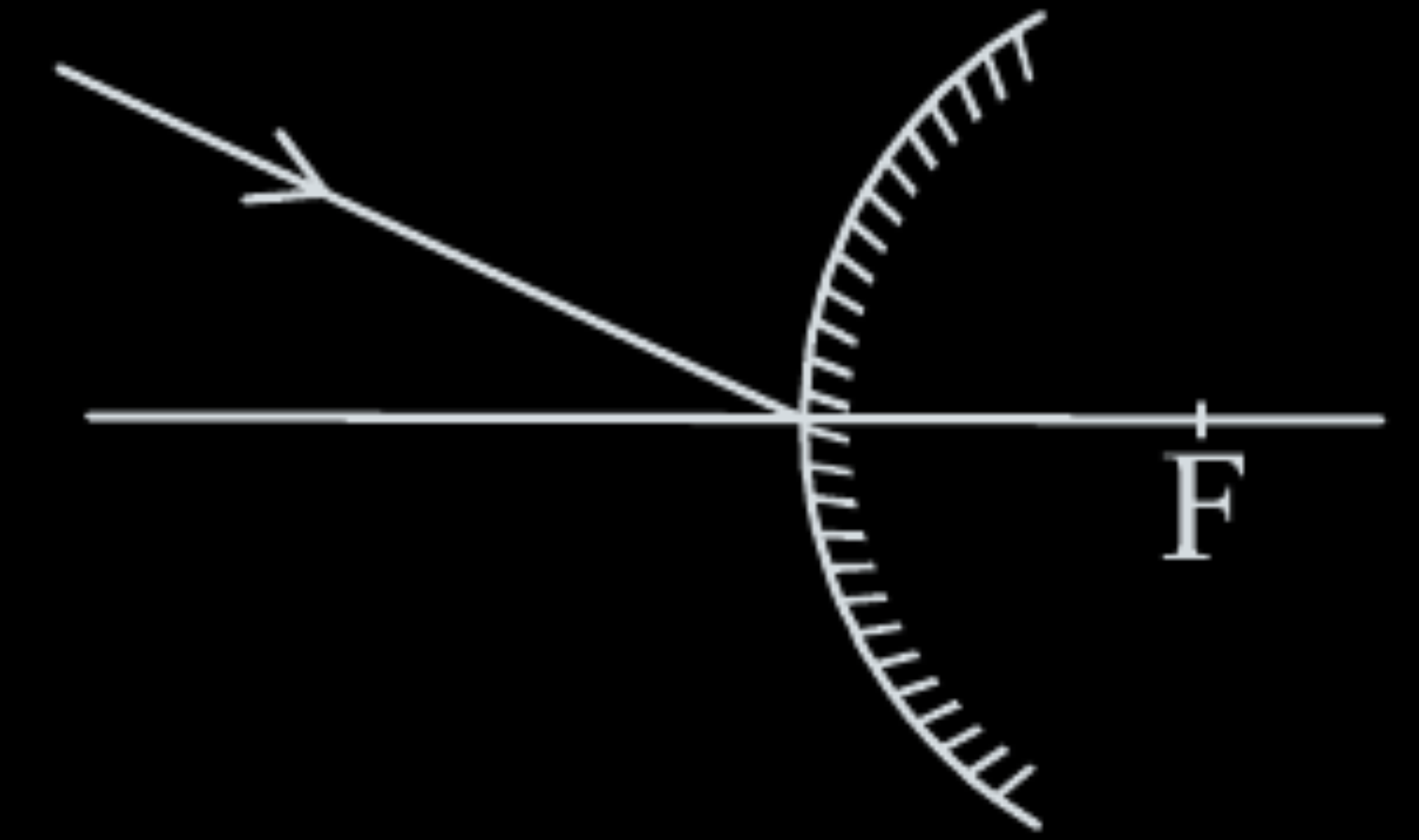
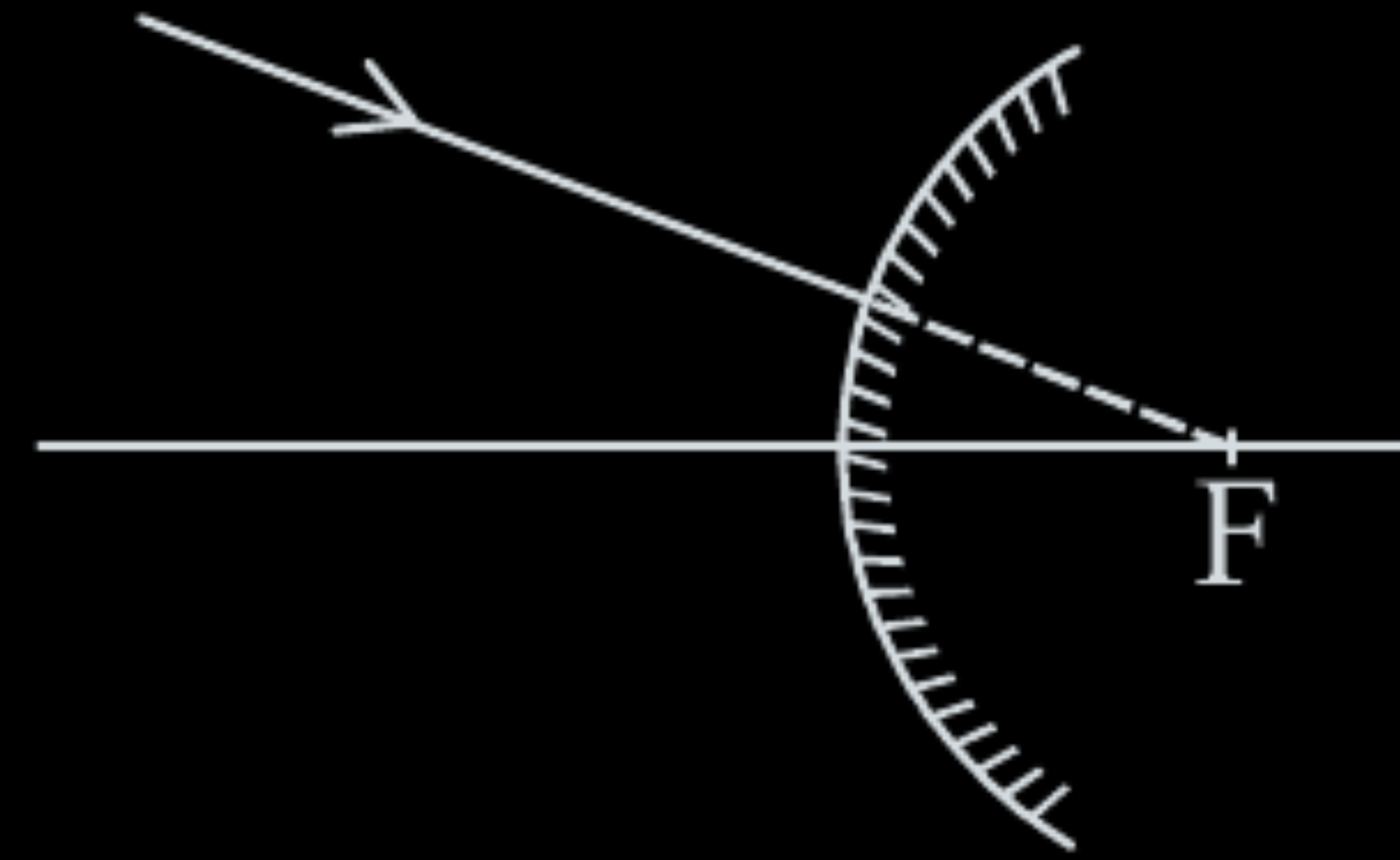
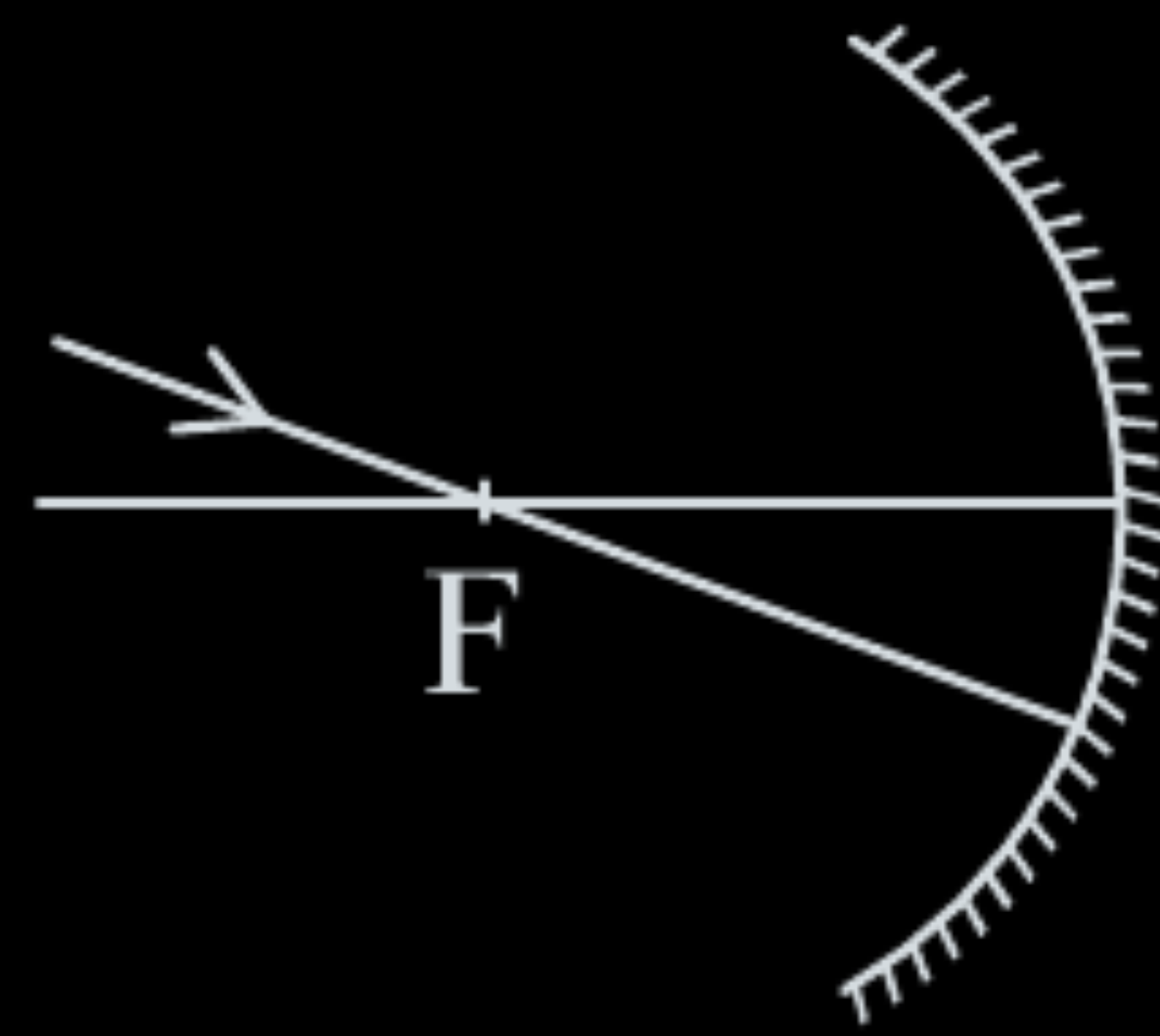


CBSE 2016

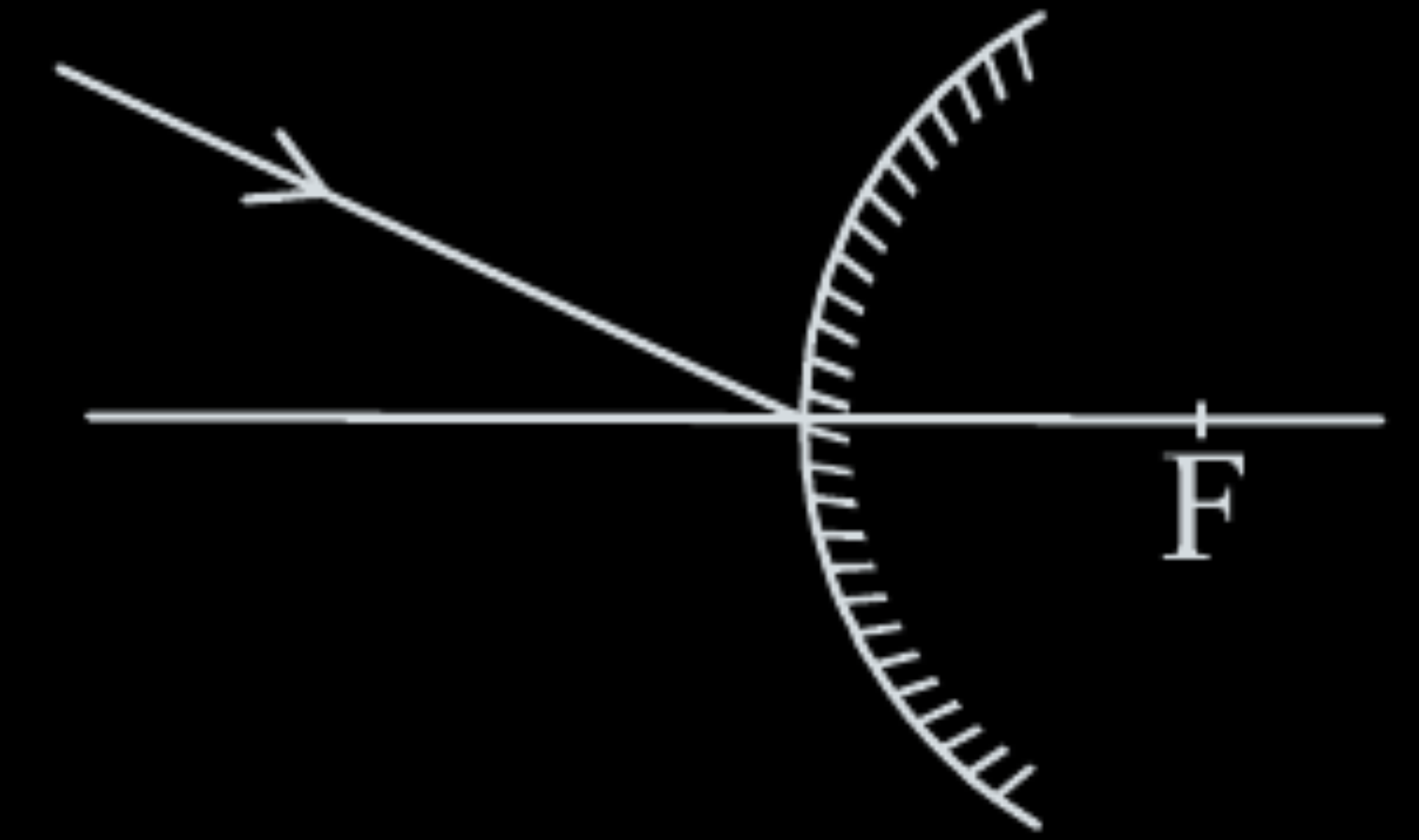
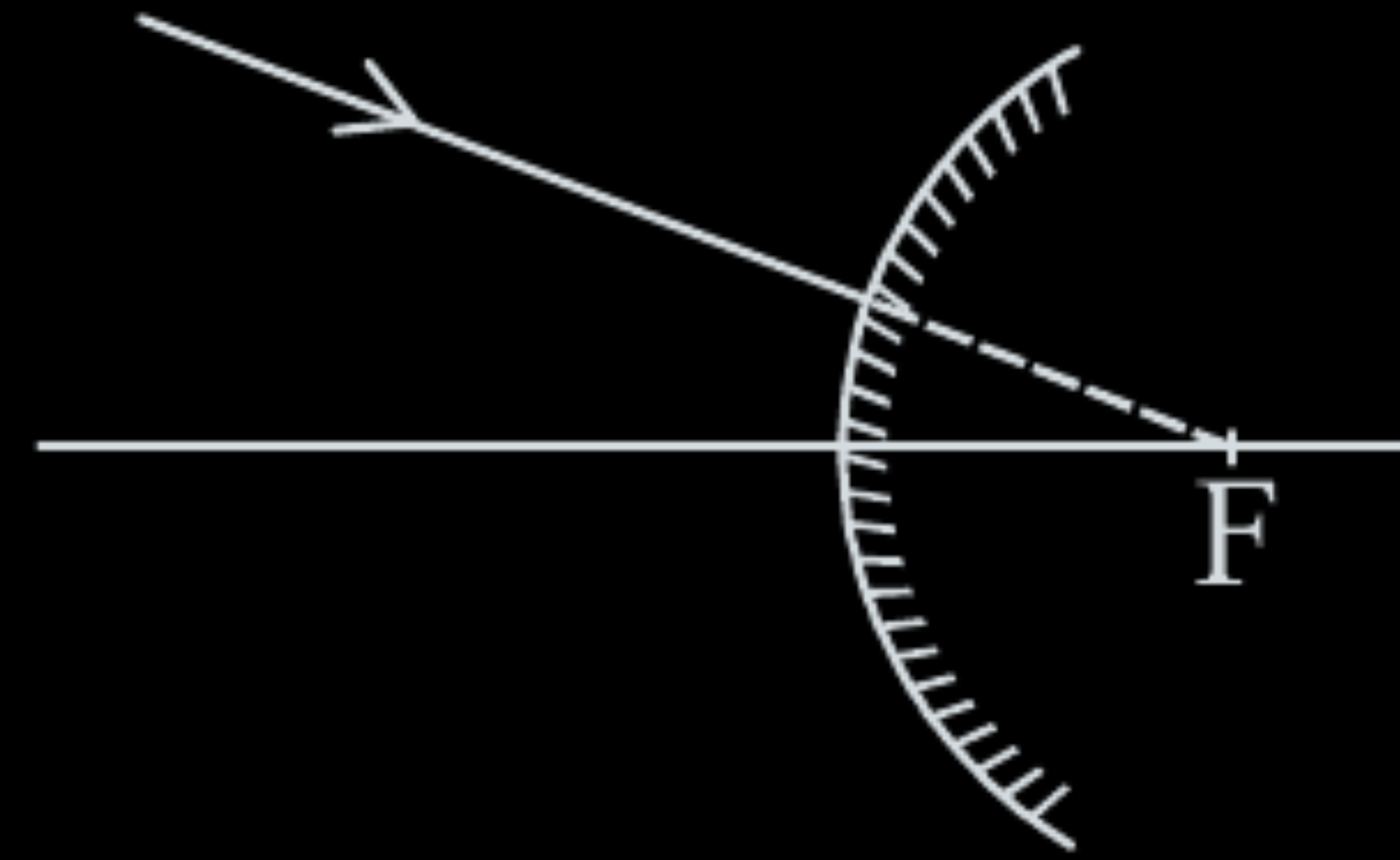
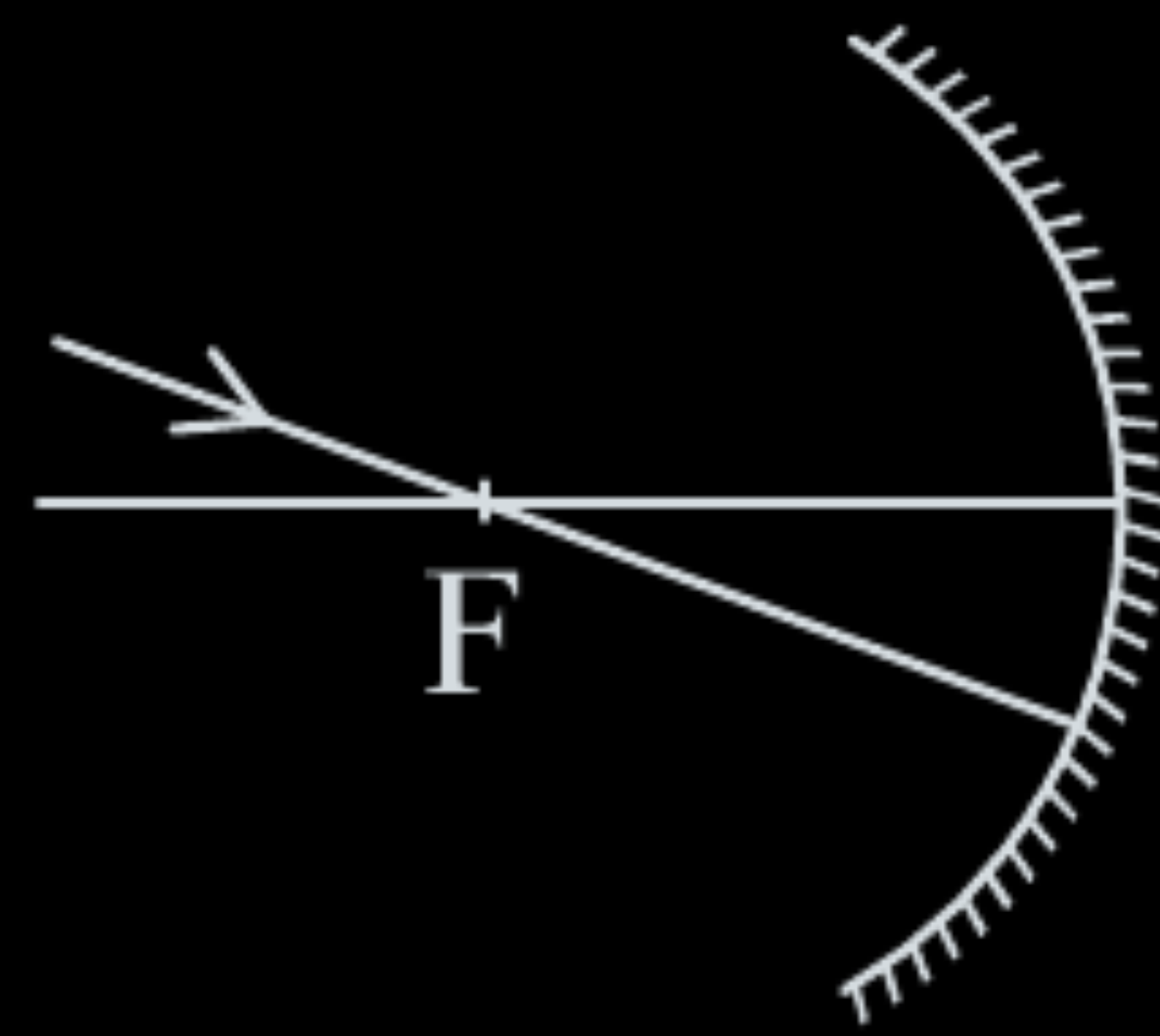


Tracing the reflected ray	1
Marking $\angle i$ & $\angle r$	$\frac{1}{2}, \frac{1}{2}$

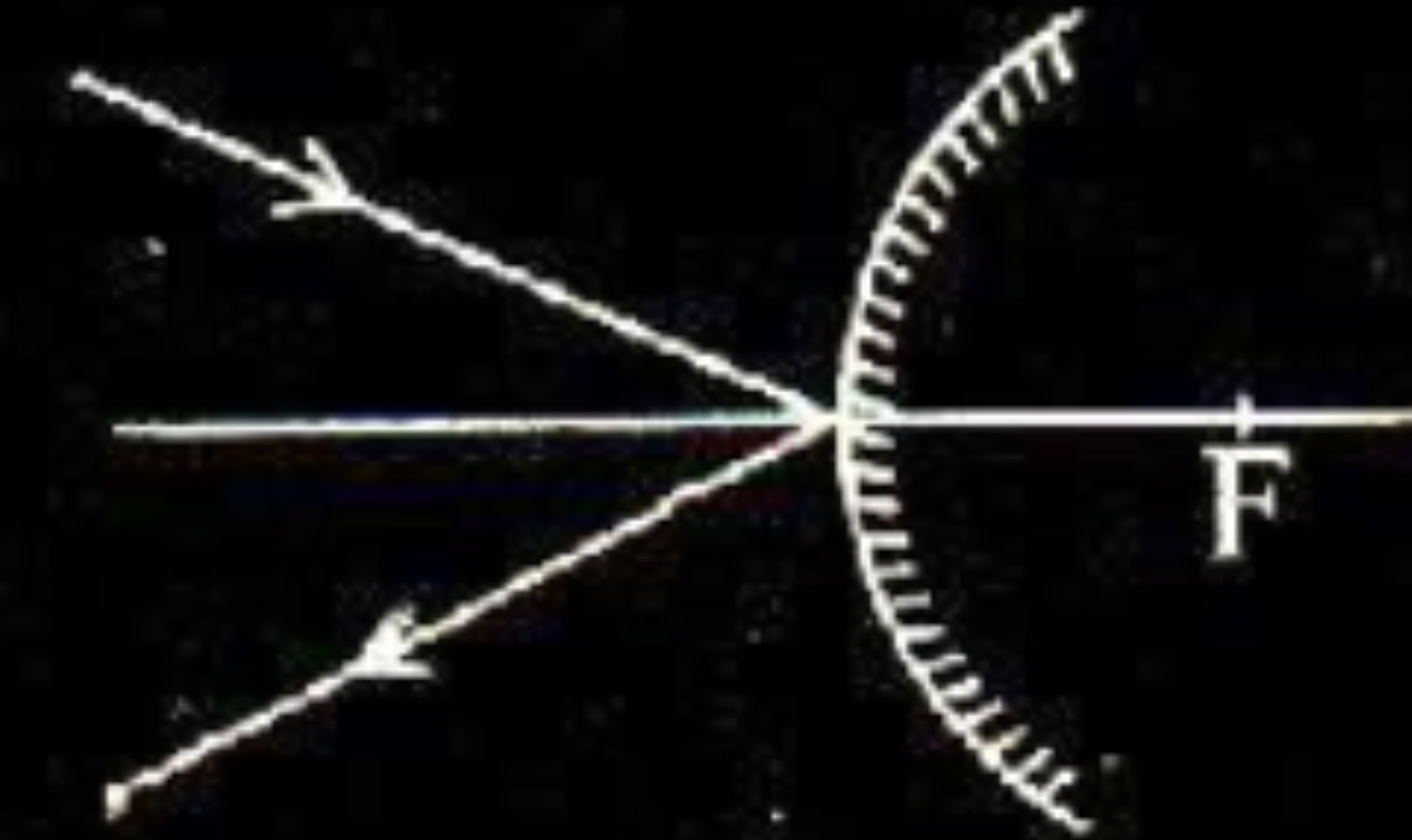
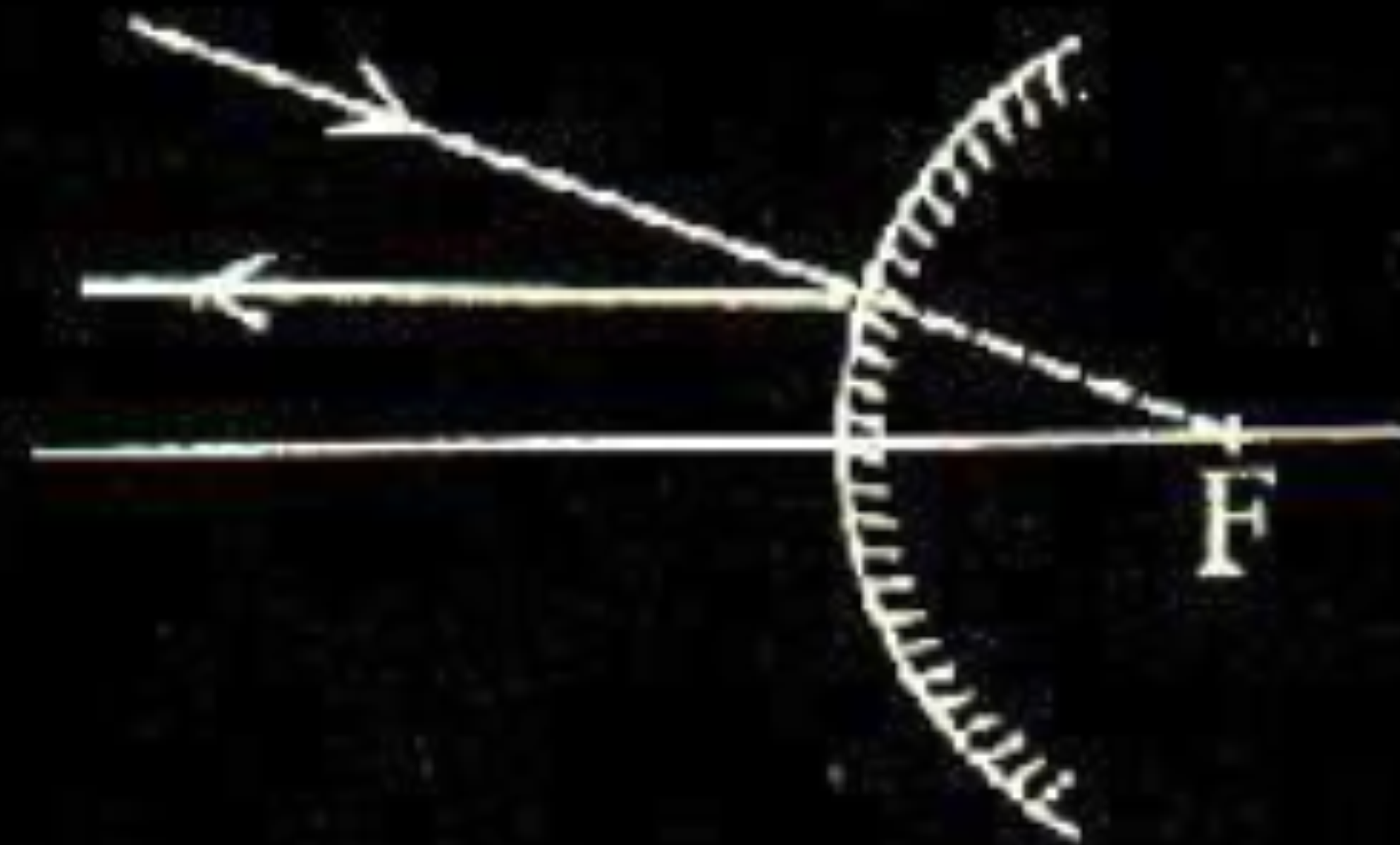
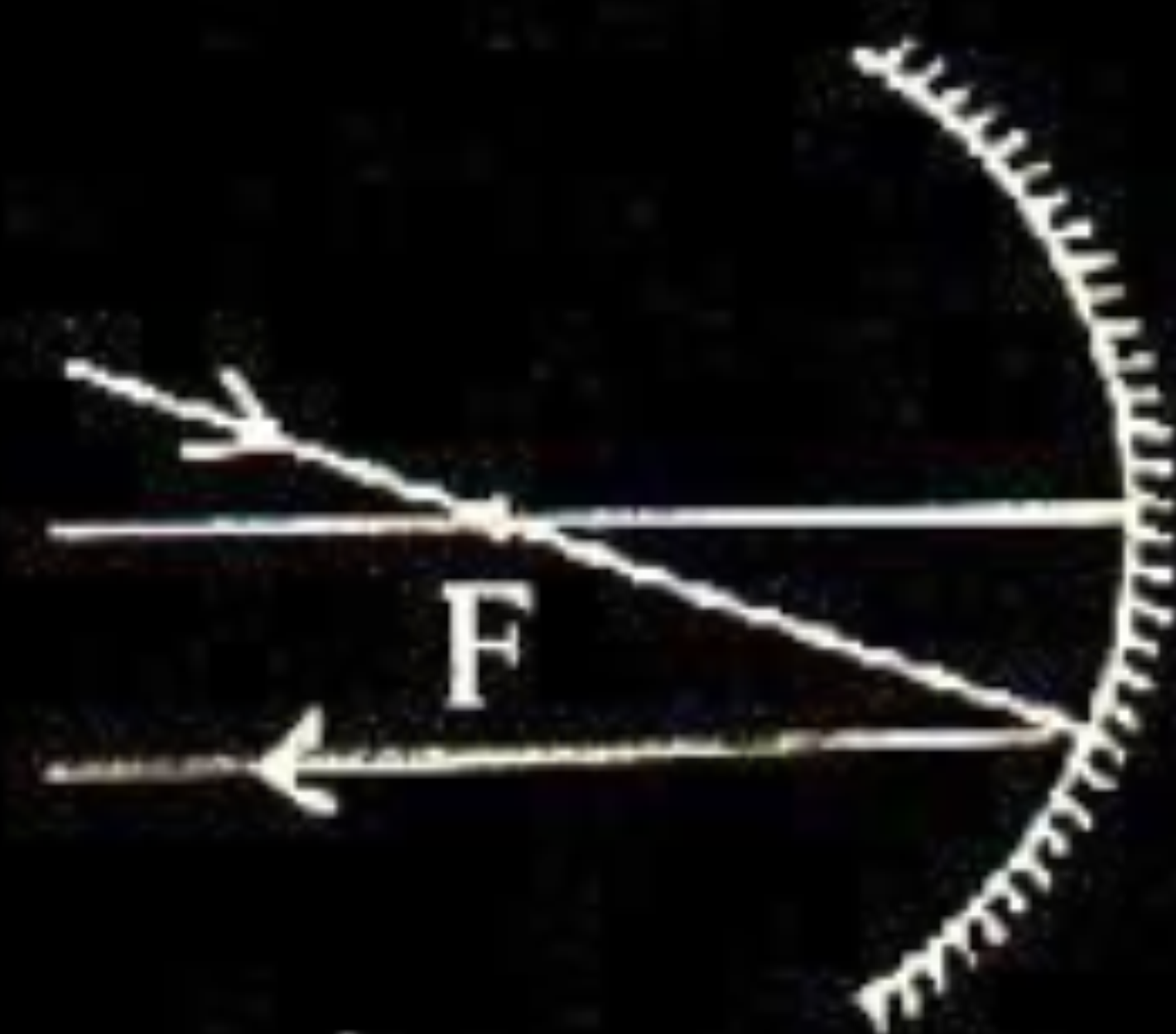
Draw the following diagram, in which a ray of light is incident on a concave/convex mirror, on your answer sheet. Show the path of this ray, after reflection, in each case.



Draw the following diagram, in which a ray of light is incident on a concave/convex mirror, on your answer sheet. Show the path of this ray, after reflection, in each case.



CBSE 2016



3x1

State two positions in which a concave mirror produces a magnified image of a given object. List two differences between the two images.

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- | | |
|--|-----|
| • When the object is placed in front of the mirror – | |
| (i) between its pole and focus | 1/2 |
| (ii) between the focus and centre of curvature | 1/2 |
| • In case (i) the image is virtual and erect | 1/2 |
| In case (ii) the image is real and inverted | 1/2 |